

Estimation and Inference in Boundary Discontinuity Designs: Distance-Based Methods Supplemental Appendix

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Abstract

This supplemental appendix presents more general theoretical results encompassing those reported in the paper, their theoretical proofs, and other technical results. It also provides additional numerical results.

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Contents

SA-1	Introduction	1
SA-1.1	Notation and Definitions	1
SA-2	Setup	2
SA-2.1	Assumptions	3
SA-2.2	Parameters	4
SA-2.3	Estimators and Other Related Quantities	5
SA-2.4	Higher-Order Signed-Distance Derivatives	6
SA-2.5	Preliminary Lemmas	7
SA-3	Boundary Average Treatment Effect Curve	9
SA-3.1	Identification	9
SA-3.2	Approximation Bias	9
SA-3.3	Point Estimation and Approximate MSE Expansions	10
SA-3.4	Distributional Approximation and Inference	11
SA-4	Weighted Boundary Average Treatment Effect	14
SA-5	Largest Boundary Average Treatment Effect	16
SA-6	Imperfect Compliance	17
SA-6.1	Second-Order Linearization	18
SA-6.2	Fuzzy BATEC	20
SA-6.3	Fuzzy WBATE	22
SA-6.4	Fuzzy LBATE	24
SA-7	Gaussian Strong Approximation	25
SA-7.1	Definitions for Function Spaces	25
SA-7.2	Multiplicative-Separable Empirical Process	26
SA-7.3	Proof of Theorem SA-25	27
SA-8	Proofs	29
SA-8.1	Proof of Lemma SA-1	29
SA-8.2	Proof of Lemma SA-2	30
SA-8.3	Proof of Lemma SA-3	32
SA-8.4	Proof of Lemma SA-4	33
SA-8.5	Proof of Lemma SA-5	35
SA-8.6	Proof of Theorem SA-1	37
SA-8.7	Proof of Theorem SA-2	37
SA-8.8	Proof of Theorem SA-3	39
SA-8.9	Proof of Theorem SA-4	41
SA-8.10	Proof of Theorem SA-5	41
SA-8.11	Proof of Theorem SA-6	42
SA-8.12	Proof of Theorem SA-7	43

SA-8.13	Proof of Theorem SA-8	45
SA-8.14	Proof of Theorem SA-9	49
SA-8.15	Proof of Theorem SA-10	50
SA-8.16	Proof of Lemma SA-6	52
SA-8.17	Proof of Lemma SA-7	52
SA-8.18	Proof of Theorem SA-11	54
SA-8.19	Proof of Theorem SA-12	54
SA-8.20	Proof of Theorem SA-13	55
SA-8.21	Proof of Theorem SA-14	55
SA-8.22	Proof of Theorem SA-15	55
SA-8.23	Proof of Theorem SA-16	55
SA-8.24	Proof of Lemma SA-8	56
SA-8.25	Proof of Theorem SA-17	56
SA-8.26	Proof of Theorem SA-18	57
SA-8.27	Proof of Theorem SA-19	57
SA-8.28	Proof of Theorem SA-20	57
SA-8.29	Proof of Lemma SA-9	58
SA-8.30	Proof of Lemma SA-10	58
SA-8.31	Proof of Theorem SA-21	59
SA-8.32	Proof of Theorem SA-22	59
SA-8.33	Proof of Theorem SA-23	60
SA-8.34	Proof of Theorem SA-24	60
SA-8.35	Proof of Theorem 6	60
SA-9	Simulation Study	63
SA-9.1	Data-Generating Process	63
SA-9.2	Calibration from the SPP Data	65
SA-9.3	Monte Carlo Implementation	67
SA-9.4	Results	68

SA-1 Introduction

This supplemental appendix considers four generalizations relative to the setup in the paper.

1. The score variable $\mathbf{X}_i$ is d -dimensional, with the genuinely multivariate boundary-discontinuity case corresponding to $d \geq 2$.
2. *Weighted Boundary Average Treatment Effect* (WBATE).
3. *Largest Boundary Average Treatment Effect* (LBATE).
4. Imperfect compliance (fuzzy) designs.

In addition, this supplement presents technical results and proofs not covered in the main text due to space limitations.

The assumptions in the paper are specialized primitive conditions; after imposing these restrictions on the supplemental results, the mapping between this supplemental appendix and the paper is as follows.

- Theorem SA-1 $\implies$ Theorem 1.
- Theorem SA-2, in its sharp level case with $d = 2$, $\implies$ Theorem 2.
- Theorem SA-3, with $d = 2$, $\implies$ Theorem 3.
- Theorem SA-4 with $d = 2$ $\implies$ Theorem 4.
- Theorem SA-5 with $d = 2$ $\implies$ Theorem 5(i).
- Theorem SA-8 with $d = 2$ $\implies$ Theorem 5(ii).
- Section SA-8.35 $\implies$ Theorem 6.

The generalizations considered in this supplemental appendix are conceptually straightforward, but technically more involved. Section SA-3 presents the main distance-based identification, estimation, and inference results. Sections SA-4 and SA-5 present the corresponding WBATE and LBATE results for sharp designs. Section SA-6 extends all the results to fuzzy designs. Section SA-7 presents a new Gaussian strong approximation result that may be of independent interest.

SA-1.1 Notation and Definitions

For textbook references on empirical process, see [van der Vaart and Wellner \[1996\]](#), [Dudley \[2014\]](#), and [Giné and Nickl \[2016\]](#). For textbook references on geometric measure theory, see [Simon et al. \[1984\]](#), [Federer \[2014\]](#), and [Folland \[2002\]](#). Throughout the appendix, $\mathbf{x}$ denotes a generic point in $\mathcal{X}$ and, when $\mathbf{x} \in \mathcal{B}$, a generic boundary point; the main paper uses $\mathbf{b}$ and $\mathbf{b}_j$ for selected or discretized evaluation points on the boundary.

- (i) *Polynomial Notation.* For scalar $u \in \mathbb{R}$, $\mathbf{r}_p(u) = (1, u, \dots, u^p)^\top$, and $\mathbf{e}_0$ denotes the intercept selector satisfying $\mathbf{e}_0^\top \mathbf{r}_p(u) = 1$. More generally, $\mathbf{e}_j$ denotes the selector satisfying $\mathbf{e}_j^\top \mathbf{r}_p(u) = u^j$ for $j = 0, \dots, p$.

- (ii) *Norms.* For a vector $\mathbf{a} \in \mathbb{R}^k$, $\|\mathbf{a}\| = (\sum_{i=1}^k \mathbf{a}_i^2)^{1/2}$ and $\|\mathbf{a}\|_\infty = \max_{1 \leq i \leq k} |\mathbf{a}_i|$. For a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\|\mathbf{A}\|_p = \sup_{\|\mathbf{x}\|_p=1} \|\mathbf{A}\mathbf{x}\|_p$, $p \in \mathbb{N} \cup \{\infty\}$, and $\lambda_{\min}(\mathbf{A})$ denotes its minimum eigenvalue when defined. For a function f on a metric space (S, d) , $\|f\|_\infty = \sup_{s \in S} |f(s)|$, and $\text{supp}(f)$ denotes the closure of $\{s \in S : f(s) \neq 0\}$. For a probability measure Q on $(\mathcal{S}, \mathcal{S})$ and $p \geq 1$, define $\|f\|_{Q,p} = (\int_{\mathcal{S}} |f|^p dQ)^{1/p}$, and $Q(f) = \int f dQ$. For a set $E \subseteq \mathbb{R}^d$, $\mathbf{m}(E)$ denotes the Lebesgue measure of E .
- (iii) *Empirical Process.* We use standard empirical-process notation: $\mathbb{E}_n[g(\mathbf{v}_i)] = n^{-1} \sum_{i=1}^n g(\mathbf{v}_i)$ and $\mathbb{G}_n[g(\mathbf{v}_i)] = n^{-1/2} \sum_{i=1}^n (g(\mathbf{v}_i) - \mathbb{E}[g(\mathbf{v}_i)])$. Let $(\mathcal{S}, d)$ be a semi-metric space. The covering number $N(\mathcal{S}, d, \varepsilon)$ is the minimal number of open d -balls of radius ε needed to cover $\mathcal{S}$. A $\mathbb{P}$ -Brownian bridge is a mean-zero Gaussian random function $W_{\mathbb{P}}(f)$, $f \in L_2(\mathcal{X}, \mathbb{P})$, with covariance $\mathbb{E}[W_{\mathbb{P}}(f)W_{\mathbb{P}}(g)] = \mathbb{P}(fg) - \mathbb{P}(f)\mathbb{P}(g)$. A class $\mathcal{F} \subseteq L_2(\mathcal{X}, \mathbb{P})$ is $\mathbb{P}$ -pregaussian if there is a version of $W_{\mathbb{P}}$ such that $W_{\mathbb{P}} \in C(\mathcal{F}; \rho_{\mathbb{P}})$ almost surely, where $\rho_{\mathbb{P}}(f, g) = (\|f - g\|_{\mathbb{P},2}^2 - (\mathbb{P}f - \mathbb{P}g)^2)^{1/2}$.
- (iv) *Geometric Measure Theory.* For a set $E \subseteq \mathcal{X}$, the *De Giorgi perimeter of E relative to $\mathcal{X}$* is $\mathcal{L}(E) = \text{TV}_{\{\mathbf{1}_E\}, \mathcal{X}}$. For $d \in \mathbb{N}$ and $0 \leq m \leq d$, $\mathfrak{H}^m$ denotes m -dimensional Hausdorff measure, and integration against $\mathfrak{H}^m$ is understood in the standard Caratheodory construction. The Hausdorff dimension of A is denoted $\dim_{\mathfrak{H}}(A)$. A set $A \subseteq \mathbb{R}^d$ is k -rectifiable if it has Hausdorff dimension k and is covered up to a $\mathfrak{H}^k$ -null set by countably many continuously differentiable maps from $\mathbb{R}^k$ to $\mathbb{R}^d$. A set $C \subseteq \mathbb{R}^d$ is a *rectifiable curve* if there exists a Lipschitz continuous function $\gamma : [0, 1] \rightarrow \mathbb{R}^d$ such that $C = \gamma([0, 1])$.
- (v) *Bounds and Asymptotics.* For real sequences, $a_n = o(b_n)$ means $a_n/b_n \rightarrow 0$, and $a_n \lesssim b_n$ means $|a_n| \leq C|b_n|$ eventually for a finite constant C . For random sequences, $a_n = o_{\mathbb{P}}(b_n)$ means $a_n/b_n \rightarrow_{\mathbb{P}} 0$, and $|a_n| \lesssim_{\mathbb{P}} |b_n|$ means $\limsup_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}[|a_n/b_n| \geq M] = 0$.
- (vi) *Distributions and Statistical Distances.* For $\boldsymbol{\mu} \in \mathbb{R}^k$ and a $k \times k$ positive definite matrix $\boldsymbol{\Sigma}$, $\text{Normal}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes the Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. $\text{Uniform}([a, b])$ denotes the uniform distribution on $[a, b]$, $\text{Bernoulli}(p)$ denotes the Bernoulli distribution with success probability p , and $\Phi(\cdot)$ denotes the standard Gaussian cumulative distribution function. For two distributions P and Q , $d_{\text{KL}}(P, Q)$ denotes the KL-distance and $d_{\chi^2}(P, Q)$ denotes the χ^2 distance.

Throughout the appendix, d and the local-polynomial order p are fixed, with $p \geq 0$. All limits are taken with $h \rightarrow 0$ as $n \rightarrow \infty$. The genuinely multivariate boundary-discontinuity results are stated for $d \geq 2$; the BATEC results (pointwise and uniform over finite boundary sets), the LBATE results, and their fuzzy analogues recover the usual one-dimensional RD case when $d = 1$ under the analogous assumptions, interpreting $\mathcal{B}$ as a finite 0-rectifiable boundary and integrating with respect to $\mathfrak{H}^0$. Most of our results hold with fixed but small enough bandwidths; we do not make this distinction explicit to avoid overly complex statements and technical discussions.

SA-2 Setup

We partition $\mathcal{X}$ into two assignment areas, $\mathcal{A}_t \subset \mathbb{R}^d$ with $t \in \{0, 1\}$, representing the control and treatment regions, respectively. That is, $\mathcal{X} = \mathcal{A}_0 \cup \mathcal{A}_1$, where $\mathcal{A}_0$ and $\mathcal{A}_1$ are disjoint regions in $\mathbb{R}^d$. The observed outcome is $Y_i = \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_0)Y_i(0) + \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_1)Y_i(1)$. The boundary is $\mathcal{B} = \text{bd}(\mathcal{A}_0) \cap \text{bd}(\mathcal{A}_1) \subseteq \mathbb{R}^d$, where $\text{bd}(\cdot)$ denotes the topological boundary of a set, and has effective dimension $d - 1$. Without loss of generality, we assume that $\mathcal{B}$ belongs to the treatment region, that is, $\mathcal{B} \subseteq \mathcal{A}_1$ and $\mathcal{B} \cap \mathcal{A}_0 = \emptyset$.

The univariate distance score induced by the multidimensional score is

$$D_i(\mathbf{x}) = [\mathbf{1}(\mathbf{X}_i \in \mathcal{A}_1) - \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_0)] \mathcal{d}(\mathbf{X}_i, \mathbf{x}), \quad \mathbf{x} \in \mathcal{B},$$

where $\mathcal{d}(\cdot, \cdot)$ denotes a distance function. For a deterministic point $\mathbf{z} \in \mathcal{X}$, write $D_{\mathbf{z}}(\mathbf{x}) = [\mathbf{1}(\mathbf{z} \in \mathcal{A}_1) - \mathbf{1}(\mathbf{z} \in \mathcal{A}_0)] \mathcal{d}(\mathbf{z}, \mathbf{x})$. Let $\mathcal{J}_0 = (-\infty, 0)$, $\mathcal{J}_1 = [0, \infty)$, and $\varsigma_t = 2t - 1$ for $t \in \{0, 1\}$. Thus, for $\mathbf{z} \in \mathcal{A}_t$, $D_{\mathbf{z}}(\mathbf{x}) = \varsigma_t \mathcal{d}(\mathbf{z}, \mathbf{x})$. Let $\mathbf{r}_p(u) = (1, u, \dots, u^p)^\top$, and $K_h(u) = K(u/h)/h^d$, where $K(\cdot)$ is a univariate kernel and h is a bandwidth parameter. No symmetry of K is imposed, so the side-specific sign remains in both the polynomial argument and the kernel weight.

For later reference and for Assumption SA-2, let

$$\Psi_{t,\mathbf{x}} = \mathbb{E} \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right)^\top K_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right], \quad t \in \{0, 1\}.$$

SA-2.1 Assumptions

The data generating process assumption from the main paper generalizes as follows.

Assumption SA-1 (Data Generating Process). *Let $t \in \{0, 1\}$.*

- (i) $(Y_1(t), \mathbf{X}_1^\top)^\top, \dots, (Y_n(t), \mathbf{X}_n^\top)^\top$ are independent and identically distributed random vectors.
- (ii) The distribution of $\mathbf{X}_i$ has compact support $\mathcal{X} \subseteq \mathbb{R}^d$ with nonempty interior and a Lebesgue density $f_X(\mathbf{x})$ that is continuous and bounded away from zero on $\mathcal{X}$.
- (iii) $\mu_t(\mathbf{x}) = \mathbb{E}[Y_i(t) | \mathbf{X}_i = \mathbf{x}]$ has a $(p+1)$ -times continuously differentiable extension to an open neighborhood of $\mathcal{X}$.
- (iv) $\sigma_t^2(\mathbf{x}) = \mathbb{V}[Y_i(t) | \mathbf{X}_i = \mathbf{x}]$ is bounded away from zero and continuous on $\mathcal{X}$.
- (v) $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[|Y_i(t)|^{2+v} | \mathbf{X}_i = \mathbf{x}] < \infty$ for some $v \geq 2$.

The boundary, distance, and kernel assumption from the main paper generalizes as follows.

Assumption SA-2 (Boundary, Distance, and Kernel). *Let $t \in \{0, 1\}$.*

- (i) $\mathcal{B} \subset \text{int}(\mathcal{X})$, and $\mathcal{B}$ is compact $(d-1)$ -rectifiable, with $\mathfrak{H}^{d-1}(\mathcal{B})$ positive and finite.
- (ii) $\mathcal{d} : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}_+$ is a metric on $\mathbb{R}^d$ equivalent to the Euclidean distance, that is, there exist positive constants C_u and C_l such that $C_l \|\mathbf{x} - \mathbf{x}'\| \leq \mathcal{d}(\mathbf{x}, \mathbf{x}') \leq C_u \|\mathbf{x} - \mathbf{x}'\|$ for all $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$.
- (iii) $K : \mathbb{R} \rightarrow [0, \infty)$ is either compactly supported and Lipschitz continuous, or $K(u) = \mathbf{1}(u \in [-1, 1])$ and the distance balls $\{\mathbf{v} \in \mathcal{X} : \mathcal{d}(\mathbf{v}, \mathbf{x}) \leq r\}$, indexed by $(\mathbf{x}, r) \in \mathcal{X} \times \mathbb{R}_+$, form a VC class.
- (iv) $\liminf_{h \downarrow 0} \inf_{\mathbf{x} \in \mathcal{B}} \lambda_{\min}(\Psi_{t,\mathbf{x}}) \gtrsim 1$.
- (v) For each $\mathbf{x} \in \mathcal{B}$ and $t \in \{0, 1\}$, the denominator in the Hausdorff-measure representation of $\theta_{t,\mathbf{x}}(r)$ below is positive and finite for all sufficiently small $|r| > 0$ with $r \in \mathcal{J}_t$.

For $d = 2$, Assumption SA-2(i) holds whenever $\mathcal{B}$ is a compact rectifiable curve contained in the interior of $\mathcal{X}$, with positive and finite length. Assumption SA-2(ii) is specific to distance-based methods and requires the distance function to be equivalent, up to constants, to the Euclidean distance. In Assumption SA-2(iii), the extra VC condition appears only for the uniform kernel because its indexing sets are distance balls rather than rectangles: Lemma SA-1 gives primitive cases where it is automatic, including the Euclidean distance.

Assumption SA-2(iv) rules out boundary–distance–kernel configurations in which, after local rescaling by h , one side of the boundary receives too little kernel-weighted mass to guarantee that the distance-based local polynomial regression is well-defined uniformly over $\mathcal{B}$. Thus, the condition ensures that the local distance

design is sufficiently rich on each side of the boundary. Lemma SA-2 gives primitive sufficient conditions for the Euclidean-distance case, and Lemma SA-3 then shows that the sample Gram matrix is full rank in large samples. Assumption SA-2(v) is specific to distance-based identification: it keeps the Hausdorff-measure ratio defining $\theta_{t,\mathbf{x}}(r)$ well-defined near the boundary and is used for identification (Theorem SA-1); under additional smoothness, the same ratio is differentiated in Theorem SA-3. These conditions are stated uniformly over $\mathcal{B}$ in anticipation of our uniform estimation and inference results, but several can be relaxed to hold only pointwise in $\mathbf{x} \in \mathcal{B}$ for pointwise results.

Assumption 3 imposes regularity conditions on a weight function along the boundary.

Assumption SA-3 (Weight Function and Boundary). *Let $w : \mathcal{B} \rightarrow (0, \infty)$ satisfy $\sup_{\mathbf{x} \in \mathcal{B}} w(\mathbf{x}) < \infty$, $\inf_{\mathbf{x} \in \mathcal{B}} w(\mathbf{x}) > 0$, and $\int_{\mathcal{B}} w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}) < \infty$.*

SA-2.2 Parameters

The *Boundary Average Treatment Effect Curve* (BATEC) is

$$\tau(\mathbf{x}) = \mathbb{E}[Y_i(1) - Y_i(0) | \mathbf{X}_i = \mathbf{x}], \quad \mathbf{x} \in \mathcal{B}.$$

For each $t \in \{0, 1\}$, the induced conditional expectation based on univariate distance is

$$\theta_{t,\mathbf{x}}(r) = \mathbb{E}[Y_i | D_i(\mathbf{x}) = r] = \mathbb{E}[Y_i | \mathcal{d}(\mathbf{X}_i, \mathbf{x}) = |r|, \mathbf{X}_i \in \mathcal{A}_t], \quad r \in \mathcal{J}_t, \quad \mathbf{x} \in \mathcal{B}.$$

More rigorously, for each $t \in \{0, 1\}$, and letting $S_{t,\mathbf{x}}(r) = \{\mathbf{v} \in \mathcal{X} : \mathcal{d}(\mathbf{v}, \mathbf{x}) = r, \mathbf{v} \in \mathcal{A}_t\}$ for $r \geq 0$ and $\mathbf{x} \in \mathcal{B}$,

$$\theta_{t,\mathbf{x}}(r) = \frac{\int_{S_{t,\mathbf{x}}(|r|)} \mu_t(\mathbf{v}) f_X(\mathbf{v}) d\mathfrak{H}^{d-1}(\mathbf{v})}{\int_{S_{t,\mathbf{x}}(|r|)} f_X(\mathbf{v}) d\mathfrak{H}^{d-1}(\mathbf{v})},$$

for $|r| > 0, \mathbf{x} \in \mathcal{B}, t \in \{0, 1\}$, and therefore (under our assumptions)

$$\theta_{t,\mathbf{x}}(0) = \lim_{r \rightarrow 0, r \in \mathcal{J}_t} \mathbb{E}[Y_i | \mathcal{d}(\mathbf{X}_i, \mathbf{x}) = |r|, \mathbf{X}_i \in \mathcal{A}_t] = \lim_{r \rightarrow 0, r \in \mathcal{J}_t} \frac{\int_{S_{t,\mathbf{x}}(|r|)} \mu_t(\mathbf{v}) f_X(\mathbf{v}) d\mathfrak{H}^{d-1}(\mathbf{v})}{\int_{S_{t,\mathbf{x}}(|r|)} f_X(\mathbf{v}) d\mathfrak{H}^{d-1}(\mathbf{v})}.$$

The induced signed-distance treatment-effect parameter is

$$\vartheta(\mathbf{x}) = \theta_{1,\mathbf{x}}(0) - \theta_{0,\mathbf{x}}(0), \quad \mathbf{x} \in \mathcal{B}.$$

Theorem SA-1 shows that $\vartheta(\mathbf{x}) = \tau(\mathbf{x})$ under Assumptions SA-1(i)-(iii) and SA-2(i), (ii), and (v).

We also consider the parameters

$$\vartheta_{\text{WBATE}} = \frac{\int_{\mathcal{B}} \vartheta(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x})}{\int_{\mathcal{B}} w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x})} \quad \text{and} \quad \vartheta_{\text{LBATE}} = \sup_{\mathbf{x} \in \mathcal{B}} \vartheta(\mathbf{x}).$$

Theorem SA-1 implies that

$$\vartheta_{\text{WBATE}} = \tau_{\text{WBATE}} = \frac{\int_{\mathcal{B}} \tau(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x})}{\int_{\mathcal{B}} w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x})}$$

is the *Weighted Boundary Average Treatment Effect* (WBATE), and

$$\vartheta_{\text{LBATE}} = \tau_{\text{LBATE}} = \sup_{\mathbf{x} \in \mathcal{B}} \tau(\mathbf{x})$$

is the *Largest Boundary Average Treatment Effect* (LBATE).

SA-2.3 Estimators and Other Related Quantities

The distance-based estimator process along the boundary is

$$\hat{\vartheta}(\mathbf{x}) = \hat{\theta}_{1,\mathbf{x}}(0) - \hat{\theta}_{0,\mathbf{x}}(0), \quad \mathbf{x} \in \mathcal{B},$$

where, for $t \in \{0, 1\}$,

$$\hat{\theta}_{t,\mathbf{x}}(0) = \mathbf{e}_0^\top \hat{\gamma}_t(\mathbf{x}), \quad \hat{\gamma}_t(\mathbf{x}) = \arg \min_{\gamma \in \mathbb{R}^{p+1}} \mathbb{E}_n \left[(Y_i - \mathbf{r}_p(D_i(\mathbf{x})/h)^\top \gamma)^2 K_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right].$$

More generally, the least squares projection is

$$\hat{\theta}_{t,\mathbf{x}}(r) = \mathbf{r}_p(r/h)^\top \hat{\gamma}_t(\mathbf{x}), \quad t \in \{0, 1\}, \quad \mathbf{x} \in \mathcal{B}.$$

The best mean square approximation is

$$\theta_{t,\mathbf{x}}^*(r) = \mathbf{r}_p(r/h)^\top \gamma_t^*(\mathbf{x}),$$

where

$$\gamma_t^*(\mathbf{x}) = \arg \min_{\gamma \in \mathbb{R}^{p+1}} \mathbb{E} \left[(Y_i - \mathbf{r}_p(D_i(\mathbf{x})/h)^\top \gamma)^2 K_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right],$$

and uniqueness will follow from the results below. Let

$$\theta_{t,\mathbf{x}}^*(0) = \mathbf{e}_0^\top \gamma_t^*(\mathbf{x}).$$

The estimation error decomposes into *linear error*, *approximation error*, and *non-linear error*: for all $t \in \{0, 1\}$ and $\mathbf{x} \in \mathcal{B}$,

$$\begin{aligned} \hat{\theta}_{t,\mathbf{x}}(0) - \theta_{t,\mathbf{x}}(0) &= \mathbf{e}_0^\top \hat{\Psi}_{t,\mathbf{x}}^{-1} \mathbb{E}_n \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) K_h(D_i(\mathbf{x})) Y_i \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right] - \theta_{t,\mathbf{x}}(0) \\ &= \mathbf{e}_0^\top \hat{\Psi}_{t,\mathbf{x}}^{-1} \mathbb{E}_n \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) K_h(D_i(\mathbf{x})) (Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right] \\ &\quad + \theta_{t,\mathbf{x}}^*(0) - \theta_{t,\mathbf{x}}(0) \\ &= \underbrace{\theta_{t,\mathbf{x}}^*(0) - \theta_{t,\mathbf{x}}(0)}_{\text{approximation error}} + \underbrace{\mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}}}_{\text{linear error}} + \underbrace{\mathbf{e}_0^\top (\hat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}) \mathbf{O}_{t,\mathbf{x}}}_{\text{non-linear error}}, \end{aligned}$$

where

$$\mathbf{O}_{t,\mathbf{x}} = \mathbb{E}_n \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) K_h(D_i(\mathbf{x})) (Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right],$$

with $\Psi_{t,\mathbf{x}}$ defined above and

$$\widehat{\Psi}_{t,\mathbf{x}} = \mathbb{E}_n \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right)^\top K_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right].$$

Therefore,

$$\widehat{\vartheta}(\mathbf{x}) - \vartheta(\mathbf{x}) = \mathfrak{B}(\mathbf{x}) + \mathfrak{L}(\mathbf{x}) + \mathfrak{Q}(\mathbf{x}). \quad (\text{SA-1})$$

The misspecification bias is

$$\mathfrak{B}(\mathbf{x}) = \mathfrak{B}_1(\mathbf{x}) - \mathfrak{B}_0(\mathbf{x}), \quad \mathfrak{B}_t(\mathbf{x}) = \theta_{t,\mathbf{x}}^*(0) - \theta_{t,\mathbf{x}}(0),$$

the linear term is

$$\mathfrak{L}(\mathbf{x}) = \mathbf{e}_0^\top \Psi_{1,\mathbf{x}}^{-1} \mathbf{O}_{1,\mathbf{x}} - \mathbf{e}_0^\top \Psi_{0,\mathbf{x}}^{-1} \mathbf{O}_{0,\mathbf{x}},$$

and the quadratic (non-linear) term is

$$\mathfrak{Q}(\mathbf{x}) = \mathbf{e}_0^\top (\widehat{\Psi}_{1,\mathbf{x}}^{-1} - \Psi_{1,\mathbf{x}}^{-1}) \mathbf{O}_{1,\mathbf{x}} - \mathbf{e}_0^\top (\widehat{\Psi}_{0,\mathbf{x}}^{-1} - \Psi_{0,\mathbf{x}}^{-1}) \mathbf{O}_{0,\mathbf{x}}.$$

The population projection defining $\theta_{t,\mathbf{x}}^*$ implies $\mathbb{E}[\mathfrak{L}(\mathbf{x})] = 0$ and $\mathbb{E}[\mathfrak{L}(\mathbf{x})^2] = \Xi_{\mathbf{x}} = \Xi_{\mathbf{x},\mathbf{x}}$, where

$$\Xi_{\mathbf{x}_1,\mathbf{x}_2} = \Xi_{0,\mathbf{x}_1,\mathbf{x}_2} + \Xi_{1,\mathbf{x}_1,\mathbf{x}_2}, \quad \Xi_{t,\mathbf{x}_1,\mathbf{x}_2} = \frac{1}{nh^d} \mathbf{e}_0^\top \Psi_{t,\mathbf{x}_1}^{-1} \Upsilon_{t,\mathbf{x}_1,\mathbf{x}_2} \Psi_{t,\mathbf{x}_2}^{-1} \mathbf{e}_0,$$

and

$$\begin{aligned} \Upsilon_{t,\mathbf{x}_1,\mathbf{x}_2} = h^d \mathbb{E} \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x}_1)}{h} \right) \mathbf{r}_p \left(\frac{D_i(\mathbf{x}_2)}{h} \right)^\top K_h(D_i(\mathbf{x}_1)) K_h(D_i(\mathbf{x}_2)) \right. \\ \left. (Y_i - \theta_{t,\mathbf{x}_1}^*(D_i(\mathbf{x}_1))) (Y_i - \theta_{t,\mathbf{x}_2}^*(D_i(\mathbf{x}_2))) \mathbf{1}(D_i(\mathbf{x}_1) \in \mathcal{J}_t) \mathbf{1}(D_i(\mathbf{x}_2) \in \mathcal{J}_t) \right]. \end{aligned}$$

Finally, for $t \in \{0, 1\}$, $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$, the sample analogues are $\widehat{\Xi}_{\mathbf{x}} = \widehat{\Xi}_{\mathbf{x},\mathbf{x}}$, where

$$\widehat{\Xi}_{\mathbf{x}_1,\mathbf{x}_2} = \widehat{\Xi}_{0,\mathbf{x}_1,\mathbf{x}_2} + \widehat{\Xi}_{1,\mathbf{x}_1,\mathbf{x}_2}, \quad \widehat{\Xi}_{t,\mathbf{x}_1,\mathbf{x}_2} = \frac{1}{nh^d} \mathbf{e}_0^\top \widehat{\Psi}_{t,\mathbf{x}_1}^{-1} \widehat{\Upsilon}_{t,\mathbf{x}_1,\mathbf{x}_2} \widehat{\Psi}_{t,\mathbf{x}_2}^{-1} \mathbf{e}_0,$$

and

$$\begin{aligned} \widehat{\Upsilon}_{t,\mathbf{x}_1,\mathbf{x}_2} = h^d \mathbb{E}_n \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x}_1)}{h} \right) \mathbf{r}_p \left(\frac{D_i(\mathbf{x}_2)}{h} \right)^\top K_h(D_i(\mathbf{x}_1)) K_h(D_i(\mathbf{x}_2)) \right. \\ \left. \cdot (Y_i - \widehat{\theta}_{t,\mathbf{x}_1}(D_i(\mathbf{x}_1))) (Y_i - \widehat{\theta}_{t,\mathbf{x}_2}(D_i(\mathbf{x}_2))) \mathbf{1}(D_i(\mathbf{x}_1) \in \mathcal{J}_t) \mathbf{1}(D_i(\mathbf{x}_2) \in \mathcal{J}_t) \right]. \end{aligned}$$

SA-2.4 Higher-Order Signed-Distance Derivatives

The main results in this appendix focus on the level parameter $\vartheta(\mathbf{x})$ because it equals the BATEC $\tau(\mathbf{x})$ under Theorem SA-1. One can also estimate derivatives of the induced signed-distance regression. For $\mathbf{x} \in \mathcal{B}$ and an integer $\nu \in \{1, \dots, p\}$, whenever the one-sided limits exist, define

$$\theta_{t,\mathbf{x}}^{(\nu)}(0) = \lim_{r \rightarrow 0, r \in \mathcal{J}_t} \frac{d^\nu}{dr^\nu} \theta_{t,\mathbf{x}}(r), \quad \vartheta^{(\nu)}(\mathbf{x}) = \theta_{1,\mathbf{x}}^{(\nu)}(0) - \theta_{0,\mathbf{x}}^{(\nu)}(0), \quad \mathbf{x} \in \mathcal{B}.$$

The corresponding local-polynomial estimator replaces the level selector by the derivative selector:

$$\hat{\vartheta}^{(\nu)}(\mathbf{x}) = \nu! h^{-\nu} (\mathbf{e}_\nu^\top \hat{\gamma}_1(\mathbf{x}) - \mathbf{e}_\nu^\top \hat{\gamma}_0(\mathbf{x})), \quad \mathbf{x} \in \mathcal{B}.$$

All level results above have direct analogues after replacing $\mathbf{e}_0$ by $\nu! h^{-\nu} \mathbf{e}_\nu$, replacing the variance rate $(nh^d)^{-1}$ by $(nh^{d+2\nu})^{-1}$, and replacing the smooth-bias rate h^m by $h^{m-\nu}$ when the induced functions $r \mapsto \theta_{t,\mathbf{x}}(r)$ are m times differentiable. We do not state those analogues theorem by theorem because their notation is substantially heavier and the resulting parameter has a narrower causal interpretation: for $\nu > 0$, $\vartheta^{(\nu)}(\mathbf{x})$ is a derivative of an induced signed-distance regression, not a derivative of the BATEC curve as a function on $\mathcal{B}$.

There is, however, one useful first-derivative interpretation for $\mathbf{x} \in \mathcal{B}$ under a smooth Euclidean geometry. Suppose $\mathcal{A}$ is Euclidean, $\mathcal{B}$ is C^2 near $\mathbf{x}$, the treatment region has a well-defined unit normal $\mathbf{n}(\mathbf{x})$ pointing from control to treatment, meaning that $\mathbf{x} + r\mathbf{n}(\mathbf{x}) \in \mathcal{A}_1$ and $\mathbf{x} - r\mathbf{n}(\mathbf{x}) \in \mathcal{A}_0$ for all sufficiently small $r > 0$, and f_X , μ_0 , and μ_1 are continuously differentiable near $\mathbf{x}$ with $f_X(\mathbf{x}) > 0$, so that the treatment-effect function τ admits the ambient gradient $\nabla\tau(\mathbf{x})$. Then,

$$\vartheta^{(1)}(\mathbf{x}) = c_d \nabla\tau(\mathbf{x})^\top \mathbf{n}(\mathbf{x}), \quad c_d = \frac{\Gamma(d/2)}{\sqrt{\pi}\Gamma((d+1)/2)}, \quad \mathbf{x} \in \mathcal{B}.$$

In the bivariate case, $c_2 = 2/\pi$. Thus, the first signed-distance derivative is proportional to the normal derivative of the treatment-effect surface. This formula uses the Euclidean metric and the local half-sphere geometry it induces; with a different metric, the relevant average is over the corresponding metric sphere and need not reduce to a Euclidean normal derivative.

SA-2.5 Preliminary Lemmas

Recall that $t \in \{0, 1\}$.

The following lemma gives a sufficient condition for Assumption SA-2(iii) when using the uniform kernel.

Lemma SA-1 (Primitive VC Condition for Distance Balls). *Let $\mathcal{D}_d = \{\{\mathbf{v} \in \mathcal{X} : d(\mathbf{v}, \mathbf{x}) \leq r\} : (\mathbf{x}, r) \in \mathcal{X} \times \mathbb{R}_+\}$. Suppose that, for some integer $J < \infty$, there are finite-dimensional linear spaces $\mathcal{V}_1, \dots, \mathcal{V}_J$ of real-valued functions on $\mathcal{X}$ and a Boolean map $\mathbf{b} : \{0, 1\}^J \rightarrow \{0, 1\}$ such that, for every $(\mathbf{x}, r) \in \mathcal{X} \times \mathbb{R}_+$, there are $f_j(\cdot; \mathbf{x}, r) \in \mathcal{V}_j$, $j = 1, \dots, J$, satisfying*

$$\mathbf{1}\{d(\mathbf{v}, \mathbf{x}) \leq r\} = \mathbf{b}(\mathbf{1}\{f_1(\mathbf{v}; \mathbf{x}, r) \geq 0\}, \dots, \mathbf{1}\{f_J(\mathbf{v}; \mathbf{x}, r) \geq 0\}), \quad \mathbf{v} \in \mathcal{X}.$$

Then, $\mathcal{D}_d$ is a VC class.

In particular, the conclusion of Lemma SA-1 holds whenever the distance balls are semialgebraic sets of uniformly bounded degree and Boolean complexity. This includes the Euclidean distance, fixed Mahalanobis distances $d(\mathbf{v}, \mathbf{x}) = ((\mathbf{v} - \mathbf{x})^\top A(\mathbf{v} - \mathbf{x}))^{1/2}$ with A positive definite, and distances generated by fixed polyhedral norms.

The following lemma gives a sufficient condition for Assumption SA-2(iv).

Lemma SA-2 (Gram Invertibility). *Suppose the following conditions hold:*

- (a) *Assumptions SA-1(i)(ii) and Assumption SA-2(iii) hold.*
- (b) *$d(\cdot, \cdot)$ is the Euclidean distance.*

- (c) *There exists a compact set $U \subseteq \mathbb{R}^d$, such that $K(\varsigma_t \|\mathbf{u}\|) > 0$ for all $\mathbf{u} \in U$, $\lambda_{\min}(\int_U \mathbf{r}_p(\|\mathbf{z}\|) \mathbf{r}_p(\|\mathbf{z}\|)^\top d\mathbf{z}) > 0$, and $\liminf_{h \downarrow 0} \inf_{\mathbf{x} \in \mathcal{B}} \int_U K(\varsigma_t \|\mathbf{u}\|) \mathbf{1}(\mathbf{x} + h\mathbf{u} \in \mathcal{A}_t) d\mathbf{u} \gtrsim 1$.*

Then, Assumption SA-2(iv) holds.

Lemma SA-3 (Gram). *Suppose Assumptions SA-1(i)(ii) and SA-2 hold. If $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then*

$$\begin{aligned} \sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\Psi}_{t,\mathbf{x}} - \Psi_{t,\mathbf{x}}\| &\lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}}, \quad 1 \lesssim \mathbb{P} \inf_{\mathbf{x} \in \mathcal{B}} \lambda_{\min}(\hat{\Psi}_{t,\mathbf{x}}) \leq \sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\Psi}_{t,\mathbf{x}}\| \lesssim \mathbb{P} 1, \\ \sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}\| &\lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}}. \end{aligned}$$

Lemma SA-4 (Stochastic Linear Approximation). *Suppose Assumptions SA-1(i)(ii)(iii)(v) and SA-2 hold. If $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then*

$$\begin{aligned} \sup_{\mathbf{x} \in \mathcal{B}} \|\mathbf{O}_{t,\mathbf{x}}\| &\lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d}, \\ \sup_{\mathbf{x} \in \mathcal{B}} |\mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}}| &\lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d}, \\ \sup_{\mathbf{x} \in \mathcal{B}} |\mathbf{e}_0^\top (\hat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}) \mathbf{O}_{t,\mathbf{x}}| &\lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}} \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right). \end{aligned}$$

Lemma SA-5 (Covariance). *Suppose Assumptions SA-1 and SA-2 hold. If $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then*

$$1 \lesssim \inf_{\mathbf{x} \in \mathcal{B}} nh^d \Xi_{t,\mathbf{x},\mathbf{x}} \leq \sup_{\mathbf{x} \in \mathcal{B}} nh^d \Xi_{t,\mathbf{x},\mathbf{x}} \lesssim 1,$$

$$\sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} \|\hat{\Upsilon}_{t,\mathbf{x}_1,\mathbf{x}_2} - \Upsilon_{t,\mathbf{x}_1,\mathbf{x}_2}\| \lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d},$$

and

$$\sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} nh^d |\hat{\Xi}_{t,\mathbf{x}_1,\mathbf{x}_2} - \Xi_{t,\mathbf{x}_1,\mathbf{x}_2}| \lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}.$$

If, in addition, $\frac{n^{\frac{v}{2+v}} h^d}{\log(1/h)} \rightarrow \infty$, then

$$\inf_{\mathbf{x} \in \mathcal{B}} \lambda_{\min}(\hat{\Upsilon}_{t,\mathbf{x},\mathbf{x}}) \gtrsim \mathbb{P} 1, \quad \inf_{\mathbf{x} \in \mathcal{B}} nh^d \hat{\Xi}_{t,\mathbf{x},\mathbf{x}} \gtrsim \mathbb{P} 1, \quad \sup_{\mathbf{x} \in \mathcal{B}} nh^d \hat{\Xi}_{t,\mathbf{x},\mathbf{x}} \lesssim \mathbb{P} 1,$$

and

$$\sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} \left| \frac{\hat{\Xi}_{t,\mathbf{x}_1,\mathbf{x}_2}}{\sqrt{\hat{\Xi}_{t,\mathbf{x}_1,\mathbf{x}_1} \hat{\Xi}_{t,\mathbf{x}_2,\mathbf{x}_2}}} - \frac{\Xi_{t,\mathbf{x}_1,\mathbf{x}_2}}{\sqrt{\Xi_{t,\mathbf{x}_1,\mathbf{x}_1} \Xi_{t,\mathbf{x}_2,\mathbf{x}_2}}} \right| \lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}.$$

SA-3 Boundary Average Treatment Effect Curve

SA-3.1 Identification

Theorem SA-1 (Distance-Based Identification). *Suppose Assumptions SA-1(i)-(iii) and SA-2(i), (ii), and (v) hold. Then,*

$$\tau(\mathbf{x}) = \vartheta(\mathbf{x}) = \lim_{r \downarrow 0} \theta_{1,\mathbf{x}}(r) - \lim_{r \uparrow 0} \theta_{0,\mathbf{x}}(r), \quad \text{for all } \mathbf{x} \in \mathcal{B}.$$

SA-3.2 Approximation Bias

The next two theorems collect the approximation-bias results corresponding to Theorems 2 and 3 in the paper. The first theorem gives a dimension-generic level upper bound and the bivariate sharpness result; the second theorem gives improved level bias bounds under smoothness of the induced signed-distance regression.

Theorem SA-2 (Approximation Bias: Minimal Level Guarantee). *Suppose Assumptions SA-1(i)-(iii) and SA-2 hold. If $h = o(1)$, then*

$$\sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})| \lesssim h,$$

for any $d \geq 1$. In particular, for $d = 2$, this bias-rate is sharp in the following sense. For some $L \geq 1$, let $\mathcal{P}$ be the class of data generating processes satisfying Assumptions SA-1(i)-(iii) with $\mathcal{X} \subseteq [-L, L]^2$, Assumption SA-2, and the following additional conditions, with (ii) and (iii) imposed for each $t \in \{0, 1\}$:

- (i) $L^{-1} \leq \inf_{\mathbf{x} \in \mathcal{X}} f_X(\mathbf{x}) \leq \sup_{\mathbf{x} \in \mathcal{X}} f_X(\mathbf{x}) \leq L$,
- (ii) $\max_{0 \leq |\nu| \leq p} \sup_{\mathbf{x} \in \mathcal{X}} |\partial^\nu \mu_t(\mathbf{x})| + \max_{0 \leq |\nu| \leq p} \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \frac{|\partial^\nu \mu_t(\mathbf{x}) - \partial^\nu \mu_t(\mathbf{y})|}{\|\mathbf{x} - \mathbf{y}\|} \leq L$, and
- (iii) $\liminf_{h \downarrow 0} \inf_{\mathbf{x} \in \mathcal{B}} \int_{\mathcal{A}_t} K_h((2t - 1)\mathcal{d}(\mathbf{u}, \mathbf{x})) d\mathbf{u} \geq L^{-1}$.

If $nh^2 \rightarrow \infty$ and $h = o(1)$, then

$$1 \lesssim \liminf_{n \rightarrow \infty} \sup_{\mathbb{P} \in \mathcal{P}} \sup_{\mathbf{x} \in \mathcal{B}} \frac{|\mathfrak{B}(\mathbf{x})|}{h} \leq \limsup_{n \rightarrow \infty} \sup_{\mathbb{P} \in \mathcal{P}} \sup_{\mathbf{x} \in \mathcal{B}} \frac{|\mathfrak{B}(\mathbf{x})|}{h} \lesssim 1.$$

The first part of Theorem SA-2 is dimension-generic. The sharpness statement is a level result: near kinked boundaries, signed-distance derivatives need not exist without additional smoothness restrictions.

Theorem SA-3 (Approximation Bias: Smooth Boundary). *Suppose Assumptions SA-1(i)-(iii) and SA-2 hold, and let $h = o(1)$.*

- (i) *If, for some integer $m \in \{1, \dots, p + 1\}$, each map $r \mapsto \theta_{t,\mathbf{x}}(r)$ is m -times continuously differentiable on $\mathcal{I}_t$ near zero with m th derivative uniformly bounded over $t \in \{0, 1\}$ and $\mathbf{x} \in \mathcal{B}$, then*

$$\sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})| \lesssim h^m.$$

The pointwise analogue holds under the same smoothness condition at a fixed $\mathbf{x} \in \mathcal{B}$.

- (ii) *Suppose $d = 2$ and $\mathcal{d}(\cdot, \cdot)$ is the Euclidean distance. For $\mathbf{x} \in \mathcal{B}$, suppose that for some $\delta, \varepsilon > 0$, $\mathcal{B} \cap \{\mathbf{y} : \|\mathbf{y} - \mathbf{x}\| \leq \varepsilon\} = \gamma([- \delta, \delta])$, where $\gamma : \mathbb{R} \rightarrow \mathbb{R}^2$ is one-to-one and belongs to $C^{\kappa+2}([- \delta, \delta], \mathbb{R}^2)$.*

Let $m_{\mathbf{x}} = \kappa \wedge (p+1)$. Then, $\theta_{0,\mathbf{x}}(\cdot)$ and $\theta_{1,\mathbf{x}}(\cdot)$ are $m_{\mathbf{x}}$ -times continuously differentiable near zero, and

$$|\mathfrak{B}(\mathbf{x})| \lesssim h^{m_{\mathbf{x}}}.$$

- (iii) Suppose $d = 2$, $\mathcal{d}(\cdot, \cdot)$ is the Euclidean distance, and $\mathcal{B} = \gamma([0, L])$, where γ is one-to-one and belongs to $C^{u+2}([0, L], \mathbb{R}^2)$ for some $L > 0$. Suppose there exist $\delta, \varepsilon > 0$ such that, for all $\mathbf{x} \in \mathcal{B}^o = \gamma([\delta, L-\delta])$, $r \in [0, \varepsilon]$, and $t \in \{0, 1\}$, the set $\{\mathbf{u} \in \mathbb{R}^2 : \mathcal{d}(\mathbf{u}, \mathbf{x}) = r\}$ intersects $\text{bd}(\mathcal{A}_t)$ at only two points in $\mathcal{B}$. Let $m = \iota \wedge (p+1)$. Then, $\theta_{0,\mathbf{x}}(\cdot)$ and $\theta_{1,\mathbf{x}}(\cdot)$ are m -times continuously differentiable near zero with derivatives uniformly bounded over $\mathbf{x} \in \mathcal{B}^o$, and

$$\sup_{\mathbf{x} \in \mathcal{B}^o} |\mathfrak{B}(\mathbf{x})| \lesssim h^m.$$

SA-3.3 Point Estimation and Approximate MSE Expansions

Theorem SA-4 (Convergence Rates). Suppose Assumptions SA-1 and SA-2 hold. Let $h = o(1)$.

- If $\vartheta(\mathbf{x})$ is well defined at a fixed $\mathbf{x} \in \mathcal{B}$, and $nh^d \rightarrow \infty$, then

$$|\hat{\vartheta}(\mathbf{x}) - \vartheta(\mathbf{x})| \lesssim_{\mathbb{P}} \frac{1}{\sqrt{nh^d}} + \frac{1}{n^{\frac{1+v}{2+v}} h^d} + |\mathfrak{B}(\mathbf{x})|.$$

- If $\vartheta(\mathbf{x})$ is well defined for all $\mathbf{x} \in \mathcal{B}$, and $\frac{nh^d}{\log(1/h)} \rightarrow \infty$, then

$$\sup_{\mathbf{x} \in \mathcal{B}} |\hat{\vartheta}(\mathbf{x}) - \vartheta(\mathbf{x})| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} + \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})|.$$

Equation (SA-1) also gives MSE-type criteria for bandwidth choice. Unlike the location-based estimator, the distance-based estimator is based on a one-dimensional projection of the full score, so a conditional-on- $\mathbf{X}$ MSE does not isolate a clean local-polynomial variance term under misspecification. Instead, we use the linearized decomposition implied by (SA-1). Because the population projection defining $\theta_{t,\mathbf{x}}^*$ implies $\mathbb{E}[\mathfrak{L}(\mathbf{x})] = 0$ and $\mathbb{E}[\mathfrak{L}(\mathbf{x})^2] = \Xi_{\mathbf{x}}$, we define the approximate MSE and approximate IMSE as

$$\text{AMSE}(\mathbf{x}) = \mathbb{E}[(\mathfrak{B}(\mathbf{x}) + \mathfrak{L}(\mathbf{x}))^2] = \mathfrak{B}(\mathbf{x})^2 + \Xi_{\mathbf{x}}, \quad \mathbf{x} \in \mathcal{B},$$

and the approximate IMSE

$$\text{AIMSE} = \int_{\mathcal{B}} \text{AMSE}(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}).$$

The results in Section SA-3.2 give different approximation and exact rate results for the bias $\mathfrak{B}(\mathbf{x})$, Lemma SA-5 implies that $V_{\mathbf{x}} = nh^d \Xi_{\mathbf{x}}$ is bounded and bounded away from zero, and Lemma SA-4 implies that

$$\sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{Q}(\mathbf{x})| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right).$$

Therefore, under appropriate rate conditions, the nonlinear term is higher order and can be ignored for first-order bandwidth calculations.

For bandwidth selection, suppose that a leading-bias expansion $\mathfrak{B}(\mathbf{x}) = h^m B_{\mathbf{x}} + o(h^m)$ is available, with $B_{\mathbf{x}}$ bounded and bounded away from zero. The approximate MSE criterion is then

$$h^{2m} B_{\mathbf{x}}^2 + \frac{1}{nh^d} V_{\mathbf{x}},$$

which gives the approximate MSE-optimal bandwidth selection rule

$$h_{\text{AMSE}, \mathbf{x}} = \left(\frac{dV_{\mathbf{x}}}{2mB_{\mathbf{x}}^2} \frac{1}{n} \right)^{\frac{1}{2m+d}}.$$

Similarly, if $\int_{\mathcal{B}} B_{\mathbf{x}}^2 w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x})$ is bounded and bounded away from zero, the approximate IMSE-optimal bandwidth choice is

$$h_{\text{AIMSE}} = \left(\frac{d \int_{\mathcal{B}} V_{\mathbf{x}} w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x})}{2m \int_{\mathcal{B}} B_{\mathbf{x}}^2 w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x})} \frac{1}{n} \right)^{\frac{1}{2m+d}}.$$

Here m is the order of the signed-distance approximation bias. Theorems SA-2 and SA-3 give primitive conditions under which such rates are available. In contrast to the location-based case in Cattaneo et al. [2026], this bias order is governed by the smoothness of the induced signed-distance regression, and therefore by the geometry of the boundary and the distance metric, not only by the polynomial order p .

SA-3.4 Distributional Approximation and Inference

Let $\mathbf{V} = ((\mathbf{X}_1^\top, Y_1), \dots, (\mathbf{X}_n^\top, Y_n))$, and recall that $t \in \{0, 1\}$. The feasible t-statistic is

$$\hat{\mathbf{T}}(\mathbf{x}) = \frac{\hat{\vartheta}(\mathbf{x}) - \vartheta(\mathbf{x})}{\sqrt{\hat{\Xi}_{\mathbf{x}, \mathbf{x}}}}, \quad \mathbf{x} \in \mathcal{B}.$$

The associated $100(1 - \alpha)\%$ confidence interval estimator is

$$\hat{\mathbf{I}}_{\alpha}(\mathbf{x}) = \left[\hat{\vartheta}(\mathbf{x}) - \mathbf{q}_{\alpha} \sqrt{\hat{\Xi}_{\mathbf{x}, \mathbf{x}}}, \hat{\vartheta}(\mathbf{x}) + \mathbf{q}_{\alpha} \sqrt{\hat{\Xi}_{\mathbf{x}, \mathbf{x}}} \right],$$

where $\mathbf{q}_{\alpha}$ denotes an appropriate quantile depending on the desired confidence level $\alpha \in (0, 1)$ and coverage objective (pointwise vs. uniform over $\mathcal{B}$). The following theorem establishes pointwise asymptotic normality and validity of confidence intervals. Let $\Phi(\cdot)$ be the cumulative distribution function of a standard univariate Gaussian random variable.

Theorem SA-5 (Confidence Intervals). *Suppose Assumptions SA-1 and SA-2 hold, and $\vartheta(\mathbf{x})$ is well defined. If $n^{\frac{\nu}{2+\nu}} h^d \rightarrow \infty$, $h = o(1)$, and $\sqrt{nh^d} |\mathfrak{B}(\mathbf{x})| \rightarrow 0$, then*

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}(\hat{\mathbf{T}}(\mathbf{x}) \leq u) - \Phi(u) \right| = o(1), \quad \mathbf{x} \in \mathcal{B},$$

and

$$\mathbb{P}(\vartheta(\mathbf{x}) \in \hat{\mathbf{I}}_{\alpha}(\mathbf{x})) = 1 - \alpha + o(1), \quad \mathbf{x} \in \mathcal{B},$$

provided that $\mathbf{q}_{\alpha} = \inf\{c > 0 : \mathbb{P}(|\hat{Z}| \geq c \mid \mathbf{V}) \leq \alpha\}$ with $\hat{Z} \mid \mathbf{V} \sim \text{Normal}(0, 1)$.

To conduct uniform inference, we rely on a new strong approximation result established in Section SA-7. First, we approximate (uniformly over $\mathbf{x} \in \mathcal{B}$) the feasible statistic $\widehat{\mathbf{T}}$ by the following linear statistic (which is a sum of independent random variables):

$$\overline{\mathbf{T}}(\mathbf{x}) = (\Xi_{\mathbf{x}, \mathbf{x}})^{-1/2} \left(\mathbf{e}_0^\top \Psi_{1, \mathbf{x}}^{-1} \mathbf{O}_{1, \mathbf{x}} - \mathbf{e}_0^\top \Psi_{0, \mathbf{x}}^{-1} \mathbf{O}_{0, \mathbf{x}} \right), \quad \mathbf{x} \in \mathcal{B}.$$

Theorem SA-6 (Stochastic Linearization). *Suppose Assumptions SA-1 and SA-2 hold, and $\vartheta(\mathbf{x})$ is well defined for all $\mathbf{x} \in \mathcal{B}$. If $\frac{n^{\frac{v}{2+v}} h^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then*

$$\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{\mathbf{T}}(\mathbf{x}) - \overline{\mathbf{T}}(\mathbf{x})| \lesssim_{\mathbb{P}} \sqrt{\log(1/h)} \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right) + \sqrt{nh^d} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})|.$$

The pointwise (in $\mathcal{B}$) analogue of this result removes the $\log(1/h)$ penalty. See the proof of Theorem SA-5 for more details. To establish a Gaussian strong approximation for $\overline{\mathbf{T}}(\mathbf{x})$, define the class of functions $\mathcal{G} = \{g_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$ and $\mathcal{M} = \{m_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$, where

$$\begin{aligned} g_{\mathbf{x}}(\mathbf{u}) &= \mathbf{1}(\mathbf{u} \in \mathcal{A}_1) \mathfrak{K}_1(\mathbf{u}; \mathbf{x}) - \mathbf{1}(\mathbf{u} \in \mathcal{A}_0) \mathfrak{K}_0(\mathbf{u}; \mathbf{x}), \\ m_{\mathbf{x}}(\mathbf{u}) &= -\mathbf{1}(\mathbf{u} \in \mathcal{A}_1) \mathfrak{K}_1(\mathbf{u}; \mathbf{x}) \theta_{1, \mathbf{x}}^*(D_{\mathbf{u}}(\mathbf{x})) + \mathbf{1}(\mathbf{u} \in \mathcal{A}_0) \mathfrak{K}_0(\mathbf{u}; \mathbf{x}) \theta_{0, \mathbf{x}}^*(D_{\mathbf{u}}(\mathbf{x})), \end{aligned} \quad (\text{SA-2})$$

with

$$\mathfrak{K}_t(\mathbf{u}; \mathbf{x}) = \frac{1}{\sqrt{n \Xi_{\mathbf{x}, \mathbf{x}}}} \mathbf{e}_0^\top \Psi_{t, \mathbf{x}}^{-1} \mathbf{r}_p \left(\frac{D_{\mathbf{u}}(\mathbf{x})}{h} \right) K_h(D_{\mathbf{u}}(\mathbf{x})),$$

for all $\mathbf{u} \in \mathcal{X}$, $\mathbf{x} \in \mathcal{B}$, and $t \in \{0, 1\}$. In addition, let $\mathcal{R}$ be the class of functions containing the singleton identity function $\text{Id} : \mathbb{R} \rightarrow \mathbb{R}$, $\text{Id}(x) = x$. Then, $\overline{\mathbf{T}}(\mathbf{x})$ can be represented as

$$\overline{\mathbf{T}}(\mathbf{x}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[g_{\mathbf{x}}(\mathbf{X}_i) \text{Id}(Y_i) + m_{\mathbf{x}}(\mathbf{X}_i) - \mathbb{E}[g_{\mathbf{x}}(\mathbf{X}_i) \text{Id}(Y_i) + m_{\mathbf{x}}(\mathbf{X}_i)] \right].$$

Define the multiplicative-separable empirical process as

$$M_n(g, r) = \frac{1}{\sqrt{n}} \sum_{i=1}^n [g(\mathbf{x}_i) r(y_i) - \mathbb{E}[g(\mathbf{x}_i) r(y_i)]], \quad g \in \mathcal{G}, r \in \mathcal{R},$$

which implies that

$$\overline{\mathbf{T}}(\mathbf{x}) = M_n(g_{\mathbf{x}}, \text{Id}) + M_n(m_{\mathbf{x}}, 1), \quad \mathbf{x} \in \mathcal{B}.$$

Theorem SA-25 gives a new Gaussian strong approximation that can be applied to $\overline{\mathbf{T}}(\mathbf{x})$. This new theorem allows polynomial moment bounds on the conditional distribution of $Y_i | \mathbf{X}_i$.

Theorem SA-7 (Gaussian Strong Approximation: $\overline{\mathbf{T}}$). *Suppose Assumptions SA-1 and SA-2 hold, and $\mathcal{X} = \prod_{l=1}^d [a_l, b_l]$, with $-\infty < a_l < b_l < \infty$ for $l = 1, \dots, d$. Suppose also that there exists a constant $C > 0$ such that for $t \in \{0, 1\}$ and for any $\mathbf{x} \in \mathcal{B}$, the De Giorgi perimeter of the localized set $E_{t, \mathbf{x}}(h) = \{\mathbf{y} \in \mathcal{A}_t : \varsigma_t \mathcal{A}(\mathbf{y}, \mathbf{x})/h \in \text{supp}(K)\}$ satisfies $\mathcal{L}(E_{t, \mathbf{x}}(h)) \leq Ch^{d-1}$. If $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$, $nh^d \rightarrow \infty$, and $h = o(1)$, then (on a possibly enlarged probability space) there exists a mean-zero Gaussian process Z indexed by $\mathcal{B}$*

with almost surely continuous sample path such that

$$\mathbb{E} \left[\sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}(\mathbf{x}) - Z(\mathbf{x})| \right] \lesssim (\log(n))^{\frac{3}{2}} \left(\frac{1}{nh^d} \right)^{\frac{1}{2d+2} \frac{v}{v+2}} + \log(n) \left(\frac{1}{n^{\frac{v}{2+v}} h^d} \right)^{\frac{1}{2}},$$

where $\lesssim$ is up to a universal constant, and Z has the same covariance structure as $\bar{T}$; that is, $\text{Cov}[\bar{T}(\mathbf{x}_1), \bar{T}(\mathbf{x}_2)] = \text{Cov}[Z(\mathbf{x}_1), Z(\mathbf{x}_2)]$ for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$.

The rectangular-support condition in Theorem SA-7 is used only to invoke the normalizing-map construction in Cattaneo and Yu [2025, Section 3.1]. After a componentwise affine rescaling, one can take $\mathcal{X} = [0, 1]^d$; because f_X is continuous and bounded away from zero on this compact rectangle, $\mathbb{P}_X$ is a valid surrogate measure and the Rosenblatt transform provides a valid normalizing transformation with finite constants. The De Giorgi perimeter condition controls the total variation of the localized distance-kernel classes.

Theorem SA-7 can be used to construct confidence bands for $(\vartheta(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$. Let $(\hat{Z}(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$ be a (conditionally on $\mathbf{V}$) mean-zero Gaussian process with feasible (conditional) covariance function

$$\text{Cov}[\hat{Z}(\mathbf{x}_1), \hat{Z}(\mathbf{x}_2) | \mathbf{V}] = \frac{\hat{\Xi}_{\mathbf{x}_1, \mathbf{x}_2}}{\sqrt{\hat{\Xi}_{\mathbf{x}_1, \mathbf{x}_1}} \sqrt{\hat{\Xi}_{\mathbf{x}_2, \mathbf{x}_2}}}, \quad \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}.$$

Theorem SA-8 (Confidence Bands). Suppose the assumptions and conditions in Theorem SA-7 hold, and $\vartheta(\mathbf{x})$ is well defined for all $\mathbf{x} \in \mathcal{B}$. If $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$, $\frac{n^{\frac{v}{2+v}} h^d}{(\log n)^3} \rightarrow \infty$, and $\sqrt{nh^d \log n} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})| \rightarrow 0$, then

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\hat{T}(\mathbf{x})| \leq u \right) - \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\hat{Z}(\mathbf{x})| \leq u | \mathbf{V} \right) \right| = o_{\mathbb{P}}(1)$$

and

$$\mathbb{P} \left[\vartheta(\mathbf{x}) \in \hat{\mathcal{I}}_{\alpha}(\mathbf{x}), \text{ for all } \mathbf{x} \in \mathcal{B} \right] = 1 - \alpha + o(1),$$

provided that $\mathbf{q}_{\alpha} = \inf \{c > 0 : \mathbb{P}(\sup_{\mathbf{x} \in \mathcal{B}} |\hat{Z}(\mathbf{x})| \geq c | \mathbf{V}) \leq \alpha\}$.

Theorem SA-7 gives a strong approximation of the entire process $\bar{T}$ by a Gaussian process. This process-level coupling is stronger than what is needed to construct confidence bands. The following theorem approximates only the supremum of the statistic. It avoids the rectangular-support and De Giorgi perimeter conditions used in the process-level result.

Theorem SA-9 (Gaussian Approximation for $\sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}(\mathbf{x})|$). Suppose Assumptions SA-1 and SA-2 hold, $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$, and $h = o(1)$. Then, for n large enough, on a possibly enlarged probability space there exists a mean-zero Gaussian process Z indexed by $\mathcal{B}$ with almost surely continuous sample paths such that

$$\text{Cov}[Z(\mathbf{x}_1), Z(\mathbf{x}_2)] = \text{Cov}[\bar{T}(\mathbf{x}_1), \bar{T}(\mathbf{x}_2)], \quad \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B},$$

and

$$\mathbb{P} \left[\left| \sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}(\mathbf{x})| - \sup_{\mathbf{x} \in \mathcal{B}} |Z(\mathbf{x})| \right| > \frac{\log n}{(nh^d)^{1/6}} + \frac{(\log n)^{5/4}}{(nh^d)^{1/4}} + \frac{(\log n)^{3/2}}{(n^{\frac{v}{2+v}} h^d)^{1/2}} \right] \lesssim \frac{1}{\log n} + \frac{\log n}{n}.$$

Using Theorem SA-9, we obtain the following confidence-band result. Relative to Theorem SA-8, it removes the process-level geometric side conditions at the cost of a stronger logarithmic rate condition.

Theorem SA-10 (Confidence Bands: Suprema Approximation). *Suppose Assumptions SA-1 and SA-2 hold, and $\vartheta(\mathbf{x})$ is well defined for all $\mathbf{x} \in \mathcal{B}$. If $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$, $\frac{n^{\frac{v}{2+v}} h^d}{(\log n)^9} \rightarrow \infty$, $h = o(1)$, and $\sqrt{nh^d \log n} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})| \rightarrow 0$, then*

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\hat{\mathbf{T}}(\mathbf{x})| \leq u \right) - \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\hat{\mathbf{Z}}(\mathbf{x})| \leq u \middle| \mathbf{V} \right) \right| = o_{\mathbb{P}}(1)$$

and

$$\mathbb{P} \left[\vartheta(\mathbf{x}) \in \hat{\mathbf{I}}_{\alpha}(\mathbf{x}), \text{ for all } \mathbf{x} \in \mathcal{B} \right] = 1 - \alpha + o(1),$$

provided that $\mathfrak{q}_{\alpha} = \inf \{ c > 0 : \mathbb{P}(\sup_{\mathbf{x} \in \mathcal{B}} |\hat{\mathbf{Z}}(\mathbf{x})| \geq c | \mathbf{V}) \leq \alpha \}$.

SA-4 Weighted Boundary Average Treatment Effect

Without loss of generality, normalize the weight function so that $\int_{\mathcal{B}} w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}) = 1$. Define the distance-based WBATE parameter and plug-in estimator by

$$\vartheta_{\text{WBATE}} = \int_{\mathcal{B}} \vartheta(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}), \quad \text{and} \quad \hat{\vartheta}_{\text{WBATE}} = \int_{\mathcal{B}} \hat{\vartheta}(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}).$$

Integrating the decomposition (SA-1) for the difference of the two side-specific estimators gives

$$\hat{\vartheta}_{\text{WBATE}} - \vartheta_{\text{WBATE}} = \mathfrak{B}_{\text{WBATE}} + \mathfrak{L}_{\text{WBATE}} + \mathfrak{Q}_{\text{WBATE}}, \tag{SA-3}$$

where

$$\mathfrak{B}_{\text{WBATE}} = \int_{\mathcal{B}} \mathfrak{B}(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}), \quad \mathfrak{L}_{\text{WBATE}} = \int_{\mathcal{B}} \mathfrak{L}(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}),$$

and

$$\mathfrak{Q}_{\text{WBATE}} = \int_{\mathcal{B}} \mathfrak{Q}(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}).$$

Define the aggregate variance of the leading term by

$$\Xi_{\text{WBATE}} = \int_{\mathcal{B}} \int_{\mathcal{B}} \Xi_{\mathbf{x}_1, \mathbf{x}_2} w(\mathbf{x}_1) w(\mathbf{x}_2) d\mathfrak{H}^{d-1}(\mathbf{x}_1) d\mathfrak{H}^{d-1}(\mathbf{x}_2).$$

Its feasible counterpart is

$$\hat{\Xi}_{\text{WBATE}} = \int_{\mathcal{B}} \int_{\mathcal{B}} \hat{\Xi}_{\mathbf{x}_1, \mathbf{x}_2} w(\mathbf{x}_1) w(\mathbf{x}_2) d\mathfrak{H}^{d-1}(\mathbf{x}_1) d\mathfrak{H}^{d-1}(\mathbf{x}_2).$$

For the aggregate stochastic rate and inference results, impose the local boundary-measure regularity con-

dition

$$\sup_{\mathbf{x} \in \mathcal{B}} \mathfrak{H}^{d-1}(\mathcal{B} \cap B(\mathbf{x}, r)) \leq C_{\mathcal{B}} r^{d-1}, \quad 0 < r \leq r_0, \quad (\text{SA-4})$$

for some constants $C_{\mathcal{B}} < \infty$ and $r_0 > 0$, where $B(\mathbf{x}, r)$ denotes a Euclidean ball. This upper local regularity condition holds, for example, for compact piecewise C^1 boundaries with finite local multiplicity, including piecewise-linear boundaries. Write $\nu_{\mathcal{B},k}$ for the k -fold product measure $(\mathfrak{H}^{d-1})^k$ on $\mathcal{B}^k$. For inference, impose the scalar aggregate nondegeneracy condition

$$\Xi_{\text{WBATE}} \gtrsim \frac{h^{d-1}}{nh^d} = \frac{1}{nh}, \quad (\text{SA-5})$$

which rules out cancellation of the integrated covariance kernel.

Lemma SA-6 (Variance Linear Term: WBATE). *Suppose Assumptions SA-1, SA-2, and SA-3 hold, and (SA-4) holds. If $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then*

$$\mathbb{V}[\mathfrak{L}_{\text{WBATE}}] = \Xi_{\text{WBATE}} \lesssim \frac{1}{nh}.$$

If, in addition, (SA-5) holds, then

$$\frac{1}{nh} \lesssim \Xi_{\text{WBATE}} \lesssim \frac{1}{nh}.$$

If $\frac{n^{v/(2+v)} h^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then

$$\hat{\Xi}_{\text{WBATE}} = \Xi_{\text{WBATE}} + o_{\mathbb{P}}\left(\frac{1}{nh}\right).$$

If, in addition, (SA-5) holds, then $\hat{\Xi}_{\text{WBATE}}/\Xi_{\text{WBATE}} \rightarrow_{\mathbb{P}} 1$.

Lemma SA-7 (Quadratic Remainder: WBATE). *Suppose Assumptions SA-1, SA-2, and SA-3 hold, and (SA-4) holds. If $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then*

$$\mathfrak{Q}_{\text{WBATE}} \lesssim_{\mathbb{P}} \frac{1}{nh^d}.$$

If, in addition, $nh^{2d-1} \rightarrow \infty$, then

$$\mathfrak{Q}_{\text{WBATE}} = o_{\mathbb{P}}\left(\frac{1}{\sqrt{nh}}\right).$$

Theorem SA-11 (Convergence Rate: WBATE). *Suppose Assumptions SA-1, SA-2, and SA-3 hold, $\vartheta(\mathbf{x})$ is well defined for all $\mathbf{x} \in \mathcal{B}$, and (SA-4) holds. If $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then*

$$|\hat{\vartheta}_{\text{WBATE}} - \vartheta_{\text{WBATE}}| \lesssim_{\mathbb{P}} |\mathfrak{B}_{\text{WBATE}}| + \frac{1}{\sqrt{nh}} + \frac{1}{nh^d}.$$

For inference, define

$$\widehat{\mathbf{T}}_{\text{WBATE}} = \frac{\widehat{\vartheta}_{\text{WBATE}} - \vartheta_{\text{WBATE}}}{\sqrt{\widehat{\Xi}_{\text{WBATE}}}},$$

and the associated confidence interval

$$\widehat{\mathbf{I}}_{\alpha, \text{WBATE}} = \left[\widehat{\vartheta}_{\text{WBATE}} - \mathbf{q}_{\alpha} \sqrt{\widehat{\Xi}_{\text{WBATE}}}, \widehat{\vartheta}_{\text{WBATE}} + \mathbf{q}_{\alpha} \sqrt{\widehat{\Xi}_{\text{WBATE}}} \right].$$

Theorem SA-12 (Asymptotic Normality: WBATE). *Suppose Assumptions SA-1, SA-2, and SA-3 hold, and (SA-5) and (SA-4) hold. If $\frac{n^{v/(2+v)} h^d}{\log(1/h)} \rightarrow \infty$, $h = o(1)$, $nh^{2d-1} \rightarrow \infty$, and $\sqrt{nh} |\mathfrak{B}_{\text{WBATE}}| \rightarrow 0$, then*

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}(\widehat{\mathbf{T}}_{\text{WBATE}} \leq u) - \Phi(u) \right| = o(1),$$

and

$$\mathbb{P}[\vartheta_{\text{WBATE}} \in \widehat{\mathbf{I}}_{\alpha, \text{WBATE}}] = 1 - \alpha + o(1)$$

provided that $\mathbf{q}_{\alpha} = \Phi^{-1}(1 - \alpha/2)$.

Condition (SA-4) controls the amount of boundary measure in bandwidth-sized neighborhoods, which gives the effective one-dimensional variance rate $1/(nh)$ for the integrated estimator.

SA-5 Largest Boundary Average Treatment Effect

The distance-based LBATE parameter and plug-in estimator are

$$\vartheta_{\text{LBATE}} = \sup_{\mathbf{x} \in \mathcal{B}} \vartheta(\mathbf{x}), \quad \widehat{\vartheta}_{\text{LBATE}} = \sup_{\mathbf{x} \in \mathcal{B}} \widehat{\vartheta}(\mathbf{x}),$$

respectively.

Theorem SA-13 (Convergence Rate: LBATE). *Suppose Assumptions SA-1 and SA-2 hold, and $\vartheta(\mathbf{x})$ is well defined for all $\mathbf{x} \in \mathcal{B}$. If $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then*

$$\left| \widehat{\vartheta}_{\text{LBATE}} - \vartheta_{\text{LBATE}} \right| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} + \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})|.$$

For uncertainty quantification, project the simultaneous confidence band for the BATEC onto the supremum functional. Define

$$\widehat{\mathbf{I}}_{\alpha, \text{LBATE}} = \left[\sup_{\mathbf{x} \in \mathcal{B}} \left(\widehat{\vartheta}(\mathbf{x}) - \mathbf{q}_{\alpha} \sqrt{\widehat{\Xi}_{\mathbf{x}, \mathbf{x}}} \right), \sup_{\mathbf{x} \in \mathcal{B}} \left(\widehat{\vartheta}(\mathbf{x}) + \mathbf{q}_{\alpha} \sqrt{\widehat{\Xi}_{\mathbf{x}, \mathbf{x}}} \right) \right],$$

where $\mathbf{q}_{\alpha} = \inf\{c > 0 : \mathbb{P}(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}(\mathbf{x})| \geq c \mid \mathbf{V}) \leq \alpha\}$.

Theorem SA-14 (Confidence Intervals: LBATE). *Suppose the assumptions and conditions in Theorem SA-*

10 hold. Then

$$\mathbb{P} \left[\vartheta_{\text{LBATE}} \in \widehat{\mathbf{I}}_{\alpha, \text{LBATE}} \right] \geq 1 - \alpha + o(1).$$

The interval is generally conservative because it is obtained by projecting a simultaneous confidence band for the entire distance-based BATEC.

SA-6 Imperfect Compliance

We extend the distance-based results to settings with imperfect compliance, where treatment assignment and treatment receipt may differ for some units. See [Hernán and Robins \[2020\]](#) for a textbook introduction.

Because $D_i(\mathbf{x}) \in \mathcal{J}_t$ if and only if $\mathbf{X}_i \in \mathcal{A}_t$, for any $\mathbf{x} \in \mathcal{B}$ let

$$W_i = \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_0)W_i(0) + \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_1)W_i(1)$$

be the observed treatment receipt, where $W_i(t) \in \{0, 1\}$ denotes the potential treatment receipt under assignment $t \in \{0, 1\}$. The observed outcome is

$$Y_i = \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_0)Y_i(0, W_i(0)) + \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_1)Y_i(1, W_i(1)).$$

For notational economy, redefine the reduced-form potential outcome as $Y_i(t) = Y_i(t, W_i(t))$. For $A \in \{Y, W\}$, write $A_i(t) = Y_i(t)$ when $A = Y$, and $A_i(t) = W_i(t)$ when $A = W$; similarly, write $A_i = Y_i$ when $A = Y$, and $A_i = W_i$ when $A = W$. The reduced-form BATEC components are

$$\tau_A(\mathbf{x}) = \mathbb{E}[A_i(1) - A_i(0) \mid \mathbf{X}_i = \mathbf{x}], \quad A \in \{Y, W\}, \quad \mathbf{x} \in \mathcal{B},$$

and the fuzzy BATEC is the Wald ratio

$$\zeta(\mathbf{x}) = \frac{\tau_Y(\mathbf{x})}{\tau_W(\mathbf{x})}, \quad \mathbf{x} \in \mathcal{B}.$$

The results below concern this reduced-form Wald target. A complier-average causal interpretation can be obtained by adding the usual exclusion and monotonicity restrictions, which are not needed for the estimation and inference theory.

The distance-based analogues are defined from the induced conditional expectations

$$\theta_{A,t,\mathbf{x}}(r) = \mathbb{E}[A_i \mid D_i(\mathbf{x}) = r] = \mathbb{E}[A_i(t) \mid \mathcal{A}(\mathbf{X}_i, \mathbf{x}) = |r|, \mathbf{X}_i \in \mathcal{A}_t], \quad r \in \mathcal{J}_t,$$

and

$$\vartheta_A(\mathbf{x}) = \theta_{A,1,\mathbf{x}}(0) - \theta_{A,0,\mathbf{x}}(0), \quad \xi(\mathbf{x}) = \frac{\vartheta_Y(\mathbf{x})}{\vartheta_W(\mathbf{x})}.$$

The estimator is

$$\widehat{\xi}(\mathbf{x}) = \frac{\widehat{\vartheta}_Y(\mathbf{x})}{\widehat{\vartheta}_W(\mathbf{x})}, \quad \widehat{\vartheta}_A(\mathbf{x}) = \mathbf{e}_0^\top \widehat{\gamma}_{A,1}(\mathbf{x}) - \mathbf{e}_0^\top \widehat{\gamma}_{A,0}(\mathbf{x}),$$

where, for $A \in \{Y, W\}$ and $t \in \{0, 1\}$,

$$\hat{\gamma}_{A,t}(\mathbf{x}) = \arg \min_{\gamma \in \mathbb{R}^{p+1}} \mathbb{E}_n \left[(A_i - \mathbf{r}_p(D_i(\mathbf{x})/h)^\top \gamma)^2 K_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right].$$

Thus, $\hat{\gamma}_{Y,t}(\mathbf{x})$ coincides with the sharp-design estimator defined above when the outcome is Y_i , while $\hat{\gamma}_{W,t}(\mathbf{x})$ is the same local polynomial fit applied to treatment receipt.

Assumption SA-4 (Fuzzy DGP). *Let $t \in \{0, 1\}$.*

- (i) *$((Y_i(t), W_i(t), \mathbf{X}_i^\top)^\top : 1 \leq i \leq n)$ are independent and identically distributed random vectors.*
- (ii) *$\mu_{W,t}(\mathbf{x}) = \mathbb{E}[W_i(t) \mid \mathbf{X}_i = \mathbf{x}]$ has a $(p+1)$ -times continuously differentiable extension to an open neighborhood of $\mathcal{X}$.*
- (iii) *The entries of $\text{Cov}[(Y_i(t), W_i(t))^\top \mid \mathbf{X}_i = \mathbf{x}]$ are continuous on $\mathcal{X}$.*

The following theorem is a natural consequence of Theorem SA-1.

Theorem SA-15 (Distance-Based Fuzzy Identification). *Suppose Assumptions SA-1(i)–(iii), SA-4, and SA-2(i), (ii), and (v) hold. Then,*

$$\vartheta_A(\mathbf{x}) = \tau_A(\mathbf{x}), \quad A \in \{Y, W\}, \quad \mathbf{x} \in \mathcal{B}.$$

If, in addition, $\tau_W(\mathbf{x}) \neq 0$, then $\xi(\mathbf{x}) = \zeta(\mathbf{x})$.

SA-6.1 Second-Order Linearization

The distance-based fuzzy analysis starts from the reduced-form distance-based decomposition for each $A \in \{Y, W\}$. Let

$$\theta_{A,t,\mathbf{x}}^*(r) = \mathbf{r}_p(r/h)^\top \gamma_{A,t}^*(\mathbf{x}),$$

where

$$\gamma_{A,t}^*(\mathbf{x}) = \arg \min_{\gamma \in \mathbb{R}^{p+1}} \mathbb{E} \left[(A_i - \mathbf{r}_p(D_i(\mathbf{x})/h)^\top \gamma)^2 K_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right].$$

The same algebra used in (SA-1) gives

$$\hat{\vartheta}_A(\mathbf{x}) - \vartheta_A(\mathbf{x}) = \mathfrak{B}_A(\mathbf{x}) + \mathfrak{L}_A(\mathbf{x}) + \mathfrak{Q}_A(\mathbf{x}), \quad A \in \{Y, W\}, \quad (\text{SA-6})$$

where

$$\begin{aligned} \mathfrak{B}_A(\mathbf{x}) &= \theta_{A,1,\mathbf{x}}^*(0) - \theta_{A,0,\mathbf{x}}^*(0) - \vartheta_A(\mathbf{x}), \\ \mathfrak{L}_A(\mathbf{x}) &= \mathbf{e}_0^\top \Psi_{1,\mathbf{x}}^{-1} \mathbf{O}_{A,1,\mathbf{x}} - \mathbf{e}_0^\top \Psi_{0,\mathbf{x}}^{-1} \mathbf{O}_{A,0,\mathbf{x}}, \\ \mathfrak{Q}_A(\mathbf{x}) &= \mathbf{e}_0^\top (\hat{\Psi}_{1,\mathbf{x}}^{-1} - \Psi_{1,\mathbf{x}}^{-1}) \mathbf{O}_{A,1,\mathbf{x}} - \mathbf{e}_0^\top (\hat{\Psi}_{0,\mathbf{x}}^{-1} - \Psi_{0,\mathbf{x}}^{-1}) \mathbf{O}_{A,0,\mathbf{x}}, \end{aligned}$$

and

$$\mathbf{O}_{A,t,\mathbf{x}} = \mathbb{E}_n \left[\mathbf{r}_p(D_i(\mathbf{x})/h) K_h(D_i(\mathbf{x})) (A_i - \theta_{A,t,\mathbf{x}}^*(D_i(\mathbf{x}))) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right].$$

The covariance kernel for the reduced-form linear terms is defined as follows. For $A, B \in \{Y, W\}$, let

$$\Xi_{A,B,\mathbf{x}_1,\mathbf{x}_2} = \Xi_{A,B,0,\mathbf{x}_1,\mathbf{x}_2} + \Xi_{A,B,1,\mathbf{x}_1,\mathbf{x}_2},$$

where

$$\Xi_{A,B,t,\mathbf{x}_1,\mathbf{x}_2} = \frac{1}{nh^d} \mathbf{e}_0^\top \Psi_{t,\mathbf{x}_1}^{-1} \Upsilon_{A,B,t,\mathbf{x}_1,\mathbf{x}_2} \Psi_{t,\mathbf{x}_2}^{-1} \mathbf{e}_0,$$

and

$$\begin{aligned} \Upsilon_{A,B,t,\mathbf{x}_1,\mathbf{x}_2} = h^d \mathbb{E} \Big[& \mathbf{r}_p(D_i(\mathbf{x}_1)/h) \mathbf{r}_p(D_i(\mathbf{x}_2)/h)^\top K_h(D_i(\mathbf{x}_1)) K_h(D_i(\mathbf{x}_2)) \\ & \times (A_i - \theta_{A,t,\mathbf{x}_1}^*(D_i(\mathbf{x}_1))) (B_i - \theta_{B,t,\mathbf{x}_2}^*(D_i(\mathbf{x}_2))) \mathbf{1}(D_i(\mathbf{x}_1) \in \mathcal{J}_t) \mathbf{1}(D_i(\mathbf{x}_2) \in \mathcal{J}_t) \Big]. \end{aligned}$$

Let $\widehat{\Xi}_{A,B,\mathbf{x}_1,\mathbf{x}_2}$ denote the sample analogue obtained by replacing Ψ , Υ , and θ^* with $\widehat{\Psi}$, $\widehat{\Upsilon}$, and $\widehat{\theta}$.

The exact second-order Wald expansion is

$$\widehat{\xi}(\mathbf{x}) - \xi(\mathbf{x}) = \mathfrak{B}_F(\mathbf{x}) + \mathfrak{L}_F(\mathbf{x}) + \mathfrak{Q}_F(\mathbf{x}) + \mathfrak{R}_F(\mathbf{x}), \quad (\text{SA-7})$$

where

$$\begin{aligned} \mathfrak{B}_F(\mathbf{x}) &= \mathbf{w}(\mathbf{x})^\top (\mathfrak{B}_Y(\mathbf{x}), \mathfrak{B}_W(\mathbf{x}))^\top, \\ \mathfrak{L}_F(\mathbf{x}) &= \mathbf{w}(\mathbf{x})^\top (\mathfrak{L}_Y(\mathbf{x}), \mathfrak{L}_W(\mathbf{x}))^\top, \\ \mathfrak{Q}_F(\mathbf{x}) &= \mathbf{w}(\mathbf{x})^\top (\mathfrak{Q}_Y(\mathbf{x}), \mathfrak{Q}_W(\mathbf{x}))^\top, \end{aligned}$$

and

$$\mathbf{w}(\mathbf{x}) = \left(\frac{1}{\vartheta_W(\mathbf{x})}, -\frac{\vartheta_Y(\mathbf{x})}{\vartheta_W(\mathbf{x})^2} \right)^\top,$$

with second-order remainder

$$\mathfrak{R}_F(\mathbf{x}) = \frac{\vartheta_Y(\mathbf{x})}{\vartheta_W(\mathbf{x})^2 \widehat{\vartheta}_W(\mathbf{x})} (\widehat{\vartheta}_W(\mathbf{x}) - \vartheta_W(\mathbf{x}))^2 - \frac{1}{\vartheta_W(\mathbf{x}) \widehat{\vartheta}_W(\mathbf{x})} (\widehat{\vartheta}_Y(\mathbf{x}) - \vartheta_Y(\mathbf{x})) (\widehat{\vartheta}_W(\mathbf{x}) - \vartheta_W(\mathbf{x})).$$

If $\inf_{\mathbf{x} \in \mathcal{B}} |\vartheta_W(\mathbf{x})| > 0$, $h = o(1)$, and $\frac{\log(1/h)}{n^{(1+v)/(2+v)} h^d} = o(1)$, then

$$\sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{R}_F(\mathbf{x})| \lesssim_{\mathbb{P}} \rho_n^2, \quad (\text{SA-8})$$

where

$$\rho_n = \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} + \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_Y(\mathbf{x})| + \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_W(\mathbf{x})|.$$

Thus, after the reduced-form distance-based decompositions are in place, the fuzzy BATEC estimator is governed by a fixed linear combination of the reduced-form outcome and first-stage components, up to the higher-order Wald remainder.

SA-6.2 Fuzzy BATEC

SA-6.2.1 Point Estimation and Approximate MSE Expansions

Theorem SA-16 (Convergence Rates: Fuzzy BATEC). *Suppose Assumptions SA-1, SA-4, and SA-2 hold, and let $h = o(1)$.*

- *If $\xi(\mathbf{x})$ is well defined at a fixed $\mathbf{x} \in \mathcal{B}$, $|\vartheta_W(\mathbf{x})| > 0$, and $n^{\frac{1+v}{2+v}} h^d \rightarrow \infty$, then*

$$|\widehat{\xi}(\mathbf{x}) - \xi(\mathbf{x})| \lesssim_{\mathbb{P}} \frac{1}{\sqrt{nh^d}} + \frac{1}{n^{\frac{1+v}{2+v}} h^d} + |\mathfrak{B}_Y(\mathbf{x})| + |\mathfrak{B}_W(\mathbf{x})|.$$

- *If $\xi(\mathbf{x})$ is well defined for all $\mathbf{x} \in \mathcal{B}$, $\inf_{\mathbf{x} \in \mathcal{B}} |\vartheta_W(\mathbf{x})| > 0$, and $\frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} = o(1)$, then*

$$\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{\xi}(\mathbf{x}) - \xi(\mathbf{x})| \lesssim_{\mathbb{P}} \rho_n.$$

Define

$$\Theta_{\mathbf{F}, \mathbf{x}_1, \mathbf{x}_2} = \begin{pmatrix} \Xi_{Y,Y,\mathbf{x}_1,\mathbf{x}_2} & \Xi_{Y,W,\mathbf{x}_1,\mathbf{x}_2} \\ \Xi_{W,Y,\mathbf{x}_1,\mathbf{x}_2} & \Xi_{W,W,\mathbf{x}_1,\mathbf{x}_2} \end{pmatrix}, \quad \Xi_{\mathbf{F}, \mathbf{x}_1, \mathbf{x}_2} = \mathbf{w}(\mathbf{x}_1)^\top \Theta_{\mathbf{F}, \mathbf{x}_1, \mathbf{x}_2} \mathbf{w}(\mathbf{x}_2).$$

Then $\mathbb{E}[\mathfrak{L}_{\mathbf{F}}(\mathbf{x})] = 0$ and $\mathbb{E}[\mathfrak{L}_{\mathbf{F}}(\mathbf{x})^2] = \Xi_{\mathbf{F}, \mathbf{x}, \mathbf{x}}$. The linearized approximate MSE and approximate IMSE are

$$\text{AMSE}_{\mathbf{F}}(\mathbf{x}) = \mathfrak{B}_{\mathbf{F}}(\mathbf{x})^2 + \Xi_{\mathbf{F}, \mathbf{x}, \mathbf{x}} \quad \text{and} \quad \text{AIMSE}_{\mathbf{F}} = \int_{\mathcal{B}} \text{AMSE}_{\mathbf{F}}(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}).$$

These criteria omit the higher-order terms $\mathfrak{Q}_{\mathbf{F}}(\mathbf{x})$ and $\mathfrak{R}_{\mathbf{F}}(\mathbf{x})$. If a leading-bias expansion $\mathfrak{B}_{\mathbf{F}}(\mathbf{x}) = h^m B_{\mathbf{F}, \mathbf{x}} + o(h^m)$ is available and $V_{\mathbf{F}, \mathbf{x}} = nh^d \Xi_{\mathbf{F}, \mathbf{x}, \mathbf{x}}$, then the approximate pointwise criterion is

$$h^{2m} B_{\mathbf{F}, \mathbf{x}}^2 + \frac{1}{nh^d} V_{\mathbf{F}, \mathbf{x}},$$

with the same bandwidth formulas as in the sharp distance-based case after replacing $(B_{\mathbf{x}}, V_{\mathbf{x}})$ by $(B_{\mathbf{F}, \mathbf{x}}, V_{\mathbf{F}, \mathbf{x}})$.

SA-6.2.2 Distributional Approximation and Inference

Let

$$\widehat{\mathbf{w}}(\mathbf{x}) = \left(\frac{1}{\widehat{\vartheta}_W(\mathbf{x})}, -\frac{\widehat{\vartheta}_Y(\mathbf{x})}{\widehat{\vartheta}_W(\mathbf{x})^2} \right)^\top,$$

and define

$$\widehat{\Theta}_{\mathbf{F}, \mathbf{x}_1, \mathbf{x}_2} = \begin{pmatrix} \widehat{\Xi}_{Y,Y,\mathbf{x}_1,\mathbf{x}_2} & \widehat{\Xi}_{Y,W,\mathbf{x}_1,\mathbf{x}_2} \\ \widehat{\Xi}_{W,Y,\mathbf{x}_1,\mathbf{x}_2} & \widehat{\Xi}_{W,W,\mathbf{x}_1,\mathbf{x}_2} \end{pmatrix}, \quad \widehat{\Xi}_{\mathbf{F}, \mathbf{x}_1, \mathbf{x}_2} = \widehat{\mathbf{w}}(\mathbf{x}_1)^\top \widehat{\Theta}_{\mathbf{F}, \mathbf{x}_1, \mathbf{x}_2} \widehat{\mathbf{w}}(\mathbf{x}_2).$$

Lemma SA-8 (Covariance: Fuzzy BATEC). *Suppose Assumptions SA-1, SA-4, and SA-2 hold, $\inf_{\mathbf{x} \in \mathcal{B}} |\vartheta_W(\mathbf{x})| > 0$, and $1 \lesssim \inf_{\mathbf{x} \in \mathcal{B}} nh^d \Xi_{\mathbf{F}, \mathbf{x}, \mathbf{x}} \leq \sup_{\mathbf{x} \in \mathcal{B}} nh^d \Xi_{\mathbf{F}, \mathbf{x}, \mathbf{x}} \lesssim 1$. If $\frac{\log(1/h)}{n^{v/(2+v)} h^d} = o(1)$ and $h = o(1)$, then*

$$\sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} nh^d |\widehat{\Xi}_{\mathbf{F}, \mathbf{x}_1, \mathbf{x}_2} - \Xi_{\mathbf{F}, \mathbf{x}_1, \mathbf{x}_2}| = o_{\mathbb{P}}(1) \quad \text{and} \quad \sup_{\mathbf{x} \in \mathcal{B}} \left| \frac{\widehat{\Xi}_{\mathbf{F}, \mathbf{x}, \mathbf{x}}}{\Xi_{\mathbf{F}, \mathbf{x}, \mathbf{x}}} - 1 \right| = o_{\mathbb{P}}(1).$$

The feasible t -statistic and confidence interval are

$$\hat{T}_F(\mathbf{x}) = \frac{\hat{\xi}(\mathbf{x}) - \xi(\mathbf{x})}{\sqrt{\hat{\Xi}_{F,\mathbf{x},\mathbf{x}}}} \quad \text{and} \quad \hat{I}_{\alpha,F}(\mathbf{x}) = \left[\hat{\xi}(\mathbf{x}) - \mathbf{q}_\alpha \sqrt{\hat{\Xi}_{F,\mathbf{x},\mathbf{x}}}, \hat{\xi}(\mathbf{x}) + \mathbf{q}_\alpha \sqrt{\hat{\Xi}_{F,\mathbf{x},\mathbf{x}}} \right].$$

For the pointwise remainder condition, define

$$\rho_{n,\mathbf{x}} = \frac{1}{\sqrt{nh^d}} + \frac{1}{n^{\frac{1+v}{2+v}} h^d} + |\mathfrak{B}_Y(\mathbf{x})| + |\mathfrak{B}_W(\mathbf{x})|.$$

Theorem SA-17 (Confidence Intervals: Fuzzy BATEC). *For a fixed boundary point $\mathbf{x} \in \mathcal{B}$, suppose Assumptions SA-1, SA-4, and SA-2 hold, $|\vartheta_W(\mathbf{x})| > 0$, and $1 \lesssim nh^d \Xi_{F,\mathbf{x},\mathbf{x}} \lesssim 1$. If $n^{v/(2+v)} h^d \rightarrow \infty$, $h = o(1)$, $\sqrt{nh^d} |\mathfrak{B}_F(\mathbf{x})| \rightarrow 0$, and $\sqrt{nh^d} \rho_{n,\mathbf{x}}^2 \rightarrow 0$, then*

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}(\hat{T}_F(\mathbf{x}) \leq u) - \Phi(u) \right| = o(1),$$

and

$$\mathbb{P}(\xi(\mathbf{x}) \in \hat{I}_{\alpha,F}(\mathbf{x})) = 1 - \alpha + o(1),$$

provided that $\mathbf{q}_\alpha = \Phi^{-1}(1 - \alpha/2)$.

For uniform inference, define the infeasible linearized statistic

$$\bar{T}_F(\mathbf{x}) = \Xi_{F,\mathbf{x},\mathbf{x}}^{-1/2} \mathfrak{L}_F(\mathbf{x}), \quad \mathbf{x} \in \mathcal{B}.$$

Theorem SA-18 (Stochastic Linearization: Fuzzy BATEC). *Suppose the assumptions of Lemma SA-8 hold. Then*

$$\begin{aligned} \sup_{\mathbf{x} \in \mathcal{B}} |\hat{T}_F(\mathbf{x}) - \bar{T}_F(\mathbf{x})| &\lesssim_{\mathbb{P}} \sqrt{\log(1/h)} \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{v/(2+v)} h^d} \right) \\ &\quad + \sqrt{nh^d} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_F(\mathbf{x})| + \sqrt{nh^d} \rho_n^2. \end{aligned}$$

Theorem SA-19 (Gaussian Approximation for $\sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}_F(\mathbf{x})|$). *Suppose Assumptions SA-1, SA-4, and SA-2 hold, $\inf_{\mathbf{x} \in \mathcal{B}} |\vartheta_W(\mathbf{x})| > 0$, $1 \lesssim \inf_{\mathbf{x} \in \mathcal{B}} nh^d \Xi_{F,\mathbf{x},\mathbf{x}}$, $\sup_{\mathbf{x} \in \mathcal{B}} nh^d \Xi_{F,\mathbf{x},\mathbf{x}} \lesssim 1$, $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$, and $h = o(1)$. Then, for n large enough, on a possibly enlarged probability space there exists a mean-zero Gaussian process Z_F indexed by $\mathcal{B}$ with almost surely continuous sample paths such that*

$$\text{Cov}[Z_F(\mathbf{x}_1), Z_F(\mathbf{x}_2)] = \text{Cov}[\bar{T}_F(\mathbf{x}_1), \bar{T}_F(\mathbf{x}_2)], \quad \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B},$$

and

$$\mathbb{P} \left[\left| \sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}_F(\mathbf{x})| - \sup_{\mathbf{x} \in \mathcal{B}} |Z_F(\mathbf{x})| \right| > \frac{\log n}{(nh^d)^{1/6}} + \frac{(\log n)^{5/4}}{(nh^d)^{1/4}} + \frac{(\log n)^{3/2}}{(n^{v/(2+v)} h^d)^{1/2}} \right] \lesssim \frac{1}{\log n} + \frac{\log n}{n}.$$

Let $(\hat{Z}_F(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$ be a mean-zero Gaussian process conditional on $\mathbf{V} = ((\mathbf{X}_i^\top, Y_i, W_i) : 1 \leq i \leq n)$ with

covariance

$$\text{Cov}[\widehat{Z}_F(\mathbf{x}_1), \widehat{Z}_F(\mathbf{x}_2) \mid \mathbf{V}] = \frac{\widehat{\Xi}_{F, \mathbf{x}_1, \mathbf{x}_2}}{\sqrt{\widehat{\Xi}_{F, \mathbf{x}_1, \mathbf{x}_1}} \sqrt{\widehat{\Xi}_{F, \mathbf{x}_2, \mathbf{x}_2}}}, \quad \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}.$$

Theorem SA-20 (Confidence Bands: Fuzzy BATEC). *Suppose the assumptions of Lemma SA-8 hold. If $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$, $\frac{n^{v/(2+v)} h^d}{(\log n)^9} \rightarrow \infty$, $h = o(1)$, $\sqrt{nh^d \log n} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_F(\mathbf{x})| \rightarrow 0$, and $\sqrt{nh^d \log n} \rho_n^2 \rightarrow 0$, then*

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{T}_F(\mathbf{x})| \leq u \right) - \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}_F(\mathbf{x})| \leq u \mid \mathbf{V} \right) \right| = o_{\mathbb{P}}(1),$$

and

$$\mathbb{P}[\xi(\mathbf{x}) \in \widehat{\mathcal{I}}_{\alpha, F}(\mathbf{x}), \text{ for all } \mathbf{x} \in \mathcal{B}] = 1 - \alpha + o(1),$$

provided that $\mathbf{q}_\alpha = \inf\{c > 0 : \mathbb{P}(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}_F(\mathbf{x})| \geq c \mid \mathbf{V}) \leq \alpha\}$.

SA-6.3 Fuzzy WBATE

We impose the normalization $\int_{\mathcal{B}} w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}) = 1$. The distance-based fuzzy WBATE parameter and plug-in estimator are

$$\xi_{\text{WBATE}} = \int_{\mathcal{B}} \xi(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}) \quad \text{and} \quad \widehat{\xi}_{\text{WBATE}} = \int_{\mathcal{B}} \widehat{\xi}(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}).$$

Under the conditions of Theorem SA-15, $\xi_{\text{WBATE}} = \zeta_{\text{WBATE}}$, where $\zeta_{\text{WBATE}} = \int_{\mathcal{B}} \zeta(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x})$.

Integrating the pointwise fuzzy expansion (SA-7) gives

$$\widehat{\xi}_{\text{WBATE}} - \xi_{\text{WBATE}} = \mathfrak{B}_{F, \text{WBATE}} + \mathfrak{L}_{F, \text{WBATE}} + \mathfrak{Q}_{F, \text{WBATE}} + \mathfrak{R}_{F, \text{WBATE}}, \quad (\text{SA-9})$$

where

$$\begin{aligned} \mathfrak{B}_{F, \text{WBATE}} &= \int_{\mathcal{B}} \mathfrak{B}_F(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}), \\ \mathfrak{L}_{F, \text{WBATE}} &= \int_{\mathcal{B}} \mathfrak{L}_F(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}), \\ \mathfrak{Q}_{F, \text{WBATE}} &= \int_{\mathcal{B}} \mathfrak{Q}_F(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}), \\ \mathfrak{R}_{F, \text{WBATE}} &= \int_{\mathcal{B}} \mathfrak{R}_F(\mathbf{x}) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}). \end{aligned}$$

The aggregate variance of the leading linearized term is

$$\Xi_{F, \text{WBATE}} = \int_{\mathcal{B}} \int_{\mathcal{B}} \Xi_{F, \mathbf{x}_1, \mathbf{x}_2} w(\mathbf{x}_1) w(\mathbf{x}_2) d\mathfrak{H}^{d-1}(\mathbf{x}_1) d\mathfrak{H}^{d-1}(\mathbf{x}_2),$$

with feasible counterpart

$$\widehat{\Xi}_{F, \text{WBATE}} = \int_{\mathcal{B}} \int_{\mathcal{B}} \widehat{\Xi}_{F, \mathbf{x}_1, \mathbf{x}_2} w(\mathbf{x}_1) w(\mathbf{x}_2) d\mathfrak{H}^{d-1}(\mathbf{x}_1) d\mathfrak{H}^{d-1}(\mathbf{x}_2).$$

For inference, impose the aggregate nondegeneracy condition

$$\Xi_{\mathbf{F}, \text{WBATE}} \gtrsim \frac{1}{nh}, \quad (\text{SA-10})$$

which rules out cancellation in the integrated fuzzy covariance kernel.

Lemma SA-9 (Variance Linear Term: Fuzzy WBATE). *Suppose Assumptions SA-1, SA-4, SA-2, and SA-3 hold, $\inf_{\mathbf{x} \in \mathcal{B}} |\vartheta_W(\mathbf{x})| > 0$, and (SA-4) holds. If $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then*

$$\mathbb{V}[\mathfrak{L}_{\mathbf{F}, \text{WBATE}}] = \Xi_{\mathbf{F}, \text{WBATE}} \lesssim \frac{1}{nh}.$$

If, in addition, (SA-10) holds, then

$$\frac{1}{nh} \lesssim \Xi_{\mathbf{F}, \text{WBATE}} \lesssim \frac{1}{nh}.$$

If $\frac{n^{v/(2+v)}h^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then

$$\widehat{\Xi}_{\mathbf{F}, \text{WBATE}} = \Xi_{\mathbf{F}, \text{WBATE}} + o_{\mathbb{P}}\left(\frac{1}{nh}\right).$$

If, in addition, (SA-10) holds, then $\widehat{\Xi}_{\mathbf{F}, \text{WBATE}}/\Xi_{\mathbf{F}, \text{WBATE}} \rightarrow_{\mathbb{P}} 1$.

Lemma SA-10 (Higher-Order Remainders: Fuzzy WBATE). *Suppose Assumptions SA-1, SA-4, SA-2, and SA-3 hold, $\inf_{\mathbf{x} \in \mathcal{B}} |\vartheta_W(\mathbf{x})| > 0$, and (SA-4) holds. If $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ and $h = o(1)$, then*

$$\mathfrak{Q}_{\mathbf{F}, \text{WBATE}} \lesssim_{\mathbb{P}} \frac{1}{nh^d}.$$

If, in addition, $\frac{\log(1/h)}{n^{(1+v)/(2+v)}h^d} = o(1)$, then

$$|\mathfrak{R}_{\mathbf{F}, \text{WBATE}}| \lesssim_{\mathbb{P}} \rho_n^2.$$

Consequently, if also $nh^{2d-1} \rightarrow \infty$ and $\sqrt{nh}\rho_n^2 \rightarrow 0$, then

$$\mathfrak{Q}_{\mathbf{F}, \text{WBATE}} + \mathfrak{R}_{\mathbf{F}, \text{WBATE}} = o_{\mathbb{P}}\left(\frac{1}{\sqrt{nh}}\right).$$

Theorem SA-21 (Convergence Rate: Fuzzy WBATE). *Suppose Assumptions SA-1, SA-4, SA-2, and SA-3 hold, $\xi(\mathbf{x})$ is well defined for all $\mathbf{x} \in \mathcal{B}$, $\inf_{\mathbf{x} \in \mathcal{B}} |\vartheta_W(\mathbf{x})| > 0$, and (SA-4) holds. If $\frac{\log(1/h)}{n^{(1+v)/(2+v)}h^d} = o(1)$ and $h = o(1)$, then*

$$\left| \widehat{\xi}_{\text{WBATE}} - \xi_{\text{WBATE}} \right| \lesssim_{\mathbb{P}} |\mathfrak{B}_{\mathbf{F}, \text{WBATE}}| + \frac{1}{\sqrt{nh}} + \frac{1}{nh^d} + \rho_n^2.$$

For inference, define

$$\widehat{\mathbf{T}}_{\mathbf{F}, \text{WBATE}} = \frac{\widehat{\xi}_{\text{WBATE}} - \xi_{\text{WBATE}}}{\sqrt{\widehat{\Xi}_{\mathbf{F}, \text{WBATE}}}},$$

and

$$\widehat{\mathbf{I}}_{\alpha, \text{F, WBATE}} = \left[\widehat{\xi}_{\text{WBATE}} - \mathbf{q}_\alpha \sqrt{\widehat{\Xi}_{\text{F, WBATE}}}, \quad \widehat{\xi}_{\text{WBATE}} + \mathbf{q}_\alpha \sqrt{\widehat{\Xi}_{\text{F, WBATE}}} \right].$$

Theorem SA-22 (Asymptotic Normality: Fuzzy WBATE). *Suppose Assumptions SA-1, SA-4, SA-2, and SA-3 hold, $\xi(\mathbf{x})$ is well defined for all $\mathbf{x} \in \mathcal{B}$, $\inf_{\mathbf{x} \in \mathcal{B}} |\vartheta_W(\mathbf{x})| > 0$, and (SA-10) and (SA-4) hold. If $\frac{\log(1/h)}{n^{v/(2+v)}h^d} = o(1)$, $h = o(1)$, $nh^{2d-1} \rightarrow \infty$, $\sqrt{nh}|\mathfrak{B}_{\text{F, WBATE}}| \rightarrow 0$, and $\sqrt{nh}\rho_n^2 \rightarrow 0$, then*

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}(\widehat{\mathbf{T}}_{\text{F, WBATE}} \leq u) - \Phi(u) \right| = o(1),$$

and

$$\mathbb{P}[\xi_{\text{WBATE}} \in \widehat{\mathbf{I}}_{\alpha, \text{F, WBATE}}] = 1 - \alpha + o(1),$$

provided that $\mathbf{q}_\alpha = \Phi^{-1}(1 - \alpha/2)$.

SA-6.4 Fuzzy LBATE

The distance-based fuzzy LBATE parameter and plug-in estimator are

$$\xi_{\text{LBATE}} = \sup_{\mathbf{x} \in \mathcal{B}} \xi(\mathbf{x}) \quad \text{and} \quad \widehat{\xi}_{\text{LBATE}} = \sup_{\mathbf{x} \in \mathcal{B}} \widehat{\xi}(\mathbf{x}).$$

Under the conditions of Theorem SA-15, $\xi_{\text{LBATE}} = \zeta_{\text{LBATE}}$, where $\zeta_{\text{LBATE}} = \sup_{\mathbf{x} \in \mathcal{B}} \zeta(\mathbf{x})$.

Theorem SA-23 (Convergence Rate: Fuzzy LBATE). *Suppose the assumptions and conditions of the uniform part of Theorem SA-16 hold. Then*

$$\left| \widehat{\xi}_{\text{LBATE}} - \xi_{\text{LBATE}} \right| \lesssim_{\mathbb{P}} \rho_n.$$

Projecting the fuzzy BATEC confidence band gives the fuzzy LBATE confidence interval

$$\widehat{\mathbf{I}}_{\alpha, \text{F, LBATE}} = \left[\sup_{\mathbf{x} \in \mathcal{B}} \left(\widehat{\xi}(\mathbf{x}) - \mathbf{q}_\alpha \sqrt{\widehat{\Xi}_{\text{F, x, x}}} \right), \quad \sup_{\mathbf{x} \in \mathcal{B}} \left(\widehat{\xi}(\mathbf{x}) + \mathbf{q}_\alpha \sqrt{\widehat{\Xi}_{\text{F, x, x}}} \right) \right],$$

where $\mathbf{q}_\alpha = \inf\{c > 0 : \mathbb{P}(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}_{\text{F}}(\mathbf{x})| \geq c \mid \mathbf{V}) \leq \alpha\}$.

Theorem SA-24 (Confidence Intervals: Fuzzy LBATE). *Suppose the assumptions and conditions in Theorem SA-20 hold. Then*

$$\mathbb{P}[\xi_{\text{LBATE}} \in \widehat{\mathbf{I}}_{\alpha, \text{F, LBATE}}] \geq 1 - \alpha + o(1).$$

As in the sharp-design LBATE case, this interval is generally conservative because it is obtained by projecting a simultaneous confidence band for the full fuzzy BATEC.

SA-7 Gaussian Strong Approximation

We present a Gaussian strong approximation theorem, which is the key technical tool behind Theorem SA-7. The theorem builds on and generalizes the results in Cattaneo and Yu [2025]. Consider the *residual-based empirical process* given by

$$M_n[g, r] = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[g(\mathbf{x}_i) r(y_i) - \mathbb{E}[g(\mathbf{x}_i) r(y_i)] \right], \quad g \in \mathcal{G}, r \in \mathcal{R},$$

where $\mathcal{G}$ and $\mathcal{R}$ are classes of functions satisfying certain regularity conditions.

SA-7.1 Definitions for Function Spaces

Let $\mathcal{F}$ be a class of measurable functions from a probability space $(\mathbb{R}^q, \mathcal{B}(\mathbb{R}^q), \mathbb{P})$ to $\mathbb{R}$. We introduce several definitions that capture properties of $\mathcal{F}$.

- (i) $\mathcal{F}$ is pointwise measurable if it contains a countable subset $\mathcal{G}$ such that for any $f \in \mathcal{F}$, there exists a sequence $(g_m : m \geq 1) \subseteq \mathcal{G}$ such that $\lim_{m \rightarrow \infty} g_m(\mathbf{u}) = f(\mathbf{u})$ for all $\mathbf{u} \in \mathbb{R}^q$.
- (ii) Let $\text{supp}(\mathcal{F}) = \cup_{f \in \mathcal{F}} \text{supp}(f)$. A probability measure $\mathbb{Q}_{\mathcal{F}}$ on $(\mathbb{R}^q, \mathcal{B}(\mathbb{R}^q))$ is a surrogate measure for $\mathbb{P}$ with respect to $\mathcal{F}$ if

- (i) $\mathbb{Q}_{\mathcal{F}}$ agrees with $\mathbb{P}$ on $\text{supp}(\mathbb{P}) \cap \text{supp}(\mathcal{F})$.
- (ii) $\mathbb{Q}_{\mathcal{F}}(\text{supp}(\mathcal{F}) \setminus \text{supp}(\mathbb{P})) = 0$.

Let $\mathcal{Q}_{\mathcal{F}} = \text{supp}(\mathbb{Q}_{\mathcal{F}})$.

- (iii) For $q = 1$ and an interval $\mathcal{J} \subseteq \mathbb{R}$, the pointwise total variation of $\mathcal{F}$ over $\mathcal{J}$ is

$$\text{pTV}_{\mathcal{F}, \mathcal{J}} = \sup_{f \in \mathcal{F}} \sup_{P \geq 1} \sup_{\mathcal{P}_P \in \mathcal{J}} \sum_{i=1}^{P-1} |f(a_{i+1}) - f(a_i)|,$$

where $\mathcal{P}_P = \{(a_1, \dots, a_P) : a_1 \leq \dots \leq a_P\}$ denotes the collection of all partitions of $\mathcal{J}$.

- (iv) For a non-empty $\mathcal{C} \subseteq \mathbb{R}^q$, the total variation of $\mathcal{F}$ over $\mathcal{C}$ is

$$\text{TV}_{\mathcal{F}, \mathcal{C}} = \inf_{\mathcal{U} \in \mathcal{O}(\mathcal{C})} \sup_{f \in \mathcal{F}} \sup_{\phi \in \mathcal{D}_q(\mathcal{U})} \int_{\mathbb{R}^q} f(\mathbf{u}) \text{div}(\phi)(\mathbf{u}) \, d\mathbf{u} / \|\phi\|_2, \infty,$$

where $\mathcal{O}(\mathcal{C})$ denotes the collection of all open sets that contains $\mathcal{C}$, and $\mathcal{D}_q(\mathcal{U})$ denotes the space of infinitely differentiable functions from $\mathbb{R}^q$ to $\mathbb{R}^q$ with compact support contained in $\mathcal{U}$.

- (v) For a non-empty $\mathcal{C} \subseteq \mathbb{R}^q$, the local total variation constant of $\mathcal{F}$ over $\mathcal{C}$, is a positive number $K_{\mathcal{F}, \mathcal{C}}$ such that for any cube $\mathcal{D} \subseteq \mathbb{R}^q$ with edges of length ℓ parallel to the coordinate axes,

$$\text{TV}_{\mathcal{F}, \mathcal{D} \cap \mathcal{C}} \leq K_{\mathcal{F}, \mathcal{C}} \ell^{d-1}.$$

- (vi) For a non-empty $\mathcal{C} \subseteq \mathbb{R}^q$, the envelopes of $\mathcal{F}$ over $\mathcal{C}$ are

$$M_{\mathcal{F}, \mathcal{C}} = \sup_{\mathbf{u} \in \mathcal{C}} M_{\mathcal{F}, \mathcal{C}}(\mathbf{u}), \quad M_{\mathcal{F}, \mathcal{C}}(\mathbf{u}) = \sup_{f \in \mathcal{F}} |f(\mathbf{u})|, \quad \mathbf{u} \in \mathcal{C}.$$

(vii) For a non-empty $\mathcal{C} \subseteq \mathbb{R}^q$, the Lipschitz constant of $\mathcal{F}$ over $\mathcal{C}$ is

$$\mathbf{L}_{\mathcal{F},\mathcal{C}} = \sup_{f \in \mathcal{F}} \sup_{\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{C}} \frac{|f(\mathbf{u}_1) - f(\mathbf{u}_2)|}{\|\mathbf{u}_1 - \mathbf{u}_2\|_\infty}.$$

(viii) For a non-empty $\mathcal{C} \subseteq \mathbb{R}^q$, the L_1 bound of $\mathcal{F}$ over $\mathcal{C}$ is

$$\mathbf{E}_{\mathcal{F},\mathcal{C}} = \sup_{f \in \mathcal{F}} \int_{\mathcal{C}} |f| \, d\mathbb{P}.$$

(ix) For a non-empty $\mathcal{C} \subseteq \mathbb{R}^q$, the uniform covering number of $\mathcal{F}$ with envelope $M_{\mathcal{F},\mathcal{C}}$ over $\mathcal{C}$ is

$$\mathbf{N}_{\mathcal{F},\mathcal{C}}(\delta, M_{\mathcal{F},\mathcal{C}}) = \sup_{\mu} N(\mathcal{F}, \|\cdot\|_{\mu,2}, \delta \|M_{\mathcal{F},\mathcal{C}}\|_{\mu,2}), \quad \delta \in (0, \infty),$$

where the supremum is taken over all finite discrete measures on $(\mathcal{C}, \mathcal{B}(\mathcal{C}))$. We assume that $M_{\mathcal{F},\mathcal{C}}(\mathbf{u})$ is finite for every $\mathbf{u} \in \mathcal{C}$.

(x) For a non-empty $\mathcal{C} \subseteq \mathbb{R}^q$, the uniform entropy integral of $\mathcal{F}$ with envelope $M_{\mathcal{F},\mathcal{C}}$ over $\mathcal{C}$ is

$$J_{\mathcal{C}}(\delta, \mathcal{F}, M_{\mathcal{F},\mathcal{C}}) = \int_0^\delta \sqrt{1 + \log \mathbf{N}_{\mathcal{F},\mathcal{C}}(\varepsilon, M_{\mathcal{F},\mathcal{C}})} \, d\varepsilon,$$

where it is assumed that $M_{\mathcal{F},\mathcal{C}}(\mathbf{u})$ is finite for every $\mathbf{u} \in \mathcal{C}$.

(xi) For a non-empty $\mathcal{C} \subseteq \mathbb{R}^q$, $\mathcal{F}$ is a VC-type class with envelope $M_{\mathcal{F},\mathcal{C}}$ over $\mathcal{C}$ if (i) $M_{\mathcal{F},\mathcal{C}}$ is measurable and $M_{\mathcal{F},\mathcal{C}}(\mathbf{u})$ is finite for every $\mathbf{u} \in \mathcal{C}$, and (ii) there exist $\mathbf{c}_{\mathcal{F},\mathcal{C}} > 0$ and $\mathbf{d}_{\mathcal{F},\mathcal{C}} > 0$ such that

$$\mathbf{N}_{\mathcal{F},\mathcal{C}}(\varepsilon, M_{\mathcal{F},\mathcal{C}}) \leq \mathbf{c}_{\mathcal{F},\mathcal{C}} \varepsilon^{-\mathbf{d}_{\mathcal{F},\mathcal{C}}}, \quad \varepsilon \in (0, 1).$$

If a surrogate measure $\mathbb{Q}_{\mathcal{F}}$ for $\mathbb{P}$ with respect to $\mathcal{F}$ has been assumed, and it is clear from the context, we drop the dependence on $\mathcal{C} = \mathcal{Q}_{\mathcal{F}}$ for all quantities in the previous definitions. That is, to save notation, we set $\text{TV}_{\mathcal{F}} = \text{TV}_{\mathcal{F},\mathcal{Q}_{\mathcal{F}}}$, $\mathbf{K}_{\mathcal{F}} = \mathbf{K}_{\mathcal{F},\mathcal{Q}_{\mathcal{F}}}$, $\mathbf{M}_{\mathcal{F}} = \mathbf{M}_{\mathcal{F},\mathcal{Q}_{\mathcal{F}}}$, $M_{\mathcal{F}}(\mathbf{u}) = M_{\mathcal{F},\mathcal{Q}_{\mathcal{F}}}(\mathbf{u})$, $\mathbf{L}_{\mathcal{F}} = \mathbf{L}_{\mathcal{F},\mathcal{Q}_{\mathcal{F}}}$, and so on, whenever there is no confusion.

SA-7.2 Multiplicative-Separable Empirical Process

The following theorem generalizes [Cattaneo and Yu \[2025, Theorem SA.1\]](#) by requiring only bounded polynomial moments for y_i conditional on $\mathbf{x}_i$.

Theorem SA-25 (Strong Approximation for $(M_n(g, r) + M_n(h, s) : g \in \mathcal{G}, r \in \mathcal{R}, h \in \mathcal{H}, s \in \mathcal{S})$). *Suppose $(\mathbf{z}_i = (\mathbf{x}_i, y_i) : 1 \leq i \leq n)$ are i.i.d. random vectors taking values in $(\mathbb{R}^{d+1}, \mathcal{B}(\mathbb{R}^{d+1}))$ with common law $\mathbb{P}_Z$, where $\mathbf{x}_i$ has distribution $\mathbb{P}_X$ supported on $\mathcal{X} \subseteq \mathbb{R}^d$, y_i has distribution $\mathbb{P}_Y$ supported on $\mathcal{Y} \subseteq \mathbb{R}$, $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[|y_i|^{2+v} | \mathbf{x}_i = \mathbf{x}] \leq 2$ for some $v > 0$, and the following conditions hold.*

- (i) $\mathcal{G}$ and $\mathcal{H}$ are real-valued pointwise measurable classes of functions on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mathbb{P}_X)$.
- (ii) There exists a surrogate measure $\mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}$ for $\mathbb{P}_X$ with respect to $\mathcal{G} \cup \mathcal{H}$ such that $\mathbb{Q}_{\mathcal{G} \cup \mathcal{H}} = \mathbf{m} \circ \phi_{\mathcal{G} \cup \mathcal{H}}$, where the normalizing transformation $\phi_{\mathcal{G} \cup \mathcal{H}} : \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}} \rightarrow [0, 1]^d$ is a diffeomorphism.

- (iii) $\mathcal{G}$ is a VC-type class with envelope $M_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}$ over $\mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}$ with $c_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} \geq e$ and $d_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} \geq 1$. $\mathcal{H}$ is a VC-type class with envelope $M_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}$ over $\mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}$ with $c_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} \geq e$ and $d_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} \geq 1$.
- (iv) $\mathcal{R}$ and $\mathcal{S}$ are real-valued pointwise measurable classes of functions on $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_Y)$.
- (v) $\mathcal{R}$ is a VC-type class with envelope $M_{\mathcal{R}, \mathcal{Y}}$ over $\mathcal{Y}$ with $c_{\mathcal{R}, \mathcal{Y}} \geq e$ and $d_{\mathcal{R}, \mathcal{Y}} \geq 1$, where $M_{\mathcal{R}, \mathcal{Y}}(y) + \text{pTV}_{\mathcal{R}, (-|y|, |y|)} \leq v(1 + |y|)$ for all $y \in \mathcal{Y}$, for some $v > 0$. $\mathcal{S}$ is a VC-type class with envelope $M_{\mathcal{S}, \mathcal{Y}}$ over $\mathcal{Y}$ with $c_{\mathcal{S}, \mathcal{Y}} \geq e$ and $d_{\mathcal{S}, \mathcal{Y}} \geq 1$, where $M_{\mathcal{S}, \mathcal{Y}}(y) + \text{pTV}_{\mathcal{S}, (-|y|, |y|)} \leq v(1 + |y|)$ for all $y \in \mathcal{Y}$, for some $v > 0$.
- (vi) There exists a constant k such that $|\log_2 E| + |\log_2 \text{TV}| + |\log_2 M| + |\log_2 L| \leq k \log_2(n)$, where $E = \max\{E_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}, E_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}\}$, $\text{TV} = \max\{\text{TV}_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}, \text{TV}_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}\}$, $M = \max\{M_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}, M_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}\}$, and $L = \max\{L_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}, L_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}\}$.

Consider the empirical process

$$A_n(g, h, r, s) = M_n(g, r) + M_n(h, s), \quad g \in \mathcal{G}, r \in \mathcal{R}, h \in \mathcal{H}, s \in \mathcal{S}.$$

Then, on a possibly enlarged probability space, there exists a sequence of mean-zero Gaussian processes $(Z_n^A(g, h, r, s) : g \in \mathcal{G}, h \in \mathcal{H}, r \in \mathcal{R}, s \in \mathcal{S})$ with almost sure continuous trajectories such that:

- $\mathbb{E}[A_n(g_1, h_1, r_1, s_1)A_n(g_2, h_2, r_2, s_2)] = \mathbb{E}[Z_n^A(g_1, h_1, r_1, s_1)Z_n^A(g_2, h_2, r_2, s_2)]$ holds for all $(g_1, h_1, r_1, s_1), (g_2, h_2, r_2, s_2) \in \mathcal{G} \times \mathcal{H} \times \mathcal{R} \times \mathcal{S}$, and
- $\mathbb{E}[\|A_n - Z_n^A\|_{\mathcal{G} \times \mathcal{H} \times \mathcal{R} \times \mathcal{S}}] \leq Cv((d \log(cn))^{\frac{3}{2}} r_n^{\frac{v}{v+2}} (\sqrt{ME})^{\frac{2}{v+2}} + d \log(cn) M n^{-\frac{v/2}{2+v}} + d \log(cn) M n^{-\frac{1}{2}} \left(\frac{\sqrt{ME}}{r_n}\right)^{\frac{2}{v+2}}),$

where C is a universal constant, $c = c_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} + c_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} + c_{\mathcal{R}, \mathcal{Y}} + c_{\mathcal{S}, \mathcal{Y}} + k$, $d = d_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} d_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} d_{\mathcal{R}, \mathcal{Y}} d_{\mathcal{S}, \mathcal{Y}} k$,

$$r_n = \min \left\{ \frac{(c_1^d M^{d+1} \text{TV}^d E)^{1/(2d+2)}}{n^{1/(2d+2)}}, \frac{(c_1^{\frac{d}{2}} c_2^{\frac{d}{2}} M \text{TV}^{\frac{d}{2}} E L^{\frac{d}{2}})^{1/(d+2)}}{n^{1/(d+2)}} \right\},$$

$$c_1 = d \sup_{\mathbf{x} \in \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} \prod_{j=1}^{d-1} \sigma_j(\nabla \phi_{\mathcal{G} \cup \mathcal{H}}(\mathbf{x})), \quad c_2 = \sup_{\mathbf{x} \in \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} \frac{1}{\sigma_d(\nabla \phi_{\mathcal{G} \cup \mathcal{H}}(\mathbf{x}))}.$$

SA-7.3 Proof of Theorem SA-25

Let $\mathcal{I}_A = \mathcal{G} \times \mathcal{H} \times \mathcal{R} \times \mathcal{S}$. For each $\iota = (g, h, r, s) \in \mathcal{I}_A$, define the combined summand

$$a_\iota(\mathbf{x}, y) = g(\mathbf{x})r(y) + h(\mathbf{x})s(y), \quad \mathcal{I}_A = \{a_\iota : \iota \in \mathcal{I}_A\}.$$

The proof below adapts the projection and coupling construction in Cattaneo and Yu [2025, Theorem SA.1]. The adaptation has two purposes. First, the two empirical processes cannot be coupled separately, because doing so would generally lose the covariance between $M_n(g, r)$ and $M_n(h, s)$. Second, a direct generic strong approximation for the ordinary class $\mathcal{I}_A$ would treat $(\mathbf{x}, y)$ symmetrically and would therefore not exploit the multiplicative-separable structure. We instead use one common normalizing transformation, one common projection, and one common Gaussian construction for the whole sum a_ι .

Reduction and common normalization. Let $\tilde{\mathcal{G}} = \{g \circ \phi_{\mathcal{G} \cup \mathcal{H}}^{-1} : g \in \mathcal{G}\}$ and $\tilde{\mathcal{H}} = \{h \circ \phi_{\mathcal{G} \cup \mathcal{H}}^{-1} : h \in \mathcal{H}\}$. By the surrogate measure and normalizing-transformation argument used in the proof of Cattaneo and Yu

[2025, Theorem SA.1], after enlarging the probability space if necessary, we may work with i.i.d. variables $\mathbf{u}_i \sim \text{Uniform}([0, 1]^d)$ such that, simultaneously for all $g \in \mathcal{G}$ and $h \in \mathcal{H}$,

$$g(\mathbf{x}_i) = g \circ \phi_{\mathcal{G} \cup \mathcal{H}}^{-1}(\mathbf{u}_i), \quad h(\mathbf{x}_i) = h \circ \phi_{\mathcal{G} \cup \mathcal{H}}^{-1}(\mathbf{u}_i).$$

This reduction is simultaneous for $\mathcal{G}$ and $\mathcal{H}$, which is why the surrogate measure is imposed on $\mathcal{G} \cup \mathcal{H}$. The constants $\mathbf{c}_1$ and $\mathbf{c}_2$ in the statement record the effect of this change of variables on the total-variation and Lipschitz constants; all displays below are written for the normalized variables and translated back through these constants.

Entropy and envelopes. The classes $\tilde{\mathcal{G}}$ and $\tilde{\mathcal{H}}$ remain pointwise measurable and VC-type after the change of variables. Products with $\mathcal{R}$ and $\mathcal{S}$ preserve VC-type entropy, and finite sums preserve the same property, so $\mathcal{F}_A$ is pointwise measurable and VC-type. For $\iota = (g, h, r, s)$,

$$|a_\iota(\mathbf{x}, y)| \leq \mathbf{v}(1 + |y|)(M_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}(\mathbf{x}) + M_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}(\mathbf{x})).$$

Consequently the entropy constants for the combined index class are bounded by fixed powers of the constants appearing in the theorem; these powers are absorbed into $\mathbf{c}$ and $\mathbf{d}$.

Total Variation. The x -side L_1 , envelope, Lipschitz, and total-variation quantities entering the projection argument are controlled by the corresponding maxima $\mathbf{E}$, $\mathbf{M}$, $\mathbf{L}$, and $\mathbf{TV}$. For example, for each fixed y and each cube $\mathcal{D} \subseteq [0, 1]^d$,

$$\begin{aligned} \mathbf{TV}_{\{a_\iota(\cdot, y) : \iota \in \mathcal{F}_A\}, \mathcal{D}} &\leq \mathbf{v}(1 + |y|)(\mathbf{TV}_{\mathcal{G}, \mathcal{D}} + \mathbf{TV}_{\mathcal{H}, \mathcal{D}}), \\ \mathbf{L}_{\{a_\iota(\cdot, y) : \iota \in \mathcal{F}_A\}, \mathcal{D}} &\leq \mathbf{v}(1 + |y|)(\mathbf{L}_{\mathcal{G}, \mathcal{D}} + \mathbf{L}_{\mathcal{H}, \mathcal{D}}). \end{aligned}$$

The analogous pointwise total-variation bound in the y coordinate follows from Assumption (v):

$$\mathbf{pTV}_{\{a_\iota(\mathbf{x}, \cdot) : \iota \in \mathcal{F}_A\}, (-|y|, |y|)} \leq \mathbf{v}(1 + |y|)(M_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}(\mathbf{x}) + M_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}(\mathbf{x})).$$

Thus the meshing, projection, and Gaussian-projection constants in the proof of Cattaneo and Yu [2025, Theorem SA.1] are bounded by the same expressions as in that theorem, with $\mathcal{G}$ replaced by the worst case between $\mathcal{G}$ and $\mathcal{H}$ and with $\mathcal{R}$ replaced by the worst case between $\mathcal{R}$ and $\mathcal{S}$.

Common projection and coupling. Let π_δ denote a δ -net projection of $\mathcal{F}_A$ under the $L_2(\mathbb{P}_Z)$ semimetric induced by a_ι , and let Π denote the common cylindered Haar projection used in the proof of Cattaneo and Yu [2025, Theorem SA.1], with main depth in the normalized $\mathbf{u}$ coordinates and multiplier depth in the y coordinate. The same projected cells are used for the two summands $g(\mathbf{x})r(y)$ and $h(\mathbf{x})s(y)$. Therefore the projected empirical vector is the projection of the sum, not the sum of two independently coupled projections. For the Gaussian process Z_n^A generated by this common construction,

$$\begin{aligned} \|A_n - Z_n^A\|_{\mathcal{F}_A} &\leq \|A_n - A_n \circ \pi_\delta\|_{\mathcal{F}_A} + \|A_n \circ \pi_\delta - \Pi A_n \circ \pi_\delta\|_{\mathcal{F}_A} \\ &\quad + \|\Pi A_n \circ \pi_\delta - \Pi Z_n^A \circ \pi_\delta\|_{\mathcal{F}_A} + \|\Pi Z_n^A \circ \pi_\delta - Z_n^A \circ \pi_\delta\|_{\mathcal{F}_A} + \|Z_n^A \circ \pi_\delta - Z_n^A\|_{\mathcal{F}_A}. \end{aligned}$$

The first and last terms are the empirical and Gaussian meshing errors. The second and fourth terms are the empirical and Gaussian projection errors. The middle term is the projected coupling error. Applying the same bounds as in the proof of Cattaneo and Yu [2025, Theorem SA.1] to this single projected vector,

and then optimizing over the quasi-dyadic projection depths, gives, in the bounded-transform version of Assumption (v),

$$\mathbb{E}[\|A_n - Z_n^A\|_{\mathcal{F}_A}] \lesssim \mathbf{v}(\mathbf{d} \log(cn))^{3/2} \mathbf{r}_n.$$

The rate $\mathbf{r}_n$ is the minimum of the two projection/coupling balances displayed in the theorem. The first balance uses total variation; the second additionally uses the Lipschitz constants and yields the improvement available for Lipschitz classes.

Polynomial moments. It remains to remove the bounded-transform simplification. For $\eta > 0$, define truncated classes $r^\eta(y) = r(y)\mathbf{1}(|y| \leq \eta)$ and $s^\eta(y) = s(y)\mathbf{1}(|y| \leq \eta)$, and let A_n^η denote the corresponding process. By Assumption (v) and $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[|y_i|^{2+v} \mid \mathbf{x}_i = \mathbf{x}] \leq 2$,

$$\sup_{r \in \mathcal{R}} \sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[|r(y_i)|^2 \mathbf{1}(|y_i| > \eta) \mid \mathbf{x}_i = \mathbf{x}] \lesssim \mathbf{v}^2 \eta^{-v},$$

and the same bound holds for $\mathcal{S}$. The empirical tail, the corresponding Gaussian tail, and the deterministic centering error are then controlled by Bernstein's inequality, Markov's inequality, and the preceding VC entropy bounds, giving

$$\begin{aligned} \mathbb{E}[\|A_n - Z_n^A\|_{\mathcal{F}_A}] &\lesssim \mathbf{v} \left\{ (\mathbf{d} \log(cn))^{3/2} (\eta \mathbf{r}_n + \sqrt{\mathbf{M}\mathbf{E}} \eta^{-v/2}) \right. \\ &\quad \left. + \mathbf{d} \log(cn) \mathbf{M} n^{-\frac{v/2}{2+v}} + \mathbf{d} \log(cn) \mathbf{M} n^{-1/2} \eta \right\}. \end{aligned}$$

Choosing

$$\eta = \left(\frac{\sqrt{\mathbf{M}\mathbf{E}}}{\mathbf{r}_n} \right)^{2/(v+2)}$$

balances the bounded-transform approximation error with the stochastic tail error, and yields exactly the bound in the theorem.

Finally, the common Gaussian construction is a Brownian-bridge construction indexed by the combined class $\mathcal{F}_A$. Hence, for any two indices $\iota_1, \iota_2 \in \mathcal{F}_A$,

$$\mathbb{E}[Z_n^A(\iota_1) Z_n^A(\iota_2)] = \mathbb{E}[A_n(\iota_1) A_n(\iota_2)].$$

This covariance identity includes the cross-covariance between $M_n(g, r)$ and $M_n(h, s)$ because the whole sum was projected and coupled jointly. Almost sure continuity of the sample paths follows from the same separability and semimetric-continuity argument used in [Cattaneo and Yu \[2025, Theorem SA.1\]](#), since the combined class is pointwise measurable and VC-type under the envelope above. This proves the theorem. $\square$

SA-8 Proofs

SA-8.1 Proof of Lemma [SA-1](#)

For each j , the class $\{\mathbf{v} \in \mathcal{X} : f(\mathbf{v}) \geq 0, f \in \mathcal{F}_j\}$ is VC because $\mathcal{F}_j$ is finite dimensional. Fixed finite Boolean combinations of VC classes are VC; see, for example, the closure properties in [van der Vaart and Wellner \[1996, Section 2.6\]](#). The displayed representation therefore implies that $\mathcal{D}_d$ is VC. If the balls are semialgebraic with uniformly bounded degree and Boolean complexity, the relevant polynomial inequalities belong to finite-dimensional spaces spanned by the monomials in $\mathbf{v}$ up to the maximal degree, and the same

argument applies. The listed distances are special cases. $\square$

SA-8.2 Proof of Lemma SA-2

Let $\varsigma_t = 2t - 1$. On $\{\mathbf{X}_i \in \mathcal{A}_t\}$, $D_i(\mathbf{x}) = \varsigma_t \|\mathbf{X}_i - \mathbf{x}\|$. Assumption SA-1(ii) implies

$$\begin{aligned} \Psi_{t,\mathbf{x}} &= \mathbb{E} \left[\mathbf{r}_p \left(\frac{\varsigma_t \|\mathbf{X}_i - \mathbf{x}\|}{h} \right) \mathbf{r}_p \left(\frac{\varsigma_t \|\mathbf{X}_i - \mathbf{x}\|}{h} \right)^\top K_h(\varsigma_t \|\mathbf{X}_i - \mathbf{x}\|) \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t) \right] \\ &= \int_{\mathcal{A}_t} \mathbf{r}_p \left(\frac{\varsigma_t \|\mathbf{u} - \mathbf{x}\|}{h} \right) \mathbf{r}_p \left(\frac{\varsigma_t \|\mathbf{u} - \mathbf{x}\|}{h} \right)^\top K_h(\varsigma_t \|\mathbf{u} - \mathbf{x}\|) f_X(\mathbf{u}) d\mathbf{u} \\ &= f_X(\mathbf{x}) \int_{\mathcal{A}_t} \mathbf{r}_p \left(\frac{\varsigma_t \|\mathbf{u} - \mathbf{x}\|}{h} \right) \mathbf{r}_p \left(\frac{\varsigma_t \|\mathbf{u} - \mathbf{x}\|}{h} \right)^\top K_h(\varsigma_t \|\mathbf{u} - \mathbf{x}\|) d\mathbf{u} + o(1), \end{aligned}$$

where in the last line we have used $\int_{\mathcal{A}_t} \left(\frac{\|\mathbf{u} - \mathbf{x}\|}{h} \right)^v K_h(\varsigma_t \|\mathbf{u} - \mathbf{x}\|) d\mathbf{u} = O(1)$ for each $0 \leq v \leq 2p$ from a standard change of variable argument.

I. Polynomial Representation of Minimum Eigenvalue

For simplicity, write

$$\mathbf{S}_{t,\mathbf{x}}(h) = \int_{\mathcal{A}_t} \mathbf{r}_p \left(\frac{\varsigma_t \|\mathbf{u} - \mathbf{x}\|}{h} \right) \mathbf{r}_p \left(\frac{\varsigma_t \|\mathbf{u} - \mathbf{x}\|}{h} \right)^\top K_h(\varsigma_t \|\mathbf{u} - \mathbf{x}\|) d\mathbf{u}.$$

A change of variable gives

$$\mathbf{S}_{t,\mathbf{x}}(h) = \int \mathbf{r}_p(\varsigma_t \|\mathbf{z}\|) \mathbf{r}_p(\varsigma_t \|\mathbf{z}\|)^\top K(\varsigma_t \|\mathbf{z}\|) \mathbf{1}(\mathbf{x} + h\mathbf{z} \in \mathcal{A}_t) d\mathbf{z}.$$

Let $\mathbf{a} \in \mathbb{R}^{p+1}$. Condition (c) in Lemma SA-2 implies the lower bound used below. If K is Lipschitz, then it is continuous, and positivity on the compact set U implies $\inf_{\mathbf{u} \in U} K(\varsigma_t \|\mathbf{u}\|) = \kappa > 0$. If $K(u) = \mathbf{1}(u \in [-1, 1])$, positivity on U implies the same conclusion with $\kappa = 1$. Then the equivalent representation of minimum eigenvalue gives

$$\begin{aligned} \lambda_{\min}(\mathbf{S}_{t,\mathbf{x}}(h)) &= \min_{\|\mathbf{a}\|=1} \int (\mathbf{a}^\top \mathbf{r}_p(\varsigma_t \|\mathbf{z}\|))^2 K(\varsigma_t \|\mathbf{z}\|) \mathbf{1}(\mathbf{x} + h\mathbf{z} \in \mathcal{A}_t) d\mathbf{z} \\ &\geq \kappa \min_{\|\mathbf{a}\|=1} \int_U (\mathbf{a}^\top \mathbf{r}_p(\varsigma_t \|\mathbf{z}\|))^2 \mathbf{1}(\mathbf{x} + h\mathbf{z} \in \mathcal{A}_t) d\mathbf{z}, \end{aligned} \tag{SA-11}$$

where the sign ς_t only premultiplies the polynomial basis by a nonsingular diagonal matrix, so the lower bound in terms of $\mathbf{r}_p(\|\mathbf{z}\|)$ below is equivalent up to fixed constants.

II. Mass Retaining Ratio in Treatment/Control Region

Denote $E_h(\mathbf{x}, t) = \{\mathbf{z} \in U : \mathbf{x} + h\mathbf{z} \in \mathcal{A}_t\}$. Assumption SA-2(iii) implies there is some upper bound $\Lambda > 0$ of $K(\cdot)$. Hence for $c_0 = 1/2 \liminf_{h \downarrow 0} \inf_{\mathbf{x} \in \mathcal{B}} \int_U K(\varsigma_t \|\mathbf{u}\|) \mathbf{1}(\mathbf{x} + h\mathbf{u} \in \mathcal{A}_t) d\mathbf{u}$, we have

$$\Lambda \mathbf{m}(E_h(\mathbf{x}, t)) \geq \int_U K(\varsigma_t \|\mathbf{u}\|) \mathbf{1}(\mathbf{x} + h\mathbf{u} \in \mathcal{A}_t) d\mathbf{u} \geq c_0$$

for small enough h , which implies

$$\mathbf{m}(E_h(\mathbf{x}, t)) \geq \alpha \mathbf{m}(U), \quad \alpha = \frac{c_0}{\Lambda \mathbf{m}(U)}. \quad (\text{SA-12})$$

III. L_2 Integral of Polynomials in Full Versus Treatment/Control Regions

Consider $S = \{q \in \mathcal{P}_p : \int_U q(\|\mathbf{u}\|)^2 d\mathbf{u} = 1\}$, where $\mathcal{P}_p$ is the collection of all polynomials of degree at most p . Equip $\mathcal{P}_p$ with the inner product induced by $\langle q_1, q_2 \rangle_U = \int_U q_1(\|\mathbf{u}\|) q_2(\|\mathbf{u}\|) d\mathbf{u}$. Let $(\phi_j, 1 \leq j \leq p+1)$ be an orthonormal basis for this finite-dimensional space. Then $T(\mathbf{a}) = \sum_{j=1}^{p+1} a_j \phi_j$ maps the Euclidean unit sphere in $\mathbb{R}^{p+1}$ onto S . Hence S is compact under this $L_2(U)$ norm, and finite-dimensional norm equivalence implies that S is also compact under the corresponding supremum norm on U . Now consider

$$\Phi_q(\varepsilon) = \mathbf{m}(\{\mathbf{u} \in U : |q(\|\mathbf{u}\|)| < \varepsilon\}), \quad q \in S, \varepsilon > 0,$$

and

$$\psi(q) = \sup \left\{ \varepsilon > 0 : \Phi_q(\varepsilon) \leq \frac{\alpha}{2} \mathbf{m}(U) \right\}.$$

Since $\int_U q(\|\mathbf{u}\|)^2 d\mathbf{u} = 1$ and q is a polynomial, $\lim_{\varepsilon \downarrow 0} \Phi_q(\varepsilon) = 0$ and $\Phi_q(\varepsilon) = \mathbf{m}(U)$ for all sufficiently large ε . Continuity of $q \in S$ implies $\psi(q) > 0$ for all $q \in S$.

Next, we want to show ψ is a lower-semicontinuous function on $(\mathcal{P}_p, \|\cdot\|_{L_\infty})$. Suppose $q_n \rightarrow q$ uniformly on U . For every $\varepsilon_0 \in (0, \psi(q))$, choose $\eta > \varepsilon_0$ such that $\Phi_q(\eta) \leq \frac{\alpha}{2} \mathbf{m}(U)$. For large enough n , uniform convergence gives $|q_n(\|\mathbf{u}\|) - q(\|\mathbf{u}\|)| \leq \eta - \varepsilon_0$ on U , so $\{|q_n(\|\cdot\|)| < \varepsilon_0\} \subseteq \{|q(\|\cdot\|)| < \eta\}$. Hence $\Phi_{q_n}(\varepsilon_0) \leq \Phi_q(\eta) \leq \frac{\alpha}{2} \mathbf{m}(U)$, which implies $\varepsilon_0 \leq \psi(q_n)$. This implies $\liminf_{n \rightarrow \infty} \psi(q_n) \geq \varepsilon_0$. Since ε_0 is arbitrary in $(0, \psi(q))$, we have $\liminf_{n \rightarrow \infty} \psi(q_n) \geq \psi(q)$.

Compactness of S and lower-semicontinuity of ψ implies ψ attains its minimum on S . Since $\psi(q) > 0$ for all $q \in S$, we know $\varepsilon_* = \inf_{q \in S} \psi(q) > 0$. Then for every $q \in S$,

$$\begin{aligned} \int_{E_h(\mathbf{x}, t)} q(\|\mathbf{u}\|)^2 d\mathbf{u} &\geq \left(\frac{\varepsilon_*}{2}\right)^2 \mathbf{m}\left(E_h(\mathbf{x}, t) \setminus \{\mathbf{u} \in U : |q(\|\mathbf{u}\|)| \leq \varepsilon_*/2\}\right) \\ &\geq \left(\frac{\varepsilon_*}{2}\right)^2 \left(\mathbf{m}(E_h(\mathbf{x}, t)) - \mathbf{m}(\{\mathbf{u} \in U : |q(\|\mathbf{u}\|)| \leq \varepsilon_*/2\})\right) \\ &\geq \frac{\varepsilon_*^2 \alpha}{8} \mathbf{m}(U). \end{aligned}$$

Scaling q from S gives

$$\int_{E_h(\mathbf{x}, t)} q(\|\mathbf{u}\|)^2 d\mathbf{u} \geq \frac{\varepsilon_*^2 \alpha}{8} \int_U q(\|\mathbf{u}\|)^2 d\mathbf{u}, \quad q \in \mathcal{P}_p. \quad (\text{SA-13})$$

IV. Lower Bound of Minimum Eigenvalue

Equations (SA-11), (SA-12) and (SA-13) together give for small enough h ,

$$\begin{aligned} \inf_{\mathbf{x} \in \mathcal{B}} \lambda_{\min}(\mathbf{S}_{t, \mathbf{x}}(h)) &\geq \kappa \inf_{\mathbf{x} \in \mathcal{B}} \min_{\|\mathbf{a}\|=1} \int_{E_h(\mathbf{x}, t)} (\mathbf{a}^\top \mathbf{r}_p(\|\mathbf{z}\|))^2 d\mathbf{z} \\ &\geq \kappa \frac{\varepsilon_*^2 \alpha}{8} \min_{\|\mathbf{a}\|=1} \int_U (\mathbf{a}^\top \mathbf{r}_p(\|\mathbf{z}\|))^2 d\mathbf{z} \end{aligned}$$

$$\geq \kappa \frac{\varepsilon_*^2 \alpha}{8} \lambda_{\min} \left(\int_U \mathbf{r}_p(\|\mathbf{z}\|) \mathbf{r}_p(\|\mathbf{z}\|)^\top d\mathbf{z} \right),$$

which implies $\liminf_{h \rightarrow 0} \inf_{\mathbf{x} \in \mathcal{B}} \lambda_{\min}(\mathbf{S}_{t,\mathbf{x}}(h)) > 0$. Because f_X is continuous and bounded away from zero on $\mathcal{X}$ and $\Psi_{t,\mathbf{x}} = f_X(\mathbf{x})\mathbf{S}_{t,\mathbf{x}}(h) + o(1)$ uniformly in $\mathbf{x} \in \mathcal{B}$, the same lower bound holds for $\Psi_{t,\mathbf{x}}$. This proves Assumption SA-2(iv). $\square$

SA-8.3 Proof of Lemma SA-3

Since $\widehat{\Psi}_{t,\mathbf{x}}$ is a finite dimensional matrix, it suffices to show the stated rate of convergence for each entry. For $0 \leq v \leq 2p$, define $\mathcal{G} = \{g_{n,t,v}(\cdot, \mathbf{x}) : \mathbf{x} \in \mathcal{X}\}$ with

$$g_{n,t,v}(\xi, \mathbf{x}) = \left(\frac{\varsigma_t \mathcal{d}(\xi, \mathbf{x})}{h} \right)^v \frac{1}{h^d} K \left(\frac{\varsigma_t \mathcal{d}(\xi, \mathbf{x})}{h} \right) \mathbf{1}(\xi \in \mathcal{A}_t), \quad \xi, \mathbf{x} \in \mathcal{X}.$$

These are the signed-distance matrix entries, up to the finite choice of v . We will show $\mathcal{G}$ is a VC-type class.

Constant Envelope Function. Since K is bounded and compactly supported, there exists a constant C_1 such that $\sup_{\mathbf{x} \in \mathcal{X}} \|g_{n,t,v}(\cdot, \mathbf{x})\|_\infty \leq C_1 h^{-d} = G$.

Diameter of $\mathcal{G}$ in L_2 . Let $R_K < \infty$ satisfy $\text{supp}(K) \subseteq [-R_K, R_K]$. For each $\mathbf{x} \in \mathcal{X}$, $g_{n,t,v}(\cdot, \mathbf{x})$ is supported on $\{\xi : \mathcal{d}(\xi, \mathbf{x}) \leq R_K h\}$. By Assumption SA-1(ii) and Assumption SA-2(i), $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{P}(\mathcal{d}(\mathbf{X}_i, \mathbf{x}) \leq R_K h) \lesssim h^d$. It follows that $\sup_{\mathbf{x} \in \mathcal{X}} \|g_{n,t,v}(\cdot, \mathbf{x})\|_{\mathbb{P},2} \leq C_2 h^{-d/2}$ for some constant C_2 . We can take C_1 large enough so that $\sigma = C_2 h^{-d/2} \leq G = C_1 h^{-d}$.

Ratio. For some constant C_3 , $\delta = \frac{\sigma}{G} = C_3 \sqrt{h^d}$.

Covering Numbers. Case 1: K is Lipschitz. Let $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$. By Assumption SA-2,

$$\begin{aligned} & \sup_{\xi \in \mathcal{X}} |g_{n,t,v}(\xi, \mathbf{x}) - g_{n,t,v}(\xi, \mathbf{x}')| \\ & \leq \sup_{\xi \in \mathcal{X}} \left| \left[\left(\frac{\varsigma_t \mathcal{d}(\xi, \mathbf{x})}{h} \right)^v - \left(\frac{\varsigma_t \mathcal{d}(\xi, \mathbf{x}')}{h} \right)^v \right] K_h(\varsigma_t \mathcal{d}(\xi, \mathbf{x})) \right| \mathbf{1}(\xi \in \mathcal{A}_t) \\ & \quad + \left| \left(\frac{\varsigma_t \mathcal{d}(\xi, \mathbf{x}')}{h} \right)^v \left[K_h(\varsigma_t \mathcal{d}(\xi, \mathbf{x})) - K_h(\varsigma_t \mathcal{d}(\xi, \mathbf{x}')) \right] \right| \mathbf{1}(\xi \in \mathcal{A}_t) \\ & \lesssim h^{-d-1} \|\mathbf{x} - \mathbf{x}'\|_\infty. \end{aligned}$$

By Lipschitz continuity property of $\mathcal{G}$, for any $\varepsilon \in (0, 1]$ and for any finitely supported measure Q and metric $\|\cdot\|_{Q,2}$ based on $L_2(Q)$,

$$N(\{g_{n,t,v}(\cdot, \mathbf{x}) : \mathbf{x} \in \mathcal{X}\}, \|\cdot\|_{Q,2}, \varepsilon \|G\|_{Q,2}) \leq N(\mathcal{X}, \|\cdot\|_\infty, \varepsilon \|G\|_{Q,2} h^{d+1}) \stackrel{(i)}{\lesssim} \left(\frac{\text{diam}(\mathcal{X})}{\varepsilon \|G\|_{Q,2} h^{d+1}} \right)^d \lesssim \left(\frac{\text{diam}(\mathcal{X})}{\varepsilon h} \right)^d,$$

where inequality (i) uses the fact that $\varepsilon \|G\|_{Q,2} h^{d+1} \lesssim \varepsilon h \lesssim 1$. Therefore, for constants C_1 and C_2 independent of Q ,

$$\sup_Q N(\mathcal{G}, \|\cdot\|_{Q,2}, \varepsilon \|G\|_{Q,2}) \leq (C_1/\varepsilon)^{C_2}, \quad \varepsilon \in (0, 1],$$

where the supremum is taken over all finite discrete measures on $\mathcal{X}$.

Case 2: $K = \mathbf{1}(\cdot \in [-1, 1])$. Consider

$$m_n(\xi, \mathbf{x}) = \left(\frac{\varsigma_t \mathcal{d}(\xi, \mathbf{x})}{h} \right)^v \frac{1}{h^d} \mathbf{1}(\xi \in \mathcal{A}_t) \mathbf{1}(\mathcal{d}(\xi, \mathbf{x}) \leq h), \quad \xi, \mathbf{x} \in \mathcal{X},$$

$\mathcal{M} = \{m_n(\cdot, \mathbf{x}) : \mathbf{x} \in \mathcal{B}\}$ and the constant envelope function $M = C_4 h^{-d}$, for some constant C_4 depending only on the diameter of $\mathcal{X}$. The polynomial factor is covered by the preceding Lipschitz argument. The class $\mathcal{L} = \{\mathbf{1}(\mathcal{d}(\cdot, \mathbf{x}) \leq h) : \mathbf{x} \in \mathcal{B}\}$ has finite VC dimension by Assumption SA-2(iii) (for the Euclidean distance, see van der Vaart and Wellner, 1996, Example 2.6.1); products with the fixed assignment indicator preserve VC-type entropy. Hence, by van der Vaart and Wellner [1996, Theorem 2.6.4],

$$\sup_Q N(\mathcal{G}, \|\cdot\|_{Q,2}, \varepsilon \|G\|_{Q,2}) \leq (C_1/\varepsilon)^{C_2}, \quad \varepsilon \in (0, 1],$$

where the supremum is taken over all finite discrete measures on $\mathcal{X}$.

Maximal Inequality. By Chernozhukov et al. [2014b, Corollary 5.1] for the empirical process on class $\mathcal{G}$,

$$\begin{aligned} \mathbb{E} \left[\sup_{l \in \mathcal{G}} |\mathbb{E}_n[l(\mathbf{X}_i)] - \mathbb{E}[l(\mathbf{X}_i)]| \right] &\lesssim \frac{\sigma}{\sqrt{n}} \sqrt{C_2 \log(C_1/\delta)} + \frac{\|G\|_{\mathbb{P},2} C_2 \log(C_1/\delta)}{n} \\ &\lesssim \frac{1}{\sqrt{nh^d}} \sqrt{d \log \left(\frac{\text{diam}(\mathcal{X})}{h^{1+d/2}} \right)} + \frac{1}{nh^d} d \log \left(\frac{\text{diam}(\mathcal{X})}{h^{1+d/2}} \right) \\ &\lesssim \sqrt{\frac{\log(1/h)}{nh^d}}. \end{aligned}$$

Thus, $\sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\Psi}_{t,\mathbf{x}} - \Psi_{t,\mathbf{x}}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}$.

By Weyl's Theorem, $\sup_{\mathbf{x} \in \mathcal{B}} |\lambda_{\min}(\hat{\Psi}_{t,\mathbf{x}}) - \lambda_{\min}(\Psi_{t,\mathbf{x}})| \leq \sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\Psi}_{t,\mathbf{x}} - \Psi_{t,\mathbf{x}}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}$. Therefore, we can lower bound the minimum eigenvalue by $\inf_{\mathbf{x} \in \mathcal{B}} \lambda_{\min}(\hat{\Psi}_{t,\mathbf{x}}) \geq \inf_{\mathbf{x} \in \mathcal{B}} \lambda_{\min}(\Psi_{t,\mathbf{x}}) - \sup_{\mathbf{x} \in \mathcal{B}} |\lambda_{\min}(\hat{\Psi}_{t,\mathbf{x}}) - \lambda_{\min}(\Psi_{t,\mathbf{x}})| \gtrsim_{\mathbb{P}} 1$.

Finally, it follows that $\sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\Psi}_{t,\mathbf{x}}^{-1}\| \lesssim_{\mathbb{P}} 1$ and hence

$$\sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}\| \leq \sup_{\mathbf{x} \in \mathcal{B}} \|\Psi_{t,\mathbf{x}}^{-1}\| \|\Psi_{t,\mathbf{x}} - \hat{\Psi}_{t,\mathbf{x}}\| \|\hat{\Psi}_{t,\mathbf{x}}^{-1}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}},$$

which completes the proof. $\square$

SA-8.4 Proof of Lemma SA-4

Consider the class $\mathcal{F} = \{(\mathbf{z}, u) \mapsto \mathbf{e}_\ell^\top g_{t,\mathbf{x}}(\mathbf{z})(u - h_\mathbf{x}(\mathbf{z})) : \mathbf{x} \in \mathcal{B}\}$, $0 \leq \ell \leq p$, where for $\mathbf{z} \in \mathcal{X}$,

$$g_{t,\mathbf{x}}(\mathbf{z}) = \mathbf{r}_p \left(\frac{D_\mathbf{z}(\mathbf{x})}{h} \right) K_h(D_\mathbf{z}(\mathbf{x})) \mathbf{1}(D_\mathbf{z}(\mathbf{x}) \in \mathcal{J}_t), \quad h_\mathbf{x}(\mathbf{z}) = \gamma_t^*(\mathbf{x})^\top \mathbf{r}_p(D_\mathbf{z}(\mathbf{x})/h).$$

By definition of $\gamma_t^*(\mathbf{x})$,

$$\gamma_t^*(\mathbf{x}) = \Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}}, \quad \mathbf{S}_{t,\mathbf{x}} = \mathbb{E} \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) K_h(D_i(\mathbf{x})) Y_i \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t) \right]. \quad (\text{SA-14})$$

Assumption SA-1 implies $\mathbf{S}_{t,\mathbf{x}}$ is continuous in $\mathbf{x}$, hence $\sup_{\mathbf{x} \in \mathcal{B}} \|\mathbf{S}_{t,\mathbf{x}}\| \lesssim 1$. And by Assumption SA-2(iv), $\inf_{\mathbf{x} \in \mathcal{B}} \lambda_{\min}(\Psi_{t,\mathbf{x}}) \gtrsim 1$. Hence

$$\sup_{\mathbf{x} \in \mathcal{B}} \|\Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}}\| \lesssim 1. \quad (\text{SA-15})$$

Now, consider properties of $\mathcal{F}$. Definition of $\gamma_t^*(\mathbf{x})$ implies $\mathbb{E}[f(\mathbf{X}_i, Y_i)] = 0$ for all $f \in \mathcal{F}$. Since K is compactly supported, there exists $C_1, C_2 > 0$ such that $F(\mathbf{z}, u) = C_1 h^{-d}(|u| + C_2)$ is an envelope function for $\mathcal{F}$. Denote $M = \max_{1 \leq i \leq n} F(\mathbf{X}_i, Y_i)$, then

$$\begin{aligned} \mathbb{E}[M^2]^{1/2} &\lesssim h^{-d} \mathbb{E} \left[\max_{1 \leq i \leq n} |Y_i|^2 + 1 \right]^{1/2} \lesssim h^{-d} \mathbb{E} \left[\max_{1 \leq i \leq n} |Y_i|^{2+v} \right]^{1/(2+v)} \\ &\lesssim h^{-d} \left[\sum_{i=1}^n \mathbb{E}[|Y_i|^{2+v}] \right]^{1/(2+v)} \lesssim h^{-d} n^{1/(2+v)}, \end{aligned}$$

where we have used the uniform $(2+v)$ moment bound in Assumption SA-1(v). Denote $\sigma = \sup_{f \in \mathcal{F}} \mathbb{E}[f(\mathbf{X}_i, Y_i)^2]^{1/2}$. Then,

$$\sigma^2 \lesssim \sup_{\mathbf{x} \in \mathcal{B}} \mathbb{E}[\|\mathbf{e}_\ell^\top g_{t,\mathbf{x}}\|_\infty^2 (|Y_i| + \|h_{\mathbf{x}}\|_\infty)^2] \lesssim h^{-d}.$$

To check the covering number of $\mathcal{F}$, note that compared to the proof of Lemma SA-3, we have one additional term $\mathbf{e}_\ell^\top g_{t,\mathbf{x}} h_{\mathbf{x}}$. All signed-distance kernel and indicator terms can be handled as in the proof of Lemma SA-3. Recall Equation (SA-14), and consider $l_{t,\mathbf{x}}(\cdot) = \mathbf{e}_\ell^\top \mathbf{r}_p(D(\cdot)\mathbf{x}/h) K_h(D(\cdot)\mathbf{x}) \mu_t(\cdot) \mathbf{1}(D(\cdot)\mathbf{x} \in \mathcal{J}_t)$ and $\mathcal{L}_t = \{l_{t,\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$. Then, for any $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$,

$$|\mathbf{S}_{t,\mathbf{x}_1} - \mathbf{S}_{t,\mathbf{x}_2}| \leq \|l_{t,\mathbf{x}_1} - l_{t,\mathbf{x}_2}\|_{\mathbb{P}_{\mathcal{X},2}},$$

and hence

$$N(\{\mathbf{e}_\ell^\top \mathbf{S}_{t,\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}, \|\cdot\|, \varepsilon h^{-d}) \leq N(\mathcal{L}_t, \|\cdot\|_{\mathbb{P}_{\mathcal{X},2}}, \varepsilon h^{-d}) \leq \sup_Q N(\mathcal{L}_t, \|\cdot\|_{Q,2}, \varepsilon h^{-d}),$$

The same argument as in the paragraph **Covering Numbers** in the proof of Lemma SA-3 then shows

$$\begin{aligned} \sup_Q N(\{g_{t,\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}, \|\cdot\|_{Q,2}, \varepsilon C_1 h^{-d}) &\leq \left(\frac{\text{diam}(\mathcal{X})}{h\varepsilon} \right)^d, \quad 0 < \varepsilon \leq 1, \\ \sup_Q N(\{g_{t,\mathbf{x}} h_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}, \|\cdot\|_{Q,2}, \varepsilon C_1 h^{-d}) &\leq \left(\frac{\text{diam}(\mathcal{X})}{h\varepsilon} \right)^d, \quad 0 < \varepsilon \leq 1, \end{aligned}$$

where sup is taken over all discrete measures on $\mathcal{X}$. Product $\{g_{t,\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$ with the singleton identity class and addition of $\{g_{t,\mathbf{x}} h_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$ preserve VC-type entropy, so

$$\sup_Q N(\mathcal{F}, \|\cdot\|_{Q,2}, \varepsilon \|F\|_{Q,2}) \leq 2 \left(\frac{2 \text{diam}(\mathcal{X})}{h\varepsilon} \right)^d, \quad 0 < \varepsilon \leq 1,$$

where sup is taken over all discrete measures on $\mathcal{X} \times \mathbb{R}$. Denote $\mathbf{C}_1 = d$, $\mathbf{C}_2 = \frac{2(2 \text{diam}(\mathcal{X}))^d}{h^d}$. Hence, by

Chernozhukov et al. [2014b, Corollary 5.1]

$$\begin{aligned}
\mathbb{E} \left[\sup_{\mathbf{x} \in \mathcal{B}} |\mathbf{e}_\ell^\top \mathbf{O}_{t,\mathbf{x}}| \right] &= \mathbb{E} \left[\sup_{f \in \mathcal{F}} |\mathbb{E}_n[f(\mathbf{X}_i, Y_i)] - \mathbb{E}[f(\mathbf{X}_i, Y_i)]| \right] \\
&\lesssim \frac{\sigma}{\sqrt{n}} \sqrt{\mathbf{C}_2 \log(\mathbf{C}_1 \|M\|_{\mathbb{P},2} / \sigma)} + \frac{\|M\|_{\mathbb{P},2} \mathbf{C}_2 \log(\mathbf{C}_1 \|M\|_{\mathbb{P},2} / \sigma)}{n} \\
&\lesssim \frac{1}{\sqrt{nh^d}} \sqrt{d \log \left(\frac{\text{diam}(\mathcal{X})}{h^{1+d/2}} \right)} + \frac{1}{n^{\frac{1+v}{2+v}} h^d} d \log \left(\frac{\text{diam}(\mathcal{X})}{h^{1+d/2}} \right) \\
&\lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d}.
\end{aligned}$$

The rest follows from finite dimensionality of $\mathbf{O}_{t,\mathbf{x}}$, and Lemma SA-3. $\square$

SA-8.5 Proof of Lemma SA-5

Denote, for $\mathbf{z} \in \mathcal{B}$, $\eta_{i,t,\mathbf{z}} = Y_i - \theta_{t,\mathbf{z}}^*(D_i(\mathbf{z}))$ and $\xi_{i,t,\mathbf{z}} = \theta_{t,\mathbf{z}}^*(D_i(\mathbf{z})) - \hat{\theta}_{t,\mathbf{z}}(D_i(\mathbf{z}))$. Then

$$\begin{aligned}
\hat{\Upsilon}_{t,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left(\frac{D_i(\mathbf{y})}{h} \right)^\top h^d K_h(D_i(\mathbf{x})) K_h(D_i(\mathbf{y})) \right. \\
&\quad \left. \cdot (\eta_{i,t,\mathbf{x}} + \xi_{i,t,\mathbf{x}})(\eta_{i,t,\mathbf{y}} + \xi_{i,t,\mathbf{y}}) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \mathbf{1}(D_i(\mathbf{y}) \in \mathcal{J}_t) \right],
\end{aligned}$$

and we decompose the error into

$$\begin{aligned}
\hat{\Upsilon}_{t,\mathbf{x},\mathbf{y}} - \Upsilon_{t,\mathbf{x},\mathbf{y}} &= \Delta_{1,t,\mathbf{x},\mathbf{y}} + \Delta_{2,t,\mathbf{x},\mathbf{y}} + \Delta_{3,t,\mathbf{x},\mathbf{y}}, \\
\Delta_{1,t,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left(\frac{D_i(\mathbf{y})}{h} \right)^\top h^d K_h(D_i(\mathbf{x})) K_h(D_i(\mathbf{y})) \xi_{i,t,\mathbf{x}} \xi_{i,t,\mathbf{y}} \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \mathbf{1}(D_i(\mathbf{y}) \in \mathcal{J}_t) \right], \\
\Delta_{2,t,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left(\frac{D_i(\mathbf{y})}{h} \right)^\top h^d K_h(D_i(\mathbf{x})) K_h(D_i(\mathbf{y})) (\eta_{i,t,\mathbf{x}} \xi_{i,t,\mathbf{y}} + \eta_{i,t,\mathbf{y}} \xi_{i,t,\mathbf{x}}) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \mathbf{1}(D_i(\mathbf{y}) \in \mathcal{J}_t) \right], \\
\Delta_{3,t,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left(\frac{D_i(\mathbf{y})}{h} \right)^\top h^d K_h(D_i(\mathbf{x})) K_h(D_i(\mathbf{y})) \eta_{i,t,\mathbf{x}} \eta_{i,t,\mathbf{y}} \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \mathbf{1}(D_i(\mathbf{y}) \in \mathcal{J}_t) \right] \\
&\quad - \mathbb{E} \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left(\frac{D_i(\mathbf{y})}{h} \right)^\top h^d K_h(D_i(\mathbf{x})) K_h(D_i(\mathbf{y})) \eta_{i,t,\mathbf{x}} \eta_{i,t,\mathbf{y}} \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \mathbf{1}(D_i(\mathbf{y}) \in \mathcal{J}_t) \right].
\end{aligned}$$

By Assumption SA-2, $K_h(D_i(\mathbf{x})) \neq 0$ implies $\|\mathbf{r}_p(D_i(\mathbf{x})/h)\|_2 \lesssim 1$. Hence by Lemma SA-3 and SA-4,

$$\begin{aligned}
&\max_{t \in \{0,1\}} \max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\xi_{i,t,\mathbf{x}}| \\
&= \max_{t \in \{0,1\}} \max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\mathbf{r}_p(D_i(\mathbf{x})/h)^\top (\hat{\gamma}_{t,\mathbf{x}} - \gamma_{t,\mathbf{x}}^*)| \mathbf{1}(K_h(D_i(\mathbf{x})) \neq 0) \\
&= \max_{t \in \{0,1\}} \max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\mathbf{r}_p(D_i(\mathbf{x})/h)^\top \hat{\Psi}_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}}| \mathbf{1}(K_h(D_i(\mathbf{x})) \neq 0) \\
&\leq \max_{t \in \{0,1\}} \sup_{\mathbf{x} \in \mathcal{B}} \left\| \hat{\Psi}_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} \right\|_2 \\
&\lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d},
\end{aligned}$$

Under the rate conditions of the lemma, a maximal inequality similar to the one used in the proof of Lemma SA-3 shows

$$\begin{aligned} \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\Delta_{1,t,\mathbf{x},\mathbf{y}}\| &\lesssim_{\mathbb{P}} \max_{t \in \{0,1\}} \max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\xi_{i,t,\mathbf{x}}|^2 \lesssim \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right)^2, \\ \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\Delta_{2,t,\mathbf{x},\mathbf{y}}\| &\lesssim_{\mathbb{P}} \max_{t \in \{0,1\}} \max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\xi_{i,t,\mathbf{x}}| \lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d}. \end{aligned} \quad (\text{SA-16})$$

Consider the (μ, ν) entry of $\Delta_{3,t,\mathbf{x},\mathbf{y}}$ and the class

$$\begin{aligned} \mathcal{F} = \left\{ (\mathbf{z}, u) \mapsto &\left(\frac{D_{\mathbf{z}}(\mathbf{x})}{h} \right)^{\mu} \left(\frac{D_{\mathbf{z}}(\mathbf{y})}{h} \right)^{\nu} h^d K_h(D_{\mathbf{z}}(\mathbf{x})) K_h(D_{\mathbf{z}}(\mathbf{y})) \right. \\ &\times (u - \mathbf{r}_p(D_{\mathbf{z}}(\mathbf{x})/h)^{\top} \boldsymbol{\gamma}_{t,\mathbf{x}}^*) (u - \mathbf{r}_p(D_{\mathbf{z}}(\mathbf{y})/h)^{\top} \boldsymbol{\gamma}_{t,\mathbf{y}}^*) \mathbf{1}(D_{\mathbf{z}}(\mathbf{x}) \in \mathcal{J}_t) \mathbf{1}(D_{\mathbf{z}}(\mathbf{y}) \in \mathcal{J}_t) : \mathbf{x}, \mathbf{y} \in \mathcal{B} \left. \right\}. \end{aligned}$$

The kernel factors, side indicators, and polynomial factors form VC-type classes by the same covering arguments used above; finite products and sums preserve VC-type entropy with constants changing only by fixed powers. By Assumptions SA-2 and SA-1(v), we have $\sup_{f \in \mathcal{F}} \mathbb{E}[f(\mathbf{X}_i, Y_i)^2]^{1/2} \lesssim h^{-d/2}$. Moreover, Assumption SA-2 and Equation (SA-15) imply there exists $C_1, C_2 > 0$ such that $F(\mathbf{z}, u) = C_1 h^{-d}(u^2 + C_2)$ is an envelope function for $\mathcal{F}$, with

$$\mathbb{E} \left[\max_{1 \leq i \leq n} F(\mathbf{X}_i, Y_i)^2 \right]^{\frac{1}{2}} \lesssim C_1 h^{-d} (\mathbb{E} \left[\max_{1 \leq i \leq n} Y_i^4 \right]^{\frac{1}{2}} + C_2) \lesssim C_1 h^{-d} (\mathbb{E} \left[\max_{1 \leq i \leq n} Y_i^{2+v} \right]^{\frac{2}{2+v}} + C_2) \lesssim h^{-d} n^{\frac{2}{2+v}}.$$

Apply Chernozhukov et al. [2014b, Corollary 5.1] similarly as in Lemma SA-4 gives

$$\mathbb{E} \left[\sup_{f \in \mathcal{F}} |\mathbb{E}_n[f(\mathbf{X}_i, Y_i)] - \mathbb{E}[f(\mathbf{X}_i, Y_i)]| \right] \lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}.$$

Finite dimensionality of $\Delta_{3,t,\mathbf{x},\mathbf{y}}$ then implies

$$\mathbb{E} \left[\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\Delta_{3,t,\mathbf{x},\mathbf{y}}\| \right] \lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}. \quad (\text{SA-17})$$

Putting together Equations (SA-16), (SA-17) and Lemma SA-3 gives the bound for $\hat{\boldsymbol{\Upsilon}}_{t,\mathbf{x},\mathbf{y}} - \boldsymbol{\Upsilon}_{t,\mathbf{x},\mathbf{y}}$. The displayed bound for $\hat{\Xi} - \Xi$ follows by pre- and post-multiplying by $\mathbf{e}_0^{\top} \hat{\boldsymbol{\Psi}}^{-1}$ and its population analogue, and using the level definition of Ξ .

It remains to verify the scale of the variance term. The population lower bound $\lambda_{\min}(\boldsymbol{\Upsilon}_{t,\mathbf{x},\mathbf{x}}) \gtrsim 1$ follows from the Gram lower bound and the conditional variance lower bound in Assumption SA-1(iv). To see why the replacement of K by K^2 is harmless, let $q_{\mathbf{a}}(u) = \mathbf{a}^{\top} \mathbf{r}_p(u)$ with $\|\mathbf{a}\| = 1$. Assumption SA-2(iv) gives a uniform lower bound on the corresponding K -weighted local second moment. Since K is fixed, bounded, and compactly supported, while $q_{\mathbf{a}}$ is uniformly bounded on $\text{supp}(K)$, choose $\delta > 0$ small enough that the contribution from $\{K \leq \delta\}$ is less than one half of this lower bound. On $\{K > \delta\}$, $K^2 \geq \delta K$, which gives the same lower bound, up to constants, for the K^2 -weighted matrix. Combining this with

$$\mathbb{E}\{(Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x})))^2 \mid \mathbf{X}_i\} \geq \sigma_t^2(\mathbf{X}_i)$$

gives $\lambda_{\min}(\boldsymbol{\Upsilon}_{t,\mathbf{x},\mathbf{x}}) \gtrsim 1$. The same local-moment calculation also gives $\sup_{\mathbf{x} \in \mathcal{B}} \|\boldsymbol{\Upsilon}_{t,\mathbf{x},\mathbf{x}}\| \lesssim 1$ and $\sup_{\mathbf{x} \in \mathcal{B}} \|\boldsymbol{\Psi}_{t,\mathbf{x}}\| \lesssim$

1. Therefore, uniformly in $\mathbf{x} \in \mathcal{B}$,

$$nh^d \Xi_{t,\mathbf{x},\mathbf{x}} = (\Psi_{t,\mathbf{x}}^{-1} \mathbf{e}_0)^\top \Upsilon_{t,\mathbf{x},\mathbf{x}} (\Psi_{t,\mathbf{x}}^{-1} \mathbf{e}_0) \lesssim \|\Psi_{t,\mathbf{x}}^{-1}\|^2 \|\Upsilon_{t,\mathbf{x},\mathbf{x}}\| \lesssim 1,$$

while

$$nh^d \Xi_{t,\mathbf{x},\mathbf{x}} \geq \lambda_{\min}(\Upsilon_{t,\mathbf{x},\mathbf{x}}) \|\Psi_{t,\mathbf{x}}^{-1} \mathbf{e}_0\|^2 \gtrsim \|\Psi_{t,\mathbf{x}}\|^{-2} \gtrsim 1.$$

Summing over $t = 0, 1$ gives the same bound for $\Xi_{\mathbf{x}} = \Xi_{\mathbf{x},\mathbf{x}}$. Combining the population bounds with the uniform consistency of $\hat{\Upsilon}_{t,\mathbf{x},\mathbf{x}}$, the bound for $\hat{\Xi} - \Xi$, and Lemma SA-3 gives $nh^d \hat{\Xi}_{t,\mathbf{x},\mathbf{x}} \gtrsim_{\mathbb{P}} 1$ and $nh^d \hat{\Xi}_{t,\mathbf{x},\mathbf{x}} \lesssim_{\mathbb{P}} 1$ uniformly in $\mathbf{x}$. The corresponding correlation bound follows from these lower bounds and the displayed uniform consistency of $\hat{\Xi}_{t,\mathbf{x}_1,\mathbf{x}_2}$. $\square$

SA-8.6 Proof of Theorem SA-1

Since $\vartheta(\mathbf{x}) = \theta_{1,\mathbf{x}}(0) - \theta_{0,\mathbf{x}}(0)$ and $\tau(\mathbf{x}) = \mu_1(\mathbf{x}) - \mu_0(\mathbf{x})$, it is enough to prove the result for one treatment assignment group $t \in \{0, 1\}$. By Assumption SA-1(iii) and Assumption SA-2(ii), for any $r \neq 0$, for any $\mathbf{x} \in \mathcal{B}$ and $\mathbf{y} \in S_{t,\mathbf{x}}(|r|)$, $|\mu_t(\mathbf{y}) - \mu_t(\mathbf{x})| \lesssim |r|$. Hence, for any $r \neq 0$, for any $\mathbf{x} \in \mathcal{B}$, $t \in \{0, 1\}$,

$$|\theta_{t,\mathbf{x}}(r) - \mu_t(\mathbf{x})| \leq \frac{\int_{S_{t,\mathbf{x}}(|r|)} |\mu_t(\mathbf{y}) - \mu_t(\mathbf{x})| f_X(\mathbf{y}) \mathfrak{H}^{d-1}(\mathrm{d}\mathbf{y})}{\int_{S_{t,\mathbf{x}}(|r|)} f_X(\mathbf{y}) \mathfrak{H}^{d-1}(\mathrm{d}\mathbf{y})} \lesssim |r|.$$

implying

$$|\theta_{t,\mathbf{x}}(0) - \mu_t(\mathbf{x})| \leq \lim_{r \rightarrow 0, r \in \mathcal{J}_t} |\theta_{t,\mathbf{x}}(r) - \mu_t(\mathbf{x})| = 0,$$

which establishes the result. $\square$

SA-8.7 Proof of Theorem SA-2

By Theorem SA-1 and Equation (SA-14), we have

$$\begin{aligned} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_t(\mathbf{x})| &= \sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}} - \mu_t(\mathbf{x}) \right| \\ &= \sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbb{E} \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) K_h(D_i(\mathbf{x})) (\mu_t(\mathbf{X}_i) - \mu_t(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right] \right| \\ &\lesssim \sup_{\mathbf{x} \in \mathcal{B}} \sup_{\mathbf{z} \in \mathcal{X}} |\mu_t(\mathbf{x}) - \mu_t(\mathbf{z})| \mathbf{1}(K_h(D_{\mathbf{z}}(\mathbf{x})) > 0) \\ &\lesssim h. \end{aligned}$$

Taking the sum over $t = 0, 1$ gives $\sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})| \lesssim h$. This result holds for all $d \geq 1$.

The same argument, uniformly over $\mathcal{P}$, gives the upper bound in the case of $d = 2$. Thus, it remains to establish the lower bound in the sharpness statement for $d = 2$. Let $R_K < \infty$ satisfy $\text{supp}(K) \subseteq [-R_K, R_K]$ and set $\bar{R}_K = \max\{1, R_K\}$. Suppose $\mathbf{X}_i \sim \text{Uniform}([-2\bar{R}_K, 2\bar{R}_K]^2)$, $\mu_0(x_1, x_2) = 0$, and $\mu_1(x_1, x_2) = x_2$ on $\mathcal{X} = [-2\bar{R}_K, 2\bar{R}_K]^2$. Let

$$\mathcal{A}_1 = \{(x, y) \in \mathcal{X} : x \geq 0, y \geq 0\}, \quad \mathcal{A}_0 = \mathcal{X} \setminus \mathcal{A}_1,$$

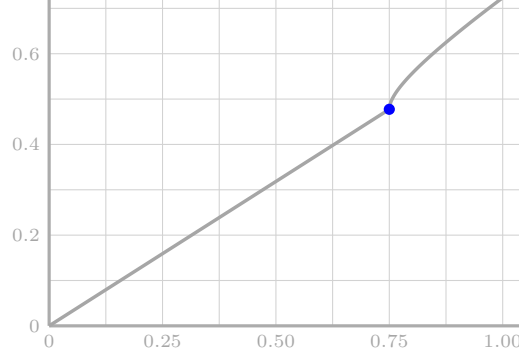


Figure SA-1: Conditional mean given distance with one kink.

and $\mathcal{B} = ([0, 2\bar{R}_K] \times \{0\}) \cup (\{0\} \times [0, 2\bar{R}_K])$. Let $\mathcal{d}$ be Euclidean distance, so $D_i(\mathbf{x}) = \|\mathbf{X}_i - \mathbf{x}\|$ on $\mathcal{A}_1$ and $D_i(\mathbf{x}) = -\|\mathbf{X}_i - \mathbf{x}\|$ on $\mathcal{A}_0$. This data-generating process belongs to the class $\mathcal{P}$ for a sufficiently large constant L . For $\mathbf{x}_s = (s, 0)$, $0 \leq s \leq 1$, the induced conditional mean on the treated side is

$$\theta_{1, \mathbf{x}_s}(r) = \begin{cases} 2r/\pi, & 0 \leq r \leq s, \\ (r+s)/(\pi - \arccos(s/r)), & r > s. \end{cases}$$

Thus the conditional mean given signed distance has a kink at $r = s$ when $s > 0$. Figure SA-1 plots this function for $s = 3/4$.

Let $\text{bias}_{n,1}(h, s)$ denote the level projection bias at $\mathbf{x}_s$ with bandwidth h . A change of variables gives the scaling identity

$$\text{bias}_{n,1}(Ch, Cs) = C \text{bias}_{n,1}(h, s), \quad 0 < C < 1, \quad (\text{SA-18})$$

because the Gram matrix is invariant under the rescaling $(u, v, s, h) \mapsto (Cu, Cv, Cs, Ch)$, while the linear conditional mean $\mu_1(u, v) = v$ scales by C . Since $\mu_0 \equiv 0$,

$$\text{bias}_{n,0}(h, s) = 0, \quad 0 < h < 1, \quad 0 < s < h.$$

Therefore it is enough to show $\sup_{0 \leq s \leq 1} |\text{bias}_{n,1}(1, s)| > 0$.

Suppressing the common density factor, write

$$\text{bias}_{n,1}(1, s) = \mathbf{e}_0^\top \mathbf{A}(s)^{-1} \mathbf{B}(s),$$

where

$$\begin{aligned} \mathbf{A}(s) &= \int_0^\infty \int_{\Theta_s(u)}^\pi \mathbf{r}_p(u) \mathbf{r}_p(u)^\top K(u) u \, d\theta \, du, \\ \mathbf{B}(s) &= \int_0^\infty \int_{\Theta_s(u)}^\pi \mathbf{r}_p(u) K(u) u^2 \sin(\theta) \, d\theta \, du, \quad \Theta_s(u) = \begin{cases} 0, & 0 \leq u \leq s, \\ \arccos(s/u), & u > s. \end{cases} \end{aligned}$$

At $s = 0$,

$$\mathbf{A}(0) = \frac{\pi}{2} \int_0^\infty u \mathbf{r}_p(u) \mathbf{r}_p(u)^\top K(u) du, \quad \mathbf{B}(0) = \int_0^\infty u^2 \mathbf{r}_p(u) K(u) du, \quad (\text{SA-19})$$

If $p = 0$, then $\mathbf{A}(0)$ and $\mathbf{B}(0)$ are scalars and $\mathbf{e}_0^\top \mathbf{A}(0)^{-1} \mathbf{B}(0) > 0$, so $\sup_{0 \leq s \leq 1} |\text{bias}_{n,1}(1, s)| > 0$ immediately. It remains to consider $p \geq 1$. Differentiating under the integral sign gives

$$\dot{\mathbf{A}}(0) = \int_0^\infty \mathbf{r}_p(u) \mathbf{r}_p(u)^\top K(u) du, \quad \dot{\mathbf{B}}(0) = \int_0^\infty u \mathbf{r}_p(u) K(u) du.$$

Using $\mathbf{A}(0)^{-1} \mathbf{B}(0) = (2/\pi) \mathbf{e}_1$, where $\mathbf{e}_1$ is the slope selector in the polynomial basis, and $\mathbf{A}(0)^{-1} \dot{\mathbf{B}}(0) = (2/\pi) \mathbf{e}_0$,

$$\begin{aligned} \left. \frac{d}{ds} \text{bias}_{n,1}(1, s) \right|_{s=0} &= -\mathbf{e}_0^\top \mathbf{A}(0)^{-1} \dot{\mathbf{A}}(0) \mathbf{A}(0)^{-1} \mathbf{B}(0) + \mathbf{e}_0^\top \mathbf{A}(0)^{-1} \dot{\mathbf{B}}(0) \\ &= -\frac{4}{\pi^2} + \frac{2}{\pi} \neq 0. \end{aligned} \quad (\text{SA-20})$$

By continuity, the same strict inequality holds for $p \geq 1$. Thus, for all $p \geq 0$, $\sup_{0 \leq s \leq 1} |\text{bias}_{n,1}(1, s)| > 0$. Combining this with (SA-18) yields

$$\inf_{0 < h < 1} \sup_{\mathbf{x} \in \mathcal{B}} \frac{|\mathfrak{B}_1(\mathbf{x}) - \mathfrak{B}_0(\mathbf{x})|}{h} \geq \inf_{0 < h < 1} \sup_{0 < s < h} \left| \text{bias}_{n,1} \left(1, \frac{s}{h} \right) \right| > 0.$$

This proves the lower bound and completes the proof. $\square$

SA-8.8 Proof of Theorem SA-3

Part (i): High-level smoothness condition.

It is enough to prove the result for one side t . Since K is compactly supported, take $R_K < \infty$ such that $\text{supp}(K) \subseteq [-R_K, R_K]$; it suffices to consider $|r| \leq R_K h$ with $r \in \mathcal{J}_t$. Taylor's theorem gives

$$\theta_{t,\mathbf{x}}(r) = \sum_{j=0}^{m-1} \frac{\theta_{t,\mathbf{x}}^{(j)}(0)}{j!} r^j + R_{t,\mathbf{x}}(r), \quad \sup_{\mathbf{x} \in \mathcal{B}} |R_{t,\mathbf{x}}(r)| \lesssim |r|^m.$$

The local polynomial fit of order p reproduces every polynomial of degree at most p . Since $m \leq p + 1$, the level bias is therefore generated only by the remainder. Using $\|\Psi_{t,\mathbf{x}}^{-1}\| \lesssim 1$ and the bounded kernel moments,

$$\sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_t(\mathbf{x})| \lesssim \sup_{\mathbf{x} \in \mathcal{B}} \mathbb{E} \left[K_h(D_i(\mathbf{x})) |D_i(\mathbf{x})|^m \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right] \lesssim h^m.$$

Taking the sum over $t = 0, 1$ gives the uniform result. The pointwise result is identical with the outer supremum removed.

Part (ii): Pointwise bivariate smooth-boundary primitive condition.

Fix $\mathbf{x} \in \mathcal{B}$ and let $m_{\mathbf{x}} = \kappa \wedge (p + 1)$. It is enough to verify the primitive smoothness condition in part (i) for one treatment side t at this fixed boundary point; the treatment-effect bias is the difference of the two side-specific biases.

After re-centering and rotating coordinates, set $\mathbf{x} = 0$ and parameterize the local curve by arc length so that $\gamma(0) = 0$ and $\gamma'(0) = (1, 0)$. For u near zero,

$$\phi(u) = \|\gamma(u) - \gamma(0)\|^2 = u^2 a(u), \quad a(0) = 1,$$

where a is C^κ and bounded away from zero in a neighborhood of zero. Therefore the two local solutions of $\phi(u) = r^2$ can be written as $u_+(r)$ and $u_-(r)$, where $u_+(r) = r q_+(r)$ and $u_-(r) = -r q_-(r)$, with q_+ and q_- being C^κ functions. This follows from the implicit function theorem applied to $u/\sqrt{\phi(u)}$ on the positive and negative sides of zero.

The polar angles at which the circle $\{\mathbf{u} : \|\mathbf{u} - \mathbf{x}\| = r\}$ intersects the local boundary are

$$\alpha_+(\mathbf{x}, r) = \arg\{\gamma(u_+(r)) - \mathbf{x}\}, \quad \alpha_-(\mathbf{x}, r) = \arg\{\gamma(u_-(r)) - \mathbf{x}\}.$$

The preceding representation and the smoothness of γ imply that $\alpha_+(\mathbf{x}, \cdot)$ and $\alpha_-(\mathbf{x}, \cdot)$ are C^κ on $[0, \varepsilon]$, after possibly shrinking ε . The local angular interval corresponding to $\mathcal{A}_t$ therefore has endpoints $\alpha_{t,1}(\mathbf{x}, r)$ and $\alpha_{t,2}(\mathbf{x}, r)$ that are C^κ in r .

For $r > 0$,

$$\theta_{t,\mathbf{x}}(r) = \frac{\int_{\alpha_{t,1}(\mathbf{x},r)}^{\alpha_{t,2}(\mathbf{x},r)} \mu_t(\mathbf{x} + r(\cos \theta, \sin \theta)) f_X(\mathbf{x} + r(\cos \theta, \sin \theta)) d\theta}{\int_{\alpha_{t,1}(\mathbf{x},r)}^{\alpha_{t,2}(\mathbf{x},r)} f_X(\mathbf{x} + r(\cos \theta, \sin \theta)) d\theta}.$$

The numerator and denominator are $C^{m_\mathbf{x}}$ in r by repeated application of Leibniz's rule, because μ_t , f_X , and the angle endpoints have the required smoothness. The denominator is bounded away from zero for small r because the angular interval length converges to π and f_X is bounded away from zero. Hence $\theta_{t,\mathbf{x}}(\cdot)$ is $m_\mathbf{x}$ -times continuously differentiable near zero. Applying the pointwise version of part (i) gives

$$|\mathfrak{B}(\mathbf{x})| \lesssim h^{m_\mathbf{x}}.$$

Part (iii): Uniform bivariate smooth-boundary primitive condition.

Let $m = \iota \wedge (p + 1)$. We verify the primitive smoothness condition in part (i) uniformly over $\mathbf{x} \in \mathcal{B}^o$.

Fix $\mathbf{x}_0 = \gamma(s_0) \in \mathcal{B}^o$ and re-center and rotate coordinates so that $\mathbf{x}_0 = 0$ and $\gamma'(s_0) = (1, 0)$ after reparameterizing the curve by arc length. Since γ is $C^{\iota+2}$, for u near zero,

$$\phi(u) = \|\gamma(s_0 + u) - \gamma(s_0)\|^2 = u^2 a(u), \quad a(0) = 1,$$

where a is C^ι and bounded away from zero uniformly for $s_0 \in [\delta, L - \delta]$. Therefore the two local solutions of $\phi(u) = r^2$ can be written as $u_+(r)$ and $u_-(r)$, where $u_+(r) = r q_+(r)$ and $u_-(r) = -r q_-(r)$ with q_+ and q_- being C^ι functions whose derivatives up to order ι are uniformly bounded. This follows from the implicit function theorem applied to $u/\sqrt{\phi(u)}$ on the positive and negative sides of zero.

The polar angles at which the circle $\{\mathbf{u} : \|\mathbf{u} - \mathbf{x}_0\| = r\}$ intersects the boundary are therefore

$$\alpha_+(\mathbf{x}_0, r) = \arg\{\gamma(s_0 + u_+(r)) - \mathbf{x}_0\}, \quad \alpha_-(\mathbf{x}_0, r) = \arg\{\gamma(s_0 + u_-(r)) - \mathbf{x}_0\}.$$

The preceding representation and the smoothness of γ imply that $\alpha_+(\mathbf{x}_0, \cdot)$ and $\alpha_-(\mathbf{x}_0, \cdot)$ are C^ι on $[0, \varepsilon]$, with derivatives uniformly bounded over $\mathbf{x}_0 \in \mathcal{B}^o$. By the assumed two-intersection condition, the angular interval

corresponding to $\mathcal{A}_t$ has endpoints given by these two angle functions, up to a fixed ordering convention.

For $r > 0$, write this interval as $[\alpha_{t,1}(\mathbf{x}_0, r), \alpha_{t,2}(\mathbf{x}_0, r)]$. Then

$$\theta_{t,\mathbf{x}_0}(r) = \frac{\int_{\alpha_{t,1}(\mathbf{x}_0, r)}^{\alpha_{t,2}(\mathbf{x}_0, r)} \mu_t(\mathbf{x}_0 + r(\cos \theta, \sin \theta)) f_X(\mathbf{x}_0 + r(\cos \theta, \sin \theta)) d\theta}{\int_{\alpha_{t,1}(\mathbf{x}_0, r)}^{\alpha_{t,2}(\mathbf{x}_0, r)} f_X(\mathbf{x}_0 + r(\cos \theta, \sin \theta)) d\theta}.$$

The numerator and denominator are C^m in r by repeated application of Leibniz's rule, because μ_t , f_X , and the angle endpoints have uniformly bounded derivatives up to order m . The denominator is bounded away from zero uniformly for small r because the angular interval length converges uniformly to π and f_X is bounded away from zero; Assumption SA-2(v) ensures that the corresponding Hausdorff-measure ratio is well defined before this local parametrization. Hence $\theta_{t,\mathbf{x}_0}$ is C^m on $[0, \varepsilon]$, and its derivatives up to order m are uniformly bounded over $\mathbf{x}_0 \in \mathcal{B}^o$.

Applying the argument in part (i) on the index set $\mathcal{B}^o$ gives

$$\sup_{\mathbf{x} \in \mathcal{B}^o} |\mathfrak{B}(\mathbf{x})| \lesssim h^m.$$

□

SA-8.9 Proof of Theorem SA-4

For a fixed $\mathbf{x} \in \mathcal{B}$, the proofs of Lemmas SA-3 and SA-4 can be done when the index set is the singleton $\{\mathbf{x}\}$ instead of $\mathcal{B}$, replacing Chernozhukov et al. [2014b, Corollary 5.1] with Bernstein's inequality and obtaining

$$\begin{aligned} |\mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}}| &\lesssim_{\mathbb{P}} \sqrt{\frac{1}{nh^d}} + \frac{1}{n^{\frac{1+v}{2+v}} h^d}, \\ |\mathbf{e}_0^\top (\hat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}) \mathbf{O}_{t,\mathbf{x}}| &\lesssim_{\mathbb{P}} \sqrt{\frac{1}{nh^d}} \left(\sqrt{\frac{1}{nh^d}} + \frac{1}{n^{\frac{1+v}{2+v}} h^d} \right). \end{aligned}$$

The pointwise convergence rate follows from decomposition (SA-1). The uniform statement follows from Lemmas SA-3 and SA-4 and the same decomposition. □

SA-8.10 Proof of Theorem SA-5

Define $\bar{\mathbf{T}}(\mathbf{x}) = \sum_{i=1}^n Z_i$, with $Z_i = Z_{1,i} - Z_{0,i}$ independent random variables ($i = 1, 2, \dots, n$),

$$Z_{t,i} = \frac{1}{n} (\Xi_{\mathbf{x},\mathbf{x}})^{-1/2} \mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) K_h(D_i(\mathbf{x})) (Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))) \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t),$$

$\mathbb{E}[Z_i] = 0$ and $\mathbb{V}[Z_i] = n^{-1}$. By the Berry-Esseen theorem,

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}(\bar{\mathbf{T}}(\mathbf{x}) \leq u) - \Phi(u) \right| \lesssim \sum_{i=1}^n \mathbb{E}[|Z_i|^3] \lesssim \sum_{i=1}^n \mathbb{E}[|Z_{1,i}|^3] + \sum_{i=1}^n \mathbb{E}[|Z_{0,i}|^3]$$

where

$$\sum_{i=1}^n \mathbb{E}[|Z_{t,i}|^3] = n^{-2} (\Xi_{\mathbf{x},\mathbf{x}})^{-3/2} \mathbb{E} \left[\left| \mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) K_h(D_i(\mathbf{x})) \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t) (Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))) \right|^3 \right]$$

$$\begin{aligned}
&\lesssim n^{-2}(\Xi_{\mathbf{x},\mathbf{x}})^{-3/2}\mathbb{E}[|K_h(D_i(\mathbf{x}))|^3|Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))|^3] \\
&\lesssim n^{-2}(\Xi_{\mathbf{x},\mathbf{x}})^{-3/2}\mathbb{E}[|K_h(D_i(\mathbf{x}))|^3(\mathbb{E}[|Y_i|^3|\mathbf{X}_i] + |\theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))|^3)] \\
&\lesssim (nh^d)^{-1/2},
\end{aligned}$$

noting that $K_h(D_i(\mathbf{x}))$ is supported on a set of probability $O(h^d)$ and satisfies $|K_h(D_i(\mathbf{x}))| \lesssim h^{-d}$, $\Xi_{\mathbf{x},\mathbf{x}} \gtrsim (nh^d)^{-1}$ by the pointwise version of the covariance lower-bound argument in Lemma SA-5, $\mathbb{E}[|Y_i|^3|\mathbf{X}_i] \lesssim 1$ by Assumption SA-1(v), and $\max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))| \lesssim 1$ because

$$\theta_{t,\mathbf{x}}^*(D_i(\mathbf{x})) = \gamma_t^*(\mathbf{x})^\top \mathbf{r}_p\left(\frac{D_i(\mathbf{x})}{h}\right) = (\Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}})^\top \mathbf{r}_p\left(\frac{D_i(\mathbf{x})}{h}\right).$$

Since

$$|\hat{\mathbf{T}}(\mathbf{x}) - \bar{\mathbf{T}}(\mathbf{x})| \lesssim_{\mathbb{P}} \frac{1}{\sqrt{nh^d}} + \frac{1}{n^{\frac{v}{2+v}} h^d} + \sqrt{nh^d} |\mathfrak{B}(\mathbf{x})|,$$

and the same pointwise covariance argument gives $\hat{\Xi}_{\mathbf{x},\mathbf{x}}/\Xi_{\mathbf{x},\mathbf{x}} \rightarrow_{\mathbb{P}} 1$ under the conditions imposed, pointwise asymptotic normality follows. Finally, validity of the confidence interval estimator is immediate. $\square$

SA-8.11 Proof of Theorem SA-6

Because $v > 0$, the condition $n^{\frac{v}{2+v}} h^d / \log(1/h) \rightarrow \infty$ implies $nh^d / \log(1/h) \rightarrow \infty$, so the preceding uniform approximation lemmas apply. We make the decomposition based on Equation (SA-1) and convergence of $\hat{\Xi}_{\mathbf{x},\mathbf{x}}$,

$$\begin{aligned}
\hat{\mathbf{T}}(\mathbf{x}) - \bar{\mathbf{T}}(\mathbf{x}) &= (\hat{\Xi}_{\mathbf{x},\mathbf{x}})^{-1/2} \left(\sum_{t \in \{0,1\}} (2t-1)(\hat{\theta}_{t,\mathbf{x}}(0) - \theta_{t,\mathbf{x}}(0)) \right) \\
&\quad - (\Xi_{\mathbf{x},\mathbf{x}})^{-1/2} \left(\sum_{t \in \{0,1\}} (2t-1) \mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} \right) \\
&= (\hat{\Xi}_{\mathbf{x},\mathbf{x}})^{-1/2} \left(\sum_{t \in \{0,1\}} (2t-1)(\hat{\theta}_{t,\mathbf{x}}(0) - \theta_{t,\mathbf{x}}(0)) \right. \\
&\quad \left. - \sum_{t \in \{0,1\}} (2t-1) \mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} \right) \quad (= \Delta_{1,\mathbf{x}}) \\
&\quad + ((\hat{\Xi}_{\mathbf{x},\mathbf{x}})^{-1/2} - (\Xi_{\mathbf{x},\mathbf{x}})^{-1/2}) \sum_{t \in \{0,1\}} (2t-1) \mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} \quad (= \Delta_{2,\mathbf{x}})
\end{aligned}$$

By Lemma SA-3 and SA-4, and the decomposition Equation (SA-1),

$$\begin{aligned}
&\sup_{\mathbf{x} \in \mathcal{B}} \left| \sum_{t \in \{0,1\}} (2t-1)(\hat{\theta}_{t,\mathbf{x}}(0) - \theta_{t,\mathbf{x}}(0)) - \sum_{t \in \{0,1\}} (2t-1) \mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} \right| \\
&\lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right) + \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})|.
\end{aligned}$$

Together with Lemma SA-5,

$$\sup_{\mathbf{x} \in \mathcal{B}} |\Delta_{1,\mathbf{x}}| \lesssim_{\mathbb{P}} \frac{\log(1/h)}{\sqrt{nh^d}} + \frac{(\log(1/h))^{\frac{3}{2}}}{n^{\frac{1+v}{2+v}} h^d} + \sqrt{nh^d} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})|. \quad (\text{SA-21})$$

By Lemma SA-3, Lemma SA-4 and Lemma SA-5, and assume $\frac{n^{\frac{v}{2+v}} h^d}{\log(1/h)} \rightarrow \infty$, then

$$\begin{aligned} \sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} \left((\Xi_{\mathbf{x},\mathbf{x}})^{-1/2} - (\widehat{\Xi}_{\mathbf{x},\mathbf{x}})^{-1/2} \right) \right| &\lesssim_{\mathbb{P}} \sqrt{nh^d} \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right) \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right) \\ &= \sqrt{\log(1/h)} \left(1 + \sqrt{\frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}} \right) \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right) \\ &\lesssim \sqrt{\log(1/h)} \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right). \end{aligned}$$

Hence

$$\sup_{\mathbf{x} \in \mathcal{B}} |\Delta_{2,\mathbf{x}}| \lesssim_{\mathbb{P}} \frac{\log(1/h)}{\sqrt{nh^d}} + \frac{(\log(1/h))^{\frac{3}{2}}}{n^{\frac{v}{2+v}} h^d}. \quad (\text{SA-22})$$

Putting together Equations (SA-21), (SA-22) give the result. $\square$

SA-8.12 Proof of Theorem SA-7

We will verify the high level conditions stated in Theorem SA-25.

By the rectangular-support condition, after a componentwise affine rescaling we can take $\mathcal{X} = [0, 1]^d$. Since f_X is continuous and bounded away from zero on this compact rectangle, Cattaneo and Yu [2025, Section 3.1] implies that $\mathcal{Q}_{\mathcal{G} \cup \mathcal{M}} = \mathbb{P}_X$ is a valid surrogate measure for $\mathbb{P}_X$ with respect to $\mathcal{G} \cup \mathcal{M}$, and that the Rosenblatt transform $\phi_{\mathcal{G} \cup \mathcal{M}} = T_{\mathbb{P}_X}$ is a valid normalizing transformation. The constants c_1 and c_2 from Theorem SA-25 are therefore finite and uniformly bounded by constants depending only on the density bounds and the support geometry, and are absorbed in the displays below.

Recall $\mathcal{G} = \{g_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$ where

$$g_{\mathbf{x}}(\mathbf{u}) = \mathbf{1}(\mathbf{u} \in \mathcal{A}_1) \mathfrak{K}_1(\mathbf{u}; \mathbf{x}) - \mathbf{1}(\mathbf{u} \in \mathcal{A}_0) \mathfrak{K}_0(\mathbf{u}; \mathbf{x}).$$

By the same signed-distance entropy and localization arguments as in Lemma SA-3, combined with [Cattaneo et al., 2024, Lemma 7] in the Lipschitz-kernel case and the VC-ball argument in the uniform-kernel case, we get properties of $\mathcal{G}$ as follows:

$$\mathbf{M}_{\mathcal{G}} \lesssim h^{-d/2}, \quad \mathbf{E}_{\mathcal{G}} \lesssim h^{d/2}, \quad \mathbf{TV}_{\mathcal{G}} \lesssim h^{d/2-1}, \quad \sup_Q N(\mathcal{G}, \|\cdot\|_{Q,2}, \varepsilon(2c+1)^{d+1} \mathbf{M}_{\mathcal{G}}) \leq 2c' \varepsilon^{-d-1} + 2.$$

By definition of $\theta_{t,\mathbf{x}}^*(\cdot)$, for each $\mathbf{x} \in \mathcal{B}$, $t \in \{0, 1\}$,

$$\theta_{t,\mathbf{x}}^*(D_{\mathbf{u}}(\mathbf{x})) = \gamma_t^*(\mathbf{x})^\top \mathbf{r}_p \left(\frac{D_{\mathbf{u}}(\mathbf{x})}{h} \right) = (\Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}})^\top \mathbf{r}_p \left(\frac{D_{\mathbf{u}}(\mathbf{x})}{h} \right),$$

recalling

$$\Psi_{t,\mathbf{x}} = \mathbb{E} \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right)^\top K_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right], \quad \mathbf{S}_{t,\mathbf{x}} = \mathbb{E} \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) K_h(D_i(\mathbf{x})) Y_i \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t) \right].$$

The preceding bounds imply that $\|\Psi_{t,\mathbf{x}}^{-1}\| \lesssim 1$, $\|\mathbf{S}_{t,\mathbf{x}}\| \lesssim 1$, and

$$\mathbf{M}_{\mathcal{M}_t} \lesssim h^{-d/2}, \quad \mathbf{E}_{\mathcal{M}_t} \lesssim h^{d/2}, \quad t \in \{0, 1\}.$$

In what follows, we verify the entropy and total variation properties of $\mathcal{M}$. Using the product rule, we have

$$\sup_{\mathbf{u} \in \mathcal{X}} \sup_{\mathbf{x}, \mathbf{x}' \in \mathcal{B}} \frac{|\theta_{t,\mathbf{x}}^*(D_{\mathbf{u}}(\mathbf{x})) - \theta_{t,\mathbf{x}}^*(D_{\mathbf{u}}(\mathbf{x}'))|}{\|\mathbf{x} - \mathbf{x}'\|} \lesssim h^{-1}.$$

Define $f_{t,\mathbf{x}}(\cdot) = \frac{h^{-d/2}}{\sqrt{n\Xi_{\mathbf{x},\mathbf{x}}}} \mathbf{e}_0^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{r}_p(\cdot) K(\cdot) (\Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}})^\top \mathbf{r}_p(\cdot)$. Then,

$$\mathfrak{K}_t(\mathbf{u}; \mathbf{x}) \theta_{t,\mathbf{x}}^*(D_{\mathbf{u}}(\mathbf{x})) = h^{-d/2} f_{t,\mathbf{x}} \left(\frac{D_{\mathbf{u}}(\mathbf{x})}{h} \right), \quad \mathbf{u} \in \mathcal{X}, \mathbf{x} \in \mathcal{B}, t \in \{0, 1\}.$$

Take $\mathcal{M}_t = \{\mathfrak{K}_t(\cdot; \mathbf{x}) \theta_{t,\mathbf{x}}^*(D_{\cdot}(\mathbf{x})) : \mathbf{x} \in \mathcal{B}\}$, $t \in \{0, 1\}$. If K is Lipschitz, then for $t \in \{0, 1\}$, $f_{t,\mathbf{x}}$ satisfies:

(i) <i>boundedness</i>	$\sup_{\mathbf{x} \in \mathcal{B}} \sup_{u \in \mathbb{R}} f_{t,\mathbf{x}}(u) \leq \mathbf{c},$
(ii) <i>compact support</i>	$\text{supp}(f_{t,\mathbf{x}}(\cdot)) \subseteq [-\mathbf{c}, \mathbf{c}], \forall \mathbf{x} \in \mathcal{B},$
(iii) <i>Lipschitz continuity</i>	$\sup_{\mathbf{x} \in \mathcal{B}} \sup_{u, u' \in \mathbb{R}} \frac{ f_{t,\mathbf{x}}(u) - f_{t,\mathbf{x}}(u') }{ u - u' } \leq \mathbf{c},$ $\sup_{u \in \mathbb{R}} \sup_{\mathbf{x}, \mathbf{x}' \in \mathcal{B}} \frac{ f_{t,\mathbf{x}}(u) - f_{t,\mathbf{x}'}(u) }{\ \mathbf{x} - \mathbf{x}'\ } \leq \mathbf{c} h^{-1},$

for some constant $\mathbf{c}$ not depending on n . Then, by an argument similar to Cattaneo et al. [2024, Lemma 7], there exists a constant $\mathbf{c}'$ depending only on $\mathbf{c}$ and d such that, for any $0 \leq \varepsilon \leq 1$,

$$\sup_Q N \left(h^{d/2} \mathcal{M}_t, \|\cdot\|_{Q,1}, (2\mathbf{c} + 1)^{d+1} \varepsilon \right) \leq \mathbf{c}' \varepsilon^{-d-1} + 1,$$

where the supremum is taken over all finite discrete measures. If $K(u) = \mathbf{1}(u \in [-1, 1])$, the same display follows by replacing the preceding Lipschitz step with the VC-ball argument used in Lemma SA-3; the polynomial factors and $\theta_{t,\mathbf{x}}^*(D_{\cdot}(\mathbf{x}))$ are uniformly bounded on the support of $K_h(D_{\cdot}(\mathbf{x}))$, and products with the fixed side indicator preserve VC-type entropy. Taking a constant envelope function $\mathbf{M}_{\mathcal{M}_t} = (2\mathbf{c} + 1)^{d+1} h^{-d/2}$, we have for any $0 < \varepsilon \leq 1$,

$$\sup_Q N \left(\mathcal{M}_t, \|\cdot\|_{Q,1}, \varepsilon \mathbf{M}_{\mathcal{M}_t} \right) \leq \mathbf{c}' \varepsilon^{-d-1} + 1.$$

By Lemma SA-11, above implies the uniform covering number for $\mathcal{M}_t$ satisfies

$$\mathbf{N}_{\mathcal{M}_t}(\varepsilon) \leq 4\mathbf{c}' (\varepsilon/2)^{-d-1}, \quad 0 < \varepsilon \leq 1.$$

Since $\mathcal{M} \subseteq \mathcal{M}_0 + \mathcal{M}_1$, here $+$ denotes the Minkowski sum, with $\mathbf{M}_{\mathcal{M}}$ taken to be $\mathbf{M}_{\mathcal{M}_0} + \mathbf{M}_{\mathcal{M}_1}$, a bound on the

uniform covering number of $\mathcal{M}$ can be given by

$$N_{\mathcal{M}}(\varepsilon) \leq 16(\mathbf{c}')^2(\varepsilon/2)^{-2d-2}, \quad 0 < \varepsilon \leq 1.$$

With the assumption that $\mathcal{L}(E_{t,\mathbf{x}}(h)) \leq Ch^{d-1}$ for $E_{t,\mathbf{x}}(h) = \{\mathbf{y} \in \mathcal{A}_t : \varsigma_t \mathcal{d}(\mathbf{y}, \mathbf{x})/h \in \text{supp}(K)\}$ for all $t \in \{0, 1\}$, $\mathbf{x} \in \mathcal{B}$, and the fact that $\text{TV}_{\mathcal{M}_t} \lesssim h^{d/2-1}$ for $t \in \{0, 1\}$, the same argument as in the paragraph **Total Variation** in the proof of Theorem SA-25 shows

$$\text{TV}_{\mathcal{M}} \lesssim h^{d/2-1}.$$

Now apply Theorem SA-25 with $\mathcal{G}$, $\mathcal{M}$ defined in Equation (SA-2), $\mathcal{R} = \{\text{Id}\}$, $\mathcal{S} = \{1\}$, noting that

$$\bar{\mathbb{T}}(\mathbf{x}) = A_n(g_{\mathbf{x}}, m_{\mathbf{x}}, \text{Id}, 1), \quad (g_{\mathbf{x}}, m_{\mathbf{x}}) \in \mathcal{G} \times \mathcal{M}, \quad \mathbf{x} \in \mathcal{B},$$

the result follows. $\square$

Lemma SA-11 (VC Class to VC2 Class). *Assume $\mathcal{F}$ is a VC class on a measure space $(\mathcal{X}, \mathcal{B})$: there exists an envelope function F and positive constants $c(\mathcal{F}), d(\mathcal{F})$ such that for all $\varepsilon \in (0, 1)$,*

$$\sup_Q N(\mathcal{F}, \|\cdot\|_{Q,1}, \varepsilon \|F\|_{Q,1}) \leq c(\mathcal{F})\varepsilon^{-d(\mathcal{F})},$$

where the supremum is taken over all finite discrete measures. Then, $\mathcal{F}$ is also VC2 class: for all $\varepsilon \in (0, 1)$,

$$\sup_Q N(\mathcal{F}, \|\cdot\|_{Q,2}, \varepsilon \|F\|_{Q,2}) \leq c(\mathcal{F})(\varepsilon^2/2)^{-d(\mathcal{F})},$$

where the supremum is taken over all finite discrete measures.

Proof of Lemma SA-11. Let Q be a finite discrete probability measure. Let $f, g \in \mathcal{F}$. Then, $\int |f-g|^2 dQ \leq 2 \int |f-g|F dQ$. Define another probability measure $\tilde{Q}(c_k) = F(c_k)Q(c_k)/\|F\|_{Q,1}$ on the support of Q , denoted by $\{c_1, \dots, c_k, \dots\}$. Then,

$$\int |f-g|^2 dQ \leq 2 \|F\|_{Q,1} \int |f-g| d\tilde{Q} \leq 2 \|F\|_{Q,1} \|f-g\|_{\tilde{Q},1}.$$

Hence, if we take an $\varepsilon^2/2$ -net in $(\mathcal{F}, \|\cdot\|_{\tilde{Q},1})$ with cardinality no greater than $c(\mathcal{F})(\varepsilon^2/2)^{-d(\mathcal{F})}$, then for any $f \in \mathcal{F}$, there exists a $g \in \mathcal{F}$ such that $\|f-g\|_{\tilde{Q},1} \leq \varepsilon^2/2 \|F\|_{\tilde{Q},1}$, and hence

$$\|f-g\|_{Q,2}^2 \leq 2\varepsilon^2/2 \|F\|_{Q,1} \|F\|_{\tilde{Q},1} \leq \varepsilon^2 \|F\|_{Q,2}^2,$$

which gives the result. $\square$

SA-8.13 Proof of Theorem SA-8

The proof is divided into three theorem-specific technical lemmas. We first record a generic comparison lemma that translates uniform process approximations into Kolmogorov-distance bounds for suprema.

Lemma SA-12 (Generic Suprema Comparison). *Let $\mathbb{T}_j = (\mathbb{T}_j(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$, $j = 1, 2$, be real-valued processes*

defined on a common probability space. Write

$$M_j = \sup_{\mathbf{x} \in \mathcal{B}} |\mathbb{T}_j(\mathbf{x})|, \quad j = 1, 2, \quad \Delta_{12} = \sup_{\mathbf{x} \in \mathcal{B}} |\mathbb{T}_1(\mathbf{x}) - \mathbb{T}_2(\mathbf{x})|.$$

For any $\eta > 0$,

$$\sup_{u \in \mathbb{R}} |\mathbb{P}(M_1 \leq u) - \mathbb{P}(M_2 \leq u)| \leq \text{AC}_2(\eta) + \mathbb{P}(\Delta_{12} > \eta),$$

where

$$\text{AC}_2(\eta) = \sup_{u \in \mathbb{R}} \mathbb{P}(|M_2 - u| \leq \eta).$$

If, in addition, $\mathbb{T}_2$ is a separable mean-zero Gaussian process and the Gaussian anti-concentration bound applies to M_2 , then

$$\sup_{u \in \mathbb{R}} |\mathbb{P}(M_1 \leq u) - \mathbb{P}(M_2 \leq u)| \leq 4\eta\{\mathbb{E}[M_2] + 1\} + \mathbb{P}(\Delta_{12} > \eta).$$

Consequently, if $\mathbb{E}[\Delta_{12}] \leq r_n$, then the last probability can be bounded by r_n/η .

Proof of Lemma SA-12. On the event $\{\Delta_{12} \leq \eta\}$, the inequalities

$$M_1 \leq u \implies M_2 \leq u + \eta, \quad M_2 \leq u - \eta \implies M_1 \leq u$$

hold for every $u \in \mathbb{R}$. Hence

$$\begin{aligned} \mathbb{P}(M_1 \leq u) - \mathbb{P}(M_2 \leq u) &\leq \mathbb{P}(u < M_2 \leq u + \eta) + \mathbb{P}(\Delta_{12} > \eta), \\ \mathbb{P}(M_2 \leq u) - \mathbb{P}(M_1 \leq u) &\leq \mathbb{P}(u - \eta < M_2 \leq u) + \mathbb{P}(\Delta_{12} > \eta). \end{aligned}$$

Taking absolute values and then the supremum over u gives the first claim. The second display follows from the Gaussian anti-concentration inequality of Chernozhukov et al. [2014a, Theorem 2.1], applied to the symmetrized Gaussian process whose supremum is M_2 . The final claim follows from Markov's inequality. $\square$

For the theorem-specific lemmas below, define

$$\mathfrak{R}_{\text{sa}} = (\log n)^{\frac{3}{2}} \left(\frac{1}{nh^d} \right)^{\frac{1}{2d+2} \cdot \frac{v}{v+2}} + \log(n) \left(\frac{1}{n^{\frac{v}{2+v}} h^d} \right)^{1/2},$$

the stochastic-linearization remainder

$$\mathfrak{R}_{\text{lin}} = \sqrt{\log(1/h)} \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right) + \sqrt{nh^d} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})|,$$

and the covariance-comparison rate

$$\mathfrak{R}_{\text{cc}} = \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}.$$

Lemma SA-13 (KS Distance Between $\bar{T}$ and Z). *Suppose the conditions of Theorem SA-8 hold. Then*

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}(\mathbf{x})| \leq u \right) - \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |Z(\mathbf{x})| \leq u \right) \right| \lesssim (\log n)^{1/4} \mathfrak{R}_{\text{sa}}^{1/2}.$$

Proof of Lemma SA-13. Let $a_n > 0$ be chosen below. Applying Lemma SA-12 with $\mathbb{T}_1 = \bar{T}$ and $\mathbb{T}_2 = Z$, and using Theorem SA-7, gives

$$\begin{aligned} \sup_{u \in \mathbb{R}} \left| \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}(\mathbf{x})| \leq u \right) - \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |Z(\mathbf{x})| \leq u \right) \right| \\ \leq 4a_n \left(\mathbb{E} \left[\sup_{\mathbf{x} \in \mathcal{B}} |Z(\mathbf{x})| \right] + 1 \right) + \frac{C \mathfrak{R}_{\text{sa}}}{a_n}. \end{aligned}$$

The Gaussian process Z has uniformly bounded pointwise variance. Moreover, by the same Lipschitz calculation for the distance-indexed local-polynomial weights used in Theorem SA-7, its canonical metric satisfies

$$\rho(\mathbf{x}, \mathbf{y}) = \mathbb{E} [\{Z(\mathbf{x}) - Z(\mathbf{y})\}^2]^{1/2} \lesssim h^{-1} \|\mathbf{x} - \mathbf{y}\|_{\infty}.$$

Since $\mathcal{B}$ is compact and $\liminf_{n \rightarrow \infty} \log h / \log n > -\infty$,

$$\log N(\varepsilon, \mathcal{B}, \rho) \lesssim d \log \left(\frac{1}{h\varepsilon} \right) \lesssim \log n + \log(1/\varepsilon), \quad 0 < \varepsilon \leq 1.$$

Dudley's entropy integral therefore gives

$$\mathbb{E} \left[\sup_{\mathbf{x} \in \mathcal{B}} |Z(\mathbf{x})| \right] \lesssim \sqrt{\log n}.$$

Choosing $a_n \asymp (\mathfrak{R}_{\text{sa}} / \sqrt{\log n})^{1/2}$ gives the result. $\square$

Lemma SA-14 (KS Distance Between $\hat{T}$ and $\bar{T}$). *Suppose the conditions of Theorem SA-8 hold. Then*

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\hat{T}(\mathbf{x})| \leq u \right) - \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}(\mathbf{x})| \leq u \right) \right| = o(1).$$

Proof of Lemma SA-14. Theorem SA-6 gives

$$\Delta_{\hat{T}, \bar{T}} := \sup_{\mathbf{x} \in \mathcal{B}} |\hat{T}(\mathbf{x}) - \bar{T}(\mathbf{x})| = O_{\mathbb{P}}(\mathfrak{R}_{\text{lin}}).$$

The side conditions imply $\mathfrak{R}_{\text{lin}} \sqrt{\log n} = o(1)$. Choose a deterministic sequence a_n such that $\mathfrak{R}_{\text{lin}} = o(a_n)$ and $a_n \sqrt{\log n} = o(1)$. Applying Lemma SA-12 with $\mathbb{T}_1 = \hat{T}$ and $\mathbb{T}_2 = \bar{T}$ gives

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}(M_{\hat{T}} \leq u) - \mathbb{P}(M_{\bar{T}} \leq u) \right| \leq \sup_{u \in \mathbb{R}} \mathbb{P}(|M_{\bar{T}} - u| \leq a_n) + \mathbb{P}(\Delta_{\hat{T}, \bar{T}} > a_n),$$

where $M_{\hat{T}} = \sup_{\mathbf{x} \in \mathcal{B}} |\hat{T}(\mathbf{x})|$ and $M_{\bar{T}} = \sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}(\mathbf{x})|$. The second term is $o(1)$. The anti-concentration term for $M_{\bar{T}}$ is bounded through Lemma SA-13: for $M_Z = \sup_{\mathbf{x} \in \mathcal{B}} |Z(\mathbf{x})|$,

$$\mathbb{P}(|M_{\bar{T}} - u| \leq a_n) \leq \mathbb{P}(|M_Z - u| \leq a_n) + 2 \sup_{v \in \mathbb{R}} |\mathbb{P}(M_{\bar{T}} \leq v) - \mathbb{P}(M_Z \leq v)|$$

$$\leq 4a_n\{\mathbb{E}[M_Z] + 1\} + 2\sup_{v \in \mathbb{R}} |\mathbb{P}(M_{\overline{T}} \leq v) - \mathbb{P}(M_Z \leq v)|.$$

The first term is $o(1)$ because $a_n\sqrt{\log n} = o(1)$, and the second is $o(1)$ by Lemma SA-13. This proves the result. $\square$

Lemma SA-15 (KS Distance Between Z and $\widehat{Z}$). *Suppose the conditions of Theorem SA-8 hold. Then*

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |Z(\mathbf{x})| \leq u \right) - \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}(\mathbf{x})| \leq u \mid \mathbf{V} \right) \right| \lesssim_{\mathbb{P}} \log(n) \mathfrak{R}_{\text{cc}}^{1/2} = o_{\mathbb{P}}(1).$$

Proof of Lemma SA-15. Let

$$\Pi_{\mathbf{x}, \mathbf{y}} = \frac{\Xi_{\mathbf{x}, \mathbf{y}}}{\sqrt{\Xi_{\mathbf{x}, \mathbf{x}} \Xi_{\mathbf{y}, \mathbf{y}}}}, \quad \widehat{\Pi}_{\mathbf{x}, \mathbf{y}} = \frac{\widehat{\Xi}_{\mathbf{x}, \mathbf{y}}}{\sqrt{\widehat{\Xi}_{\mathbf{x}, \mathbf{x}} \widehat{\Xi}_{\mathbf{y}, \mathbf{y}}}}.$$

Lemma SA-5 gives

$$\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} |\widehat{\Pi}_{\mathbf{x}, \mathbf{y}} - \Pi_{\mathbf{x}, \mathbf{y}}| \lesssim_{\mathbb{P}} \mathfrak{R}_{\text{cc}},$$

because $\inf_{\mathbf{x} \in \mathcal{B}} \Xi_{\mathbf{x}, \mathbf{x}} \gtrsim (nh^d)^{-1}$ and $\inf_{\mathbf{x} \in \mathcal{B}} \widehat{\Xi}_{\mathbf{x}, \mathbf{x}} \gtrsim_{\mathbb{P}} (nh^d)^{-1}$.

Let $\mathcal{B}_{\delta_n}$ be a δ_n -net of $\mathcal{B}$ under $\|\cdot\|_{\infty}$ with cardinality $M \lesssim \delta_n^{-d}$, and write $\mathbf{Z}^{\delta_n}$ and $\widehat{\mathbf{Z}}^{\delta_n}$ for the restrictions of Z and $\widehat{Z}$ to this net. By Chernozhukov et al. [2022, Theorem 2.1],

$$\sup_{\mathbf{y} \in \mathbb{R}^M} \left| \mathbb{P}(\mathbf{Z}^{\delta_n} \leq \mathbf{y}) - \mathbb{P}(\widehat{\mathbf{Z}}^{\delta_n} \leq \mathbf{y} \mid \mathbf{V}) \right| \lesssim_{\mathbb{P}} \log(M) \mathfrak{R}_{\text{cc}}^{1/2}.$$

Hence the same bound applies to the distribution functions of $\|\mathbf{Z}^{\delta_n}\|_{\infty}$ and $\|\widehat{\mathbf{Z}}^{\delta_n}\|_{\infty}$.

Choose $\delta_n = n^{-s}$ with $s > 0$ large enough that

$$\sqrt{\log n} h^{-1} \delta_n = o(\mathfrak{R}_{\text{cc}}^{1/2}).$$

Such an s exists because h is bounded below by a polynomial in n . The same Gaussian-metric calculation used in Lemma SA-13 and its feasible analogue imply

$$\begin{aligned} \mathbb{E} \left[\sup_{\substack{\mathbf{x}, \mathbf{y} \in \mathcal{B} \\ \|\mathbf{x} - \mathbf{y}\|_{\infty} \leq \delta_n}} |Z(\mathbf{x}) - Z(\mathbf{y})| \right] &\lesssim \sqrt{\log n} h^{-1} \delta_n, \\ \mathbb{E} \left[\sup_{\substack{\mathbf{x}, \mathbf{y} \in \mathcal{B} \\ \|\mathbf{x} - \mathbf{y}\|_{\infty} \leq \delta_n}} |\widehat{Z}(\mathbf{x}) - \widehat{Z}(\mathbf{y})| \mid \mathbf{V} \right] &\lesssim_{\mathbb{P}} \sqrt{\log n} h^{-1} \delta_n. \end{aligned}$$

Markov's inequality makes both modulus errors $o_{\mathbb{P}}(\mathfrak{R}_{\text{cc}}^{1/2})$. Combining the finite-net comparison, the two modulus bounds, and Gaussian anti-concentration gives the displayed Kolmogorov bound. The side conditions imply $\log(n) \mathfrak{R}_{\text{cc}}^{1/2} = o_{\mathbb{P}}(1)$. $\square$

The proof of Theorem SA-8 now follows from the triangle inequality applied to the three comparison lemmas above. Furthermore, by definition of $\widehat{\mathbf{I}}_{\alpha}(\mathbf{x})$,

$$\mathbb{P} \left[\vartheta(\mathbf{x}) \in \widehat{\mathbf{I}}_{\alpha}(\mathbf{x}), \text{ for all } \mathbf{x} \in \mathcal{B} \right] = \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{\mathbf{T}}(\mathbf{x})| \leq \mathfrak{q}_{\alpha} \right)$$

$$\begin{aligned}
&= \mathbb{E} \left[\mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}(\mathbf{x})| \leq q_\alpha \mid \mathbf{V} \right) \right] + o(1) \\
&= 1 - \alpha + o(1).
\end{aligned}$$

This completes the proof. $\square$

SA-8.14 Proof of Theorem SA-9

We employ the following result established by Chernozhukov et al. [2014b, Corollary 2.2].

Lemma SA-16 (Gaussian Approximation of Suprema). *Let $X_1, \dots, X_n$ be i.i.d. random variables in a measurable space $(S, \mathcal{S})$ with law P , where $n \geq 3$. Let $\mathcal{F}$ be a class of P -centered measurable functions $f : S \rightarrow \mathbb{R}$, and define $\mathbb{G}_n f = n^{-1/2} \sum_{i=1}^n f(X_i)$. Suppose that:*

- (i) $\mathcal{F}$ is pointwise measurable.
- (ii) There exists a measurable envelope F such that $|f| \leq F$ for all $f \in \mathcal{F}$, and constants $A \geq e$ and $v \geq 1$ such that

$$\sup_Q N(\mathcal{F}, \|\cdot\|_{Q,2}, \epsilon \|F\|_{Q,2}) \leq (A/\epsilon)^v, \quad 0 < \epsilon \leq 1,$$

where the supremum is over all finite discrete probability measures on $(S, \mathcal{S})$.

- (iii) For some $b \geq \sigma > 0$ and $q \in [4, \infty]$,

$$\sup_{f \in \mathcal{F}} P|f|^k \leq \sigma^2 b^{k-2}, \quad k = 2, 3, 4, \quad \text{and} \quad \|F\|_{P,q} \leq b.$$

Let $Z = \sup_{f \in \mathcal{F}} \mathbb{G}_n f$. Then, for every $\gamma \in (0, 1)$, there exists, on a possibly enlarged probability space, a tight mean-zero Gaussian process $(\mathbb{G}_P f : f \in \mathcal{F})$ with covariance function $\mathbb{E}[\mathbb{G}_P(f) \mathbb{G}_P(g)] = P(fg)$ and almost surely uniformly $\|\cdot\|_{P,2}$ -continuous trajectories such that, for $\tilde{Z} = \sup_{f \in \mathcal{F}} \mathbb{G}_P f$,

$$\mathbb{P} \left\{ |Z - \tilde{Z}| > \frac{bK_n}{\gamma^{1/q} n^{1/2-1/q}} + \frac{(b\sigma)^{1/2} K_n^{3/4}}{\gamma^{1/2} n^{1/4}} + \frac{(b\sigma^2 K_n^2)^{1/3}}{\gamma^{1/3} n^{1/6}} \right\} \leq C \left(\gamma + \frac{\log n}{n} \right),$$

where $K_n = v(\log n \vee \log(Ab/\sigma))$, and $C > 0$ depends only on q .

Let $\mathbf{z}_i = (\mathbf{X}_i, Y_i)$, $1 \leq i \leq n$. For each $\mathbf{x} \in \mathcal{B}$, define

$$\tilde{f}_{\mathbf{x}}(\mathbf{u}, u) = g_{\mathbf{x}}(\mathbf{u})u + m_{\mathbf{x}}(\mathbf{u}), \quad f_{\mathbf{x}}(\mathbf{u}, u) = \tilde{f}_{\mathbf{x}}(\mathbf{u}, u) - \mathbb{E}[\tilde{f}_{\mathbf{x}}(\mathbf{X}_i, Y_i)].$$

Let $\mathcal{F}_{\pm} = \{f_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\} \cup \{-f_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$. By the representation preceding Theorem SA-7,

$$\sup_{\mathbf{x} \in \mathcal{B}} |\overline{T}(\mathbf{x})| = \sup_{f \in \mathcal{F}_{\pm}} \frac{1}{\sqrt{n}} \sum_{i=1}^n f(\mathbf{z}_i).$$

We apply Lemma SA-16 to $\mathcal{F}_{\pm}$.

First, $\mathcal{F}_{\pm}$ is pointwise measurable by the compactness and separability of $\mathcal{B}$ and the measurability of the signed-distance kernels. The entropy calculation used in the proof of Theorem SA-7, but without the total-variation step, gives a VC-type bound for $\mathcal{G}$ and $\mathcal{M}$. The fixed side indicators $\mathbf{1}(\mathbf{u} \in \mathcal{A}_t)$ only contract

$L_2(Q)$ distances. Thus, with envelope

$$F(\mathbf{u}, u) = Ch^{-d/2}(|u| + 1),$$

for a sufficiently large constant C , there are constants $A_h \geq e$ and $v \geq 1$, with A_h at most polynomial in $1/h$, such that

$$\sup_Q N(\mathcal{F}_\pm, \|\cdot\|_{Q,2}, \epsilon \|F\|_{Q,2}) \leq (A_h/\epsilon)^v, \quad 0 < \epsilon \leq 1.$$

Centering changes the covering constants only by a universal factor because $|P(\tilde{f}_\mathbf{x} - \tilde{f}_\mathbf{y})| \leq \|\tilde{f}_\mathbf{x} - \tilde{f}_\mathbf{y}\|_{P,2}$.

For the moment condition, take $q = 2 + v \geq 4$. Assumption SA-1(v) and the boundedness of $\theta_{t,\mathbf{x}}^*(D_\mathbf{u}(\mathbf{x}))$ on the kernel support give

$$\|F\|_{P,2+v} \lesssim h^{-d/2}.$$

The same local-polynomial moment calculation used in the proofs of Lemmas SA-4 and SA-5 gives

$$\sup_{f \in \mathcal{F}_\pm} P|f|^k \lesssim h^{-d(k-2)/2}, \quad k = 2, 3, 4.$$

Hence Lemma SA-16 applies with $\sigma = 1$ and $b \asymp h^{-d/2}$. Since $\liminf_{n \rightarrow \infty} \log h / \log n > -\infty$ and A_h is at most polynomial in $1/h$, $K_n \lesssim \log n$. Taking $\gamma = (\log n)^{-1}$ yields

$$\mathbb{P} \left[\left| \sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}(\mathbf{x})| - \sup_{\mathbf{x} \in \mathcal{B}} |Z(\mathbf{x})| \right| > \frac{\log n}{(nh^d)^{1/6}} + \frac{(\log n)^{5/4}}{(nh^d)^{1/4}} + \frac{(\log n)^{3/2}}{(n^{\frac{v}{2+v}} h^d)^{1/2}} \right] \lesssim \frac{1}{\log n} + \frac{\log n}{n}.$$

The covariance of the Gaussian process delivered by Lemma SA-16 is $P(f_{\mathbf{x}_1} f_{\mathbf{x}_2})$, which equals $\text{Cov}[\bar{T}(\mathbf{x}_1), \bar{T}(\mathbf{x}_2)]$. This proves the theorem. $\square$

SA-8.15 Proof of Theorem SA-10

Define

$$\mathfrak{R}_{\text{sup}} = \frac{\log n}{(nh^d)^{1/6}} + \frac{(\log n)^{5/4}}{(nh^d)^{1/4}} + \frac{(\log n)^{3/2}}{(n^{\frac{v}{2+v}} h^d)^{1/2}}.$$

The entropy calculation in the proof of Theorem SA-9 implies

$$\mathbb{E} \left[\sup_{\mathbf{x} \in \mathcal{B}} |Z(\mathbf{x})| \right] \lesssim \sqrt{\log n}.$$

The condition $n^{\frac{v}{2+v}} h^d / (\log n)^9 \rightarrow \infty$ implies $\mathfrak{R}_{\text{sup}} \sqrt{\log n} = o(1)$. Let

$$M_{\bar{T}} = \sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}(\mathbf{x})|, \quad M_Z = \sup_{\mathbf{x} \in \mathcal{B}} |Z(\mathbf{x})|.$$

Theorem SA-9 and Gaussian anti-concentration give

$$\sup_{u \in \mathbb{R}} |\mathbb{P}(M_{\overline{T}} \leq u) - \mathbb{P}(M_Z \leq u)| = o(1). \quad (\text{SA-23})$$

Next, Theorem SA-6 gives

$$\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{\mathbf{T}}(\mathbf{x}) - \overline{\mathbf{T}}(\mathbf{x})| = O_{\mathbb{P}}(\mathfrak{R}_{1\text{in}}),$$

where

$$\mathfrak{R}_{1\text{in}} = \sqrt{\log(1/h)} \left(\sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right) + \sqrt{nh^d} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}(\mathbf{x})|.$$

The polynomial lower bound on h , the condition $n^{\frac{v}{2+v}} h^d / (\log n)^9 \rightarrow \infty$, and the stated small-bias condition imply $\mathfrak{R}_{1\text{in}} \sqrt{\log n} = o(1)$. Combining this with Equation (SA-23) and another Gaussian anti-concentration step yields

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{\mathbf{T}}(\mathbf{x})| \leq u \right) - \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\overline{\mathbf{T}}(\mathbf{x})| \leq u \right) \right| = o(1). \quad (\text{SA-24})$$

Finally, Lemma SA-5 gives the feasible covariance-comparison rate

$$\mathfrak{R}_{\text{cc}} = \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}.$$

The same finite-net Gaussian comparison and anti-concentration argument used in the proof of Theorem SA-8 gives

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |Z(\mathbf{x})| \leq u \right) - \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}(\mathbf{x})| \leq u \mid \mathbf{V} \right) \right| \lesssim_{\mathbb{P}} \log(n) \mathfrak{R}_{\text{cc}}^{1/2} = o_{\mathbb{P}}(1), \quad (\text{SA-25})$$

where the last equality follows from $n^{\frac{v}{2+v}} h^d / (\log n)^9 \rightarrow \infty$. The displayed Kolmogorov approximation follows from the triangle inequality applied to Equations (SA-23), (SA-24), and (SA-25).

The simultaneous coverage statement follows from the definition of $\widehat{\mathbf{I}}_{\alpha}(\mathbf{x})$ and the conditional critical value $\mathfrak{q}_{\alpha}$:

$$\begin{aligned} \mathbb{P} \left[\vartheta(\mathbf{x}) \in \widehat{\mathbf{I}}_{\alpha}(\mathbf{x}), \text{ for all } \mathbf{x} \in \mathcal{B} \right] &= \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{\mathbf{T}}(\mathbf{x})| \leq \mathfrak{q}_{\alpha} \right) \\ &= \mathbb{E} \left[\mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}(\mathbf{x})| \leq \mathfrak{q}_{\alpha} \mid \mathbf{V} \right) \right] + o(1) \\ &= 1 - \alpha + o(1). \end{aligned}$$

□

SA-8.16 Proof of Lemma SA-6

Let $\mathfrak{L}_{t,\text{WBATE}}$ and $\Xi_{t,\text{WBATE}}$ denote the side-specific analogues of $\mathfrak{L}_{\text{WBATE}}$ and Ξ_{WBATE} . Fubini's theorem and the definition of $\Xi_{t,\mathbf{x}_1,\mathbf{x}_2}$ give

$$\mathbb{V}(\mathfrak{L}_{t,\text{WBATE}}) = \Xi_{t,\text{WBATE}}.$$

The covariance between the two side-specific linear terms is zero because the treatment and control assignment indicators are disjoint observation by observation. Hence $\mathbb{V}(\mathfrak{L}_{\text{WBATE}}) = \Xi_{\text{WBATE}}$.

Let $R_K < \infty$ satisfy $\text{supp}(K) \subseteq [-R_K, R_K]$. By equivalence of $\mathcal{d}$ and the Euclidean distance, if $\Xi_{\mathbf{x}_1,\mathbf{x}_2} \neq 0$ then $\|\mathbf{x}_1 - \mathbf{x}_2\| \lesssim h$. Therefore

$$\mathcal{E}(h) = \{(\mathbf{x}_1, \mathbf{x}_2) \in \mathcal{B}^2 : \|\mathbf{x}_1 - \mathbf{x}_2\| \leq Ch\}$$

has product boundary measure $\nu_{\mathcal{B},2}(\mathcal{E}(h)) \lesssim h^{d-1}$ by (SA-4). Uniform boundedness of $\Psi_{t,\mathbf{x}}^{-1}$ and $\Upsilon_{t,\mathbf{x}_1,\mathbf{x}_2}$ gives $|\Xi_{\mathbf{x}_1,\mathbf{x}_2}| \lesssim (nh^d)^{-1}$, and hence

$$\Xi_{\text{WBATE}} \lesssim \frac{h^{d-1}}{nh^d} = \frac{1}{nh}.$$

Under (SA-5), this upper bound is complemented by the matching lower bound.

The same support restriction applies to $\widehat{\Xi}_{\mathbf{x}_1,\mathbf{x}_2}$. Hence, using Lemma SA-5,

$$\begin{aligned} \left| \widehat{\Xi}_{\text{WBATE}} - \Xi_{\text{WBATE}} \right| &\leq \int_{\mathcal{E}(h)} \left| \widehat{\Xi}_{\mathbf{x}_1,\mathbf{x}_2} - \Xi_{\mathbf{x}_1,\mathbf{x}_2} \right| |w(\mathbf{x}_1)w(\mathbf{x}_2)| d\nu_{\mathcal{B},2}(\mathbf{x}_1, \mathbf{x}_2) \\ &\lesssim_{\mathbb{P}} h^{d-1} \sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} \left| \widehat{\Xi}_{\mathbf{x}_1,\mathbf{x}_2} - \Xi_{\mathbf{x}_1,\mathbf{x}_2} \right| \\ &= o_{\mathbb{P}}\left(\frac{1}{nh}\right), \end{aligned}$$

which proves the plug-in variance claim. The ratio claim follows by dividing this additive bound by the lower bound in (SA-5). $\square$

SA-8.17 Proof of Lemma SA-7

It is enough to consider one side $t \in \{0, 1\}$. Define the side-specific nonlinear WBATE remainder

$$\mathfrak{Q}_{t,\text{WBATE}} = \int_{\mathcal{B}} \mathbf{e}_0^\top (\widehat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}) \mathbf{O}_{t,\mathbf{x}} w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}), \quad \mathfrak{Q}_{\text{WBATE}} = \mathfrak{Q}_{1,\text{WBATE}} - \mathfrak{Q}_{0,\text{WBATE}}.$$

Let

$$\mathbf{A}_{t,\mathbf{x}} = \widehat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1} = -\widehat{\Psi}_{t,\mathbf{x}}^{-1} (\widehat{\Psi}_{t,\mathbf{x}} - \Psi_{t,\mathbf{x}}) \Psi_{t,\mathbf{x}}^{-1}.$$

On the event where $\sup_{\mathbf{x} \in \mathcal{B}} \|\widehat{\Psi}_{t,\mathbf{x}}^{-1}\| \lesssim 1$, which has probability approaching one by Lemma SA-3,

$$|\mathfrak{Q}_{t,\text{WBATE}}| \lesssim \left| \int_{\mathcal{B}} \mathbf{A}_{t,\mathbf{x}} \mathbf{O}_{t,\mathbf{x}} w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}) \right|.$$

We suppress fixed finite-dimensional vectors and matrices in the bounds below.

Decompose $\mathbf{O}_{t,\mathbf{x}}$ into a conditional-mean-zero part and a conditional-mean part:

$$\mathbf{O}_{t,\mathbf{x}} = \mathbf{Q}_{t,\mathbf{x}} + \mathbf{M}_{t,\mathbf{x}},$$

where

$$\begin{aligned}\mathbf{Q}_{t,\mathbf{x}} &= \mathbb{E}_n \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) K_h(D_i(\mathbf{x})) \varepsilon_{t,i} \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t) \right], \\ \mathbf{M}_{t,\mathbf{x}} &= \mathbb{E}_n \left[\mathbf{r}_p \left(\frac{D_i(\mathbf{x})}{h} \right) K_h(D_i(\mathbf{x})) \{ \mu_t(\mathbf{X}_i) - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x})) \} \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t) \right],\end{aligned}$$

with $\varepsilon_{t,i} = Y_i(t) - \mu_t(\mathbf{X}_i)$. The first component is mean zero conditional on $\mathbf{X}_1, \dots, \mathbf{X}_n$. The second component is not conditionally mean zero; this is the additional distance-based term absent from the location-based analysis.

For the first component, conditional on $\mathbf{X}_1, \dots, \mathbf{X}_n$, write $\psi_{t,\mathbf{x}}(\mathbf{z}) = \mathbf{r}_p(D_{\mathbf{z}}(\mathbf{x})/h) K_h(D_{\mathbf{z}}(\mathbf{x})) \mathbf{1}(\mathbf{z} \in \mathcal{A}_t)$. The summand inside the boundary integral is nonzero only when $\mathbf{z}$ is within an $O(h)$ neighborhood of $\mathbf{x}$. Therefore, after applying Fubini to the conditional second moment, the only pairs $(\mathbf{x}_1, \mathbf{x}_2)$ that contribute are those satisfying $\|\mathbf{x}_1 - \mathbf{x}_2\| \lesssim h$. The local boundary-measure condition (SA-4) gives product Hausdorff measure $O(h^{d-1})$ for this effective pair set, while each localized kernel product has conditional expectation of order h^{-d} . Hence

$$\mathbb{E} \left[\left| \int_{\mathcal{B}} \mathbf{A}_{t,\mathbf{x}} \mathbf{Q}_{t,\mathbf{x}} w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}) \right|^2 \middle| \mathbf{X}_1, \dots, \mathbf{X}_n \right] \lesssim \frac{\sup_{\mathbf{x} \in \mathcal{B}} \|\mathbf{A}_{t,\mathbf{x}}\|^2}{nh}.$$

Lemma SA-3 therefore implies

$$\int_{\mathcal{B}} \mathbf{A}_{t,\mathbf{x}} \mathbf{Q}_{t,\mathbf{x}} w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}) = O_{\mathbb{P}} \left(\frac{\sqrt{\log(1/h)}}{nh^{(d+1)/2}} \right) = o_{\mathbb{P}} \left(\frac{1}{nh^d} \right),$$

because $d \geq 2$, $h = o(1)$, and $h^{d-1} \log(1/h) \rightarrow 0$.

It remains to bound the conditional-mean component. Write each coordinate of $\int_{\mathcal{B}} \mathbf{A}_{t,\mathbf{x}} \mathbf{M}_{t,\mathbf{x}} w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x})$ as

$$\frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n H_{ij}, \quad H_{ij} = \int_{\mathcal{B}} a_{\mathbf{x}}(\mathbf{X}_i) m_{\mathbf{x}}(\mathbf{X}_j) w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}),$$

where $a_{\mathbf{x}}$ is a centered coordinate of $\widehat{\Psi}_{t,\mathbf{x}} - \Psi_{t,\mathbf{x}}$ and $m_{\mathbf{x}}$ is the corresponding conditional-mean coordinate of $\mathbf{M}_{t,\mathbf{x}}$. The localized part of $a_{\mathbf{x}}$ and the whole of $m_{\mathbf{x}}$ are supported on $O(h)$ -neighborhoods of $\mathbf{x}$, with envelopes of order h^{-d} . The centering constant in $a_{\mathbf{x}}$ is only $O(1)$ and therefore gives smaller contributions than the localized part. Thus the dominant part of H_{ij} is nonzero only when the relevant observations are within distance $O(h)$ of the same boundary point.

For the diagonal term, the boundary integral has size at most $h^{-2d} \mathfrak{H}^{d-1}(\mathcal{B} \cap B(\mathbf{X}_i, Ch)) \lesssim h^{-d-1}$ on the event that $\mathbf{X}_i$ lies in an $O(h)$ tube around $\mathcal{B}$, and is zero otherwise up to the smaller centering-constant contribution. Since this tube has probability $O(h)$,

$$\mathbb{E}|H_{ii}| \lesssim h^{-d}.$$

For the off-diagonal term, independence and Fubini give

$$\mathbb{E}[H_{ij}^2] = \int_{\mathcal{B}} \int_{\mathcal{B}} \mathbb{E}[a_{\mathbf{x}_1}(\mathbf{X}_i)a_{\mathbf{x}_2}(\mathbf{X}_i)]\mathbb{E}[m_{\mathbf{x}_1}(\mathbf{X}_j)m_{\mathbf{x}_2}(\mathbf{X}_j)]w(\mathbf{x}_1)w(\mathbf{x}_2) d\mathfrak{H}^{d-1}(\mathbf{x}_1) d\mathfrak{H}^{d-1}(\mathbf{x}_2).$$

The leading localized contribution to the two expectations in the integrand is zero unless $\|\mathbf{x}_1 - \mathbf{x}_2\| \lesssim h$, and the terms containing the centering constants are of the same order or smaller. On this effective set each expectation is bounded by $O(h^{-d})$, and the effective set has product Hausdorff measure $O(h^{d-1})$. Hence

$$\mathbb{E}[H_{ij}^2] \lesssim h^{-d-1}, \quad i \neq j.$$

The off-diagonal kernel is degenerate because $\mathbb{E}[a_{\mathbf{x}}(\mathbf{X}_i)] = 0$ and $\mathbb{E}[m_{\mathbf{x}}(\mathbf{X}_j)] = 0$ for every $\mathbf{x}$. Thus,

$$\frac{1}{n^2} \sum_{i=1}^n H_{ii} = O_{\mathbb{P}}\left(\frac{1}{nh^d}\right), \quad \frac{1}{n^2} \sum_{i \neq j} H_{ij} = O_{\mathbb{P}}\left(\frac{1}{nh^{(d+1)/2}}\right) = o_{\mathbb{P}}\left(\frac{1}{nh^d}\right).$$

Combining the two components and summing over $t \in \{0, 1\}$ gives $\mathfrak{Q}_{\text{WBATE}} = O_{\mathbb{P}}(1/(nh^d))$. Finally,

$$\frac{1/(nh^d)}{1/\sqrt{nh}} = \frac{1}{\sqrt{nh^{2d-1}}},$$

which is $o(1)$ under $nh^{2d-1} \rightarrow \infty$. □

SA-8.18 Proof of Theorem SA-11

The decomposition (SA-3), Lemma SA-6, and Lemma SA-7 give

$$\left| \widehat{\vartheta}_{\text{WBATE}} - \vartheta_{\text{WBATE}} \right| \leq |\mathfrak{B}_{\text{WBATE}}| + O_{\mathbb{P}}\left(\frac{1}{\sqrt{nh}}\right) + O_{\mathbb{P}}\left(\frac{1}{nh^d}\right),$$

which is the stated rate. □

SA-8.19 Proof of Theorem SA-12

By (SA-3), Lemma SA-7, the small-bias condition, and the aggregate nondegeneracy condition,

$$\frac{\widehat{\vartheta}_{\text{WBATE}} - \vartheta_{\text{WBATE}} - \mathfrak{L}_{\text{WBATE}}}{\sqrt{\Xi_{\text{WBATE}}}} = o_{\mathbb{P}}(1).$$

Lemma SA-6 gives $\widehat{\Xi}_{\text{WBATE}}/\Xi_{\text{WBATE}} \rightarrow_{\mathbb{P}} 1$.

Write $\mathfrak{L}_{\text{WBATE}} = \sum_{i=1}^n \ell_i$, where the ℓ_i are independent, mean-zero summands obtained by integrating the side-specific linear influence functions over $\mathcal{B}$. By construction, $\sum_{i=1}^n \mathbb{E}[\ell_i^2] = \Xi_{\text{WBATE}}$. To verify Lyapunov's condition, let

$$\mathcal{E}_3(h) = \{(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) \in \mathcal{B}^3 : \|\mathbf{x}_j - \mathbf{x}_k\| \leq Ch, 1 \leq j, k \leq 3\}.$$

Compact support of K , metric equivalence, boundedness of w , and (SA-4) imply $\nu_{\mathcal{B},3}(\mathcal{E}_3(h)) \lesssim h^{2d-2}$. The

same local-moment calculation used in Lemma SA-4 gives

$$\sum_{i=1}^n \mathbb{E} \left[\left| \frac{\ell_i}{\sqrt{\Xi_{\text{WBATE}}}} \right|^3 \right] \lesssim \frac{1}{\sqrt{nh}} \rightarrow 0,$$

where the last step follows from $nh^{2d-1} \rightarrow \infty$ and $d \geq 2$. Lyapunov's central limit theorem gives

$$\frac{\mathcal{L}_{\text{WBATE}}}{\sqrt{\Xi_{\text{WBATE}}}} \rightsquigarrow \text{Normal}(0, 1).$$

Slutsky's theorem gives the feasible statistic result, and the confidence interval statement follows from the standard normal critical value. $\square$

SA-8.20 Proof of Theorem SA-13

The supremum functional is one-Lipschitz under the sup-norm:

$$\left| \sup_{\mathbf{x} \in \mathcal{B}} \hat{\vartheta}(\mathbf{x}) - \sup_{\mathbf{x} \in \mathcal{B}} \vartheta(\mathbf{x}) \right| \leq \sup_{\mathbf{x} \in \mathcal{B}} \left| \hat{\vartheta}(\mathbf{x}) - \vartheta(\mathbf{x}) \right|.$$

The theorem follows immediately from Theorem SA-4. $\square$

SA-8.21 Proof of Theorem SA-14

On the event

$$\vartheta(\mathbf{x}) \in \left[\hat{\vartheta}(\mathbf{x}) - \mathfrak{q}_\alpha \sqrt{\hat{\Xi}_{\mathbf{x}, \mathbf{x}}}, \hat{\vartheta}(\mathbf{x}) + \mathfrak{q}_\alpha \sqrt{\hat{\Xi}_{\mathbf{x}, \mathbf{x}}} \right] \quad \text{for all } \mathbf{x} \in \mathcal{B},$$

taking suprema over $\mathbf{x} \in \mathcal{B}$ implies

$$\vartheta_{\text{LBATE}} \in \hat{\mathcal{I}}_{\alpha, \text{LBATE}}.$$

Theorem SA-10 gives the first event probability $1 - \alpha + o(1)$, which proves the result. $\square$

SA-8.22 Proof of Theorem SA-15

For $A = Y$, Theorem SA-1 applies directly after the reduced-form redefinition $Y_i(t) = Y_i(t, W_i(t))$. For $A = W$, the same proof applies with $W_i(t)$ in place of $Y_i(t)$. Thus the sharp-design identification argument is applied separately to the reduced-form outcome component and the first-stage treatment-receipt component: Assumption SA-4(i)–(ii) gives the reduced-form smoothness and sampling conditions, while Assumption SA-2(i), (ii), and (v) gives the same distance-based identification conditions. Therefore $\vartheta_A(\mathbf{x}) = \tau_A(\mathbf{x})$ for $A \in \{Y, W\}$. If $\tau_W(\mathbf{x}) \neq 0$, then the ratio identity $\xi(\mathbf{x}) = \zeta(\mathbf{x})$ follows immediately. $\square$

SA-8.23 Proof of Theorem SA-16

Theorem SA-4 applies to the reduced-form outcome estimator $\hat{\vartheta}_Y(\mathbf{x})$. The same proof applies to $\hat{\vartheta}_W(\mathbf{x})$ because Assumption SA-4(i)–(ii) gives the corresponding smoothness and moment conditions; boundedness

of $W_i(t)$ gives the required moment bounds automatically. Thus, pointwise and uniformly,

$$\widehat{\vartheta}_A(\mathbf{x}) - \vartheta_A(\mathbf{x}) = O_{\mathbb{P}}\left(\frac{1}{\sqrt{nh^d}} + \frac{1}{n^{\frac{1+v}{2+v}}h^d} + |\mathfrak{B}_A(\mathbf{x})|\right)$$

with the uniform version replacing the first two terms by their logarithmic analogues. Because $v \geq 2$, the pointwise condition $(n^{(1+v)/(2+v)}h^d)^{-1} = o(1)$ implies $(nh^d)^{-1/2} = o(1)$. Uniformly, $\log(1/h)/(n^{(1+v)/(2+v)}h^d) = o(1)$ implies the corresponding logarithmic stochastic terms are $o(1)$. The condition $h = o(1)$ supplies the shrinking-neighborhood requirement in the reduced-form distance-based rates, and Theorem SA-2 implies the reduced-form approximation biases are $o(1)$. Together, these restrictions imply that $\widehat{\vartheta}_W(\mathbf{x})$ is bounded away from zero with probability approaching one, pointwise or uniformly as appropriate. The exact Wald expansion (SA-7) and the bound (SA-8) then give the stated rates. $\square$

SA-8.24 Proof of Lemma SA-8

Fix $A, B \in \{Y, W\}$. The proof of Lemma SA-5 applies to the cross-covariance kernel $\Xi_{A,B,\mathbf{x}_1,\mathbf{x}_2}$ with

$$(A_i - \theta_{A,t,\mathbf{x}_1}^*(D_i(\mathbf{x}_1)))(B_i - \theta_{B,t,\mathbf{x}_2}^*(D_i(\mathbf{x}_2)))$$

in place of the squared residual product. The moment bounds needed for $A = B = Y$ follow from Assumption SA-1(v); those involving W follow from $W_i(t) \in \{0, 1\}$ and Cauchy–Schwarz. Assumption SA-4(iii) provides continuity of the reduced-form conditional covariance entries. Hence, uniformly over $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$,

$$\sup_{A,B \in \{Y,W\}} nh^d |\widehat{\Xi}_{A,B,\mathbf{x}_1,\mathbf{x}_2} - \Xi_{A,B,\mathbf{x}_1,\mathbf{x}_2}| = o_{\mathbb{P}}(1).$$

Theorem SA-16 gives $\sup_{\mathbf{x} \in \mathcal{B}} \|\widehat{\mathfrak{w}}(\mathbf{x}) - \mathfrak{w}(\mathbf{x})\| = o_{\mathbb{P}}(1)$ under the denominator condition and the stated bandwidth restrictions. Combining the last two displays with the definitions of $\Xi_{\mathbf{F}}$ and $\widehat{\Xi}_{\mathbf{F}}$ gives

$$\sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} nh^d |\widehat{\Xi}_{\mathbf{F},\mathbf{x}_1,\mathbf{x}_2} - \Xi_{\mathbf{F},\mathbf{x}_1,\mathbf{x}_2}| = o_{\mathbb{P}}(1).$$

The relative consistency statement follows from the imposed lower bound $\inf_{\mathbf{x} \in \mathcal{B}} nh^d \Xi_{\mathbf{F},\mathbf{x},\mathbf{x}} \gtrsim 1$. $\square$

SA-8.25 Proof of Theorem SA-17

Because $v \geq 2$, the condition $n^{v/(2+v)}h^d \rightarrow \infty$ implies both $nh^d \rightarrow \infty$ and $(n^{(1+v)/(2+v)}h^d)^{-1} \rightarrow 0$. Therefore the reduced-form stochastic terms for Y and W are consistent. The condition $h = o(1)$ gives shrinking local neighborhoods, and Theorem SA-2 makes the first-stage approximation bias vanish.

For the fixed $\mathbf{x}$, the exact expansion (SA-7) gives

$$\widehat{\xi}(\mathbf{x}) - \xi(\mathbf{x}) = \mathfrak{B}_{\mathbf{F}}(\mathbf{x}) + \mathfrak{L}_{\mathbf{F}}(\mathbf{x}) + o_{\mathbb{P}}((nh^d)^{-1/2}),$$

because the reduced-form nonlinear remainders are $o_{\mathbb{P}}((nh^d)^{-1/2})$ by the pointwise analogue of Lemma SA-4, and the pointwise reduced-form rates give $\widehat{\vartheta}_A(\mathbf{x}) - \vartheta_A(\mathbf{x}) = O_{\mathbb{P}}(\rho_{n,\mathbf{x}})$ for $A \in \{Y, W\}$. Since $|\vartheta_W(\mathbf{x})| > 0$ and $\rho_{n,\mathbf{x}} = o(1)$, $\widehat{\vartheta}_W(\mathbf{x})$ is bounded away from zero with probability approaching one. Therefore the exact Wald remainder satisfies $\mathfrak{R}_{\mathbf{F}}(\mathbf{x}) = O_{\mathbb{P}}(\rho_{n,\mathbf{x}}^2) = o_{\mathbb{P}}((nh^d)^{-1/2})$ by the condition $\sqrt{nh^d}\rho_{n,\mathbf{x}}^2 \rightarrow 0$. The separate small-bias condition makes $\mathfrak{B}_{\mathbf{F}}(\mathbf{x}) = o((nh^d)^{-1/2})$.

The leading term $\mathfrak{L}_F(\mathbf{x})$ is a sum of independent mean-zero variables with variance $\Xi_{F,\mathbf{x},\mathbf{x}}$. The moment assumptions and the bound $nh^d\Xi_{F,\mathbf{x},\mathbf{x}} \asymp 1$ verify Lyapunov's condition, giving

$$\frac{\mathfrak{L}_F(\mathbf{x})}{\sqrt{\Xi_{F,\mathbf{x},\mathbf{x}}}} \rightsquigarrow \text{Normal}(0, 1).$$

The pointwise version of Lemma SA-8 gives $\widehat{\Xi}_{F,\mathbf{x},\mathbf{x}}/\Xi_{F,\mathbf{x},\mathbf{x}} \rightarrow_{\mathbb{P}} 1$. Slutsky's theorem therefore gives the normal approximation for $\widehat{T}_F(\mathbf{x})$ and the confidence interval coverage. $\square$

SA-8.26 Proof of Theorem SA-18

Start from (SA-7). Lemma SA-4, applied to $A = Y$ and $A = W$, gives the uniform bound for the reduced-form nonlinear terms. Lemma SA-8 gives uniform consistency of the feasible standard error. The contribution of the reduced-form approximation biases is

$$\sqrt{nh^d} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_F(\mathbf{x})|.$$

Finally, (SA-8) gives the Wald-ratio contribution $\sqrt{nh^d}\rho_n^2$. Combining these bounds gives the displayed result. $\square$

SA-8.27 Proof of Theorem SA-19

The proof is the fuzzy analogue of the proof of Theorem SA-9. The linearized fuzzy statistic can be written as an empirical process indexed by $\mathbf{x} \in \mathcal{B}$:

$$\overline{T}_F(\mathbf{x}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n f_{F,\mathbf{x}}(\mathbf{X}_i, Y_i, W_i),$$

where $f_{F,\mathbf{x}}$ is centered and equals a bounded linear combination, with coefficients $\mathfrak{w}(\mathbf{x})$, of the reduced-form distance-kernel terms for Y_i and W_i . The denominator condition $\inf_{\mathbf{x} \in \mathcal{B}} |\vartheta_W(\mathbf{x})| > 0$, the smooth reduced-form conditional means, and the metric equivalence in Assumption SA-2(ii) imply that $\mathfrak{w}(\mathbf{x})$ is bounded and Lipschitz on $\mathcal{B}$. The entropy calculation for Theorem SA-9 is therefore unchanged up to constants: adding the bounded treatment-receipt coordinate and bounded Lipschitz weights preserves the same VC-type covering bound. The envelope can be taken as $Ch^{-d/2}(|Y_i| + 1)$, because W_i is binary. The moment bounds follow from Assumption SA-1(v) and boundedness of W_i .

Applying Lemma SA-16 to $\{f_{F,\mathbf{x}} : \mathbf{x} \in \mathcal{B}\} \cup \{-f_{F,\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$ gives the displayed Gaussian approximation rate. The covariance of the constructed Gaussian process is, by construction, the covariance of $\overline{T}_F$. $\square$

SA-8.28 Proof of Theorem SA-20

Theorem SA-18 and the stated rate conditions imply

$$\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{T}_F(\mathbf{x}) - \overline{T}_F(\mathbf{x})| = o_{\mathbb{P}}((\log n)^{-1/2}).$$

Theorem SA-19 approximates the supremum of the infeasible linearized statistic by the supremum of a Gaussian process. Lemma SA-8, together with the same finite-net Gaussian comparison and anti-concentration

argument used in the proof of Theorem SA-10, replaces this infeasible Gaussian process by the feasible conditional process $\widehat{Z}_F$. Thus,

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{T}_F(\mathbf{x})| \leq u \right) - \mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}_F(\mathbf{x})| \leq u \mid \mathbf{V} \right) \right| = o_{\mathbb{P}}(1).$$

The coverage statement follows from the definition of $\widehat{I}_{\alpha, F}$ and the conditional quantile $\mathbf{q}_{\alpha}$. $\square$

SA-8.29 Proof of Lemma SA-9

By Fubini's theorem and the definition of $\Xi_{F, \mathbf{x}_1, \mathbf{x}_2}$,

$$\begin{aligned} \mathbb{V}[\mathcal{L}_{F, \text{WBATE}}] &= \int_{\mathcal{B}} \int_{\mathcal{B}} \text{Cov}[\mathcal{L}_F(\mathbf{x}_1), \mathcal{L}_F(\mathbf{x}_2)] w(\mathbf{x}_1) w(\mathbf{x}_2) d\mathfrak{H}^{d-1}(\mathbf{x}_1) d\mathfrak{H}^{d-1}(\mathbf{x}_2) \\ &= \Xi_{F, \text{WBATE}}. \end{aligned}$$

The denominator condition implies that $\mathfrak{w}(\cdot)$ is uniformly bounded on $\mathcal{B}$. Compact support of K and equivalence of $\mathcal{d}$ and the Euclidean distance imply that $\Xi_{F, \mathbf{x}_1, \mathbf{x}_2} = 0$ unless $\|\mathbf{x}_1 - \mathbf{x}_2\| \lesssim h$. Therefore, as in the proof of Lemma SA-6, the effective integration set has product boundary measure $O(h^{d-1})$ under (SA-4). The reduced-form covariance kernels are uniformly of order $(nh^d)^{-1}$, and hence

$$\Xi_{F, \text{WBATE}} \lesssim \frac{h^{d-1}}{nh^d} = \frac{1}{nh}.$$

The matching lower bound follows from (SA-10).

For the plug-in variance, the proof of Lemma SA-8 gives the uniform covariance approximation

$$\sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} nh^d \left| \widehat{\Xi}_{F, \mathbf{x}_1, \mathbf{x}_2} - \Xi_{F, \mathbf{x}_1, \mathbf{x}_2} \right| = o_{\mathbb{P}}(1),$$

using only the denominator nondegeneracy and the stated rate conditions for the consistency of $\widehat{\mathfrak{w}}(\cdot)$. The same compact-support restriction applies to the feasible covariance kernel. Integrating the uniform error over the effective set of measure $O(h^{d-1})$ gives

$$\left| \widehat{\Xi}_{F, \text{WBATE}} - \Xi_{F, \text{WBATE}} \right| = o_{\mathbb{P}} \left(\frac{h^{d-1}}{nh^d} \right) = o_{\mathbb{P}} \left(\frac{1}{nh} \right).$$

The relative consistency statement follows after dividing by the lower bound in (SA-10). $\square$

SA-8.30 Proof of Lemma SA-10

Because $\inf_{\mathbf{x} \in \mathcal{B}} |\vartheta_W(\mathbf{x})| > 0$ and the reduced-form conditional means are continuous, the weight vector $\mathfrak{w}(\mathbf{x})$ is uniformly bounded over $\mathcal{B}$. The fuzzy nonlinear term can be written as a finite linear combination of reduced-form WBATE nonlinear terms:

$$\mathcal{Q}_{F, \text{WBATE}} = \int_{\mathcal{B}} \mathfrak{w}(\mathbf{x})^\top (\mathcal{Q}_Y(\mathbf{x}), \mathcal{Q}_W(\mathbf{x}))^\top w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}).$$

The proof of Lemma SA-7 applies to each reduced-form component, with the bounded deterministic weights $w(\mathbf{x})\mathfrak{w}_Y(\mathbf{x})$ and $w(\mathbf{x})\mathfrak{w}_W(\mathbf{x})$ in place of $w(\mathbf{x})$. The moment bounds for terms involving Y follow from

Assumption SA-1(v), and those involving W are immediate from $W_i(t) \in \{0, 1\}$. Hence

$$\Omega_{F, \text{WBATE}} = O_{\mathbb{P}}\left(\frac{1}{nh^d}\right).$$

For the Wald-ratio remainder, (SA-8) and the integrability of w give

$$|\mathfrak{R}_{F, \text{WBATE}}| \leq \int_{\mathcal{B}} |\mathfrak{R}_F(\mathbf{x})| w(\mathbf{x}) d\mathfrak{H}^{d-1}(\mathbf{x}) \lesssim_{\mathbb{P}} \rho_n^2.$$

The final negligibility claim follows because

$$\frac{1/(nh^d)}{1/\sqrt{nh}} = \frac{1}{\sqrt{nh^{2d-1}}},$$

which is $o(1)$ under $nh^{2d-1} \rightarrow \infty$, while $\sqrt{nh}\rho_n^2 \rightarrow 0$ by assumption. $\square$

SA-8.31 Proof of Theorem SA-21

The decomposition (SA-9), Lemma SA-9, and Lemma SA-10 give

$$\left| \widehat{\xi}_{\text{WBATE}} - \xi_{\text{WBATE}} \right| \leq |\mathfrak{B}_{F, \text{WBATE}}| + O_{\mathbb{P}}\left(\frac{1}{\sqrt{nh}}\right) + O_{\mathbb{P}}\left(\frac{1}{nh^d}\right) + O_{\mathbb{P}}(\rho_n^2),$$

which is the stated rate. $\square$

SA-8.32 Proof of Theorem SA-22

By (SA-9), Lemma SA-10, the small-bias condition, the condition $\sqrt{nh}\rho_n^2 \rightarrow 0$, and the aggregate nondegeneracy condition,

$$\frac{\widehat{\xi}_{\text{WBATE}} - \xi_{\text{WBATE}} - \mathfrak{L}_{F, \text{WBATE}}}{\sqrt{\Xi_{F, \text{WBATE}}}} = o_{\mathbb{P}}(1).$$

Lemma SA-9 gives $\widehat{\Xi}_{F, \text{WBATE}}/\Xi_{F, \text{WBATE}} \rightarrow_{\mathbb{P}} 1$.

Write $\mathfrak{L}_{F, \text{WBATE}} = \sum_{i=1}^n \ell_{i, F}$, where the $\ell_{i, F}$ are the independent mean-zero summands obtained by integrating the reduced-form fuzzy linear influence functions over $\mathcal{B}$. By construction, $\sum_{i=1}^n \mathbb{E}[\ell_{i, F}^2] = \Xi_{F, \text{WBATE}}$. The same localization calculation used in the proof of Theorem SA-12, now with envelope proportional to $|Y_i| + 1$ because W_i is binary and $\mathfrak{w}(\cdot)$ is bounded, gives

$$\sum_{i=1}^n \mathbb{E} \left[\left| \frac{\ell_{i, F}}{\sqrt{\Xi_{F, \text{WBATE}}}} \right|^3 \right] \lesssim \frac{1}{\sqrt{nh}} \rightarrow 0.$$

Thus Lyapunov's central limit theorem yields

$$\frac{\mathfrak{L}_{F, \text{WBATE}}}{\sqrt{\Xi_{F, \text{WBATE}}}} \rightsquigarrow \text{Normal}(0, 1).$$

Slutsky's theorem gives the feasible statistic result, and the confidence interval coverage follows from the standard normal critical value. $\square$

SA-8.33 Proof of Theorem SA-23

The supremum functional is one-Lipschitz under the sup-norm:

$$\left| \widehat{\xi}_{\text{LBATE}} - \xi_{\text{LBATE}} \right| \leq \sup_{\mathbf{x} \in \mathcal{B}} |\widehat{\xi}(\mathbf{x}) - \xi(\mathbf{x})|.$$

The result follows from the uniform part of Theorem SA-16. $\square$

SA-8.34 Proof of Theorem SA-24

On the event

$$\xi(\mathbf{x}) \in \left[\widehat{\xi}(\mathbf{x}) - \mathfrak{q}_\alpha \sqrt{\widehat{\Xi}_{\mathbf{F}, \mathbf{x}, \mathbf{x}}}, \widehat{\xi}(\mathbf{x}) + \mathfrak{q}_\alpha \sqrt{\widehat{\Xi}_{\mathbf{F}, \mathbf{x}, \mathbf{x}}} \right] \quad \text{for all } \mathbf{x} \in \mathcal{B},$$

taking suprema over $\mathbf{x} \in \mathcal{B}$ implies

$$\xi_{\text{LBATE}} \in \widehat{\mathcal{I}}_{\alpha, \mathbf{F}, \text{LBATE}}.$$

Theorem SA-20 gives the first event probability $1 - \alpha + o(1)$, which proves the result. $\square$

SA-8.35 Proof of Theorem 6

Let $0 < s < 1/6$ be a parameter that is chosen later. Consider a C^1 function $\Theta_s : [0, 1] \rightarrow [\underline{\theta}, \pi]$, for some fixed $\underline{\theta} > 0$, satisfying

$$\Theta_s(r) = \pi \quad (0 \leq r \leq 2s), \quad \frac{\sin(\Theta_s(r))}{\Theta_s(r)} = \frac{2}{\pi} - \frac{s}{r} \quad (3s \leq r \leq 1), \quad \sup_{0 < r \leq 1} r |\Theta_s'(r)| \leq C,$$

with a smooth interpolation on $[2s, 3s]$. Such a function exists because $\theta \mapsto \sin(\theta)/\theta$ maps $(0, \pi]$ continuously and monotonically onto $[0, 1)$.

Data Generating Process $\mathbb{P}_0$.

Let

$$\mathcal{X}_0 = \{r(\cos \theta, \sin \theta) : 0 \leq r \leq 1, 0 \leq \theta \leq \Theta_s(r)\}.$$

Suppose $\mathbf{X}_i$ has Lebesgue density

$$f_{0,X}(r(\cos \theta, \sin \theta)) = \frac{2}{\Theta_s(r)}, \quad 0 \leq r \leq 1, \quad 0 \leq \theta \leq \Theta_s(r),$$

and define

$$\mu_0(x_1, x_2) = \frac{1}{2} + \frac{x_1}{100}, \quad (x_1, x_2) \in \mathbb{R}^2.$$

Let $Y_i = \mathbf{1}(\varepsilon_i \leq \mu_0(\mathbf{X}_i))$, where $(\varepsilon_i : i = 1, \dots, n)$ are i.i.d. $\text{Uniform}([0, 1])$ random variables independent of $(\mathbf{X}_i : i = 1, \dots, n)$. Let $\eta_0(r) = \mathbb{E}_{\mathbb{P}_0}[Y_i | \|\mathbf{X}_i\| = r]$. The boundary $\text{bd}(\mathcal{X}_0)$ is rectifiable with uniformly bounded length.

Data Generating Process $\mathbb{P}_1$.

Let

$$\mathcal{X}_1 = \{r(\cos \theta, \sin \theta) : 0 \leq r \leq 1, 0 \leq \theta \leq \pi/2\},$$

let $\mathbf{X}_i$ have density $f_{1,X} = 4/\pi$ on $\mathcal{X}_1$, and define

$$\mu_1(x_1, x_2) = \frac{1}{2} + \frac{x_1 - s}{100}, \quad (x_1, x_2) \in \mathbb{R}^2.$$

Let $Y_i = \mathbf{1}(\varepsilon_i \leq \mu_1(\mathbf{X}_i))$ with the same independent uniform construction as above. Let $\eta_1(r) = \mathbb{E}_{\mathbb{P}_1}[Y_i | \|\mathbf{X}_i\| = r]$. The boundary $\text{bd}(\mathcal{X}_1)$ is rectifiable.

Figure SA-2 displays the two support sets used in the minimax lower-bound construction.

Minimax Lower Bound.

First, under both models, $\mathbb{P}_j(\|\mathbf{X}_i\| \leq r) = r^2$ for all $0 \leq r \leq 1$:

$$\mathbb{P}_0(\|\mathbf{X}_i\| \leq r) = \int_0^r \int_0^{\Theta_s(\rho)} \frac{2}{\Theta_s(\rho)} \rho \, d\theta \, d\rho = r^2 = \int_0^r \int_0^{\pi/2} \frac{4}{\pi} \rho \, d\theta \, d\rho = \mathbb{P}_1(\|\mathbf{X}_i\| \leq r).$$

The two data-generating processes belong to $\mathcal{P}_{\text{MP}}$ for a fixed sufficiently large L : the supports are compact subsets of $[-L, L]^2$ with rectifiable boundaries of uniformly bounded length, the densities are continuous and uniformly bounded above and below on their supports, and the Bernoulli conditional means and variances satisfy the required smoothness and boundedness conditions. Hence, choosing $(0, 0)$ as the point of evaluation in both $\mathbb{P}_0$ and $\mathbb{P}_1$, the one-observation KL distance for the distance-based experiment based on $(\|\mathbf{X}_i\|, Y_i)$ is

$$d_{\text{KL}}(\mathbb{P}_0(\|\mathbf{X}_i\|, Y_i), \mathbb{P}_1(\|\mathbf{X}_i\|, Y_i)) = 2 \int_0^1 d_{\text{KL}}(\text{Bernoulli}(\eta_0(r)), \text{Bernoulli}(\eta_1(r))) r \, dr.$$

Under $\mathbb{P}_0$ and $\mathbb{P}_1$,

$$\eta_0(r) = \frac{1}{2} + \frac{r}{100} \frac{\sin(\Theta_s(r))}{\Theta_s(r)}, \quad \eta_1(r) = \frac{1}{2} + \frac{1}{100} \left(\frac{2r}{\pi} - s \right).$$

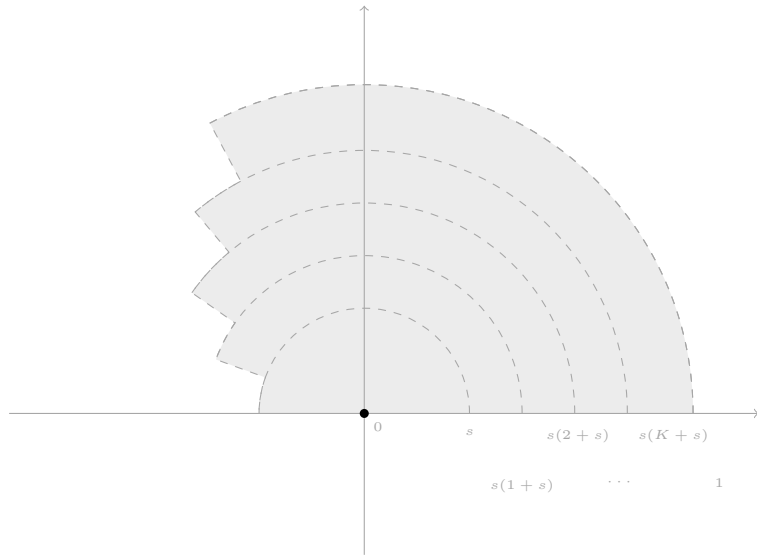
Thus $\eta_0(r) = \eta_1(r)$ for all $r \in [3s, 1]$. For $r \in [0, 3s]$, both probabilities are bounded away from zero and one, and $|\eta_0(r) - \eta_1(r)| \leq Cs$. Therefore,

$$\int_0^1 d_{\text{KL}}(\text{Bernoulli}(\eta_0(r)), \text{Bernoulli}(\eta_1(r))) r \, dr \leq C \int_0^{3s} (\eta_0(r) - \eta_1(r))^2 r \, dr \leq Cs^4.$$

Moreover, $|\mu_0(0, 0) - \mu_1(0, 0)| = s/100$. Hence, by [Tsybakov \[2008, Theorem 2.2 \(iii\)\]](#), take $Cs_*^4 = \log(2)/n$, and conclude that the pointwise risk at $\mathbf{x} = 0$ is bounded below by a constant multiple of s_* . Since the uniform loss dominates the pointwise loss at any boundary point,

$$\inf_{T_n \in \mathcal{T}} \sup_{\mathbb{P} \in \mathcal{P}_{\text{MP}}} \mathbb{E}_{\mathbb{P}} \left[\sup_{\mathbf{x} \in \mathcal{B}(\mathbb{P})} |T_n(\mathbf{U}_n(\mathbf{x})) - \mu(\mathbf{x})| \right] \geq \frac{1}{1600} s_* \gtrsim n^{-\frac{1}{4}}.$$

This concludes the proof. □



$\mathcal{X}_0$ from DGP $\mathbb{P}_0$



$\mathcal{X}_1$ from DGP $\mathbb{P}_1$

Figure SA-2: Support sets in the two-point minimax lower-bound construction.

SA-9 Simulation Study

The score $\mathbf{X}_i = (X_{1i}, X_{2i})^\top$ is generated as $(100B_{1i} - 25, 100B_{2i} - 25)^\top$, where B_{1i} and B_{2i} are independent $\text{Beta}(3, 4)$ random variables, a choice that approximately reproduces salient features of the SPP dataset. The treatment assignment region is the same L-shaped region used in the empirical application,

$$\mathcal{A}_1 = \{\mathbf{x} = (x_1, x_2)^\top : x_1 \geq 0, x_2 \geq 0\}, \quad \mathcal{A}_0 = \mathcal{X} \setminus \mathcal{A}_1,$$

with boundary $\mathcal{B}$ given by the two coordinate-axis segments and a kink at $\mathbf{x} = (0, 0)^\top$. The simulation uses 21 evaluation points $\{\mathbf{b}_1, \dots, \mathbf{b}_{21}\} \subset \mathcal{B}$, with $\mathbf{b}_{11} = (0, 0)^\top$ corresponding to the kink.

The simulation design sets $n = 20,000$ and uses 5,000 Monte Carlo replications. We evaluate performance in terms of point estimation and uncertainty quantification, both pointwise and uniformly over the evaluation grid. We consider two related Monte Carlo exercises. The first is a sharp-design exercise, which uses only the reduced-form or intention-to-treat (ITT) outcome equation and therefore reports sharp distance-based results for the BATEC. The second is a fuzzy-design exercise, which jointly generates the reduced-form outcome and treatment-receipt status so that the ITT, first-stage (FS), and fuzzy Wald analyses can all be performed on the same simulated samples.

Throughout the simulation study, the distance-based estimator uses a local linear polynomial, $p = 1$. The DGP varies the polynomial order s of the underlying bivariate conditional mean surfaces. We consider $s \in \{1, 2\}$, corresponding to linear and quadratic specifications. In addition, each mean specification is paired with either homoskedastic or heteroskedastic outcome disturbances.

SA-9.1 Data-Generating Process

For each Monte Carlo replication, draw $\mathbf{X}_i = (X_{1i}, X_{2i})^\top$ independently from the distribution described above and set $T_i = \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_1)$. Let $\mathbf{m}_s(\cdot)$ be a monomial vector of degree s , that is,

$$\mathbf{m}_0(\mathbf{x}) = 1, \quad \mathbf{m}_1(\mathbf{x}) = (1, x_1, x_2)^\top, \quad \text{and} \quad \mathbf{m}_2(\mathbf{x}) = (1, x_1, x_2, x_1^2, x_1x_2, x_2^2)^\top.$$

We consider DGP polynomial specifications of order $s \in \{1, 2\}$, using the monomial vector evaluated at the unscaled score.

Sharp Design: ITT Equation.

The sharp-design simulation uses an outcome-only DGP. For each $t \in \{0, 1\}$ and $i = 1, \dots, n$, define

$$\mu_{Y,t}(\mathbf{X}_i) = \mathbf{m}_s(\mathbf{X}_i)^\top \boldsymbol{\beta}_{Y,t,s}, \quad \sigma_{Y,t,v}^2(\mathbf{X}_i) = \exp(\mathbf{m}_v(\mathbf{X}_i)^\top \boldsymbol{\delta}_{Y,t,v}),$$

where $v \in \{0, 2\}$ indexes the homoskedastic and heteroskedastic specifications. Draw $u_{Y,t,i} \sim \text{Normal}(0, 1)$ independently across i and t and independently of all score variables, and set

$$\varepsilon_{Y,t,i} = \sigma_{Y,t,v}(\mathbf{X}_i) u_{Y,t,i}.$$

Then

$$Y_i(t) = \mu_{Y,t}(\mathbf{X}_i) + \varepsilon_{Y,t,i}.$$

The observed sharp-design outcome is

$$Y_i = Y_i(0)(1 - T_i) + Y_i(1)T_i.$$

The target at each boundary point $\mathbf{b} \in \mathcal{B}$ is

$$\tau(\mathbf{b}) = \mathbb{E}[Y_i(1) - Y_i(0) \mid \mathbf{X}_i = \mathbf{b}] = \mu_{Y,1}(\mathbf{b}) - \mu_{Y,0}(\mathbf{b}).$$

This is the design presented in the paper, where we suppress the outcome subscript and write μ_t , σ_t , and $u_{t,i}$ in place of $\mu_{Y,t}$, $\sigma_{Y,t,v}$, and $u_{Y,t,i}$ for notation simplicity.

Fuzzy Design: ITT, FS, and Wald Equations.

For each $t \in \{0, 1\}$ and $i = 1, \dots, n$, define

$$\begin{aligned} \mu_{Y,t}(\mathbf{X}_i) &= \mathbf{m}_s(\mathbf{X}_i)^\top \boldsymbol{\beta}_{Y,t,s}, \\ \mu_{W,t}(\mathbf{X}_i) &= \Lambda(\eta_{W,t}(\mathbf{X}_i)), \quad \eta_{W,t}(\mathbf{X}_i) = \mathbf{m}_s(\mathbf{X}_i)^\top \boldsymbol{\beta}_{W,t,s}, \end{aligned}$$

where $\Lambda(a) = 1/(1 + \exp(-a))$. We also consider homoskedastic and heteroskedastic specifications for the residualized outcome innovation. Let $v \in \{0, 2\}$ index the variance specification, where $v = 0$ is homoskedastic and $v = 2$ is heteroskedastic:

$$\sigma_{Y,t,v}^2(\mathbf{x}) = \exp(\mathbf{m}_v(\mathbf{x})^\top \boldsymbol{\delta}_{Y,t,v}).$$

Given these conditional mean and variance functions, draw $U_{W,t,i} \sim \text{Uniform}(0, 1)$ and $u_{Y,t,i} \sim \text{Normal}(0, 1)$ mutually independently across i and t and independently of all score variables, and set

$$\varepsilon_{Y,t,i} = \sigma_{Y,t,v}(\mathbf{X}_i)u_{Y,t,i}.$$

The potential treatment statuses and treatment-indexed potential outcomes are

$$\begin{aligned} W_i(t) &= \mathbf{1}\{U_{W,t,i} \leq \mu_{W,t}(\mathbf{X}_i)\}, \\ Y_i(t, w) &= \mu_{Y,t}(\mathbf{X}_i) + \lambda_t\{w - \mu_{W,t}(\mathbf{X}_i)\} + \varepsilon_{Y,t,i}, \quad w \in \{0, 1\}. \end{aligned}$$

The reduced-form potential outcome induced by assignment t is $Y_i(t, W_i(t))$. Equivalently,

$$Y_i(t, W_i(t)) = \mu_{Y,t}(\mathbf{X}_i) + \lambda_t\{W_i(t) - \mu_{W,t}(\mathbf{X}_i)\} + \varepsilon_{Y,t,i}.$$

The observed outcomes and treatment status are

$$\begin{aligned} Y_i &= Y_i(0, W_i(0))(1 - T_i) + Y_i(1, W_i(1))T_i, \\ W_i &= W_i(0)(1 - T_i) + W_i(1)T_i, \quad T_i = \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_1), \end{aligned}$$

for $i = 1, 2, \dots, n$.

In the reduced-form notation used for sharp estimation and inference, $Y_i(t)$ is shorthand for $Y_i(t, W_i(t))$. This centered specification preserves the reduced-form outcome surface because $\mathbb{E}[Y_i(t, W_i(t)) \mid \mathbf{X}_i = \mathbf{x}] =$

$\mu_{Y,t}(\mathbf{x})$, while allowing the potential outcome and potential treatment status equations to be correlated:

$$\begin{aligned}\mathbb{E}[W_i(t)|\mathbf{X}_i = \mathbf{x}] &= \mu_{W,t}(\mathbf{x}), \\ \text{Cov}(Y_i(t, W_i(t)), W_i(t)|\mathbf{X}_i = \mathbf{x}) &= \lambda_t \mu_{W,t}(\mathbf{x}) \{1 - \mu_{W,t}(\mathbf{x})\}.\end{aligned}$$

The heteroskedastic specification changes the conditional variance of the residualized outcome innovation $\varepsilon_{Y,t,i}$, but it does not change the conditional mean surfaces or the first-stage equation. In the fuzzy design, the conditional variance of the reduced-form outcome also includes the treatment-receipt component induced by $\lambda_t \{W_i(t) - \mu_{W,t}(\mathbf{X}_i)\}$. Consequently, for each boundary point $\mathbf{b} \in \mathcal{B}$,

$$\begin{aligned}\tau_W(\mathbf{b}) &= \mathbb{E}[W_i(1) - W_i(0)|\mathbf{X}_i = \mathbf{b}] = \mu_{W,1}(\mathbf{b}) - \mu_{W,0}(\mathbf{b}), \\ \tau_Y(\mathbf{b}) &= \mathbb{E}[Y_i(1, W_i(1)) - Y_i(0, W_i(0))|\mathbf{X}_i = \mathbf{b}] = \mu_{Y,1}(\mathbf{b}) - \mu_{Y,0}(\mathbf{b}) = \mathbf{m}_s(\mathbf{b})^\top (\boldsymbol{\beta}_{Y,1,s} - \boldsymbol{\beta}_{Y,0,s}), \\ \zeta(\mathbf{b}) &= \frac{\tau_Y(\mathbf{b})}{\tau_W(\mathbf{b})}.\end{aligned}$$

Within each fuzzy-design Monte Carlo replication, the sharp estimator applied to Y_i targets the reduced-form or ITT curve $\tau_Y(\mathbf{b})$, the same estimator applied to W_i targets the first-stage curve $\tau_W(\mathbf{b})$, and the fuzzy estimator targets the Wald curve $\zeta(\mathbf{b})$. Thus, the ITT, first-stage, and fuzzy simulation results are different summaries of the same simulated samples rather than results from separate fuzzy DGPs. The fuzzy design therefore extends the sharp ITT equation by adding a treatment-receipt equation and allowing the reduced-form outcome and treatment-receipt shocks to be correlated through λ_t .

SA-9.2 Calibration from the SPP Data

The DGP parameters are calibrated once from the SPP data and then held fixed across Monte Carlo replications. We use the following steps separately for each DGP polynomial order $s \in \{1, 2\}$. The conditional mean calibration follows the JASA simulation design; the additional calibration determines the homoskedastic or heteroskedastic variance of the innovation entering the outcome equation. The sharp-design exercise uses the calibrated ITT objects $\boldsymbol{\beta}_{Y,t,s}$ and $\boldsymbol{\delta}_{Y,t,v}$; the fuzzy-design exercise additionally uses the treatment-receipt coefficients $\boldsymbol{\beta}_{W,t,s}$ and the residual-dependence parameters λ_t .

Step 1. Fix the DGP polynomial order $s \in \{1, 2\}$ and define the side-specific calibration samples $\mathcal{J}_t = \{i : T_i = t\}$ and $N_t = |\mathcal{J}_t|$, for $t \in \{0, 1\}$.

Step 2. For each side t , estimate the outcome regression coefficients by side-specific least squares:

$$\hat{\boldsymbol{\beta}}_{Y,t,s} = \arg \min_{\boldsymbol{\beta}} \sum_{i \in \mathcal{J}_t} \{Y_i - \mathbf{m}_s(\mathbf{X}_i)^\top \boldsymbol{\beta}\}^2.$$

This gives the fitted reduced-form outcome surface $\hat{\mu}_{Y,t}(\mathbf{x}) = \mathbf{m}_s(\mathbf{x})^\top \hat{\boldsymbol{\beta}}_{Y,t,s}$.

Step 3. For each side t , estimate the treatment-status regression coefficients by side-specific logit:

$$\hat{\boldsymbol{\beta}}_{W,t,s} = \arg \max_{\boldsymbol{\beta}} \sum_{i \in \mathcal{J}_t} [W_i \mathbf{m}_s(\mathbf{X}_i)^\top \boldsymbol{\beta} - \log\{1 + \exp(\mathbf{m}_s(\mathbf{X}_i)^\top \boldsymbol{\beta})\}].$$

This gives the fitted first-stage surface $\hat{\mu}_{W,t}(\mathbf{x}) = \Lambda(\mathbf{m}_s(\mathbf{x})^\top \hat{\boldsymbol{\beta}}_{W,t,s})$.

Step 4. Construct side-specific residuals in the calibration sample:

$$\widehat{\varepsilon}_{Y,i} = Y_i - \widehat{\mu}_{Y,T_i}(\mathbf{X}_i), \quad \widehat{\varepsilon}_{W,i} = W_i - \widehat{\mu}_{W,T_i}(\mathbf{X}_i).$$

Step 5. Estimate the residual dependence parameter on each side:

$$\widehat{\lambda}_t = \frac{\sum_{i \in \mathcal{J}_t} \widehat{\varepsilon}_{Y,i} \widehat{\varepsilon}_{W,i}}{\sum_{i \in \mathcal{J}_t} \widehat{\varepsilon}_{W,i}^2}, \quad t \in \{0, 1\}.$$

Define the estimated residualized outcome innovation, distinct from the standardized simulation draw $u_{Y,t,i}$,

$$\widehat{\omega}_{Y,i} = \widehat{\varepsilon}_{Y,i} - \widehat{\lambda}_{T_i} \widehat{\varepsilon}_{W,i}.$$

For the homoskedastic specification, estimate the intercept-only log-variance coefficient

$$\widehat{\delta}_{Y,t,0} = \log \left(\frac{1}{N_t} \sum_{i \in \mathcal{J}_t} \widehat{\omega}_{Y,i}^2 \right), \quad t \in \{0, 1\}.$$

For the heteroskedastic specification, estimate the side-specific log-variance regression for the residualized outcome innovation:

$$\widehat{\delta}_{Y,t,2} = \arg \min_{\boldsymbol{\delta}} \sum_{i \in \mathcal{J}_t} [\log \{ \max(\widehat{\omega}_{Y,i}^2, \epsilon_0) \} - \mathbf{m}_2(\mathbf{X}_i)^\top \boldsymbol{\delta}]^2,$$

where $\epsilon_0 > 0$ is a small truncation constant used only to avoid taking logs of zero.

Step 6. Set the Monte Carlo DGP parameters equal to the calibration estimates: $\boldsymbol{\beta}_{Y,t,s} = \widehat{\boldsymbol{\beta}}_{Y,t,s}$, $\boldsymbol{\beta}_{W,t,s} = \widehat{\boldsymbol{\beta}}_{W,t,s}$, $\lambda_t = \widehat{\lambda}_t$, $\boldsymbol{\delta}_{Y,t,0} = \widehat{\boldsymbol{\delta}}_{Y,t,0}$, and $\boldsymbol{\delta}_{Y,t,2} = \widehat{\boldsymbol{\delta}}_{Y,t,2}$ for $t \in \{0, 1\}$.

Step 7. Before running the Monte Carlo replications, compute the true target functions analytically from the calibrated conditional mean surfaces. For each evaluation point $\mathbf{b} \in \mathcal{B}$, the sharp-design BATEC is

$$\tau(\mathbf{b}) = \mathbf{m}_s(\mathbf{b})^\top (\boldsymbol{\beta}_{Y,1,s} - \boldsymbol{\beta}_{Y,0,s}).$$

For the fuzzy-design exercise, the reduced-form/ITT, first-stage, and Wald targets are

$$\tau_Y(\mathbf{b}) = \mathbf{m}_s(\mathbf{b})^\top (\boldsymbol{\beta}_{Y,1,s} - \boldsymbol{\beta}_{Y,0,s}), \quad \tau_W(\mathbf{b}) = \Lambda(\mathbf{m}_s(\mathbf{b})^\top \boldsymbol{\beta}_{W,1,s}) - \Lambda(\mathbf{m}_s(\mathbf{b})^\top \boldsymbol{\beta}_{W,0,s}),$$

and $\zeta(\mathbf{b}) = \frac{\tau_Y(\mathbf{b})}{\tau_W(\mathbf{b})}$. The same calibrated outcome surface is used in both exercises, so $\tau(\mathbf{b})$ and $\tau_Y(\mathbf{b})$ have the same algebraic form under this calibration; they are kept notationally distinct because they refer to different Monte Carlo exercises and different causal objects.

This calibration reproduces the empirical residual covariance between the outcome and take-up equations on each side of the boundary, while keeping the marginal reduced-form and first-stage surfaces fixed by the fitted SPP models. The homoskedastic and heteroskedastic DGPs share the same target curves for a given mean specification; they differ only in the conditional variance of the residualized outcome innovation.

SA-9.3 Monte Carlo Implementation

The following steps describe how to simulate from the calibrated model. They are repeated separately for each pair $(s, v) \in \{1, 2\} \times \{0, 2\}$. For the sharp-design exercise, the same score draws, assignment rule, evaluation grid, polynomial order, and bandwidth choices are used, but Steps 4–6 are simplified by keeping only the outcome equation Y_i and dropping $W_i(t)$, $\mu_{W,t}$, λ_t , and the Wald ratio.

Step 1. Fix the Monte Carlo inputs: the sample size n , the number of replications M , the evaluation grid $\{\mathbf{b}_1, \dots, \mathbf{b}_{21}\}$ with $\mathbf{b}_{11}$ at the kink, the assignment regions $\mathcal{A}_0$ and $\mathcal{A}_1$, the local polynomial order $p = 1$, and the calibrated parameters. The sharp-design exercise uses only $\{\beta_{Y,t,s}, \delta_{Y,t,0}, \delta_{Y,t,2} : t \in \{0, 1\}\}$, while the fuzzy-design exercise uses the full set

$$\{\beta_{Y,t,s}, \beta_{W,t,s}, \lambda_t, \delta_{Y,t,0}, \delta_{Y,t,2} : t \in \{0, 1\}\}.$$

Step 2. Before entering the Monte Carlo loop, evaluate the true target functions on the grid. For the sharp-design exercise this is only the BATEC target $\tau(\mathbf{b}_j)$. For the fuzzy-design exercise the targets are

$$\tau_Y(\mathbf{b}_j), \quad \tau_W(\mathbf{b}_j), \quad \zeta(\mathbf{b}_j), \quad j = 1, \dots, 21.$$

These target values are fixed across Monte Carlo replications and are common to the homoskedastic and heteroskedastic versions of a given mean specification.

Step 3. For replication $m = 1, \dots, M$, draw independent score vectors $\mathbf{X}_i = (100B_{1i} - 25, 100B_{2i} - 25)^\top$, with B_{1i} and B_{2i} independent $\text{Beta}(3, 4)$ random variables, and set $T_i = \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_1)$.

Step 4. Form the design vectors $\mathbf{m}_s(\mathbf{X}_i)$ and compute the side-specific conditional outcome means and variances

$$\mu_{Y,t}(\mathbf{X}_i), \quad \sigma_{Y,t,v}^2(\mathbf{X}_i), \quad t \in \{0, 1\}.$$

Under homoskedasticity, $\sigma_{Y,t,v}^2(\mathbf{X}_i)$ is constant within side t . Under heteroskedasticity, it changes with $\mathbf{X}_i$ through the fitted log-variance model. For the fuzzy-design exercise, also compute $\mu_{W,t}(\mathbf{X}_i)$.

Step 5. For the sharp-design exercise, draw $u_{Y,t,i} \sim \text{Normal}(0, 1)$ independently across i and t and independently of all score variables, and set

$$\varepsilon_{Y,t,i} = \sigma_{Y,t,v}(\mathbf{X}_i) u_{Y,t,i}.$$

Then construct $Y_i(t) = \mu_{Y,t}(\mathbf{X}_i) + \varepsilon_{Y,t,i}$. For the fuzzy-design exercise, draw $U_{W,t,i} \sim \text{Uniform}(0, 1)$ and $u_{Y,t,i} \sim \text{Normal}(0, 1)$ mutually independently across i and t and independently of all score variables, define $\varepsilon_{Y,t,i}$ as above, and construct

$$\begin{aligned} W_i(t) &= \mathbf{1}\{U_{W,t,i} \leq \mu_{W,t}(\mathbf{X}_i)\}, \\ Y_i(t, W_i(t)) &= \mu_{Y,t}(\mathbf{X}_i) + \lambda_t\{W_i(t) - \mu_{W,t}(\mathbf{X}_i)\} + \varepsilon_{Y,t,i}. \end{aligned}$$

Step 6. Select observed variables using the assignment indicator. For the sharp-design exercise, $Y_i =$

$Y_i(0)(1 - T_i) + Y_i(1)T_i$. For the fuzzy-design exercise,

$$\begin{aligned} W_i &= W_i(0)(1 - T_i) + W_i(1)T_i, \\ Y_i &= Y_i(0, W_i(0))(1 - T_i) + Y_i(1, W_i(1))T_i. \end{aligned}$$

The simulated dataset for replication m is therefore $\{Y_i, T_i, \mathbf{X}_i : i = 1, \dots, n\}$ in the sharp-design exercise and $\{Y_i, W_i, T_i, \mathbf{X}_i : i = 1, \dots, n\}$ in the fuzzy-design exercise.

- Step 7. For each evaluation point $\mathbf{b}_j$, compute the signed distance score $D_i(\mathbf{b}_j)$ using the Euclidean distance and the assignment side. Estimate the target curves using the distance-based local linear estimator.
- Step 8. **Sharp/ITT Analysis.** In the sharp-design exercise, apply the sharp distance-based estimator to Y_i for the target $\tau(\mathbf{b}_j)$. In the fuzzy-design exercise, apply the same sharp estimator to the reduced-form outcome Y_i for the ITT target $\tau_Y(\mathbf{b}_j)$.
- Step 9. **FS Analysis.** In the fuzzy-design exercise, apply the sharp distance-based estimator to W_i for the target $\tau_W(\mathbf{b}_j)$.
- Step 10. **Fuzzy Analysis.** In the fuzzy-design exercise, apply the fuzzy distance-based estimator using Y_i as the outcome and W_i as the treatment-receipt status for the Wald target $\zeta(\mathbf{b}_j)$.
- Step 11. Repeat the analysis using the bandwidth choices considered in the main text: the smooth-boundary selector, the kink-adaptive selector using the known kink location, the unknown-kink bandwidth rule, and the bandwidth induced by the univariate `rdrobust` selector.
- Step 12. After all M replications, compute Monte Carlo summaries by comparing the stored estimates and confidence sets with the fixed analytic targets. For each target and each grid point, report bias, standard deviation, root mean squared error, pointwise coverage, and average interval length. For uniform inference, report simultaneous coverage and average band length over the grid. Finally, also recover WBATE and LBATE summaries and report their associated performance.

We use the sharp simulations to study the finite-sample behavior of the sharp BATEC estimator for $\tau(\mathbf{b})$. The fuzzy simulations additionally study the reduced-form/ITT estimators, the first-stage estimators, and the fuzzy Wald estimator. A complier-average causal interpretation of $\zeta(\mathbf{b})$ would require additional assumptions that are not imposed in the simulations. In particular, one would need a local exclusion restriction ruling out a direct effect of assignment on the outcome except through treatment take-up, a monotonicity condition ruling out defiers near the boundary, and a nonzero first-stage effect. Under those conditions, the Wald curve $\zeta(\mathbf{b})$ can be interpreted as a local average treatment effect for compliers at the boundary point $\mathbf{b}$.

SA-9.4 Results

The following tables report Monte Carlo summaries separately for the sharp estimator and, in the fuzzy-design simulations, for the reduced-form/ITT, first-stage, and fuzzy estimators. Results are reported separately by DGP polynomial order $s \in \{1, 2\}$ and by homoskedastic or heteroskedastic outcome disturbances. We first report the sharp-design results. We then report two sets of fuzzy-design results: one using the ITT bandwidth selector, matching the main empirical specification, and one using the fuzzy bandwidth selector.

The sharp-design simulation results are reported first. Specifically:

- Tables [SA-1–SA-4](#) report the sharp simulation results for the linear homoskedastic specification.
- Tables [SA-5–SA-8](#) report the sharp simulation results for the linear heteroskedastic specification.
- Tables [SA-9–SA-12](#) report the sharp simulation results for the quadratic homoskedastic specification.
- Tables [SA-13–SA-16](#) report the sharp simulation results for the quadratic heteroskedastic specification.

The first set of fuzzy-design simulation results uses the ITT bandwidth selector. Specifically:

- Tables [SA-17–SA-20](#) report the reduced-form (ITT) simulation results for the linear homoskedastic specification.
- Tables [SA-21–SA-24](#) report the reduced-form (ITT) simulation results for the linear heteroskedastic specification.
- Tables [SA-25–SA-28](#) report the reduced-form (ITT) simulation results for the quadratic homoskedastic specification.
- Tables [SA-29–SA-32](#) report the reduced-form (ITT) simulation results for the quadratic heteroskedastic specification.
- Tables [SA-33–SA-36](#) report the take-up (FS) simulation results for the linear homoskedastic specification.
- Tables [SA-37–SA-40](#) report the take-up (FS) simulation results for the linear heteroskedastic specification.
- Tables [SA-41–SA-44](#) report the take-up (FS) simulation results for the quadratic homoskedastic specification.
- Tables [SA-45–SA-48](#) report the take-up (FS) simulation results for the quadratic heteroskedastic specification.
- Tables [SA-49–SA-52](#) report the fuzzy simulation results for the linear homoskedastic specification.
- Tables [SA-53–SA-56](#) report the fuzzy simulation results for the linear heteroskedastic specification.
- Tables [SA-57–SA-60](#) report the fuzzy simulation results for the quadratic homoskedastic specification.
- Tables [SA-61–SA-64](#) report the fuzzy simulation results for the quadratic heteroskedastic specification.

The second set of fuzzy-design simulation results uses the fuzzy bandwidth selector. Specifically:

- Tables [SA-65–SA-68](#) report the reduced-form (ITT) simulation results for the linear homoskedastic specification.
- Tables [SA-69–SA-72](#) report the reduced-form (ITT) simulation results for the linear heteroskedastic specification.
- Tables [SA-73–SA-76](#) report the reduced-form (ITT) simulation results for the quadratic homoskedastic specification.

- Tables [SA-77–SA-80](#) report the reduced-form (ITT) simulation results for the quadratic heteroskedastic specification.
- Tables [SA-81–SA-84](#) report the take-up (FS) simulation results for the linear homoskedastic specification.
- Tables [SA-85–SA-88](#) report the take-up (FS) simulation results for the linear heteroskedastic specification.
- Tables [SA-89–SA-92](#) report the take-up (FS) simulation results for the quadratic homoskedastic specification.
- Tables [SA-93–SA-96](#) report the take-up (FS) simulation results for the quadratic heteroskedastic specification.
- Tables [SA-97–SA-100](#) report the fuzzy simulation results for the linear homoskedastic specification.
- Tables [SA-101–SA-104](#) report the fuzzy simulation results for the linear heteroskedastic specification.
- Tables [SA-105–SA-108](#) report the fuzzy simulation results for the quadratic homoskedastic specification.
- Tables [SA-109–SA-112](#) report the fuzzy simulation results for the quadratic heteroskedastic specification.

The columns report the average bandwidths, average bias, standard deviation, root mean squared error, empirical coverage, and average interval or band length. For the Uniform row, empirical coverage and average length refer to uniform coverage and average uniform band length.

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.531	19.531	2120	4259	-0.001	0.033	0.033	0.958	0.240
b₂	19.133	19.133	2135	4107	-0.001	0.034	0.034	0.956	0.243
b₃	18.879	18.879	2182	3953	-0.000	0.035	0.035	0.955	0.245
b₄	18.784	18.784	2267	3798	0.002	0.036	0.036	0.952	0.246
b₅	18.826	18.826	2383	3637	0.004	0.035	0.036	0.947	0.246
b₆	18.978	18.978	2512	3469	0.007	0.035	0.035	0.953	0.245
b₇	19.090	19.090	2617	3257	0.008	0.034	0.035	0.960	0.247
b₈	19.093	19.093	2675	2989	0.008	0.035	0.036	0.959	0.252
b₉	19.128	19.128	2711	2714	0.006	0.036	0.037	0.953	0.258
b₁₀	19.376	19.376	2758	2482	0.003	0.038	0.038	0.955	0.268
b₁₁	21.264	21.264	3050	2652	0.000	0.038	0.038	0.951	0.279
b₁₂	19.240	19.240	2736	2567	-0.006	0.037	0.038	0.955	0.263
b₁₃	19.204	19.204	2705	2964	-0.012	0.036	0.038	0.954	0.252
b₁₄	19.259	19.259	2637	3357	-0.014	0.035	0.038	0.951	0.245
b₁₅	19.026	19.026	2464	3617	-0.011	0.035	0.037	0.953	0.244
b₁₆	18.641	18.641	2239	3751	-0.004	0.035	0.036	0.955	0.248
b₁₇	18.403	18.403	2065	3814	0.000	0.035	0.035	0.950	0.251
b₁₈	18.665	18.665	2013	3929	0.002	0.033	0.034	0.951	0.250
b₁₉	19.455	19.455	2039	4153	0.002	0.034	0.034	0.953	0.245
b₂₀	20.632	20.632	2094	4449	0.001	0.033	0.033	0.948	0.237
b₂₁	21.791	21.791	2110	4669	0.000	0.033	0.033	0.950	0.232
Uniform								0.946	0.373
WBATE					-0.000	0.017	0.017	0.948	0.093
LBATE					0.043	0.027	0.050	0.984	0.390

Table SA-1: Sharp BATEC Effects (Simulations, Linear Homoskedastic, Smooth-Boundary Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.845	18.845	2020	3988	-0.000	0.034	0.034	0.958	0.247
b₂	17.539	17.539	1876	3510	-0.000	0.035	0.035	0.953	0.261
b₃	15.909	15.909	1661	2913	0.000	0.038	0.038	0.948	0.285
b₄	13.996	13.996	1382	2256	0.000	0.043	0.043	0.950	0.322
b₅	12.000	12.000	1081	1641	-0.000	0.050	0.050	0.947	0.375
b₆	10.000	10.000	788	1113	-0.000	0.060	0.060	0.947	0.451
b₇	8.547	8.547	594	781	0.000	0.071	0.071	0.942	0.532
b₈	8.365	8.365	597	678	0.001	0.075	0.075	0.939	0.559
b₉	8.380	8.380	641	580	0.002	0.077	0.077	0.942	0.572
b₁₀	8.489	8.489	701	479	0.002	0.080	0.080	0.942	0.586
b₁₁	9.316	9.316	871	442	-0.000	0.086	0.086	0.931	0.636
b₁₂	8.429	8.429	675	519	-0.005	0.078	0.078	0.939	0.580
b₁₃	8.414	8.414	612	667	-0.005	0.075	0.075	0.940	0.557
b₁₄	8.724	8.724	615	822	-0.002	0.070	0.070	0.942	0.522
b₁₅	11.200	11.200	961	1419	-0.001	0.054	0.054	0.941	0.402
b₁₆	13.996	13.996	1382	2258	-0.000	0.043	0.043	0.942	0.322
b₁₇	16.487	16.487	1740	3124	0.000	0.037	0.037	0.946	0.275
b₁₈	18.113	18.113	1931	3712	0.001	0.034	0.034	0.951	0.256
b₁₉	19.403	19.403	2033	4132	0.002	0.034	0.034	0.953	0.245
b₂₀	20.632	20.632	2094	4449	0.001	0.033	0.033	0.948	0.237
b₂₁	21.791	21.791	2110	4669	0.000	0.033	0.033	0.950	0.232
Uniform								0.934	0.616
WBATE					-0.000	0.021	0.021	0.952	0.120
LBATE					0.089	0.046	0.100	0.973	0.757

Table SA-2: Sharp BATEC Effects (Simulations, Linear Homoskedastic, Kink-Adaptive Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.557	8.557	609	823	0.000	0.069	0.069	0.957	0.276
b₂	8.382	8.382	597	801	-0.000	0.072	0.072	0.950	0.280
b₃	8.271	8.271	588	786	-0.001	0.074	0.074	0.949	0.282
b₄	8.229	8.229	582	777	-0.001	0.073	0.073	0.956	0.284
b₅	8.248	8.248	579	772	-0.002	0.074	0.074	0.952	0.283
b₆	8.314	8.314	576	770	-0.000	0.073	0.073	0.952	0.283
b₇	8.364	8.364	574	750	-0.000	0.073	0.073	0.951	0.285
b₈	8.365	8.365	597	678	0.001	0.075	0.075	0.951	0.291
b₉	8.380	8.380	641	580	0.002	0.077	0.077	0.950	0.298
b₁₀	8.489	8.489	701	479	0.002	0.080	0.080	0.951	0.309
b₁₁	9.316	9.316	871	442	-0.000	0.086	0.086	0.947	0.330
b₁₂	8.429	8.429	675	519	-0.005	0.078	0.078	0.950	0.303
b₁₃	8.414	8.414	612	667	-0.005	0.075	0.075	0.953	0.291
b₁₄	8.437	8.437	582	773	-0.002	0.074	0.074	0.949	0.282
b₁₅	8.335	8.335	585	784	-0.001	0.074	0.074	0.945	0.281
b₁₆	8.167	8.167	574	765	0.001	0.075	0.075	0.944	0.286
b₁₇	8.063	8.063	561	746	0.001	0.075	0.075	0.947	0.289
b₁₈	8.177	8.177	565	755	0.000	0.075	0.075	0.947	0.288
b₁₉	8.523	8.523	588	795	0.002	0.073	0.073	0.947	0.282
b₂₀	9.039	9.039	621	853	0.002	0.071	0.071	0.947	0.272
b₂₁	9.547	9.547	640	896	0.001	0.069	0.069	0.945	0.266
Uniform								0.944	0.438
WBATE					-0.000	0.024	0.024	0.953	0.095
LBATE					0.123	0.049	0.132	0.981	0.457

Table SA-3: Sharp BATEC Effects (Simulations, Linear Homoskedastic, Unknown-Kink Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.346	19.346	2103	4191	-0.002	0.035	0.035	0.960	0.244
b₂	19.187	19.187	2155	4130	-0.002	0.036	0.036	0.957	0.245
b₃	19.183	19.183	2247	4066	-0.001	0.037	0.037	0.958	0.244
b₄	19.301	19.301	2372	3985	0.001	0.037	0.037	0.959	0.244
b₅	19.499	19.499	2514	3876	0.004	0.037	0.038	0.953	0.242
b₆	19.771	19.771	2662	3744	0.006	0.037	0.037	0.953	0.241
b₇	20.043	20.043	2791	3576	0.008	0.036	0.037	0.960	0.241
b₈	20.290	20.290	2888	3372	0.007	0.036	0.037	0.958	0.243
b₉	20.611	20.611	2967	3165	0.006	0.037	0.037	0.955	0.246
b₁₀	21.431	21.431	3097	3076	0.003	0.037	0.038	0.957	0.248
b₁₁	23.256	23.256	3339	3247	0.000	0.038	0.038	0.956	0.260
b₁₂	21.129	21.129	3057	3118	-0.006	0.037	0.038	0.957	0.246
b₁₃	20.504	20.504	2936	3371	-0.012	0.037	0.039	0.956	0.242
b₁₄	20.155	20.155	2800	3662	-0.014	0.037	0.039	0.953	0.240
b₁₅	19.764	19.764	2605	3882	-0.010	0.038	0.039	0.956	0.241
b₁₆	19.102	19.102	2335	3923	-0.004	0.038	0.038	0.957	0.247
b₁₇	18.429	18.429	2083	3831	0.001	0.037	0.037	0.956	0.254
b₁₈	18.250	18.250	1960	3778	0.003	0.036	0.036	0.954	0.258
b₁₉	18.549	18.549	1927	3806	0.003	0.036	0.036	0.952	0.257
b₂₀	19.074	19.074	1909	3854	0.002	0.037	0.038	0.949	0.256
b₂₁	19.668	19.668	1873	3873	0.001	0.038	0.038	0.949	0.258
Uniform								0.949	0.371
WBATE					-0.000	0.017	0.017	0.945	0.092
LBATE					0.047	0.031	0.057	0.983	0.409

Table SA-4: Sharp BATEC Effects (Simulations, Linear Homoskedastic, **rdrobust** Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.917	19.917	2174	4411	-0.002	0.044	0.044	0.956	0.312
b₂	19.329	19.329	2166	4181	-0.002	0.044	0.045	0.958	0.315
b₃	18.839	18.839	2175	3939	-0.001	0.045	0.045	0.957	0.318
b₄	18.488	18.488	2211	3699	0.001	0.046	0.046	0.953	0.320
b₅	18.264	18.264	2270	3459	0.003	0.046	0.046	0.954	0.321
b₆	18.133	18.133	2338	3215	0.006	0.046	0.046	0.958	0.323
b₇	18.010	18.010	2394	2947	0.008	0.046	0.046	0.960	0.325
b₈	17.712	17.712	2394	2613	0.008	0.047	0.047	0.956	0.332
b₉	17.533	17.533	2395	2303	0.007	0.048	0.048	0.949	0.339
b₁₀	17.461	17.461	2393	2015	0.005	0.048	0.049	0.954	0.343
b₁₁	18.552	18.552	2578	1984	0.002	0.048	0.048	0.949	0.340
b₁₂	17.414	17.414	2382	2113	-0.004	0.050	0.050	0.953	0.351
b₁₃	17.561	17.561	2372	2520	-0.010	0.050	0.051	0.959	0.351
b₁₄	17.865	17.865	2349	2953	-0.011	0.051	0.052	0.953	0.349
b₁₅	17.984	17.984	2251	3296	-0.008	0.052	0.053	0.957	0.354
b₁₆	18.195	18.195	2153	3603	-0.003	0.053	0.053	0.956	0.361
b₁₇	18.627	18.627	2099	3896	0.000	0.054	0.054	0.954	0.367
b₁₈	19.308	19.308	2100	4180	0.002	0.054	0.054	0.952	0.372
b₁₉	20.440	20.440	2160	4548	0.002	0.053	0.053	0.952	0.371
b₂₀	21.899	21.899	2232	4967	0.001	0.053	0.053	0.951	0.370
b₂₁	23.484	23.484	2276	5358	0.000	0.054	0.054	0.957	0.371
Uniform								0.950	0.516
WBATE					0.000	0.022	0.022	0.952	0.121
LBATE					0.066	0.038	0.076	0.982	0.553

Table SA-5: Sharp BATEC Effects (Simulations, Linear Heteroskedastic, Smooth-Boundary Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.062	19.062	2049	4073	-0.002	0.045	0.045	0.954	0.324
b₂	17.606	17.606	1887	3536	-0.001	0.046	0.046	0.951	0.342
b₃	15.908	15.908	1661	2914	-0.000	0.050	0.050	0.948	0.371
b₄	13.995	13.995	1382	2256	-0.000	0.055	0.055	0.945	0.414
b₅	12.000	12.000	1082	1641	0.000	0.063	0.063	0.943	0.477
b₆	10.000	10.000	789	1112	-0.000	0.075	0.075	0.944	0.571
b₇	8.241	8.241	553	727	-0.001	0.090	0.090	0.937	0.688
b₈	7.762	7.762	511	593	0.002	0.099	0.099	0.942	0.736
b₉	7.681	7.681	537	496	0.004	0.103	0.103	0.934	0.751
b₁₀	7.650	7.650	574	392	0.006	0.102	0.102	0.938	0.769
b₁₁	8.128	8.128	681	328	0.003	0.101	0.101	0.938	0.759
b₁₂	7.629	7.629	554	431	-0.004	0.105	0.105	0.937	0.776
b₁₃	7.694	7.694	508	569	-0.003	0.104	0.105	0.942	0.775
b₁₄	8.450	8.450	578	772	0.001	0.095	0.095	0.941	0.710
b₁₅	11.200	11.200	962	1418	-0.001	0.074	0.074	0.946	0.551
b₁₆	13.992	13.992	1382	2256	-0.001	0.062	0.062	0.943	0.459
b₁₇	16.564	16.564	1751	3153	-0.001	0.056	0.056	0.949	0.407
b₁₈	18.582	18.582	1993	3894	0.001	0.054	0.054	0.950	0.384
b₁₉	20.310	20.310	2144	4494	0.001	0.053	0.053	0.952	0.373
b₂₀	21.890	21.890	2231	4963	0.001	0.053	0.053	0.951	0.370
b₂₁	23.483	23.483	2276	5358	0.000	0.054	0.054	0.957	0.371
Uniform								0.927	0.829
WBATE					0.000	0.027	0.027	0.949	0.153
LBATE					0.127	0.061	0.141	0.967	1.012

Table SA-6: Sharp BATEC Effects (Simulations, Linear Heteroskedastic, Kink-Adaptive Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.726	8.726	630	856	-0.003	0.093	0.093	0.950	0.356
b₂	8.468	8.468	608	818	-0.002	0.093	0.093	0.948	0.360
b₃	8.253	8.253	585	781	0.000	0.094	0.094	0.949	0.365
b₄	8.100	8.100	566	751	0.001	0.095	0.095	0.947	0.369
b₅	8.002	8.002	549	725	0.000	0.096	0.096	0.949	0.371
b₆	7.944	7.944	532	701	-0.001	0.096	0.096	0.954	0.374
b₇	7.890	7.890	514	670	-0.001	0.096	0.096	0.957	0.377
b₈	7.760	7.760	511	593	0.002	0.099	0.100	0.951	0.384
b₉	7.681	7.681	537	496	0.004	0.103	0.103	0.941	0.393
b₁₀	7.650	7.650	574	392	0.006	0.102	0.102	0.951	0.397
b₁₁	8.128	8.128	681	328	0.003	0.101	0.101	0.954	0.395
b₁₂	7.629	7.629	554	431	-0.004	0.105	0.105	0.949	0.406
b₁₃	7.693	7.693	508	569	-0.003	0.105	0.105	0.955	0.405
b₁₄	7.827	7.827	507	665	0.001	0.104	0.104	0.953	0.403
b₁₅	7.879	7.879	530	698	0.000	0.105	0.105	0.952	0.406
b₁₆	7.971	7.971	550	726	-0.003	0.106	0.106	0.947	0.411
b₁₇	8.161	8.161	573	762	-0.003	0.107	0.107	0.951	0.416
b₁₈	8.459	8.459	600	807	-0.000	0.109	0.109	0.945	0.419
b₁₉	8.955	8.955	641	878	-0.001	0.109	0.109	0.947	0.416
b₂₀	9.594	9.594	688	963	-0.002	0.107	0.107	0.946	0.412
b₂₁	10.288	10.288	727	1044	-0.001	0.106	0.106	0.953	0.411
Uniform								0.942	0.600
WBATE					-0.000	0.032	0.032	0.952	0.124
LBATE					0.175	0.069	0.188	0.979	0.636

Table SA-7: Sharp BATEC Effects (Simulations, Linear Heteroskedastic, Unknown-Kink Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	19.141	19.141	2074	4119	-0.003	0.049	0.049	0.958	0.328
b ₂	18.748	18.748	2087	3973	-0.003	0.049	0.049	0.955	0.329
b ₃	18.546	18.546	2138	3846	-0.002	0.050	0.050	0.959	0.328
b ₄	18.409	18.409	2206	3689	-0.001	0.051	0.051	0.954	0.327
b ₅	18.352	18.352	2291	3510	0.002	0.051	0.051	0.954	0.326
b ₆	18.342	18.342	2380	3306	0.004	0.050	0.051	0.964	0.325
b ₇	17.995	17.995	2387	2973	0.006	0.050	0.050	0.963	0.331
b ₈	17.794	17.794	2404	2669	0.007	0.051	0.052	0.957	0.337
b ₉	17.707	17.707	2421	2387	0.006	0.053	0.053	0.954	0.343
b ₁₀	17.845	17.845	2456	2148	0.004	0.053	0.053	0.957	0.344
b ₁₁	18.868	18.868	2619	2107	0.001	0.052	0.052	0.957	0.341
b ₁₂	17.862	17.862	2459	2264	-0.004	0.054	0.054	0.956	0.350
b ₁₃	18.205	18.205	2495	2731	-0.009	0.055	0.055	0.960	0.347
b ₁₄	18.657	18.657	2506	3220	-0.010	0.055	0.056	0.954	0.343
b ₁₅	18.922	18.922	2442	3620	-0.007	0.057	0.057	0.960	0.347
b ₁₆	18.870	18.870	2297	3851	-0.002	0.058	0.058	0.960	0.359
b ₁₇	18.966	18.966	2179	4030	0.001	0.060	0.060	0.959	0.370
b ₁₈	19.187	19.187	2096	4149	0.003	0.060	0.061	0.958	0.381
b ₁₉	19.676	19.676	2065	4268	0.004	0.061	0.061	0.957	0.390
b ₂₀	20.508	20.508	2065	4446	0.003	0.063	0.063	0.953	0.399
b ₂₁	21.347	21.347	2041	4551	0.004	0.065	0.065	0.959	0.411
Uniform								0.949	0.528
WBATE					0.000	0.023	0.023	0.947	0.122
LBATE					0.080	0.052	0.095	0.980	0.616

Table SA-8: Sharp BATEC Effects (Simulations, Linear Heteroskedastic, **rdrobust** Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.917	17.917	1899	3638	-0.002	0.036	0.036	0.948	0.260
b₂	17.386	17.386	1869	3461	-0.002	0.037	0.037	0.950	0.266
b₃	17.186	17.186	1889	3364	-0.001	0.038	0.038	0.957	0.268
b₄	17.352	17.352	1993	3331	0.002	0.038	0.039	0.957	0.265
b₅	17.854	17.854	2186	3338	0.009	0.038	0.039	0.953	0.259
b₆	18.432	18.432	2398	3310	0.016	0.037	0.041	0.950	0.253
b₇	18.885	18.885	2573	3201	0.021	0.036	0.042	0.958	0.250
b₈	19.002	19.002	2655	2966	0.022	0.035	0.041	0.952	0.253
b₉	19.131	19.131	2712	2716	0.017	0.036	0.040	0.950	0.258
b₁₀	19.388	19.388	2760	2485	0.008	0.038	0.039	0.951	0.267
b₁₁	21.252	21.252	3048	2650	-0.002	0.038	0.038	0.947	0.279
b₁₂	19.198	19.198	2728	2557	-0.010	0.037	0.039	0.947	0.263
b₁₃	19.022	19.022	2668	2915	-0.017	0.036	0.040	0.954	0.254
b₁₄	18.697	18.697	2520	3197	-0.018	0.037	0.041	0.951	0.252
b₁₅	17.741	17.741	2201	3228	-0.012	0.039	0.041	0.948	0.261
b₁₆	16.747	16.747	1880	3135	-0.004	0.040	0.040	0.956	0.274
b₁₇	16.235	16.235	1717	3042	-0.001	0.039	0.039	0.947	0.282
b₁₈	16.400	16.400	1699	3076	-0.001	0.038	0.038	0.945	0.282
b₁₉	17.082	17.082	1739	3239	-0.002	0.037	0.037	0.946	0.276
b₂₀	17.979	17.979	1779	3429	-0.003	0.037	0.037	0.947	0.269
b₂₁	18.924	18.924	1793	3580	-0.005	0.036	0.037	0.952	0.265
Uniform								0.948	0.398
WBATE					0.001	0.017	0.017	0.950	0.095
LBATE					0.031	0.029	0.043	0.990	0.417

Table SA-9: Sharp BATEC Effects (Simulations, Quadratic Homoskedastic, Smooth-Boundary Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.644	17.644	1860	3529	-0.002	0.036	0.036	0.948	0.263
b₂	16.703	16.703	1759	3205	-0.001	0.038	0.038	0.948	0.274
b₃	15.611	15.611	1617	2813	-0.001	0.039	0.039	0.951	0.290
b₄	13.979	13.979	1378	2252	-0.001	0.043	0.043	0.949	0.322
b₅	12.000	12.000	1080	1642	-0.001	0.050	0.050	0.945	0.375
b₆	10.000	10.000	787	1114	-0.001	0.059	0.059	0.947	0.450
b₇	8.509	8.509	589	775	-0.001	0.070	0.070	0.944	0.535
b₈	8.325	8.325	592	674	0.003	0.075	0.075	0.942	0.559
b₉	8.381	8.381	641	581	0.009	0.076	0.077	0.940	0.571
b₁₀	8.494	8.494	702	480	0.007	0.080	0.080	0.942	0.585
b₁₁	9.311	9.311	870	442	-0.001	0.086	0.086	0.933	0.636
b₁₂	8.411	8.411	672	517	-0.007	0.079	0.079	0.942	0.580
b₁₃	8.334	8.334	601	656	-0.004	0.076	0.076	0.941	0.562
b₁₄	8.644	8.644	605	807	-0.000	0.070	0.070	0.943	0.524
b₁₅	11.200	11.200	962	1419	-0.001	0.053	0.053	0.945	0.401
b₁₆	13.955	13.955	1376	2245	-0.002	0.044	0.044	0.947	0.322
b₁₇	15.651	15.651	1620	2831	-0.001	0.040	0.040	0.944	0.290
b₁₈	16.293	16.293	1684	3034	-0.001	0.038	0.038	0.945	0.283
b₁₉	17.074	17.074	1738	3236	-0.002	0.037	0.037	0.946	0.276
b₂₀	17.979	17.979	1779	3429	-0.003	0.037	0.037	0.947	0.269
b₂₁	18.924	18.924	1793	3580	-0.005	0.036	0.037	0.952	0.265
Uniform								0.930	0.629
WBATE					-0.001	0.021	0.021	0.947	0.121
LBATE					0.085	0.054	0.100	0.981	0.792

Table SA-10: Sharp BATEC Effects (Simulations, Quadratic Homoskedastic, Kink-Adaptive Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	7.849	7.849	524	693	0.001	0.078	0.078	0.945	0.300
b₂	7.617	7.617	504	662	0.002	0.079	0.079	0.949	0.307
b₃	7.529	7.529	498	651	0.001	0.079	0.079	0.948	0.310
b₄	7.602	7.602	506	663	-0.001	0.080	0.080	0.946	0.307
b₅	7.822	7.822	527	696	-0.002	0.078	0.078	0.945	0.300
b₆	8.075	8.075	547	729	-0.001	0.075	0.075	0.954	0.292
b₇	8.274	8.274	563	736	-0.001	0.073	0.073	0.954	0.288
b₈	8.325	8.325	592	674	0.003	0.075	0.075	0.950	0.292
b₉	8.381	8.381	641	581	0.009	0.076	0.077	0.951	0.298
b₁₀	8.494	8.494	702	480	0.007	0.080	0.080	0.946	0.308
b₁₁	9.311	9.311	870	442	-0.001	0.086	0.086	0.947	0.329
b₁₂	8.411	8.411	672	517	-0.007	0.079	0.079	0.946	0.303
b₁₃	8.333	8.333	601	656	-0.004	0.076	0.076	0.951	0.293
b₁₄	8.191	8.191	553	729	-0.000	0.075	0.075	0.947	0.290
b₁₅	7.772	7.772	519	682	0.000	0.078	0.078	0.949	0.302
b₁₆	7.337	7.337	476	616	0.001	0.082	0.082	0.945	0.317
b₁₇	7.113	7.113	449	577	0.001	0.086	0.086	0.944	0.327
b₁₈	7.185	7.185	449	579	0.001	0.084	0.084	0.945	0.327
b₁₉	7.484	7.484	468	609	-0.001	0.084	0.084	0.943	0.319
b₂₀	7.877	7.877	488	645	-0.002	0.080	0.080	0.947	0.311
b₂₁	8.291	8.291	502	673	-0.001	0.078	0.078	0.948	0.306
Uniform								0.937	0.467
WBATE					0.000	0.025	0.025	0.948	0.098
LBATE					0.114	0.052	0.126	0.988	0.488

Table SA-11: Sharp BATEC Effects (Simulations, Quadratic Homoskedastic, Unknown-Kink Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.226	17.226	1805	3381	-0.004	0.038	0.038	0.951	0.271
b₂	16.871	16.871	1794	3282	-0.004	0.039	0.039	0.955	0.275
b₃	16.770	16.770	1829	3228	-0.003	0.040	0.040	0.963	0.277
b₄	16.999	16.999	1940	3229	-0.000	0.041	0.041	0.961	0.275
b₅	17.517	17.517	2124	3259	0.006	0.042	0.043	0.956	0.270
b₆	18.251	18.251	2356	3293	0.013	0.042	0.044	0.955	0.262
b₇	18.950	18.950	2571	3265	0.018	0.040	0.044	0.956	0.256
b₈	19.610	19.610	2754	3184	0.019	0.038	0.043	0.953	0.252
b₉	20.292	20.292	2907	3079	0.016	0.037	0.040	0.950	0.249
b₁₀	21.262	21.262	3066	3032	0.008	0.038	0.039	0.954	0.250
b₁₁	23.045	23.045	3306	3188	-0.002	0.039	0.039	0.953	0.262
b₁₂	20.859	20.859	3010	3043	-0.011	0.038	0.039	0.956	0.248
b₁₃	19.881	19.881	2820	3194	-0.017	0.038	0.041	0.954	0.248
b₁₄	18.689	18.689	2509	3233	-0.017	0.041	0.044	0.951	0.258
b₁₅	17.167	17.167	2091	3088	-0.009	0.045	0.045	0.956	0.276
b₁₆	15.870	15.870	1740	2869	-0.001	0.044	0.044	0.957	0.294
b₁₇	15.296	15.296	1578	2731	0.002	0.043	0.043	0.948	0.302
b₁₈	15.414	15.414	1561	2731	0.001	0.041	0.041	0.947	0.300
b₁₉	15.959	15.959	1591	2833	0.000	0.041	0.041	0.949	0.294
b₂₀	16.585	16.585	1605	2925	-0.001	0.041	0.041	0.955	0.290
b₂₁	17.091	17.091	1577	2934	-0.003	0.042	0.042	0.955	0.292
Uniform								0.947	0.409
WBATE					0.001	0.018	0.018	0.949	0.096
LBATE					0.037	0.034	0.050	0.990	0.447

Table SA-12: Sharp BATEC Effects (Simulations, Quadratic Homoskedastic, **rdrobust** Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.550	18.550	1986	3881	-0.002	0.042	0.042	0.949	0.302
b₂	17.907	17.907	1948	3654	-0.001	0.042	0.042	0.955	0.304
b₃	17.562	17.562	1954	3494	0.000	0.042	0.042	0.960	0.303
b₄	17.483	17.483	2019	3373	0.003	0.043	0.043	0.960	0.299
b₅	17.596	17.596	2134	3256	0.009	0.042	0.043	0.958	0.293
b₆	17.768	17.768	2262	3110	0.016	0.042	0.045	0.958	0.287
b₇	17.881	17.881	2367	2912	0.021	0.041	0.046	0.951	0.283
b₈	17.656	17.656	2382	2600	0.021	0.041	0.046	0.950	0.287
b₉	17.529	17.529	2394	2304	0.017	0.041	0.045	0.949	0.291
b₁₀	17.409	17.409	2382	2004	0.009	0.042	0.043	0.952	0.294
b₁₁	18.492	18.492	2566	1971	-0.001	0.042	0.042	0.951	0.293
b₁₂	17.353	17.353	2369	2100	-0.009	0.043	0.044	0.956	0.300
b₁₃	17.438	17.438	2346	2490	-0.016	0.043	0.046	0.954	0.298
b₁₄	17.577	17.577	2288	2874	-0.017	0.043	0.046	0.952	0.296
b₁₅	17.244	17.244	2100	3076	-0.010	0.045	0.046	0.957	0.305
b₁₆	16.769	16.769	1884	3142	-0.003	0.046	0.046	0.962	0.318
b₁₇	16.620	16.620	1777	3181	-0.000	0.046	0.046	0.954	0.328
b₁₈	17.013	17.013	1784	3306	-0.001	0.046	0.046	0.947	0.329
b₁₉	17.833	17.833	1836	3526	-0.002	0.045	0.045	0.953	0.325
b₂₀	18.869	18.869	1887	3768	-0.004	0.045	0.045	0.954	0.321
b₂₁	20.049	20.049	1920	4002	-0.006	0.045	0.045	0.945	0.318
Uniform								0.952	0.458
WBATE					0.001	0.019	0.019	0.950	0.105
LBATE					0.042	0.033	0.053	0.988	0.481

Table SA-13: Sharp BATEC Effects (Simulations, Quadratic Heteroskedastic, Smooth-Boundary Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.127	18.127	1925	3713	-0.001	0.043	0.043	0.950	0.307
b₂	16.987	16.987	1800	3309	-0.000	0.044	0.043	0.950	0.317
b₃	15.693	15.693	1630	2842	0.001	0.045	0.045	0.955	0.333
b₄	13.973	13.973	1379	2251	0.001	0.048	0.048	0.948	0.366
b₅	12.000	12.000	1082	1642	0.000	0.055	0.055	0.945	0.418
b₆	10.000	10.000	788	1113	-0.001	0.065	0.065	0.942	0.494
b₇	8.216	8.216	549	723	0.001	0.081	0.081	0.940	0.595
b₈	7.739	7.739	508	590	0.005	0.088	0.088	0.934	0.634
b₉	7.680	7.680	536	496	0.011	0.088	0.088	0.943	0.641
b₁₀	7.627	7.627	570	390	0.008	0.087	0.087	0.946	0.651
b₁₁	8.101	8.101	677	326	-0.001	0.090	0.090	0.937	0.648
b₁₂	7.602	7.602	550	428	-0.006	0.091	0.091	0.937	0.659
b₁₃	7.640	7.640	500	562	-0.003	0.089	0.089	0.944	0.651
b₁₄	8.433	8.433	576	770	-0.002	0.080	0.080	0.943	0.590
b₁₅	11.200	11.200	962	1419	-0.001	0.060	0.060	0.946	0.452
b₁₆	13.943	13.943	1374	2241	-0.001	0.049	0.049	0.951	0.373
b₁₇	15.831	15.831	1646	2896	-0.001	0.047	0.047	0.951	0.340
b₁₈	16.848	16.848	1760	3241	-0.001	0.046	0.046	0.948	0.331
b₁₉	17.819	17.819	1834	3520	-0.002	0.045	0.045	0.953	0.326
b₂₀	18.869	18.869	1887	3768	-0.004	0.045	0.045	0.954	0.321
b₂₁	20.049	20.049	1920	4002	-0.006	0.045	0.045	0.945	0.318
Uniform								0.932	0.713
WBATE					-0.000	0.023	0.023	0.950	0.131
LBATE					0.102	0.058	0.117	0.979	0.888

Table SA-14: Sharp BATEC Effects (Simulations, Quadratic Heteroskedastic, Kink-Adaptive Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.127	8.127	558	743	0.001	0.090	0.090	0.948	0.347
b₂	7.845	7.845	533	703	0.001	0.090	0.090	0.946	0.350
b₃	7.694	7.694	518	680	0.001	0.088	0.088	0.953	0.349
b₄	7.659	7.659	513	673	0.000	0.088	0.088	0.947	0.345
b₅	7.709	7.709	514	674	-0.002	0.087	0.087	0.952	0.338
b₆	7.785	7.785	513	674	-0.002	0.086	0.086	0.951	0.332
b₇	7.834	7.834	507	661	0.000	0.086	0.086	0.947	0.328
b₈	7.735	7.735	508	590	0.005	0.088	0.088	0.947	0.331
b₉	7.680	7.680	536	496	0.011	0.088	0.088	0.948	0.335
b₁₀	7.627	7.627	570	390	0.008	0.087	0.087	0.950	0.338
b₁₁	8.101	8.101	677	326	-0.001	0.090	0.090	0.946	0.339
b₁₂	7.602	7.602	550	428	-0.006	0.091	0.091	0.945	0.345
b₁₃	7.640	7.640	500	562	-0.003	0.089	0.089	0.950	0.342
b₁₄	7.701	7.701	493	645	-0.002	0.088	0.088	0.951	0.341
b₁₅	7.555	7.555	492	643	-0.000	0.089	0.089	0.948	0.348
b₁₆	7.347	7.347	476	618	-0.001	0.093	0.093	0.946	0.362
b₁₇	7.281	7.281	468	606	0.001	0.096	0.096	0.945	0.372
b₁₈	7.453	7.453	480	625	-0.001	0.097	0.097	0.941	0.372
b₁₉	7.813	7.813	505	667	-0.002	0.094	0.094	0.950	0.367
b₂₀	8.267	8.267	533	714	0.000	0.093	0.093	0.947	0.361
b₂₁	8.784	8.784	556	759	0.000	0.094	0.094	0.941	0.357
Uniform								0.943	0.531
WBATE					0.000	0.028	0.028	0.947	0.107
LBATE					0.137	0.059	0.149	0.986	0.556

Table SA-15: Sharp BATEC Effects (Simulations, Quadratic Heteroskedastic, Unknown-Kink Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.834	17.834	1890	3614	-0.003	0.045	0.045	0.951	0.315
b₂	17.392	17.392	1874	3473	-0.003	0.045	0.045	0.953	0.315
b₃	17.123	17.123	1890	3353	-0.002	0.046	0.046	0.959	0.314
b₄	17.070	17.070	1956	3254	0.001	0.047	0.047	0.964	0.311
b₅	17.206	17.206	2067	3162	0.006	0.048	0.048	0.959	0.306
b₆	17.450	17.450	2199	3053	0.011	0.048	0.049	0.958	0.299
b₇	17.514	17.514	2288	2849	0.016	0.047	0.050	0.949	0.296
b₈	17.563	17.563	2356	2616	0.018	0.047	0.050	0.948	0.295
b₉	17.567	17.567	2393	2354	0.015	0.046	0.048	0.954	0.296
b₁₀	17.679	17.679	2426	2107	0.008	0.045	0.046	0.956	0.296
b₁₁	18.549	18.549	2564	2029	-0.002	0.046	0.046	0.958	0.298
b₁₂	17.600	17.600	2410	2197	-0.009	0.047	0.048	0.957	0.302
b₁₃	17.641	17.641	2383	2579	-0.014	0.047	0.050	0.956	0.301
b₁₄	17.575	17.575	2288	2913	-0.013	0.049	0.051	0.953	0.303
b₁₅	16.976	16.976	2057	3031	-0.006	0.051	0.052	0.962	0.318
b₁₆	16.166	16.166	1798	2969	0.001	0.050	0.050	0.961	0.337
b₁₇	15.779	15.779	1656	2908	0.004	0.051	0.051	0.960	0.349
b₁₈	15.962	15.962	1637	2940	0.003	0.051	0.051	0.950	0.351
b₁₉	16.531	16.531	1665	3054	0.001	0.050	0.050	0.955	0.350
b₂₀	17.189	17.189	1679	3159	0.000	0.051	0.051	0.954	0.351
b₂₁	17.845	17.845	1664	3220	-0.000	0.053	0.053	0.947	0.356
Uniform								0.954	0.480
WBATE					0.001	0.020	0.020	0.947	0.107
LBATE					0.052	0.042	0.067	0.991	0.540

Table SA-16: Sharp BATEC Effects (Simulations, Quadratic Heteroskedastic, **rdrobust** Bandwidth).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.534	19.534	2120	4260	0.000	0.033	0.033	0.950	0.241
b₂	19.152	19.152	2138	4114	0.000	0.034	0.034	0.952	0.244
b₃	18.902	18.902	2185	3961	0.001	0.035	0.035	0.950	0.246
b₄	18.805	18.805	2271	3805	0.002	0.035	0.036	0.952	0.247
b₅	18.832	18.832	2383	3639	0.004	0.035	0.036	0.955	0.247
b₆	19.003	19.003	2517	3478	0.006	0.035	0.036	0.958	0.247
b₇	19.115	19.115	2621	3265	0.008	0.035	0.036	0.955	0.248
b₈	19.116	19.116	2679	2996	0.008	0.036	0.037	0.953	0.253
b₉	19.218	19.218	2729	2738	0.007	0.037	0.038	0.952	0.259
b₁₀	19.480	19.480	2777	2508	0.004	0.038	0.038	0.952	0.268
b₁₁	21.345	21.345	3064	2673	0.001	0.039	0.039	0.954	0.280
b₁₂	19.319	19.319	2751	2587	-0.005	0.038	0.038	0.954	0.264
b₁₃	19.199	19.199	2704	2962	-0.011	0.037	0.038	0.949	0.254
b₁₄	19.204	19.204	2625	3342	-0.012	0.036	0.038	0.946	0.247
b₁₅	18.993	18.993	2458	3608	-0.009	0.036	0.037	0.950	0.245
b₁₆	18.625	18.625	2237	3746	-0.003	0.037	0.037	0.955	0.249
b₁₇	18.421	18.421	2068	3821	0.001	0.035	0.035	0.959	0.252
b₁₈	18.678	18.678	2014	3935	0.001	0.034	0.034	0.957	0.251
b₁₉	19.490	19.490	2043	4167	0.001	0.033	0.033	0.949	0.244
b₂₀	20.668	20.668	2098	4464	0.001	0.033	0.033	0.956	0.236
b₂₁	21.765	21.765	2107	4658	-0.000	0.033	0.033	0.953	0.232
Uniform								0.950	0.374
WBATE					0.000	0.017	0.017	0.949	0.093
LBATE					0.044	0.027	0.052	0.982	0.391

Table SA-17: Intention-to-Treat Effects (Simulations, Linear Homoskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.862	18.862	2022	3994	0.001	0.034	0.034	0.950	0.248
b₂	17.546	17.546	1877	3512	0.001	0.036	0.036	0.947	0.262
b₃	15.909	15.909	1661	2914	0.001	0.039	0.039	0.948	0.286
b₄	13.997	13.997	1382	2257	0.001	0.044	0.044	0.949	0.324
b₅	12.000	12.000	1081	1641	-0.000	0.050	0.050	0.943	0.378
b₆	10.000	10.000	788	1113	-0.002	0.059	0.059	0.943	0.454
b₇	8.562	8.562	596	783	-0.002	0.071	0.071	0.947	0.535
b₈	8.375	8.375	599	680	0.001	0.076	0.076	0.938	0.560
b₉	8.420	8.420	647	585	0.004	0.077	0.077	0.944	0.574
b₁₀	8.534	8.534	708	483	0.003	0.079	0.079	0.940	0.586
b₁₁	9.351	9.351	877	445	-0.000	0.086	0.086	0.936	0.639
b₁₂	8.464	8.464	680	522	-0.006	0.077	0.078	0.943	0.582
b₁₃	8.411	8.411	612	667	-0.004	0.076	0.076	0.948	0.561
b₁₄	8.717	8.717	614	821	0.001	0.071	0.071	0.939	0.524
b₁₅	11.200	11.200	961	1419	0.002	0.054	0.054	0.943	0.403
b₁₆	13.996	13.996	1382	2258	0.001	0.043	0.043	0.946	0.323
b₁₇	16.483	16.483	1739	3123	0.000	0.037	0.037	0.949	0.277
b₁₈	18.112	18.112	1930	3712	0.001	0.034	0.034	0.954	0.257
b₁₉	19.440	19.440	2038	4146	0.001	0.033	0.033	0.949	0.245
b₂₀	20.668	20.668	2098	4464	0.001	0.033	0.033	0.956	0.236
b₂₁	21.765	21.765	2107	4658	-0.000	0.033	0.033	0.953	0.232
Uniform								0.929	0.618
WBATE					0.000	0.021	0.021	0.951	0.120
LBATE					0.091	0.048	0.103	0.967	0.752

Table SA-18: Intention-to-Treat Effects (Simulations, Linear Homoskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.558	8.558	609	823	0.001	0.072	0.072	0.944	0.277
b₂	8.391	8.391	598	803	0.001	0.073	0.073	0.942	0.281
b₃	8.281	8.281	589	788	0.001	0.073	0.073	0.950	0.283
b₄	8.239	8.239	583	779	0.001	0.074	0.074	0.948	0.285
b₅	8.251	8.251	579	773	-0.001	0.073	0.073	0.948	0.285
b₆	8.325	8.325	577	772	-0.002	0.073	0.073	0.951	0.285
b₇	8.375	8.375	575	752	-0.002	0.073	0.073	0.952	0.286
b₈	8.375	8.375	599	680	0.001	0.076	0.076	0.947	0.292
b₉	8.420	8.420	647	585	0.004	0.077	0.077	0.951	0.299
b₁₀	8.534	8.534	708	483	0.003	0.079	0.079	0.950	0.309
b₁₁	9.351	9.351	877	445	-0.000	0.086	0.086	0.947	0.331
b₁₂	8.464	8.464	680	522	-0.006	0.077	0.078	0.951	0.304
b₁₃	8.411	8.411	612	667	-0.004	0.076	0.076	0.952	0.293
b₁₄	8.413	8.413	579	768	0.001	0.075	0.075	0.945	0.284
b₁₅	8.321	8.321	583	781	0.002	0.074	0.074	0.947	0.283
b₁₆	8.160	8.160	573	764	0.001	0.074	0.074	0.950	0.287
b₁₇	8.070	8.070	561	747	-0.000	0.075	0.075	0.947	0.291
b₁₈	8.183	8.183	566	756	-0.001	0.074	0.074	0.952	0.289
b₁₉	8.539	8.539	590	798	-0.000	0.074	0.074	0.942	0.281
b₂₀	9.055	9.055	623	856	-0.000	0.070	0.070	0.952	0.271
b₂₁	9.535	9.535	639	893	-0.001	0.069	0.069	0.950	0.266
Uniform								0.943	0.439
WBATE					-0.000	0.024	0.024	0.951	0.095
LBATE					0.124	0.049	0.134	0.980	0.457

Table SA-19: Intention-to-Treat Effects (Simulations, Linear Homoskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.325	19.325	2100	4182	-0.001	0.036	0.036	0.953	0.246
b₂	19.171	19.171	2152	4123	-0.000	0.036	0.036	0.952	0.246
b₃	19.127	19.127	2235	4046	0.000	0.037	0.037	0.953	0.246
b₄	19.198	19.198	2351	3949	0.001	0.037	0.037	0.955	0.246
b₅	19.404	19.404	2495	3845	0.003	0.037	0.038	0.958	0.245
b₆	19.638	19.638	2635	3703	0.005	0.037	0.038	0.962	0.244
b₇	19.901	19.901	2763	3534	0.007	0.037	0.038	0.957	0.244
b₈	20.219	20.219	2874	3351	0.007	0.038	0.039	0.954	0.245
b₉	20.701	20.701	2983	3189	0.006	0.038	0.038	0.953	0.246
b₁₀	21.556	21.556	3119	3110	0.004	0.038	0.038	0.953	0.249
b₁₁	23.400	23.400	3362	3287	0.001	0.039	0.039	0.956	0.261
b₁₂	21.216	21.216	3072	3143	-0.006	0.038	0.038	0.957	0.246
b₁₃	20.499	20.499	2936	3369	-0.011	0.037	0.039	0.953	0.243
b₁₄	20.118	20.118	2793	3651	-0.012	0.038	0.039	0.950	0.241
b₁₅	19.695	19.695	2591	3862	-0.008	0.039	0.040	0.957	0.243
b₁₆	19.024	19.024	2321	3897	-0.002	0.039	0.039	0.958	0.249
b₁₇	18.414	18.414	2081	3827	0.002	0.038	0.038	0.962	0.256
b₁₈	18.243	18.243	1959	3776	0.003	0.036	0.036	0.961	0.259
b₁₉	18.530	18.530	1924	3797	0.002	0.036	0.036	0.950	0.257
b₂₀	19.104	19.104	1912	3865	0.001	0.037	0.037	0.953	0.256
b₂₁	19.617	19.617	1867	3854	0.000	0.038	0.038	0.953	0.258
Uniform								0.948	0.373
WBATE					0.000	0.017	0.017	0.946	0.093
LBATE					0.049	0.032	0.059	0.982	0.410

Table SA-20: Intention-to-Treat Effects (Simulations, Linear Homoskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.822	19.822	2159	4374	-0.001	0.042	0.042	0.955	0.300
b₂	19.210	19.210	2146	4137	-0.000	0.043	0.043	0.958	0.303
b₃	18.737	18.737	2155	3903	0.000	0.043	0.043	0.959	0.306
b₄	18.364	18.364	2185	3658	0.002	0.043	0.043	0.957	0.307
b₅	18.091	18.091	2234	3405	0.003	0.044	0.044	0.954	0.309
b₆	17.917	17.917	2293	3150	0.006	0.044	0.045	0.954	0.311
b₇	17.759	17.759	2342	2876	0.007	0.045	0.046	0.952	0.313
b₈	17.432	17.432	2336	2540	0.007	0.046	0.047	0.957	0.321
b₉	17.290	17.290	2346	2244	0.006	0.047	0.047	0.950	0.327
b₁₀	17.153	17.153	2333	1945	0.004	0.047	0.047	0.951	0.330
b₁₁	18.116	18.116	2499	1885	0.001	0.046	0.046	0.950	0.322
b₁₂	17.198	17.198	2339	2062	-0.005	0.048	0.048	0.953	0.340
b₁₃	17.364	17.364	2332	2471	-0.010	0.049	0.050	0.954	0.342
b₁₄	17.720	17.720	2319	2912	-0.011	0.050	0.052	0.952	0.340
b₁₅	17.870	17.870	2228	3261	-0.008	0.052	0.053	0.947	0.347
b₁₆	18.113	18.113	2137	3576	-0.003	0.052	0.052	0.952	0.354
b₁₇	18.540	18.540	2085	3865	0.001	0.052	0.052	0.955	0.362
b₁₈	19.281	19.281	2097	4170	0.002	0.053	0.053	0.951	0.367
b₁₉	20.431	20.431	2159	4545	0.002	0.053	0.053	0.954	0.367
b₂₀	21.957	21.957	2239	4992	0.001	0.053	0.053	0.955	0.366
b₂₁	23.561	23.561	2283	5391	0.000	0.054	0.054	0.957	0.368
Uniform								0.946	0.502
WBATE					0.000	0.021	0.021	0.950	0.116
LBATE					0.065	0.038	0.075	0.979	0.539

Table SA-21: Intention-to-Treat Effects (Simulations, Linear Heteroskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.027	19.027	2044	4060	-0.000	0.043	0.043	0.952	0.311
b₂	17.579	17.579	1882	3526	0.000	0.045	0.045	0.954	0.328
b₃	15.902	15.902	1660	2912	0.001	0.047	0.047	0.953	0.354
b₄	13.995	13.995	1382	2257	0.001	0.052	0.052	0.948	0.395
b₅	12.000	12.000	1081	1641	-0.001	0.061	0.061	0.946	0.456
b₆	10.000	10.000	788	1112	-0.002	0.072	0.072	0.945	0.542
b₇	8.181	8.181	545	717	-0.003	0.089	0.089	0.945	0.660
b₈	7.642	7.642	495	577	0.000	0.096	0.096	0.945	0.714
b₉	7.575	7.575	521	483	0.003	0.101	0.101	0.934	0.725
b₁₀	7.515	7.515	554	378	0.002	0.100	0.100	0.940	0.742
b₁₁	7.937	7.937	653	311	-0.001	0.098	0.098	0.932	0.718
b₁₂	7.534	7.534	540	421	-0.006	0.101	0.101	0.945	0.755
b₁₃	7.608	7.608	496	557	-0.002	0.102	0.102	0.942	0.755
b₁₄	8.434	8.434	576	769	0.000	0.093	0.093	0.938	0.688
b₁₅	11.200	11.200	962	1419	-0.001	0.074	0.074	0.939	0.535
b₁₆	13.991	13.991	1382	2256	-0.000	0.061	0.061	0.941	0.448
b₁₇	16.545	16.545	1748	3147	0.000	0.055	0.055	0.949	0.399
b₁₈	18.559	18.559	1990	3886	0.001	0.054	0.054	0.949	0.378
b₁₉	20.288	20.288	2142	4486	0.001	0.053	0.053	0.953	0.369
b₂₀	21.947	21.947	2238	4988	0.001	0.053	0.053	0.955	0.366
b₂₁	23.560	23.560	2283	5391	0.000	0.054	0.054	0.957	0.368
Uniform								0.924	0.802
WBATE					-0.000	0.026	0.026	0.946	0.147
LBATE					0.123	0.060	0.137	0.966	0.974

Table SA-22: Intention-to-Treat Effects (Simulations, Linear Heteroskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.684	8.684	625	848	-0.001	0.088	0.088	0.951	0.341
b₂	8.416	8.416	601	807	0.000	0.088	0.088	0.951	0.346
b₃	8.209	8.209	580	773	0.001	0.090	0.090	0.952	0.350
b₄	8.045	8.045	559	740	-0.000	0.092	0.092	0.950	0.353
b₅	7.926	7.926	539	711	-0.002	0.093	0.093	0.952	0.357
b₆	7.850	7.850	521	684	-0.003	0.093	0.093	0.952	0.360
b₇	7.780	7.780	501	651	-0.002	0.094	0.094	0.951	0.365
b₈	7.637	7.637	495	576	0.000	0.096	0.096	0.952	0.373
b₉	7.575	7.575	521	483	0.003	0.101	0.101	0.948	0.379
b₁₀	7.515	7.515	554	378	0.002	0.100	0.100	0.944	0.383
b₁₁	7.937	7.937	653	311	-0.001	0.098	0.098	0.948	0.373
b₁₂	7.534	7.534	540	421	-0.006	0.101	0.101	0.952	0.394
b₁₃	7.607	7.607	496	557	-0.002	0.102	0.102	0.950	0.395
b₁₄	7.763	7.763	500	654	-0.000	0.103	0.103	0.947	0.394
b₁₅	7.829	7.829	524	689	-0.002	0.105	0.105	0.945	0.397
b₁₆	7.935	7.935	546	720	0.000	0.105	0.105	0.944	0.403
b₁₇	8.122	8.122	568	755	0.001	0.105	0.105	0.952	0.409
b₁₈	8.447	8.447	598	805	-0.000	0.108	0.108	0.946	0.412
b₁₉	8.951	8.951	640	877	-0.000	0.107	0.107	0.947	0.411
b₂₀	9.620	9.620	691	969	0.002	0.105	0.105	0.948	0.408
b₂₁	10.322	10.322	731	1051	0.001	0.106	0.106	0.951	0.408
Uniform								0.938	0.583
WBATE					-0.001	0.031	0.031	0.946	0.120
LBATE					0.170	0.068	0.184	0.975	0.618

Table SA-23: Intention-to-Treat Effects (Simulations, Linear Heteroskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.920	18.920	2042	4036	-0.002	0.048	0.048	0.954	0.318
b₂	18.614	18.614	2065	3924	-0.001	0.047	0.047	0.963	0.317
b₃	18.397	18.397	2110	3794	-0.001	0.048	0.048	0.960	0.316
b₄	18.284	18.284	2182	3647	-0.000	0.048	0.048	0.958	0.314
b₅	18.172	18.172	2256	3454	0.002	0.049	0.049	0.955	0.314
b₆	18.104	18.104	2332	3236	0.004	0.049	0.049	0.957	0.314
b₇	17.715	17.715	2330	2898	0.005	0.051	0.051	0.954	0.321
b₈	17.474	17.474	2340	2587	0.006	0.051	0.052	0.958	0.327
b₉	17.306	17.306	2344	2288	0.005	0.053	0.053	0.955	0.334
b₁₀	17.385	17.385	2370	2040	0.003	0.052	0.053	0.954	0.334
b₁₁	18.282	18.282	2516	1968	0.000	0.051	0.051	0.954	0.326
b₁₂	17.522	17.522	2395	2180	-0.005	0.052	0.053	0.955	0.341
b₁₃	17.903	17.903	2435	2652	-0.010	0.054	0.055	0.958	0.340
b₁₄	18.374	18.374	2449	3140	-0.010	0.055	0.056	0.959	0.337
b₁₅	18.805	18.805	2421	3586	-0.006	0.057	0.058	0.955	0.340
b₁₆	18.919	18.919	2307	3867	-0.002	0.058	0.058	0.956	0.350
b₁₇	18.942	18.942	2175	4021	0.002	0.057	0.057	0.960	0.363
b₁₈	19.136	19.136	2090	4131	0.003	0.060	0.060	0.958	0.376
b₁₉	19.695	19.695	2068	4278	0.003	0.061	0.061	0.955	0.386
b₂₀	20.557	20.557	2072	4464	0.004	0.061	0.062	0.955	0.395
b₂₁	21.470	21.470	2054	4601	0.005	0.064	0.064	0.959	0.407
Uniform								0.949	0.515
WBATE					0.000	0.022	0.022	0.942	0.118
LBATE					0.079	0.052	0.094	0.979	0.604

Table SA-24: Intention-to-Treat Effects (Simulations, Linear Heteroskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.945	17.945	1903	3648	-0.002	0.035	0.035	0.948	0.262
b₂	17.380	17.380	1868	3459	-0.002	0.036	0.036	0.950	0.268
b₃	17.202	17.202	1891	3370	-0.001	0.037	0.037	0.959	0.269
b₄	17.380	17.380	1997	3341	0.002	0.039	0.039	0.956	0.267
b₅	17.857	17.857	2186	3339	0.009	0.039	0.040	0.954	0.260
b₆	18.430	18.430	2397	3310	0.016	0.038	0.042	0.951	0.254
b₇	18.889	18.889	2574	3202	0.021	0.037	0.043	0.948	0.251
b₈	18.988	18.988	2653	2961	0.022	0.036	0.042	0.948	0.254
b₉	19.108	19.108	2707	2710	0.017	0.036	0.040	0.947	0.259
b₁₀	19.393	19.393	2761	2487	0.009	0.037	0.038	0.951	0.268
b₁₁	21.303	21.303	3057	2662	-0.000	0.038	0.038	0.953	0.279
b₁₂	19.219	19.219	2732	2561	-0.008	0.037	0.038	0.953	0.263
b₁₃	19.004	19.004	2665	2911	-0.015	0.037	0.040	0.948	0.254
b₁₄	18.692	18.692	2519	3196	-0.017	0.037	0.041	0.947	0.252
b₁₅	17.796	17.796	2212	3246	-0.012	0.039	0.041	0.952	0.260
b₁₆	16.806	16.806	1891	3154	-0.004	0.039	0.039	0.953	0.273
b₁₇	16.262	16.262	1722	3052	-0.001	0.038	0.038	0.954	0.282
b₁₈	16.380	16.380	1697	3070	-0.001	0.038	0.038	0.952	0.282
b₁₉	17.047	17.047	1735	3229	-0.001	0.038	0.038	0.942	0.276
b₂₀	17.934	17.934	1774	3415	-0.002	0.038	0.038	0.944	0.269
b₂₁	18.901	18.901	1791	3574	-0.005	0.037	0.038	0.949	0.264
Uniform								0.947	0.398
WBATE					0.001	0.017	0.018	0.950	0.096
LBATE					0.032	0.029	0.043	0.988	0.416

Table SA-25: Intention-to-Treat Effects (Simulations, Quadratic Homoskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.655	17.655	1862	3533	-0.002	0.036	0.036	0.949	0.265
b₂	16.710	16.710	1760	3207	-0.001	0.037	0.037	0.950	0.275
b₃	15.625	15.625	1619	2818	-0.001	0.039	0.039	0.954	0.291
b₄	13.979	13.979	1379	2252	-0.000	0.043	0.043	0.948	0.324
b₅	12.000	12.000	1080	1642	0.000	0.050	0.050	0.950	0.377
b₆	10.000	10.000	787	1114	0.000	0.060	0.060	0.950	0.453
b₇	8.509	8.509	589	775	-0.000	0.073	0.073	0.940	0.538
b₈	8.319	8.319	591	673	0.004	0.077	0.077	0.944	0.563
b₉	8.372	8.372	639	580	0.010	0.077	0.077	0.940	0.572
b₁₀	8.496	8.496	702	480	0.009	0.078	0.079	0.942	0.586
b₁₁	9.333	9.333	873	444	0.002	0.085	0.085	0.932	0.637
b₁₂	8.420	8.420	673	518	-0.004	0.078	0.078	0.945	0.580
b₁₃	8.326	8.326	600	655	-0.001	0.076	0.076	0.944	0.562
b₁₄	8.643	8.643	604	807	0.001	0.070	0.070	0.947	0.525
b₁₅	11.200	11.200	962	1419	-0.001	0.054	0.054	0.945	0.401
b₁₆	13.958	13.958	1377	2246	-0.002	0.043	0.043	0.946	0.322
b₁₇	15.667	15.667	1623	2837	-0.002	0.039	0.039	0.952	0.290
b₁₈	16.274	16.274	1681	3028	-0.001	0.038	0.038	0.953	0.283
b₁₉	17.039	17.039	1734	3226	-0.001	0.038	0.038	0.942	0.276
b₂₀	17.934	17.934	1774	3415	-0.002	0.038	0.038	0.944	0.269
b₂₁	18.901	18.901	1791	3574	-0.005	0.037	0.038	0.949	0.264
Uniform								0.930	0.631
WBATE					0.000	0.021	0.021	0.949	0.121
LBATE					0.086	0.052	0.101	0.979	0.791

Table SA-26: Intention-to-Treat Effects (Simulations, Quadratic Homoskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	7.862	7.862	525	695	0.001	0.079	0.079	0.944	0.302
b₂	7.614	7.614	504	661	0.001	0.082	0.082	0.941	0.310
b₃	7.536	7.536	498	652	-0.000	0.080	0.080	0.949	0.312
b₄	7.614	7.614	507	666	-0.001	0.080	0.080	0.947	0.309
b₅	7.823	7.823	527	697	-0.000	0.077	0.077	0.953	0.301
b₆	8.074	8.074	547	729	-0.000	0.076	0.076	0.951	0.294
b₇	8.275	8.275	563	736	-0.000	0.076	0.076	0.950	0.290
b₈	8.319	8.319	591	673	0.004	0.077	0.077	0.945	0.293
b₉	8.372	8.372	639	580	0.010	0.077	0.077	0.948	0.299
b₁₀	8.496	8.496	702	480	0.009	0.078	0.079	0.954	0.309
b₁₁	9.333	9.333	873	444	0.002	0.085	0.085	0.950	0.330
b₁₂	8.420	8.420	673	518	-0.004	0.078	0.078	0.948	0.304
b₁₃	8.326	8.326	600	655	-0.001	0.076	0.076	0.951	0.293
b₁₄	8.189	8.189	553	729	0.001	0.075	0.075	0.947	0.290
b₁₅	7.797	7.797	522	687	-0.001	0.078	0.078	0.946	0.301
b₁₆	7.363	7.363	479	620	-0.001	0.083	0.083	0.943	0.316
b₁₇	7.124	7.124	450	579	-0.004	0.084	0.084	0.944	0.327
b₁₈	7.176	7.176	448	577	-0.001	0.082	0.082	0.949	0.326
b₁₉	7.468	7.468	466	607	0.003	0.084	0.084	0.940	0.319
b₂₀	7.857	7.857	486	643	0.000	0.082	0.082	0.940	0.311
b₂₁	8.281	8.281	501	672	-0.001	0.078	0.078	0.946	0.305
Uniform								0.938	0.468
WBATE					0.001	0.025	0.025	0.946	0.098
LBATE					0.115	0.052	0.126	0.989	0.486

Table SA-27: Intention-to-Treat Effects (Simulations, Quadratic Homoskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	17.303	17.303	1816	3409	-0.003	0.038	0.038	0.950	0.271
b ₂	16.910	16.910	1799	3295	-0.004	0.039	0.039	0.951	0.276
b ₃	16.825	16.825	1837	3247	-0.003	0.040	0.040	0.959	0.278
b ₄	17.058	17.058	1950	3248	0.000	0.042	0.042	0.963	0.276
b ₅	17.578	17.578	2135	3278	0.006	0.043	0.044	0.958	0.270
b ₆	18.313	18.313	2368	3312	0.013	0.043	0.045	0.954	0.262
b ₇	19.002	19.002	2581	3280	0.019	0.041	0.045	0.950	0.256
b ₈	19.652	19.652	2762	3196	0.020	0.039	0.044	0.944	0.252
b ₉	20.340	20.340	2915	3094	0.016	0.037	0.041	0.946	0.250
b ₁₀	21.331	21.331	3079	3051	0.008	0.037	0.038	0.954	0.250
b ₁₁	23.114	23.114	3319	3205	-0.001	0.038	0.038	0.957	0.262
b ₁₂	20.904	20.904	3020	3053	-0.009	0.037	0.038	0.958	0.248
b ₁₃	19.854	19.854	2815	3187	-0.015	0.039	0.042	0.954	0.249
b ₁₄	18.695	18.695	2510	3236	-0.015	0.041	0.044	0.951	0.258
b ₁₅	17.220	17.220	2101	3104	-0.008	0.044	0.045	0.958	0.275
b ₁₆	15.920	15.920	1748	2886	-0.000	0.044	0.044	0.956	0.293
b ₁₇	15.326	15.326	1584	2741	0.001	0.041	0.041	0.952	0.301
b ₁₈	15.400	15.400	1560	2726	0.001	0.041	0.041	0.954	0.299
b ₁₉	15.907	15.907	1584	2816	0.001	0.042	0.042	0.946	0.294
b ₂₀	16.537	16.537	1600	2910	-0.001	0.043	0.043	0.949	0.290
b ₂₁	17.049	17.049	1573	2921	-0.003	0.043	0.043	0.955	0.291
Uniform								0.948	0.409
WBATE					0.001	0.018	0.018	0.946	0.096
LBATE					0.037	0.033	0.050	0.989	0.444

Table SA-28: Intention-to-Treat Effects (Simulations, Quadratic Homoskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.213	18.213	1939	3752	-0.004	0.040	0.040	0.953	0.286
b₂	17.557	17.557	1894	3525	-0.003	0.040	0.040	0.956	0.289
b₃	17.203	17.203	1893	3370	-0.002	0.041	0.041	0.958	0.289
b₄	17.159	17.159	1958	3268	0.002	0.042	0.042	0.953	0.285
b₅	17.317	17.317	2078	3171	0.008	0.042	0.043	0.950	0.279
b₆	17.528	17.528	2213	3041	0.015	0.041	0.044	0.951	0.274
b₇	17.616	17.616	2312	2838	0.020	0.040	0.045	0.952	0.272
b₈	17.424	17.424	2334	2539	0.021	0.040	0.046	0.950	0.278
b₉	17.357	17.357	2359	2261	0.017	0.040	0.044	0.949	0.283
b₁₀	17.274	17.274	2356	1973	0.009	0.040	0.041	0.951	0.288
b₁₁	18.406	18.406	2551	1951	-0.000	0.040	0.040	0.954	0.285
b₁₂	17.297	17.297	2359	2086	-0.009	0.042	0.043	0.953	0.294
b₁₃	17.369	17.369	2333	2472	-0.015	0.042	0.045	0.955	0.293
b₁₄	17.534	17.534	2280	2862	-0.016	0.043	0.046	0.958	0.291
b₁₅	17.147	17.147	2080	3047	-0.009	0.045	0.046	0.952	0.302
b₁₆	16.682	16.682	1869	3114	-0.001	0.046	0.046	0.955	0.315
b₁₇	16.578	16.578	1771	3166	0.001	0.046	0.046	0.949	0.325
b₁₈	16.978	16.978	1779	3293	0.000	0.045	0.045	0.951	0.327
b₁₉	17.884	17.884	1843	3543	-0.002	0.045	0.045	0.949	0.323
b₂₀	18.952	18.952	1897	3799	-0.003	0.046	0.046	0.946	0.319
b₂₁	20.130	20.130	1928	4034	-0.005	0.045	0.046	0.951	0.318
Uniform								0.948	0.447
WBATE					0.001	0.018	0.018	0.951	0.101
LBATE					0.041	0.032	0.052	0.985	0.473

Table SA-29: Intention-to-Treat Effects (Simulations, Quadratic Heteroskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.887	17.887	1892	3622	-0.003	0.040	0.040	0.952	0.290
b₂	16.808	16.808	1774	3244	-0.002	0.041	0.041	0.956	0.299
b₃	15.591	15.591	1615	2807	-0.001	0.043	0.043	0.954	0.313
b₄	13.963	13.963	1377	2248	0.000	0.047	0.047	0.945	0.342
b₅	12.000	12.000	1082	1642	0.001	0.053	0.053	0.939	0.392
b₆	10.000	10.000	788	1113	0.002	0.062	0.062	0.940	0.466
b₇	8.168	8.168	543	715	0.001	0.077	0.077	0.941	0.568
b₈	7.637	7.637	494	577	0.003	0.084	0.084	0.940	0.614
b₉	7.604	7.604	526	487	0.010	0.086	0.086	0.940	0.627
b₁₀	7.568	7.568	562	384	0.008	0.086	0.087	0.942	0.641
b₁₁	8.064	8.064	672	322	0.001	0.085	0.085	0.937	0.638
b₁₂	7.578	7.578	547	425	-0.005	0.088	0.089	0.940	0.651
b₁₃	7.610	7.610	496	558	-0.003	0.086	0.086	0.944	0.644
b₁₄	8.430	8.430	576	769	-0.001	0.078	0.078	0.942	0.582
b₁₅	11.200	11.200	962	1418	-0.000	0.061	0.061	0.942	0.447
b₁₆	13.926	13.926	1372	2236	0.000	0.050	0.050	0.947	0.369
b₁₇	15.794	15.794	1641	2883	0.001	0.047	0.047	0.945	0.338
b₁₈	16.820	16.820	1756	3230	-0.000	0.046	0.046	0.952	0.329
b₁₉	17.866	17.866	1841	3536	-0.002	0.045	0.045	0.950	0.323
b₂₀	18.951	18.951	1897	3799	-0.003	0.046	0.046	0.946	0.319
b₂₁	20.130	20.130	1928	4034	-0.005	0.045	0.046	0.951	0.318
Uniform								0.921	0.694
WBATE					0.000	0.022	0.022	0.944	0.127
LBATE					0.099	0.057	0.115	0.973	0.863

Table SA-30: Intention-to-Treat Effects (Simulations, Quadratic Heteroskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	7.979	7.979	540	716	-0.003	0.084	0.084	0.950	0.327
b ₂	7.692	7.692	514	675	-0.001	0.084	0.084	0.954	0.331
b ₃	7.537	7.537	499	652	-0.000	0.084	0.084	0.955	0.332
b ₄	7.517	7.517	497	648	-0.000	0.086	0.086	0.944	0.328
b ₅	7.587	7.587	500	653	0.001	0.085	0.085	0.945	0.322
b ₆	7.679	7.679	501	656	0.001	0.083	0.083	0.949	0.317
b ₇	7.718	7.718	493	643	0.001	0.083	0.083	0.951	0.316
b ₈	7.634	7.634	494	576	0.003	0.084	0.084	0.951	0.321
b ₉	7.604	7.604	526	487	0.010	0.086	0.086	0.947	0.327
b ₁₀	7.568	7.568	562	384	0.008	0.086	0.087	0.949	0.332
b ₁₁	8.064	8.064	672	322	0.001	0.085	0.085	0.952	0.332
b ₁₂	7.578	7.578	547	425	-0.005	0.088	0.089	0.952	0.340
b ₁₃	7.609	7.609	496	558	-0.003	0.086	0.086	0.953	0.337
b ₁₄	7.682	7.682	490	642	-0.000	0.086	0.086	0.951	0.336
b ₁₅	7.512	7.512	488	636	0.000	0.091	0.091	0.942	0.346
b ₁₆	7.309	7.309	472	612	0.005	0.094	0.094	0.946	0.360
b ₁₇	7.263	7.263	466	603	0.004	0.098	0.098	0.943	0.370
b ₁₈	7.438	7.438	478	622	0.001	0.096	0.096	0.947	0.371
b ₁₉	7.835	7.835	508	670	-0.002	0.094	0.094	0.950	0.364
b ₂₀	8.303	8.303	537	720	-0.000	0.093	0.093	0.946	0.360
b ₂₁	8.819	8.819	560	766	0.000	0.092	0.092	0.947	0.357
Uniform								0.933	0.519
WBATE					0.001	0.027	0.027	0.949	0.104
LBATE					0.134	0.058	0.146	0.983	0.545

Table SA-31: Intention-to-Treat Effects (Simulations, Quadratic Heteroskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	17.349	17.349	1823	3432	-0.005	0.044	0.044	0.957	0.301
b ₂	16.879	16.879	1796	3288	-0.004	0.044	0.044	0.959	0.302
b ₃	16.600	16.600	1802	3173	-0.004	0.044	0.044	0.960	0.302
b ₄	16.606	16.606	1872	3104	-0.001	0.046	0.046	0.958	0.299
b ₅	16.893	16.893	2007	3068	0.005	0.048	0.048	0.951	0.293
b ₆	17.267	17.267	2163	3001	0.011	0.047	0.049	0.952	0.286
b ₇	17.265	17.265	2238	2782	0.016	0.047	0.050	0.952	0.285
b ₈	17.338	17.338	2311	2557	0.018	0.046	0.049	0.949	0.286
b ₉	17.379	17.379	2356	2307	0.015	0.044	0.047	0.956	0.289
b ₁₀	17.542	17.542	2399	2077	0.008	0.044	0.045	0.956	0.290
b ₁₁	18.603	18.603	2573	2042	-0.001	0.043	0.043	0.959	0.288
b ₁₂	17.562	17.562	2403	2187	-0.009	0.045	0.046	0.956	0.296
b ₁₃	17.605	17.605	2376	2569	-0.014	0.046	0.048	0.958	0.295
b ₁₄	17.447	17.447	2262	2877	-0.012	0.049	0.050	0.957	0.300
b ₁₅	16.875	16.875	2039	3004	-0.005	0.053	0.053	0.960	0.316
b ₁₆	16.081	16.081	1785	2944	0.003	0.052	0.052	0.959	0.335
b ₁₇	15.755	15.755	1653	2902	0.005	0.051	0.052	0.950	0.346
b ₁₈	15.961	15.961	1637	2941	0.003	0.051	0.051	0.953	0.349
b ₁₉	16.555	16.555	1669	3063	0.001	0.051	0.051	0.952	0.347
b ₂₀	17.249	17.249	1687	3182	0.000	0.052	0.052	0.951	0.349
b ₂₁	17.956	17.956	1676	3259	-0.000	0.053	0.053	0.954	0.355
Uniform								0.946	0.469
WBATE					0.002	0.019	0.019	0.948	0.104
LBATE					0.053	0.042	0.067	0.985	0.532

Table SA-32: Intention-to-Treat Effects (Simulations, Quadratic Heteroskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.534	19.534	2120	4260	-0.002	0.027	0.028	0.949	0.207
b₂	19.152	19.152	2138	4114	-0.002	0.028	0.028	0.945	0.210
b₃	18.902	18.902	2185	3961	-0.002	0.028	0.028	0.944	0.212
b₄	18.805	18.805	2271	3805	-0.002	0.029	0.029	0.945	0.214
b₅	18.832	18.832	2383	3639	-0.002	0.029	0.029	0.945	0.215
b₆	19.003	19.003	2517	3478	-0.002	0.029	0.029	0.945	0.216
b₇	19.115	19.115	2621	3265	-0.002	0.030	0.030	0.944	0.219
b₈	19.116	19.116	2679	2996	-0.001	0.031	0.031	0.948	0.224
b₉	19.218	19.218	2729	2738	0.000	0.032	0.032	0.945	0.232
b₁₀	19.480	19.480	2777	2508	0.000	0.035	0.035	0.945	0.250
b₁₁	21.345	21.345	3064	2673	-0.001	0.036	0.036	0.942	0.281
b₁₂	19.319	19.319	2751	2587	-0.004	0.033	0.033	0.946	0.240
b₁₃	19.199	19.199	2704	2962	-0.006	0.030	0.031	0.942	0.224
b₁₄	19.204	19.204	2625	3342	-0.007	0.029	0.029	0.944	0.216
b₁₅	18.993	18.993	2458	3608	-0.005	0.029	0.029	0.948	0.213
b₁₆	18.625	18.625	2237	3746	-0.002	0.029	0.029	0.950	0.214
b₁₇	18.421	18.421	2068	3821	0.000	0.029	0.029	0.948	0.215
b₁₈	18.678	18.678	2014	3935	0.001	0.028	0.028	0.944	0.213
b₁₉	19.490	19.490	2043	4167	0.002	0.027	0.027	0.948	0.206
b₂₀	20.668	20.668	2098	4464	0.003	0.026	0.026	0.948	0.198
b₂₁	21.765	21.765	2107	4658	0.004	0.026	0.026	0.949	0.193
Uniform								0.935	0.328
WBATE					-0.001	0.015	0.015	0.949	0.085
LBATE					0.011	0.024	0.026	0.993	0.324

Table SA-33: First-Stage Effects (Simulations, Linear Homoskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.862	18.862	2022	3994	-0.002	0.028	0.028	0.948	0.213
b₂	17.546	17.546	1877	3512	-0.001	0.030	0.030	0.948	0.226
b₃	15.909	15.909	1661	2914	-0.001	0.033	0.033	0.943	0.248
b₄	13.997	13.997	1382	2257	-0.000	0.038	0.038	0.942	0.282
b₅	12.000	12.000	1081	1641	-0.001	0.044	0.044	0.940	0.331
b₆	10.000	10.000	788	1113	-0.001	0.053	0.053	0.948	0.401
b₇	8.562	8.562	596	783	-0.002	0.062	0.063	0.941	0.477
b₈	8.375	8.375	599	680	-0.002	0.068	0.068	0.935	0.501
b₉	8.420	8.420	647	585	-0.000	0.070	0.070	0.932	0.519
b₁₀	8.534	8.534	708	483	-0.000	0.075	0.075	0.931	0.539
b₁₁	9.351	9.351	877	445	-0.001	0.088	0.088	0.919	0.659
b₁₂	8.464	8.464	680	522	-0.004	0.072	0.072	0.933	0.531
b₁₃	8.411	8.411	612	667	-0.003	0.068	0.068	0.935	0.501
b₁₄	8.717	8.717	614	821	-0.002	0.062	0.062	0.940	0.463
b₁₅	11.200	11.200	961	1419	-0.000	0.047	0.047	0.936	0.353
b₁₆	13.996	13.996	1382	2258	-0.000	0.037	0.037	0.942	0.279
b₁₇	16.483	16.483	1739	3123	0.000	0.031	0.031	0.949	0.237
b₁₈	18.112	18.112	1930	3712	0.001	0.029	0.029	0.945	0.218
b₁₉	19.440	19.440	2038	4146	0.002	0.027	0.027	0.948	0.207
b₂₀	20.668	20.668	2098	4464	0.003	0.026	0.026	0.948	0.198
b₂₁	21.765	21.765	2107	4658	0.004	0.026	0.026	0.949	0.193
Uniform								0.884	0.551
WBATE					-0.001	0.019	0.019	0.952	0.110
LBATE					0.034	0.035	0.049	0.979	0.637

Table SA-34: First-Stage Effects (Simulations, Linear Homoskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.558	8.558	609	823	-0.001	0.064	0.064	0.948	0.244
b₂	8.391	8.391	598	803	-0.001	0.064	0.064	0.946	0.247
b₃	8.281	8.281	589	788	-0.001	0.065	0.065	0.943	0.250
b₄	8.239	8.239	583	779	-0.000	0.066	0.066	0.939	0.251
b₅	8.251	8.251	579	773	-0.001	0.066	0.066	0.941	0.251
b₆	8.325	8.325	577	772	-0.002	0.065	0.065	0.946	0.251
b₇	8.375	8.375	575	752	-0.003	0.065	0.065	0.950	0.253
b₈	8.375	8.375	599	680	-0.002	0.068	0.068	0.943	0.260
b₉	8.420	8.420	647	585	-0.000	0.070	0.070	0.942	0.269
b₁₀	8.534	8.534	708	483	-0.000	0.075	0.075	0.941	0.289
b₁₁	9.351	9.351	877	445	-0.001	0.088	0.088	0.940	0.339
b₁₂	8.464	8.464	680	522	-0.004	0.072	0.072	0.944	0.278
b₁₃	8.411	8.411	612	667	-0.003	0.068	0.068	0.946	0.260
b₁₄	8.413	8.413	579	768	-0.001	0.065	0.065	0.939	0.249
b₁₅	8.321	8.321	583	781	-0.000	0.065	0.065	0.939	0.247
b₁₆	8.160	8.160	573	764	0.001	0.065	0.065	0.943	0.251
b₁₇	8.070	8.070	561	747	0.000	0.065	0.065	0.948	0.254
b₁₈	8.183	8.183	566	756	-0.001	0.066	0.066	0.939	0.252
b₁₉	8.539	8.539	590	798	-0.002	0.063	0.063	0.944	0.244
b₂₀	9.055	9.055	623	856	-0.000	0.060	0.060	0.946	0.234
b₂₁	9.535	9.535	639	893	0.001	0.059	0.059	0.943	0.228
Uniform								0.921	0.391
WBATE					-0.001	0.022	0.022	0.955	0.087
LBATE					0.069	0.043	0.082	0.989	0.386

Table SA-35: First-Stage Effects (Simulations, Linear Homoskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	19.325	19.325	2100	4182	-0.002	0.029	0.029	0.949	0.212
b ₂	19.171	19.171	2152	4123	-0.002	0.029	0.029	0.948	0.213
b ₃	19.127	19.127	2235	4046	-0.002	0.029	0.029	0.948	0.213
b ₄	19.198	19.198	2351	3949	-0.003	0.029	0.030	0.948	0.213
b ₅	19.404	19.404	2495	3845	-0.003	0.030	0.030	0.945	0.213
b ₆	19.638	19.638	2635	3703	-0.002	0.030	0.030	0.946	0.214
b ₇	19.901	19.901	2763	3534	-0.002	0.030	0.030	0.945	0.215
b ₈	20.219	20.219	2874	3351	-0.001	0.031	0.031	0.949	0.217
b ₉	20.701	20.701	2983	3189	0.000	0.031	0.031	0.949	0.220
b ₁₀	21.556	21.556	3119	3110	0.000	0.033	0.033	0.950	0.232
b ₁₁	23.400	23.400	3362	3287	-0.001	0.035	0.035	0.947	0.261
b ₁₂	21.216	21.216	3072	3143	-0.004	0.032	0.032	0.947	0.225
b ₁₃	20.499	20.499	2936	3369	-0.007	0.030	0.031	0.946	0.215
b ₁₄	20.118	20.118	2793	3651	-0.007	0.029	0.030	0.945	0.211
b ₁₅	19.695	19.695	2591	3862	-0.005	0.030	0.030	0.946	0.210
b ₁₆	19.024	19.024	2321	3897	-0.001	0.030	0.030	0.951	0.215
b ₁₇	18.414	18.414	2081	3827	0.001	0.030	0.030	0.951	0.219
b ₁₈	18.243	18.243	1959	3776	0.001	0.030	0.030	0.943	0.220
b ₁₉	18.530	18.530	1924	3797	0.002	0.030	0.030	0.950	0.218
b ₂₀	19.104	19.104	1912	3865	0.002	0.030	0.030	0.948	0.216
b ₂₁	19.617	19.617	1867	3854	0.002	0.030	0.030	0.952	0.217
Uniform								0.935	0.326
WBATE					-0.002	0.015	0.015	0.948	0.084
LBATE					0.013	0.027	0.030	0.991	0.343

Table SA-36: First-Stage Effects (Simulations, Linear Homoskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.822	19.822	2159	4374	-0.001	0.027	0.027	0.949	0.204
b₂	19.210	19.210	2146	4137	-0.001	0.027	0.028	0.951	0.209
b₃	18.737	18.737	2155	3903	-0.001	0.028	0.028	0.950	0.214
b₄	18.364	18.364	2185	3658	-0.001	0.029	0.029	0.949	0.218
b₅	18.091	18.091	2234	3405	-0.002	0.030	0.030	0.948	0.223
b₆	17.917	17.917	2293	3150	-0.002	0.031	0.031	0.942	0.229
b₇	17.759	17.759	2342	2876	-0.001	0.032	0.032	0.945	0.235
b₈	17.432	17.432	2336	2540	-0.001	0.034	0.034	0.945	0.246
b₉	17.290	17.290	2346	2244	0.000	0.036	0.036	0.942	0.257
b₁₀	17.153	17.153	2333	1945	0.000	0.040	0.040	0.939	0.281
b₁₁	18.116	18.116	2499	1885	-0.000	0.044	0.044	0.942	0.334
b₁₂	17.198	17.198	2339	2062	-0.003	0.038	0.038	0.943	0.268
b₁₃	17.364	17.364	2332	2471	-0.005	0.033	0.034	0.946	0.247
b₁₄	17.720	17.720	2319	2912	-0.005	0.031	0.032	0.945	0.233
b₁₅	17.870	17.870	2228	3261	-0.004	0.031	0.031	0.941	0.225
b₁₆	18.113	18.113	2137	3576	-0.001	0.030	0.030	0.939	0.219
b₁₇	18.540	18.540	2085	3865	0.001	0.029	0.029	0.945	0.213
b₁₈	19.281	19.281	2097	4170	0.002	0.028	0.028	0.948	0.206
b₁₉	20.431	20.431	2159	4545	0.003	0.026	0.026	0.950	0.196
b₂₀	21.957	21.957	2239	4992	0.004	0.025	0.025	0.945	0.186
b₂₁	23.561	23.561	2283	5391	0.005	0.024	0.024	0.948	0.178
Uniform								0.928	0.344
WBATE					-0.001	0.015	0.016	0.950	0.087
LBATE					0.011	0.023	0.025	0.992	0.336

Table SA-37: First-Stage Effects (Simulations, Linear Heteroskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.027	19.027	2044	4060	-0.001	0.028	0.028	0.951	0.211
b₂	17.579	17.579	1882	3526	-0.001	0.029	0.029	0.951	0.226
b₃	15.902	15.902	1660	2912	-0.000	0.032	0.032	0.952	0.248
b₄	13.995	13.995	1382	2257	-0.000	0.037	0.037	0.946	0.282
b₅	12.000	12.000	1081	1641	-0.000	0.043	0.043	0.945	0.331
b₆	10.000	10.000	788	1112	-0.001	0.052	0.052	0.941	0.401
b₇	8.181	8.181	545	717	-0.000	0.066	0.066	0.941	0.498
b₈	7.642	7.642	495	577	-0.000	0.073	0.073	0.936	0.550
b₉	7.575	7.575	521	483	-0.002	0.077	0.077	0.936	0.574
b₁₀	7.515	7.515	554	378	-0.002	0.085	0.085	0.925	0.611
b₁₁	7.937	7.937	653	311	-0.001	0.107	0.107	0.908	0.777
b₁₂	7.534	7.534	540	421	-0.003	0.080	0.080	0.927	0.593
b₁₃	7.608	7.608	496	557	-0.000	0.073	0.073	0.935	0.553
b₁₄	8.434	8.434	576	769	0.001	0.063	0.063	0.934	0.477
b₁₅	11.200	11.200	962	1419	0.000	0.047	0.047	0.940	0.353
b₁₆	13.991	13.991	1382	2256	0.000	0.037	0.037	0.941	0.279
b₁₇	16.545	16.545	1748	3147	0.001	0.032	0.032	0.938	0.236
b₁₈	18.559	18.559	1990	3886	0.001	0.028	0.028	0.947	0.212
b₁₉	20.288	20.288	2142	4486	0.002	0.026	0.027	0.950	0.197
b₂₀	21.947	21.947	2238	4988	0.004	0.025	0.025	0.945	0.186
b₂₁	23.560	23.560	2283	5391	0.005	0.024	0.024	0.948	0.178
Uniform								0.873	0.580
WBATE					0.000	0.020	0.020	0.946	0.113
LBATE					0.043	0.042	0.060	0.973	0.721

Table SA-38: First-Stage Effects (Simulations, Linear Heteroskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.684	8.684	625	848	-0.001	0.062	0.061	0.949	0.241
b₂	8.416	8.416	601	807	-0.000	0.062	0.062	0.953	0.247
b₃	8.209	8.209	580	773	-0.000	0.065	0.065	0.947	0.252
b₄	8.045	8.045	559	740	-0.000	0.067	0.067	0.941	0.256
b₅	7.926	7.926	539	711	-0.001	0.068	0.068	0.944	0.261
b₆	7.850	7.850	521	684	-0.001	0.068	0.068	0.946	0.266
b₇	7.780	7.780	501	651	-0.000	0.070	0.070	0.942	0.272
b₈	7.637	7.637	495	576	-0.000	0.073	0.073	0.948	0.285
b₉	7.575	7.575	521	483	-0.002	0.077	0.077	0.945	0.298
b₁₀	7.515	7.515	554	378	-0.002	0.085	0.085	0.937	0.325
b₁₁	7.937	7.937	653	311	-0.001	0.107	0.107	0.930	0.402
b₁₂	7.534	7.534	540	421	-0.003	0.080	0.080	0.943	0.310
b₁₃	7.607	7.607	496	557	-0.000	0.073	0.073	0.946	0.287
b₁₄	7.763	7.763	500	654	0.001	0.070	0.070	0.943	0.270
b₁₅	7.829	7.829	524	689	0.000	0.069	0.069	0.942	0.263
b₁₆	7.935	7.935	546	720	0.001	0.067	0.067	0.944	0.257
b₁₇	8.122	8.122	568	755	0.002	0.066	0.066	0.942	0.251
b₁₈	8.447	8.447	598	805	0.000	0.063	0.063	0.944	0.243
b₁₉	8.951	8.951	640	877	0.001	0.059	0.059	0.949	0.232
b₂₀	9.620	9.620	691	969	0.002	0.057	0.057	0.944	0.220
b₂₁	10.322	10.322	731	1051	0.002	0.055	0.055	0.942	0.210
Uniform								0.910	0.410
WBATE					-0.000	0.023	0.023	0.947	0.089
LBATE					0.074	0.044	0.086	0.985	0.405

Table SA-39: First-Stage Effects (Simulations, Linear Heteroskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.920	18.920	2042	4036	-0.002	0.029	0.029	0.947	0.217
b₂	18.614	18.614	2065	3924	-0.002	0.029	0.029	0.952	0.220
b₃	18.397	18.397	2110	3794	-0.002	0.030	0.030	0.949	0.222
b₄	18.284	18.284	2182	3647	-0.002	0.031	0.031	0.949	0.224
b₅	18.172	18.172	2256	3454	-0.002	0.031	0.031	0.950	0.227
b₆	18.104	18.104	2332	3236	-0.002	0.032	0.032	0.945	0.232
b₇	17.715	17.715	2330	2898	-0.002	0.034	0.034	0.945	0.241
b₈	17.474	17.474	2340	2587	-0.001	0.035	0.035	0.947	0.251
b₉	17.306	17.306	2344	2288	0.000	0.038	0.038	0.946	0.263
b₁₀	17.385	17.385	2370	2040	0.000	0.041	0.041	0.939	0.284
b₁₁	18.282	18.282	2516	1968	-0.000	0.046	0.046	0.943	0.339
b₁₂	17.522	17.522	2395	2180	-0.003	0.038	0.039	0.943	0.269
b₁₃	17.903	17.903	2435	2652	-0.005	0.034	0.034	0.950	0.246
b₁₄	18.374	18.374	2449	3140	-0.005	0.032	0.032	0.945	0.231
b₁₅	18.805	18.805	2421	3586	-0.004	0.031	0.031	0.941	0.221
b₁₆	18.919	18.919	2307	3867	-0.001	0.031	0.031	0.941	0.217
b₁₇	18.942	18.942	2175	4021	0.001	0.030	0.030	0.944	0.214
b₁₈	19.136	19.136	2090	4131	0.001	0.029	0.029	0.946	0.212
b₁₉	19.695	19.695	2068	4278	0.003	0.028	0.028	0.948	0.208
b₂₀	20.557	20.557	2072	4464	0.003	0.027	0.028	0.948	0.204
b₂₁	21.470	21.470	2054	4601	0.004	0.028	0.028	0.947	0.201
Uniform								0.926	0.354
WBATE					-0.001	0.016	0.016	0.945	0.088
LBATE					0.014	0.026	0.029	0.991	0.366

Table SA-40: First-Stage Effects (Simulations, Linear Heteroskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.945	17.945	1903	3648	-0.005	0.030	0.030	0.947	0.226
b₂	17.380	17.380	1868	3459	-0.004	0.030	0.031	0.950	0.232
b₃	17.202	17.202	1891	3370	-0.004	0.031	0.031	0.947	0.234
b₄	17.380	17.380	1997	3341	-0.002	0.031	0.031	0.947	0.232
b₅	17.857	17.857	2186	3339	0.000	0.031	0.031	0.947	0.227
b₆	18.430	18.430	2397	3310	0.003	0.030	0.030	0.946	0.223
b₇	18.889	18.889	2574	3202	0.006	0.030	0.030	0.947	0.221
b₈	18.988	18.988	2653	2961	0.006	0.030	0.031	0.947	0.225
b₉	19.108	19.108	2707	2710	0.005	0.032	0.032	0.944	0.231
b₁₀	19.393	19.393	2761	2487	0.001	0.034	0.034	0.947	0.248
b₁₁	21.303	21.303	3057	2662	-0.006	0.036	0.037	0.947	0.278
b₁₂	19.219	19.219	2732	2561	-0.007	0.033	0.034	0.949	0.238
b₁₃	19.004	19.004	2665	2911	-0.007	0.030	0.031	0.943	0.223
b₁₄	18.692	18.692	2519	3196	-0.006	0.030	0.030	0.939	0.219
b₁₅	17.796	17.796	2212	3246	-0.003	0.030	0.031	0.947	0.224
b₁₆	16.806	16.806	1891	3154	-0.002	0.031	0.031	0.943	0.234
b₁₇	16.262	16.262	1722	3052	-0.001	0.032	0.032	0.946	0.240
b₁₈	16.380	16.380	1697	3070	-0.000	0.031	0.031	0.944	0.239
b₁₉	17.047	17.047	1735	3229	-0.000	0.031	0.031	0.943	0.233
b₂₀	17.934	17.934	1774	3415	0.000	0.030	0.030	0.950	0.226
b₂₁	18.901	18.901	1791	3574	0.000	0.029	0.029	0.947	0.221
Uniform								0.924	0.348
WBATE					-0.001	0.015	0.015	0.947	0.087
LBATE					0.017	0.026	0.031	0.985	0.354

Table SA-41: First-Stage Effects (Simulations, Quadratic Homoskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.655	17.655	1862	3533	-0.005	0.030	0.030	0.946	0.229
b₂	16.710	16.710	1760	3207	-0.004	0.031	0.031	0.951	0.239
b₃	15.625	15.625	1619	2818	-0.003	0.033	0.033	0.947	0.253
b₄	13.979	13.979	1379	2252	-0.002	0.037	0.037	0.944	0.283
b₅	12.000	12.000	1080	1642	-0.001	0.044	0.044	0.942	0.331
b₆	10.000	10.000	787	1114	-0.000	0.053	0.053	0.943	0.400
b₇	8.509	8.509	589	775	-0.000	0.064	0.064	0.935	0.477
b₈	8.319	8.319	591	673	0.001	0.067	0.067	0.938	0.503
b₉	8.372	8.372	639	580	0.004	0.068	0.068	0.932	0.516
b₁₀	8.496	8.496	702	480	0.003	0.073	0.073	0.933	0.536
b₁₁	9.333	9.333	873	444	-0.002	0.086	0.086	0.918	0.652
b₁₂	8.420	8.420	673	518	-0.002	0.070	0.070	0.933	0.525
b₁₃	8.326	8.326	600	655	-0.000	0.067	0.067	0.931	0.498
b₁₄	8.643	8.643	604	807	0.000	0.062	0.062	0.935	0.460
b₁₅	11.200	11.200	962	1419	-0.001	0.046	0.046	0.945	0.348
b₁₆	13.958	13.958	1377	2246	-0.002	0.037	0.037	0.940	0.276
b₁₇	15.667	15.667	1623	2837	-0.001	0.033	0.033	0.945	0.247
b₁₈	16.274	16.274	1681	3028	-0.001	0.032	0.032	0.944	0.240
b₁₉	17.039	17.039	1734	3226	-0.000	0.031	0.031	0.943	0.233
b₂₀	17.934	17.934	1774	3415	0.000	0.030	0.030	0.950	0.226
b₂₁	18.901	18.901	1791	3574	0.000	0.029	0.029	0.947	0.221
Uniform								0.882	0.560
WBATE					-0.001	0.019	0.019	0.947	0.110
LBATE					0.044	0.038	0.059	0.967	0.646

Table SA-42: First-Stage Effects (Simulations, Quadratic Homoskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	7.862	7.862	525	695	0.000	0.069	0.069	0.943	0.268
b₂	7.614	7.614	504	661	0.000	0.071	0.071	0.944	0.274
b₃	7.536	7.536	498	652	-0.000	0.072	0.072	0.942	0.276
b₄	7.614	7.614	507	666	-0.001	0.070	0.070	0.946	0.273
b₅	7.823	7.823	527	697	-0.000	0.069	0.069	0.944	0.266
b₆	8.074	8.074	547	729	-0.000	0.067	0.067	0.944	0.259
b₇	8.275	8.275	563	736	-0.000	0.067	0.067	0.944	0.255
b₈	8.319	8.319	591	673	0.001	0.067	0.067	0.942	0.260
b₉	8.372	8.372	639	580	0.004	0.068	0.068	0.942	0.268
b₁₀	8.496	8.496	702	480	0.003	0.073	0.073	0.945	0.287
b₁₁	9.333	9.333	873	444	-0.002	0.086	0.086	0.943	0.335
b₁₂	8.420	8.420	673	518	-0.002	0.070	0.070	0.947	0.275
b₁₃	8.326	8.326	600	655	-0.000	0.067	0.067	0.944	0.259
b₁₄	8.189	8.189	553	729	0.000	0.067	0.067	0.940	0.253
b₁₅	7.797	7.797	522	687	-0.000	0.068	0.068	0.945	0.262
b₁₆	7.363	7.363	479	620	-0.000	0.072	0.072	0.934	0.275
b₁₇	7.124	7.124	450	579	-0.001	0.073	0.073	0.942	0.283
b₁₈	7.176	7.176	448	577	0.001	0.074	0.074	0.938	0.283
b₁₉	7.468	7.468	466	607	0.003	0.071	0.071	0.941	0.275
b₂₀	7.857	7.857	486	643	0.001	0.070	0.070	0.945	0.267
b₂₁	8.281	8.281	501	672	-0.000	0.067	0.067	0.942	0.261
Uniform								0.909	0.415
WBATE					0.000	0.023	0.023	0.947	0.089
LBATE					0.086	0.047	0.099	0.979	0.414

Table SA-43: First-Stage Effects (Simulations, Quadratic Homoskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.303	17.303	1816	3409	-0.005	0.031	0.032	0.946	0.235
b₂	16.910	16.910	1799	3295	-0.005	0.032	0.032	0.952	0.240
b₃	16.825	16.825	1837	3247	-0.004	0.032	0.032	0.950	0.242
b₄	17.058	17.058	1950	3248	-0.003	0.032	0.033	0.948	0.240
b₅	17.578	17.578	2135	3278	-0.000	0.032	0.032	0.949	0.236
b₆	18.313	18.313	2368	3312	0.003	0.032	0.032	0.945	0.230
b₇	19.002	19.002	2581	3280	0.005	0.032	0.032	0.944	0.226
b₈	19.652	19.652	2762	3196	0.006	0.031	0.031	0.945	0.223
b₉	20.340	20.340	2915	3094	0.004	0.031	0.032	0.943	0.223
b₁₀	21.331	21.331	3079	3051	-0.000	0.033	0.033	0.948	0.232
b₁₁	23.114	23.114	3319	3205	-0.007	0.035	0.036	0.947	0.261
b₁₂	20.904	20.904	3020	3053	-0.009	0.032	0.033	0.949	0.225
b₁₃	19.854	19.854	2815	3187	-0.008	0.031	0.032	0.942	0.218
b₁₄	18.695	18.695	2510	3236	-0.006	0.032	0.032	0.939	0.225
b₁₅	17.220	17.220	2101	3104	-0.003	0.033	0.033	0.945	0.238
b₁₆	15.920	15.920	1748	2886	-0.001	0.034	0.034	0.941	0.251
b₁₇	15.326	15.326	1584	2741	-0.000	0.034	0.034	0.942	0.257
b₁₈	15.400	15.400	1560	2726	0.000	0.034	0.034	0.943	0.255
b₁₉	15.907	15.907	1584	2816	0.000	0.033	0.033	0.943	0.249
b₂₀	16.537	16.537	1600	2910	-0.000	0.033	0.033	0.950	0.245
b₂₁	17.049	17.049	1573	2921	-0.001	0.033	0.033	0.948	0.245
Uniform								0.926	0.357
WBATE					-0.002	0.016	0.016	0.944	0.087
LBATE					0.021	0.029	0.036	0.984	0.380

Table SA-44: First-Stage Effects (Simulations, Quadratic Homoskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.213	18.213	1939	3752	-0.005	0.029	0.030	0.951	0.223
b₂	17.557	17.557	1894	3525	-0.005	0.030	0.030	0.947	0.230
b₃	17.203	17.203	1893	3370	-0.004	0.030	0.030	0.950	0.234
b₄	17.159	17.159	1958	3268	-0.002	0.031	0.031	0.949	0.235
b₅	17.317	17.317	2078	3171	0.000	0.031	0.031	0.945	0.234
b₆	17.528	17.528	2213	3041	0.003	0.031	0.031	0.943	0.234
b₇	17.616	17.616	2312	2838	0.006	0.031	0.031	0.948	0.236
b₈	17.424	17.424	2334	2539	0.007	0.032	0.033	0.944	0.244
b₉	17.357	17.357	2359	2261	0.006	0.035	0.035	0.938	0.253
b₁₀	17.274	17.274	2356	1973	0.002	0.038	0.038	0.941	0.276
b₁₁	18.406	18.406	2551	1951	-0.004	0.042	0.042	0.944	0.325
b₁₂	17.297	17.297	2359	2086	-0.007	0.037	0.037	0.946	0.263
b₁₃	17.369	17.369	2333	2472	-0.007	0.032	0.033	0.947	0.244
b₁₄	17.534	17.534	2280	2862	-0.006	0.031	0.031	0.951	0.233
b₁₅	17.147	17.147	2080	3047	-0.004	0.031	0.031	0.946	0.232
b₁₆	16.682	16.682	1869	3114	-0.002	0.031	0.031	0.947	0.235
b₁₇	16.578	16.578	1771	3166	-0.002	0.031	0.031	0.952	0.236
b₁₈	16.978	16.978	1779	3293	-0.002	0.030	0.030	0.950	0.231
b₁₉	17.884	17.884	1843	3543	-0.001	0.029	0.029	0.948	0.223
b₂₀	18.952	18.952	1897	3799	-0.001	0.028	0.028	0.942	0.214
b₂₁	20.130	20.130	1928	4034	0.000	0.027	0.027	0.946	0.208
Uniform								0.935	0.361
WBATE					-0.001	0.015	0.015	0.948	0.088
LBATE					0.016	0.025	0.029	0.990	0.363

Table SA-45: First-Stage Effects (Simulations, Quadratic Heteroskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.887	17.887	1892	3622	-0.005	0.030	0.030	0.952	0.226
b₂	16.808	16.808	1774	3244	-0.004	0.031	0.031	0.948	0.238
b₃	15.591	15.591	1615	2807	-0.003	0.033	0.033	0.952	0.254
b₄	13.963	13.963	1377	2248	-0.002	0.037	0.037	0.946	0.283
b₅	12.000	12.000	1082	1642	-0.000	0.043	0.043	0.943	0.331
b₆	10.000	10.000	788	1113	0.001	0.053	0.053	0.939	0.400
b₇	8.168	8.168	543	715	0.001	0.066	0.066	0.935	0.495
b₈	7.637	7.637	494	577	0.003	0.074	0.074	0.923	0.545
b₉	7.604	7.604	526	487	0.006	0.078	0.078	0.925	0.566
b₁₀	7.568	7.568	562	384	0.005	0.083	0.083	0.925	0.598
b₁₁	8.064	8.064	672	322	-0.000	0.102	0.102	0.913	0.756
b₁₂	7.578	7.578	547	425	-0.002	0.079	0.079	0.925	0.582
b₁₃	7.610	7.610	496	558	0.000	0.073	0.073	0.929	0.543
b₁₄	8.430	8.430	576	769	-0.000	0.062	0.062	0.932	0.471
b₁₅	11.200	11.200	962	1418	-0.001	0.046	0.046	0.936	0.348
b₁₆	13.926	13.926	1372	2236	-0.001	0.036	0.036	0.943	0.277
b₁₇	15.794	15.794	1641	2883	-0.002	0.032	0.032	0.951	0.245
b₁₈	16.820	16.820	1756	3230	-0.002	0.030	0.030	0.950	0.233
b₁₉	17.866	17.866	1841	3536	-0.001	0.029	0.029	0.948	0.223
b₂₀	18.951	18.951	1897	3799	-0.001	0.028	0.028	0.942	0.214
b₂₁	20.130	20.130	1928	4034	0.000	0.027	0.027	0.946	0.208
Uniform								0.872	0.585
WBATE					-0.000	0.020	0.020	0.942	0.113
LBATE					0.053	0.046	0.070	0.963	0.718

Table SA-46: First-Stage Effects (Simulations, Quadratic Heteroskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	7.979	7.979	540	716	-0.003	0.069	0.069	0.942	0.264
b ₂	7.692	7.692	514	675	-0.002	0.070	0.070	0.945	0.272
b ₃	7.537	7.537	499	652	-0.002	0.070	0.070	0.952	0.276
b ₄	7.517	7.517	497	648	-0.000	0.071	0.071	0.945	0.276
b ₅	7.587	7.587	500	653	0.002	0.071	0.071	0.944	0.274
b ₆	7.679	7.679	501	656	0.002	0.070	0.070	0.945	0.272
b ₇	7.718	7.718	493	643	0.001	0.071	0.071	0.944	0.273
b ₈	7.634	7.634	494	576	0.003	0.074	0.075	0.936	0.283
b ₉	7.604	7.604	526	487	0.006	0.078	0.078	0.938	0.295
b ₁₀	7.568	7.568	562	384	0.005	0.083	0.083	0.936	0.319
b ₁₁	8.064	8.064	672	322	-0.000	0.102	0.102	0.942	0.390
b ₁₂	7.578	7.578	547	425	-0.002	0.079	0.079	0.940	0.304
b ₁₃	7.609	7.609	496	558	0.000	0.073	0.073	0.942	0.283
b ₁₄	7.682	7.682	490	642	0.000	0.069	0.069	0.947	0.269
b ₁₅	7.512	7.512	488	636	0.001	0.071	0.071	0.939	0.272
b ₁₆	7.309	7.309	472	612	0.002	0.072	0.072	0.940	0.277
b ₁₇	7.263	7.263	466	603	-0.001	0.072	0.072	0.947	0.278
b ₁₈	7.438	7.438	478	622	-0.001	0.070	0.070	0.948	0.273
b ₁₉	7.835	7.835	508	670	-0.001	0.068	0.068	0.945	0.263
b ₂₀	8.303	8.303	537	720	-0.001	0.067	0.067	0.943	0.253
b ₂₁	8.819	8.819	560	766	-0.001	0.063	0.063	0.944	0.245
Uniform								0.911	0.430
WBATE					0.000	0.023	0.023	0.948	0.090
LBATE					0.088	0.048	0.100	0.977	0.427

Table SA-47: First-Stage Effects (Simulations, Quadratic Heteroskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.349	17.349	1823	3432	-0.006	0.031	0.032	0.952	0.236
b₂	16.879	16.879	1796	3288	-0.005	0.032	0.032	0.949	0.241
b₃	16.600	16.600	1802	3173	-0.004	0.032	0.032	0.956	0.246
b₄	16.606	16.606	1872	3104	-0.003	0.033	0.033	0.949	0.248
b₅	16.893	16.893	2007	3068	-0.001	0.034	0.034	0.946	0.246
b₆	17.267	17.267	2163	3001	0.002	0.034	0.034	0.946	0.244
b₇	17.265	17.265	2238	2782	0.005	0.034	0.034	0.951	0.248
b₈	17.338	17.338	2311	2557	0.006	0.035	0.035	0.947	0.252
b₉	17.379	17.379	2356	2307	0.005	0.036	0.037	0.940	0.260
b₁₀	17.542	17.542	2399	2077	0.002	0.039	0.039	0.944	0.278
b₁₁	18.603	18.603	2573	2042	-0.004	0.045	0.045	0.946	0.329
b₁₂	17.562	17.562	2403	2187	-0.007	0.038	0.038	0.948	0.265
b₁₃	17.605	17.605	2376	2569	-0.007	0.034	0.034	0.949	0.246
b₁₄	17.447	17.447	2262	2877	-0.005	0.033	0.033	0.948	0.241
b₁₅	16.875	16.875	2039	3004	-0.003	0.033	0.034	0.946	0.244
b₁₆	16.081	16.081	1785	2944	-0.001	0.034	0.034	0.945	0.251
b₁₇	15.755	15.755	1653	2902	-0.001	0.033	0.033	0.950	0.253
b₁₈	15.961	15.961	1637	2941	-0.001	0.032	0.032	0.948	0.248
b₁₉	16.555	16.555	1669	3063	-0.001	0.032	0.032	0.949	0.242
b₂₀	17.249	17.249	1687	3182	-0.001	0.032	0.032	0.942	0.237
b₂₁	17.956	17.956	1676	3259	-0.001	0.032	0.032	0.945	0.235
Uniform								0.929	0.380
WBATE					-0.001	0.016	0.016	0.945	0.090
LBATE					0.021	0.029	0.036	0.986	0.400

Table SA-48: First-Stage Effects (Simulations, Quadratic Heteroskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.534	19.534	2120	4260	0.003	0.055	0.055	0.960	0.397
b₂	19.152	19.152	2138	4114	0.003	0.056	0.056	0.956	0.400
b₃	18.902	18.902	2185	3961	0.004	0.057	0.057	0.959	0.403
b₄	18.805	18.805	2271	3805	0.007	0.058	0.058	0.957	0.404
b₅	18.832	18.832	2383	3639	0.011	0.058	0.059	0.960	0.403
b₆	19.003	19.003	2517	3478	0.015	0.057	0.059	0.958	0.402
b₇	19.115	19.115	2621	3265	0.017	0.057	0.059	0.957	0.406
b₈	19.116	19.116	2679	2996	0.016	0.058	0.060	0.956	0.412
b₉	19.218	19.218	2729	2738	0.012	0.059	0.060	0.957	0.418
b₁₀	19.480	19.480	2777	2508	0.007	0.060	0.060	0.965	0.427
b₁₁	21.345	21.345	3064	2673	0.002	0.059	0.060	0.965	0.436
b₁₂	19.319	19.319	2751	2587	-0.005	0.059	0.059	0.961	0.419
b₁₃	19.199	19.199	2704	2962	-0.012	0.058	0.059	0.957	0.400
b₁₄	19.204	19.204	2625	3342	-0.014	0.055	0.057	0.954	0.381
b₁₅	18.993	18.993	2458	3608	-0.010	0.054	0.055	0.956	0.372
b₁₆	18.625	18.625	2237	3746	-0.003	0.054	0.054	0.961	0.373
b₁₇	18.421	18.421	2068	3821	0.001	0.052	0.052	0.963	0.375
b₁₈	18.678	18.678	2014	3935	0.001	0.049	0.049	0.959	0.370
b₁₉	19.490	19.490	2043	4167	0.000	0.048	0.048	0.952	0.356
b₂₀	20.668	20.668	2098	4464	-0.002	0.047	0.047	0.954	0.340
b₂₁	21.765	21.765	2107	4658	-0.004	0.046	0.046	0.951	0.329
Uniform								0.970	0.587
WBATE					0.002	0.026	0.026	0.952	0.146
LBATE					0.053	0.044	0.069	0.996	0.652

Table SA-49: Fuzzy Effects (Simulations, Linear Homoskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.862	18.862	2022	3994	0.004	0.056	0.056	0.957	0.408
b₂	17.546	17.546	1877	3512	0.004	0.058	0.058	0.953	0.432
b₃	15.909	15.909	1661	2914	0.004	0.063	0.063	0.953	0.471
b₄	13.997	13.997	1382	2257	0.003	0.070	0.070	0.958	0.535
b₅	12.000	12.000	1081	1641	0.001	0.081	0.081	0.961	0.629
b₆	10.000	10.000	788	1113	-0.000	0.097	0.097	0.957	0.774
b₇	8.562	8.562	596	783	0.001	0.118	0.118	0.967	0.956
b₈	8.375	8.375	599	680	0.005	0.124	0.124	0.966	1.027
b₉	8.420	8.420	647	585	0.009	0.124	0.124	0.975	1.048
b₁₀	8.534	8.534	708	483	0.008	0.127	0.127	0.970	1.364
b₁₁	9.351	9.351	877	445	0.004	0.136	0.136	0.977	22.769
b₁₂	8.464	8.464	680	522	-0.005	0.126	0.126	0.970	1.213
b₁₃	8.411	8.411	612	667	-0.003	0.122	0.122	0.970	0.960
b₁₄	8.717	8.717	614	821	0.004	0.110	0.110	0.967	0.867
b₁₅	11.200	11.200	961	1419	0.004	0.083	0.083	0.958	0.631
b₁₆	13.996	13.996	1382	2258	0.002	0.064	0.064	0.954	0.491
b₁₇	16.483	16.483	1739	3123	0.001	0.055	0.055	0.957	0.413
b₁₈	18.112	18.112	1930	3712	0.001	0.050	0.050	0.959	0.379
b₁₉	19.440	19.440	2038	4146	-0.000	0.048	0.048	0.951	0.357
b₂₀	20.668	20.668	2098	4464	-0.002	0.047	0.047	0.954	0.340
b₂₁	21.765	21.765	2107	4658	-0.004	0.046	0.046	0.951	0.329
Uniform								0.978	2.660
WBATE					0.002	0.033	0.033	0.959	1.263
LBATE					0.135	0.087	0.160	0.997	18.630

Table SA-50: Fuzzy Effects (Simulations, Linear Homoskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	8.558	8.558	609	823	0.005	0.118	0.119	0.956	0.458
b ₂	8.391	8.391	598	803	0.005	0.120	0.120	0.952	0.463
b ₃	8.281	8.281	589	788	0.004	0.120	0.120	0.958	0.467
b ₄	8.239	8.239	583	779	0.003	0.122	0.122	0.959	0.468
b ₅	8.251	8.251	579	773	0.001	0.120	0.120	0.959	0.468
b ₆	8.325	8.325	577	772	-0.001	0.121	0.121	0.957	0.467
b ₇	8.375	8.375	575	752	0.001	0.122	0.122	0.956	0.471
b ₈	8.375	8.375	599	680	0.005	0.124	0.124	0.957	0.478
b ₉	8.420	8.420	647	585	0.009	0.124	0.124	0.960	0.486
b ₁₀	8.534	8.534	708	483	0.008	0.127	0.127	0.962	0.496
b ₁₁	9.351	9.351	877	445	0.004	0.136	0.136	0.961	0.519
b ₁₂	8.464	8.464	680	522	-0.005	0.126	0.126	0.960	0.486
b ₁₃	8.411	8.411	612	667	-0.003	0.122	0.122	0.960	0.462
b ₁₄	8.413	8.413	579	768	0.004	0.115	0.115	0.957	0.442
b ₁₅	8.321	8.321	583	781	0.005	0.114	0.114	0.956	0.432
b ₁₆	8.160	8.160	573	764	0.001	0.113	0.113	0.954	0.433
b ₁₇	8.070	8.070	561	747	-0.000	0.110	0.110	0.955	0.432
b ₁₈	8.183	8.183	566	756	0.001	0.108	0.108	0.956	0.426
b ₁₉	8.539	8.539	590	798	0.002	0.107	0.107	0.951	0.409
b ₂₀	9.055	9.055	623	856	-0.000	0.100	0.100	0.952	0.388
b ₂₁	9.535	9.535	639	893	-0.002	0.097	0.097	0.951	0.375
Uniform								0.972	0.691
WBATE					0.002	0.039	0.039	0.953	0.151
LBATE					0.179	0.088	0.199	0.997	0.793

Table SA-51: Fuzzy Effects (Simulations, Linear Homoskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	19.325	19.325	2100	4182	0.002	0.058	0.058	0.961	0.406
b ₂	19.171	19.171	2152	4123	0.003	0.059	0.059	0.960	0.406
b ₃	19.127	19.127	2235	4046	0.004	0.059	0.060	0.960	0.404
b ₄	19.198	19.198	2351	3949	0.006	0.061	0.061	0.959	0.403
b ₅	19.404	19.404	2495	3845	0.010	0.061	0.061	0.963	0.400
b ₆	19.638	19.638	2635	3703	0.013	0.060	0.061	0.959	0.399
b ₇	19.901	19.901	2763	3534	0.015	0.059	0.061	0.957	0.400
b ₈	20.219	20.219	2874	3351	0.014	0.060	0.062	0.958	0.400
b ₉	20.701	20.701	2983	3189	0.011	0.059	0.060	0.959	0.398
b ₁₀	21.556	21.556	3119	3110	0.007	0.058	0.059	0.961	0.397
b ₁₁	23.400	23.400	3362	3287	0.002	0.058	0.058	0.961	0.406
b ₁₂	21.216	21.216	3072	3143	-0.005	0.059	0.059	0.963	0.392
b ₁₃	20.499	20.499	2936	3369	-0.012	0.058	0.060	0.958	0.385
b ₁₄	20.118	20.118	2793	3651	-0.013	0.058	0.059	0.955	0.373
b ₁₅	19.695	19.695	2591	3862	-0.009	0.058	0.058	0.958	0.369
b ₁₆	19.024	19.024	2321	3897	-0.003	0.057	0.057	0.964	0.375
b ₁₇	18.414	18.414	2081	3827	0.002	0.055	0.055	0.966	0.380
b ₁₈	18.243	18.243	1959	3776	0.003	0.053	0.053	0.961	0.382
b ₁₉	18.530	18.530	1924	3797	0.002	0.052	0.052	0.954	0.376
b ₂₀	19.104	19.104	1912	3865	0.000	0.052	0.052	0.957	0.369
b ₂₁	19.617	19.617	1867	3854	-0.002	0.053	0.053	0.956	0.368
Uniform								0.971	0.585
WBATE					0.002	0.027	0.027	0.948	0.145
LBATE					0.056	0.046	0.072	0.997	0.668

Table SA-52: Fuzzy Effects (Simulations, Linear Homoskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.822	19.822	2159	4374	0.001	0.073	0.073	0.962	0.522
b₂	19.210	19.210	2146	4137	0.001	0.074	0.074	0.962	0.525
b₃	18.737	18.737	2155	3903	0.002	0.074	0.074	0.963	0.526
b₄	18.364	18.364	2185	3658	0.005	0.074	0.074	0.957	0.528
b₅	18.091	18.091	2234	3405	0.009	0.075	0.076	0.957	0.530
b₆	17.917	17.917	2293	3150	0.013	0.076	0.077	0.961	0.531
b₇	17.759	17.759	2342	2876	0.015	0.077	0.078	0.960	0.535
b₈	17.432	17.432	2336	2540	0.015	0.078	0.079	0.958	0.547
b₉	17.290	17.290	2346	2244	0.012	0.078	0.079	0.957	0.553
b₁₀	17.153	17.153	2333	1945	0.007	0.076	0.076	0.959	0.547
b₁₁	18.116	18.116	2499	1885	0.002	0.070	0.070	0.965	0.497
b₁₂	17.198	17.198	2339	2062	-0.005	0.078	0.078	0.962	0.567
b₁₃	17.364	17.364	2332	2471	-0.012	0.082	0.083	0.958	0.570
b₁₄	17.720	17.720	2319	2912	-0.014	0.083	0.084	0.955	0.559
b₁₅	17.870	17.870	2228	3261	-0.010	0.084	0.085	0.954	0.564
b₁₆	18.113	18.113	2137	3576	-0.003	0.083	0.083	0.957	0.572
b₁₇	18.540	18.540	2085	3865	0.001	0.082	0.082	0.955	0.581
b₁₈	19.281	19.281	2097	4170	0.001	0.083	0.083	0.954	0.585
b₁₉	20.431	20.431	2159	4545	-0.000	0.083	0.083	0.954	0.581
b₂₀	21.957	21.957	2239	4992	-0.002	0.082	0.082	0.957	0.574
b₂₁	23.561	23.561	2283	5391	-0.004	0.083	0.083	0.959	0.571
Uniform								0.971	0.830
WBATE					0.001	0.034	0.034	0.946	0.191
LBATE					0.083	0.059	0.102	0.996	0.913

Table SA-53: Fuzzy Effects (Simulations, Linear Heteroskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.027	19.027	2044	4060	0.001	0.074	0.074	0.962	0.540
b₂	17.579	17.579	1882	3526	0.002	0.077	0.077	0.959	0.568
b₃	15.902	15.902	1660	2912	0.003	0.081	0.081	0.959	0.613
b₄	13.995	13.995	1382	2257	0.002	0.090	0.090	0.954	0.687
b₅	12.000	12.000	1081	1641	0.000	0.103	0.103	0.957	0.798
b₆	10.000	10.000	788	1112	-0.002	0.123	0.123	0.961	0.966
b₇	8.181	8.181	545	717	-0.003	0.152	0.152	0.966	1.360
b₈	7.642	7.642	495	577	0.002	0.165	0.165	0.974	1.454
b₉	7.575	7.575	521	483	0.009	0.174	0.174	0.971	1.570
b₁₀	7.515	7.515	554	378	0.009	0.171	0.171	0.973	9.809
b₁₁	7.937	7.937	653	311	0.004	0.155	0.155	0.984	1619.334
b₁₂	7.534	7.534	540	421	-0.005	0.173	0.173	0.974	2.352
b₁₃	7.608	7.608	496	557	-0.002	0.173	0.173	0.971	1.508
b₁₄	8.434	8.434	576	769	-0.000	0.155	0.155	0.959	4.695
b₁₅	11.200	11.200	962	1419	-0.002	0.120	0.120	0.950	0.897
b₁₆	13.991	13.991	1382	2256	-0.001	0.098	0.098	0.952	0.733
b₁₇	16.545	16.545	1748	3147	-0.001	0.087	0.087	0.951	0.644
b₁₈	18.559	18.559	1990	3886	-0.000	0.085	0.085	0.952	0.604
b₁₉	20.288	20.288	2142	4486	-0.000	0.083	0.083	0.953	0.585
b₂₀	21.947	21.947	2238	4988	-0.002	0.082	0.082	0.957	0.574
b₂₁	23.560	23.560	2283	5391	-0.004	0.083	0.083	0.959	0.571
Uniform								0.978	119.194
WBATE					0.000	0.043	0.043	0.960	77.948
LBATE					0.194	0.120	0.228	0.998	1248.167

Table SA-54: Fuzzy Effects (Simulations, Linear Heteroskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.684	8.684	625	848	0.001	0.153	0.153	0.959	0.593
b₂	8.416	8.416	601	807	0.002	0.154	0.154	0.958	0.599
b₃	8.209	8.209	580	773	0.003	0.156	0.156	0.958	0.605
b₄	8.045	8.045	559	740	0.001	0.160	0.160	0.955	0.610
b₅	7.926	7.926	539	711	-0.002	0.160	0.160	0.955	0.616
b₆	7.850	7.850	521	684	-0.004	0.161	0.161	0.962	0.619
b₇	7.780	7.780	501	651	-0.003	0.162	0.162	0.958	0.625
b₈	7.637	7.637	495	576	0.002	0.165	0.165	0.961	0.637
b₉	7.575	7.575	521	483	0.009	0.174	0.174	0.954	0.647
b₁₀	7.515	7.515	554	378	0.009	0.171	0.171	0.956	0.643
b₁₁	7.937	7.937	653	311	0.004	0.155	0.155	0.966	0.582
b₁₂	7.534	7.534	540	421	-0.005	0.173	0.173	0.958	0.664
b₁₃	7.607	7.607	496	557	-0.002	0.174	0.174	0.953	0.660
b₁₄	7.763	7.763	500	654	-0.000	0.172	0.172	0.956	0.652
b₁₅	7.829	7.829	524	689	-0.003	0.172	0.172	0.953	0.651
b₁₆	7.935	7.935	546	720	-0.000	0.171	0.171	0.955	0.654
b₁₇	8.122	8.122	568	755	0.000	0.169	0.169	0.954	0.657
b₁₈	8.447	8.447	598	805	-0.001	0.174	0.174	0.946	0.658
b₁₉	8.951	8.951	640	877	-0.001	0.168	0.168	0.952	0.649
b₂₀	9.620	9.620	691	969	0.000	0.162	0.162	0.954	0.637
b₂₁	10.322	10.322	731	1051	-0.000	0.163	0.163	0.956	0.631
Uniform								0.970	0.967
WBATE					0.001	0.052	0.052	0.947	0.199
LBATE					0.261	0.126	0.290	0.996	1.095

Table SA-55: Fuzzy Effects (Simulations, Linear Heteroskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	18.920	18.920	2042	4036	-0.000	0.084	0.084	0.962	0.554
b ₂	18.614	18.614	2065	3924	-0.000	0.083	0.083	0.961	0.551
b ₃	18.397	18.397	2110	3794	0.000	0.083	0.083	0.963	0.548
b ₄	18.284	18.284	2182	3647	0.003	0.083	0.083	0.959	0.543
b ₅	18.172	18.172	2256	3454	0.006	0.084	0.084	0.960	0.541
b ₆	18.104	18.104	2332	3236	0.010	0.084	0.085	0.966	0.540
b ₇	17.715	17.715	2330	2898	0.011	0.086	0.086	0.962	0.551
b ₈	17.474	17.474	2340	2587	0.012	0.086	0.087	0.965	0.558
b ₉	17.306	17.306	2344	2288	0.010	0.087	0.087	0.959	0.570
b ₁₀	17.385	17.385	2370	2040	0.005	0.084	0.084	0.964	0.556
b ₁₁	18.282	18.282	2516	1968	0.001	0.076	0.076	0.969	0.509
b ₁₂	17.522	17.522	2395	2180	-0.005	0.086	0.086	0.966	0.570
b ₁₃	17.903	17.903	2435	2652	-0.011	0.090	0.090	0.961	0.569
b ₁₄	18.374	18.374	2449	3140	-0.011	0.091	0.092	0.964	0.556
b ₁₅	18.805	18.805	2421	3586	-0.007	0.093	0.093	0.960	0.555
b ₁₆	18.919	18.919	2307	3867	-0.002	0.092	0.092	0.962	0.567
b ₁₇	18.942	18.942	2175	4021	0.002	0.091	0.092	0.962	0.583
b ₁₈	19.136	19.136	2090	4131	0.003	0.096	0.096	0.964	0.600
b ₁₉	19.695	19.695	2068	4278	0.003	0.097	0.097	0.958	0.611
b ₂₀	20.557	20.557	2072	4464	0.004	0.097	0.097	0.961	0.619
b ₂₁	21.470	21.470	2054	4601	0.004	0.100	0.100	0.963	0.635
Uniform								0.976	0.854
WBATE					0.002	0.037	0.037	0.941	0.195
LBATE					0.102	0.077	0.127	0.997	1.032

Table SA-56: Fuzzy Effects (Simulations, Linear Heteroskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.945	17.945	1903	3648	0.002	0.060	0.060	0.957	0.443
b₂	17.380	17.380	1868	3459	0.001	0.061	0.061	0.953	0.450
b₃	17.202	17.202	1891	3370	0.002	0.062	0.063	0.960	0.448
b₄	17.380	17.380	1997	3341	0.007	0.063	0.064	0.960	0.439
b₅	17.857	17.857	2186	3339	0.016	0.063	0.065	0.956	0.424
b₆	18.430	18.430	2397	3310	0.026	0.060	0.065	0.950	0.407
b₇	18.889	18.889	2574	3202	0.032	0.057	0.065	0.958	0.394
b₈	18.988	18.988	2653	2961	0.031	0.055	0.063	0.956	0.389
b₉	19.108	19.108	2707	2710	0.025	0.054	0.059	0.957	0.387
b₁₀	19.393	19.393	2761	2487	0.015	0.054	0.056	0.959	0.390
b₁₁	21.303	21.303	3057	2662	0.006	0.053	0.054	0.959	0.391
b₁₂	19.219	19.219	2732	2561	-0.006	0.054	0.054	0.959	0.379
b₁₃	19.004	19.004	2665	2911	-0.018	0.053	0.056	0.953	0.365
b₁₄	18.692	18.692	2519	3196	-0.023	0.052	0.057	0.951	0.358
b₁₅	17.796	17.796	2212	3246	-0.016	0.054	0.056	0.957	0.368
b₁₆	16.806	16.806	1891	3154	-0.005	0.054	0.055	0.958	0.385
b₁₇	16.262	16.262	1722	3052	-0.001	0.053	0.053	0.954	0.395
b₁₈	16.380	16.380	1697	3070	-0.002	0.052	0.052	0.956	0.392
b₁₉	17.047	17.047	1735	3229	-0.002	0.052	0.052	0.952	0.380
b₂₀	17.934	17.934	1774	3415	-0.004	0.051	0.051	0.951	0.369
b₂₁	18.901	18.901	1791	3574	-0.008	0.050	0.051	0.957	0.360
Uniform								0.966	0.596
WBATE					0.004	0.026	0.026	0.951	0.143
LBATE					0.063	0.043	0.076	0.995	0.648

Table SA-57: Fuzzy Effects (Simulations, Quadratic Homoskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.655	17.655	1862	3533	0.003	0.061	0.061	0.957	0.448
b₂	16.710	16.710	1760	3207	0.002	0.063	0.063	0.952	0.463
b₃	15.625	15.625	1619	2818	0.002	0.065	0.065	0.958	0.485
b₄	13.979	13.979	1379	2252	0.002	0.070	0.070	0.956	0.536
b₅	12.000	12.000	1080	1642	0.002	0.081	0.081	0.958	0.622
b₆	10.000	10.000	787	1114	0.001	0.096	0.096	0.962	0.752
b₇	8.509	8.509	589	775	0.000	0.114	0.114	0.967	0.910
b₈	8.319	8.319	591	673	0.006	0.118	0.118	0.971	0.952
b₉	8.372	8.372	639	580	0.014	0.117	0.118	0.964	0.960
b₁₀	8.496	8.496	702	480	0.013	0.118	0.119	0.971	0.954
b₁₁	9.333	9.333	873	444	0.006	0.123	0.123	0.968	14.087
b₁₂	8.420	8.420	673	518	-0.004	0.115	0.115	0.965	0.905
b₁₃	8.326	8.326	600	655	-0.002	0.110	0.110	0.969	0.871
b₁₄	8.643	8.643	604	807	0.001	0.102	0.102	0.962	0.787
b₁₅	11.200	11.200	962	1419	-0.001	0.076	0.076	0.955	0.581
b₁₆	13.958	13.958	1377	2246	-0.002	0.060	0.060	0.953	0.458
b₁₇	15.667	15.667	1623	2837	-0.002	0.054	0.054	0.954	0.407
b₁₈	16.274	16.274	1681	3028	-0.002	0.053	0.053	0.957	0.394
b₁₉	17.039	17.039	1734	3226	-0.002	0.052	0.052	0.952	0.380
b₂₀	17.934	17.934	1774	3415	-0.004	0.051	0.051	0.951	0.369
b₂₁	18.901	18.901	1791	3574	-0.008	0.050	0.051	0.957	0.360
Uniform								0.976	1.936
WBATE					0.001	0.032	0.032	0.953	0.819
LBATE					0.133	0.079	0.155	0.997	11.582

Table SA-58: Fuzzy Effects (Simulations, Quadratic Homoskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	7.862	7.862	525	695	0.002	0.133	0.133	0.952	0.512
b₂	7.614	7.614	504	661	0.001	0.139	0.139	0.952	0.522
b₃	7.536	7.536	498	652	0.001	0.135	0.135	0.960	0.520
b₄	7.614	7.614	507	666	-0.001	0.134	0.134	0.955	0.511
b₅	7.823	7.823	527	697	0.001	0.128	0.127	0.958	0.491
b₆	8.074	8.074	547	729	0.001	0.123	0.123	0.954	0.471
b₇	8.275	8.275	563	736	-0.000	0.120	0.120	0.956	0.458
b₈	8.319	8.319	591	673	0.006	0.118	0.118	0.956	0.454
b₉	8.372	8.372	639	580	0.014	0.117	0.118	0.955	0.451
b₁₀	8.496	8.496	702	480	0.013	0.118	0.119	0.954	0.453
b₁₁	9.333	9.333	873	444	0.006	0.123	0.123	0.956	0.466
b₁₂	8.420	8.420	673	518	-0.004	0.115	0.115	0.951	0.440
b₁₃	8.326	8.326	600	655	-0.002	0.110	0.110	0.955	0.423
b₁₄	8.189	8.189	553	729	0.002	0.110	0.110	0.952	0.416
b₁₅	7.797	7.797	522	687	-0.001	0.110	0.110	0.951	0.428
b₁₆	7.363	7.363	479	620	-0.001	0.117	0.117	0.953	0.447
b₁₇	7.124	7.124	450	579	-0.006	0.118	0.119	0.953	0.459
b₁₈	7.176	7.176	448	577	-0.003	0.115	0.115	0.955	0.454
b₁₉	7.468	7.468	466	607	0.001	0.114	0.114	0.952	0.439
b₂₀	7.857	7.857	486	643	-0.001	0.112	0.112	0.951	0.426
b₂₁	8.281	8.281	501	672	-0.003	0.106	0.106	0.956	0.415
Uniform								0.966	0.702
WBATE					0.001	0.038	0.038	0.945	0.148
LBATE					0.181	0.085	0.200	0.995	0.783

Table SA-59: Fuzzy Effects (Simulations, Quadratic Homoskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	17.303	17.303	1816	3409	0.000	0.064	0.064	0.960	0.461
b ₂	16.910	16.910	1799	3295	-0.001	0.065	0.065	0.956	0.466
b ₃	16.825	16.825	1837	3247	-0.000	0.067	0.067	0.965	0.464
b ₄	17.058	17.058	1950	3248	0.004	0.069	0.069	0.965	0.457
b ₅	17.578	17.578	2135	3278	0.012	0.070	0.071	0.963	0.441
b ₆	18.313	18.313	2368	3312	0.021	0.068	0.071	0.958	0.422
b ₇	19.002	19.002	2581	3280	0.028	0.063	0.069	0.957	0.404
b ₈	19.652	19.652	2762	3196	0.028	0.059	0.065	0.955	0.388
b ₉	20.340	20.340	2915	3094	0.023	0.055	0.060	0.957	0.374
b ₁₀	21.331	21.331	3079	3051	0.014	0.054	0.056	0.961	0.364
b ₁₁	23.114	23.114	3319	3205	0.005	0.053	0.054	0.960	0.368
b ₁₂	20.904	20.904	3020	3053	-0.006	0.053	0.054	0.960	0.357
b ₁₃	19.854	19.854	2815	3187	-0.017	0.054	0.057	0.956	0.358
b ₁₄	18.695	18.695	2510	3236	-0.020	0.057	0.060	0.952	0.368
b ₁₅	17.220	17.220	2101	3104	-0.011	0.061	0.062	0.962	0.390
b ₁₆	15.920	15.920	1748	2886	-0.000	0.061	0.061	0.963	0.414
b ₁₇	15.326	15.326	1584	2741	0.002	0.058	0.058	0.959	0.422
b ₁₈	15.400	15.400	1560	2726	0.001	0.056	0.056	0.958	0.416
b ₁₉	15.907	15.907	1584	2816	0.000	0.057	0.057	0.956	0.406
b ₂₀	16.537	16.537	1600	2910	-0.002	0.057	0.057	0.959	0.399
b ₂₁	17.049	17.049	1573	2921	-0.004	0.058	0.058	0.958	0.399
Uniform								0.971	0.613
WBATE					0.004	0.027	0.027	0.949	0.145
LBATE					0.066	0.043	0.079	0.995	0.678

Table SA-60: Fuzzy Effects (Simulations, Quadratic Homoskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.213	18.213	1939	3752	-0.000	0.070	0.070	0.957	0.500
b₂	17.557	17.557	1894	3525	0.000	0.070	0.070	0.955	0.499
b₃	17.203	17.203	1893	3370	0.001	0.069	0.069	0.957	0.491
b₄	17.159	17.159	1958	3268	0.006	0.069	0.069	0.962	0.477
b₅	17.317	17.317	2078	3171	0.014	0.068	0.070	0.963	0.459
b₆	17.528	17.528	2213	3041	0.024	0.066	0.070	0.958	0.442
b₇	17.616	17.616	2312	2838	0.030	0.063	0.070	0.954	0.429
b₈	17.424	17.424	2334	2539	0.030	0.062	0.069	0.956	0.426
b₉	17.357	17.357	2359	2261	0.024	0.060	0.065	0.954	0.423
b₁₀	17.274	17.274	2356	1973	0.014	0.058	0.059	0.962	0.413
b₁₁	18.406	18.406	2551	1951	0.004	0.053	0.054	0.959	0.369
b₁₂	17.297	17.297	2359	2086	-0.008	0.060	0.060	0.962	0.426
b₁₃	17.369	17.369	2333	2472	-0.019	0.062	0.065	0.960	0.428
b₁₄	17.534	17.534	2280	2862	-0.021	0.063	0.067	0.959	0.424
b₁₅	17.147	17.147	2080	3047	-0.012	0.067	0.068	0.954	0.441
b₁₆	16.682	16.682	1869	3114	0.000	0.067	0.067	0.962	0.460
b₁₇	16.578	16.578	1771	3166	0.004	0.067	0.067	0.952	0.476
b₁₈	16.978	16.978	1779	3293	0.001	0.066	0.066	0.953	0.476
b₁₉	17.884	17.884	1843	3543	-0.002	0.066	0.066	0.951	0.468
b₂₀	18.952	18.952	1897	3799	-0.005	0.066	0.066	0.946	0.462
b₂₁	20.130	20.130	1928	4034	-0.008	0.066	0.066	0.950	0.457
Uniform								0.969	0.680
WBATE					0.004	0.028	0.028	0.949	0.153
LBATE					0.074	0.047	0.087	0.995	0.729

Table SA-61: Fuzzy Effects (Simulations, Quadratic Heteroskedastic, Smooth-Boundary Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.887	17.887	1892	3622	0.000	0.071	0.071	0.958	0.506
b₂	16.808	16.808	1774	3244	0.001	0.071	0.071	0.953	0.516
b₃	15.591	15.591	1615	2807	0.002	0.073	0.073	0.954	0.534
b₄	13.963	13.963	1377	2248	0.002	0.077	0.077	0.956	0.576
b₅	12.000	12.000	1082	1642	0.002	0.086	0.086	0.961	0.653
b₆	10.000	10.000	788	1113	0.002	0.099	0.099	0.958	0.776
b₇	8.168	8.168	543	715	0.001	0.122	0.122	0.964	0.970
b₈	7.637	7.637	494	577	0.004	0.131	0.131	0.968	1.065
b₉	7.604	7.604	526	487	0.011	0.130	0.130	0.970	1.065
b₁₀	7.568	7.568	562	384	0.010	0.125	0.126	0.970	1.300
b₁₁	8.064	8.064	672	322	0.003	0.112	0.112	0.982	11.299
b₁₂	7.578	7.578	547	425	-0.006	0.129	0.129	0.966	2.008
b₁₃	7.610	7.610	496	558	-0.005	0.128	0.128	0.967	1.014
b₁₄	8.430	8.430	576	769	-0.001	0.115	0.115	0.958	0.896
b₁₅	11.200	11.200	962	1418	0.000	0.089	0.089	0.950	0.666
b₁₆	13.926	13.926	1372	2236	0.002	0.074	0.074	0.955	0.542
b₁₇	15.794	15.794	1641	2883	0.003	0.069	0.069	0.951	0.494
b₁₈	16.820	16.820	1756	3230	0.001	0.067	0.067	0.954	0.480
b₁₉	17.866	17.866	1841	3536	-0.002	0.066	0.066	0.951	0.468
b₂₀	18.951	18.951	1897	3799	-0.005	0.066	0.066	0.946	0.462
b₂₁	20.130	20.130	1928	4034	-0.008	0.066	0.066	0.950	0.457
Uniform								0.975	1.943
WBATE					0.001	0.034	0.034	0.961	0.759
LBATE					0.149	0.086	0.173	0.997	10.563

Table SA-62: Fuzzy Effects (Simulations, Quadratic Heteroskedastic, Kink-Adaptive Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	7.979	7.979	540	716	-0.001	0.149	0.149	0.955	0.573
b₂	7.692	7.692	514	675	0.002	0.149	0.149	0.954	0.573
b₃	7.537	7.537	499	652	0.002	0.144	0.144	0.957	0.566
b₄	7.517	7.517	497	648	0.000	0.144	0.144	0.953	0.549
b₅	7.587	7.587	500	653	0.000	0.139	0.139	0.955	0.529
b₆	7.679	7.679	501	656	0.001	0.132	0.132	0.956	0.512
b₇	7.718	7.718	493	643	0.000	0.132	0.132	0.954	0.502
b₈	7.634	7.634	494	576	0.004	0.131	0.131	0.958	0.498
b₉	7.604	7.604	526	487	0.011	0.130	0.130	0.955	0.494
b₁₀	7.568	7.568	562	384	0.010	0.125	0.126	0.959	0.481
b₁₁	8.064	8.064	672	322	0.003	0.112	0.112	0.963	0.425
b₁₂	7.578	7.578	547	425	-0.006	0.129	0.129	0.955	0.497
b₁₃	7.609	7.609	496	558	-0.005	0.127	0.128	0.959	0.495
b₁₄	7.682	7.682	490	642	-0.000	0.128	0.128	0.955	0.494
b₁₅	7.512	7.512	488	636	-0.000	0.135	0.135	0.949	0.507
b₁₆	7.309	7.309	472	612	0.005	0.135	0.135	0.954	0.524
b₁₇	7.263	7.263	466	603	0.007	0.142	0.142	0.948	0.540
b₁₈	7.438	7.438	478	622	0.001	0.139	0.139	0.949	0.539
b₁₉	7.835	7.835	508	670	-0.003	0.135	0.135	0.953	0.527
b₂₀	8.303	8.303	537	720	-0.000	0.135	0.135	0.947	0.519
b₂₁	8.819	8.819	560	766	-0.000	0.132	0.132	0.951	0.512
Uniform								0.964	0.791
WBATE					0.001	0.041	0.041	0.954	0.160
LBATE					0.205	0.092	0.225	0.995	0.873

Table SA-63: Fuzzy Effects (Simulations, Quadratic Heteroskedastic, Unknown-Kink Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.349	17.349	1823	3432	-0.002	0.078	0.078	0.961	0.527
b₂	16.879	16.879	1796	3288	-0.002	0.076	0.076	0.959	0.524
b₃	16.600	16.600	1802	3173	-0.002	0.076	0.076	0.964	0.516
b₄	16.606	16.606	1872	3104	0.002	0.078	0.078	0.963	0.505
b₅	16.893	16.893	2007	3068	0.009	0.078	0.078	0.966	0.487
b₆	17.267	17.267	2163	3001	0.018	0.076	0.078	0.959	0.465
b₇	17.265	17.265	2238	2782	0.023	0.074	0.078	0.955	0.453
b₈	17.338	17.338	2311	2557	0.024	0.071	0.075	0.956	0.442
b₉	17.379	17.379	2356	2307	0.020	0.067	0.070	0.960	0.435
b₁₀	17.542	17.542	2399	2077	0.012	0.063	0.064	0.966	0.418
b₁₁	18.603	18.603	2573	2042	0.003	0.057	0.057	0.964	0.377
b₁₂	17.562	17.562	2403	2187	-0.008	0.065	0.065	0.965	0.431
b₁₃	17.605	17.605	2376	2569	-0.016	0.067	0.069	0.962	0.431
b₁₄	17.447	17.447	2262	2877	-0.015	0.072	0.073	0.961	0.438
b₁₅	16.875	16.875	2039	3004	-0.005	0.077	0.077	0.962	0.461
b₁₆	16.081	16.081	1785	2944	0.006	0.076	0.076	0.968	0.488
b₁₇	15.755	15.755	1653	2902	0.009	0.075	0.075	0.953	0.507
b₁₈	15.961	15.961	1637	2941	0.006	0.073	0.074	0.956	0.508
b₁₉	16.555	16.555	1669	3063	0.003	0.074	0.074	0.954	0.504
b₂₀	17.249	17.249	1687	3182	0.001	0.076	0.076	0.949	0.506
b₂₁	17.956	17.956	1676	3259	0.000	0.077	0.077	0.957	0.512
Uniform								0.977	0.717
WBATE					0.004	0.030	0.030	0.949	0.158
LBATE					0.083	0.055	0.100	0.997	0.814

Table SA-64: Fuzzy Effects (Simulations, Quadratic Heteroskedastic, **rdrobust** Bandwidth, ITT Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.940	18.940	2038	4030	0.000	0.034	0.034	0.947	0.248
b₂	18.560	18.560	2046	3894	0.000	0.035	0.035	0.953	0.251
b₃	18.341	18.341	2087	3764	0.000	0.035	0.035	0.948	0.253
b₄	18.247	18.247	2164	3620	0.002	0.036	0.036	0.951	0.254
b₅	18.265	18.265	2269	3460	0.004	0.036	0.036	0.955	0.255
b₆	18.404	18.404	2394	3295	0.006	0.036	0.036	0.956	0.254
b₇	18.492	18.492	2494	3084	0.007	0.036	0.037	0.954	0.256
b₈	18.409	18.409	2536	2801	0.007	0.037	0.038	0.952	0.262
b₉	18.445	18.445	2577	2534	0.006	0.038	0.038	0.951	0.269
b₁₀	18.570	18.570	2606	2280	0.004	0.039	0.040	0.952	0.280
b₁₁	20.183	20.183	2867	2376	0.000	0.040	0.040	0.954	0.297
b₁₂	18.471	18.471	2589	2371	-0.005	0.039	0.039	0.952	0.275
b₁₃	18.461	18.461	2555	2759	-0.011	0.037	0.039	0.950	0.263
b₁₄	18.567	18.567	2494	3154	-0.012	0.036	0.038	0.947	0.255
b₁₅	18.485	18.485	2354	3449	-0.008	0.036	0.037	0.948	0.251
b₁₆	18.323	18.323	2178	3645	-0.003	0.036	0.036	0.954	0.252
b₁₇	18.304	18.304	2048	3779	0.000	0.035	0.035	0.956	0.253
b₁₈	18.620	18.620	2006	3912	0.001	0.034	0.034	0.956	0.252
b₁₉	19.300	19.300	2021	4091	0.001	0.033	0.033	0.949	0.247
b₂₀	20.308	20.308	2057	4323	0.000	0.033	0.033	0.955	0.241
b₂₁	21.402	21.402	2067	4519	-0.000	0.033	0.033	0.950	0.236
Uniform								0.949	0.386
WBATE					-0.000	0.017	0.017	0.948	0.095
LBATE					0.044	0.026	0.052	0.981	0.399

Table SA-65: Intention-to-Treat Effects (Simulations, Linear Homoskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.446	18.446	1967	3834	0.001	0.034	0.034	0.947	0.253
b₂	17.301	17.301	1843	3422	0.001	0.036	0.036	0.948	0.266
b₃	15.833	15.833	1650	2887	0.001	0.039	0.039	0.949	0.288
b₄	13.993	13.993	1382	2255	0.001	0.044	0.044	0.950	0.324
b₅	12.000	12.000	1081	1641	-0.000	0.050	0.050	0.943	0.378
b₆	10.000	10.000	788	1113	-0.002	0.059	0.059	0.943	0.454
b₇	8.375	8.375	570	751	-0.002	0.072	0.072	0.945	0.546
b₈	8.066	8.066	554	636	0.000	0.079	0.079	0.936	0.580
b₉	8.081	8.081	595	543	0.003	0.080	0.080	0.942	0.597
b₁₀	8.136	8.136	646	441	0.003	0.083	0.083	0.940	0.614
b₁₁	8.842	8.842	794	394	-0.001	0.091	0.091	0.930	0.677
b₁₂	8.092	8.092	623	480	-0.006	0.081	0.081	0.943	0.608
b₁₃	8.088	8.088	564	622	-0.004	0.079	0.079	0.946	0.582
b₁₄	8.563	8.563	593	793	0.001	0.072	0.072	0.939	0.533
b₁₅	11.200	11.200	961	1419	0.002	0.054	0.054	0.943	0.403
b₁₆	13.993	13.993	1381	2257	0.001	0.043	0.043	0.945	0.323
b₁₇	16.467	16.467	1737	3117	0.000	0.037	0.037	0.949	0.277
b₁₈	18.090	18.090	1927	3704	0.000	0.034	0.034	0.955	0.257
b₁₉	19.259	19.259	2016	4074	0.001	0.033	0.033	0.949	0.247
b₂₀	20.308	20.308	2057	4323	0.000	0.033	0.033	0.955	0.241
b₂₁	21.402	21.402	2067	4519	-0.000	0.033	0.033	0.950	0.236
Uniform								0.925	0.633
WBATE					-0.000	0.021	0.021	0.950	0.122
LBATE					0.094	0.049	0.106	0.966	0.783

Table SA-66: Intention-to-Treat Effects (Simulations, Linear Homoskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.298	8.298	578	774	0.001	0.075	0.075	0.942	0.286
b₂	8.131	8.131	566	754	0.001	0.076	0.076	0.942	0.290
b₃	8.035	8.035	559	741	0.001	0.076	0.076	0.951	0.293
b₄	7.994	7.994	553	733	0.000	0.076	0.076	0.946	0.294
b₅	8.002	8.002	549	726	-0.001	0.075	0.075	0.947	0.295
b₆	8.063	8.063	546	723	-0.003	0.075	0.075	0.949	0.294
b₇	8.101	8.101	540	706	-0.002	0.075	0.075	0.951	0.296
b₈	8.065	8.065	554	636	0.000	0.079	0.079	0.947	0.303
b₉	8.081	8.081	595	543	0.003	0.080	0.080	0.949	0.311
b₁₀	8.136	8.136	646	441	0.003	0.083	0.083	0.950	0.324
b₁₁	8.842	8.842	794	394	-0.001	0.091	0.091	0.944	0.351
b₁₂	8.092	8.092	623	480	-0.006	0.081	0.081	0.951	0.318
b₁₃	8.088	8.088	564	622	-0.004	0.079	0.079	0.950	0.304
b₁₄	8.134	8.134	544	719	0.001	0.077	0.077	0.946	0.294
b₁₅	8.098	8.098	556	739	0.002	0.076	0.076	0.949	0.290
b₁₆	8.028	8.028	556	738	0.000	0.075	0.075	0.950	0.292
b₁₇	8.019	8.019	555	737	-0.001	0.075	0.075	0.948	0.292
b₁₈	8.157	8.157	562	751	-0.001	0.074	0.074	0.951	0.290
b₁₉	8.456	8.456	580	782	-0.001	0.074	0.074	0.945	0.284
b₂₀	8.897	8.897	604	827	-0.001	0.071	0.071	0.953	0.277
b₂₁	9.376	9.376	621	865	-0.001	0.070	0.070	0.948	0.271
Uniform								0.942	0.454
WBATE					-0.000	0.025	0.025	0.951	0.097
LBATE					0.127	0.048	0.136	0.979	0.468

Table SA-67: Intention-to-Treat Effects (Simulations, Linear Homoskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.446	18.446	1976	3846	-0.001	0.036	0.036	0.951	0.257
b₂	18.337	18.337	2020	3819	-0.001	0.037	0.037	0.951	0.257
b₃	18.348	18.348	2101	3775	-0.000	0.037	0.037	0.951	0.257
b₄	18.488	18.488	2219	3716	0.001	0.038	0.038	0.952	0.256
b₅	18.737	18.737	2365	3637	0.002	0.038	0.038	0.954	0.255
b₆	18.966	18.966	2501	3501	0.004	0.038	0.038	0.955	0.253
b₇	19.098	19.098	2604	3300	0.006	0.038	0.039	0.953	0.255
b₈	19.288	19.288	2695	3089	0.006	0.039	0.040	0.953	0.257
b₉	19.613	19.613	2782	2892	0.005	0.039	0.039	0.953	0.260
b₁₀	20.205	20.205	2884	2747	0.003	0.039	0.039	0.952	0.265
b₁₁	21.806	21.806	3110	2852	-0.000	0.040	0.040	0.954	0.281
b₁₂	19.941	19.941	2845	2797	-0.006	0.039	0.040	0.955	0.262
b₁₃	19.520	19.520	2749	3093	-0.010	0.039	0.040	0.953	0.255
b₁₄	19.298	19.298	2631	3407	-0.011	0.039	0.040	0.947	0.251
b₁₅	19.009	19.009	2456	3648	-0.007	0.039	0.040	0.954	0.252
b₁₆	18.515	18.515	2227	3730	-0.002	0.039	0.039	0.955	0.257
b₁₇	18.115	18.115	2033	3722	0.002	0.038	0.038	0.960	0.261
b₁₈	18.047	18.047	1931	3705	0.002	0.036	0.036	0.957	0.262
b₁₉	18.280	18.280	1892	3705	0.002	0.036	0.036	0.948	0.261
b₂₀	18.825	18.825	1878	3762	0.002	0.037	0.037	0.954	0.260
b₂₁	19.374	19.374	1839	3766	0.001	0.038	0.038	0.952	0.261
Uniform								0.947	0.389
WBATE					-0.000	0.017	0.017	0.948	0.095
LBATE					0.050	0.031	0.058	0.978	0.424

Table SA-68: Intention-to-Treat Effects (Simulations, Linear Homoskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.488	19.488	2114	4244	-0.001	0.042	0.042	0.952	0.305
b₂	18.846	18.846	2089	4002	-0.001	0.043	0.043	0.957	0.309
b₃	18.345	18.345	2087	3766	-0.000	0.043	0.043	0.958	0.312
b₄	17.972	17.972	2110	3530	0.001	0.044	0.044	0.957	0.314
b₅	17.684	17.684	2151	3279	0.003	0.045	0.045	0.953	0.316
b₆	17.475	17.475	2202	3021	0.005	0.045	0.045	0.953	0.318
b₇	17.293	17.293	2244	2747	0.007	0.046	0.046	0.951	0.321
b₈	16.916	16.916	2229	2406	0.007	0.047	0.048	0.956	0.330
b₉	16.732	16.732	2233	2109	0.006	0.048	0.048	0.950	0.337
b₁₀	16.448	16.448	2194	1787	0.003	0.049	0.049	0.950	0.343
b₁₁	16.857	16.857	2265	1614	0.000	0.049	0.049	0.948	0.345
b₁₂	16.613	16.613	2223	1927	-0.005	0.049	0.049	0.952	0.351
b₁₃	16.893	16.893	2234	2351	-0.010	0.050	0.051	0.953	0.351
b₁₄	17.334	17.334	2239	2803	-0.011	0.051	0.052	0.952	0.347
b₁₅	17.565	17.565	2165	3168	-0.008	0.052	0.053	0.947	0.352
b₁₆	17.929	17.929	2101	3515	-0.003	0.052	0.052	0.951	0.357
b₁₇	18.505	18.505	2078	3852	0.000	0.051	0.051	0.956	0.362
b₁₈	19.329	19.329	2103	4188	0.001	0.052	0.052	0.951	0.366
b₁₉	20.492	20.492	2167	4569	0.001	0.052	0.052	0.953	0.366
b₂₀	21.968	21.968	2239	4998	0.001	0.053	0.053	0.953	0.366
b₂₁	23.598	23.598	2285	5410	0.000	0.054	0.054	0.955	0.368
Uniform								0.946	0.512
WBATE					-0.000	0.021	0.021	0.948	0.118
LBATE					0.065	0.037	0.075	0.979	0.542

Table SA-69: Intention-to-Treat Effects (Simulations, Linear Heteroskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.811	18.811	2015	3976	-0.000	0.043	0.043	0.951	0.314
b₂	17.441	17.441	1863	3475	0.000	0.045	0.045	0.954	0.330
b₃	15.849	15.849	1652	2894	0.001	0.047	0.047	0.954	0.355
b₄	13.990	13.990	1381	2255	0.001	0.053	0.053	0.948	0.396
b₅	12.000	12.000	1081	1641	-0.001	0.061	0.061	0.946	0.456
b₆	10.000	10.000	788	1112	-0.002	0.072	0.072	0.945	0.542
b₇	8.099	8.099	534	703	-0.003	0.090	0.090	0.944	0.666
b₈	7.421	7.421	465	547	0.000	0.099	0.099	0.946	0.734
b₉	7.330	7.330	487	456	0.002	0.104	0.104	0.931	0.748
b₁₀	7.206	7.206	510	349	0.002	0.104	0.104	0.937	0.772
b₁₁	7.385	7.385	572	266	-0.001	0.106	0.106	0.932	0.769
b₁₂	7.278	7.278	504	395	-0.006	0.104	0.104	0.943	0.780
b₁₃	7.402	7.402	469	531	-0.002	0.105	0.105	0.940	0.775
b₁₄	8.414	8.414	574	766	0.000	0.094	0.094	0.938	0.690
b₁₅	11.200	11.200	962	1419	-0.001	0.074	0.074	0.939	0.535
b₁₆	13.989	13.989	1381	2256	-0.000	0.061	0.061	0.941	0.448
b₁₇	16.557	16.557	1750	3151	-0.000	0.054	0.054	0.949	0.399
b₁₈	18.615	18.615	1998	3908	0.001	0.053	0.053	0.949	0.377
b₁₉	20.332	20.332	2147	4503	0.001	0.053	0.053	0.952	0.369
b₂₀	21.947	21.947	2236	4989	0.001	0.053	0.053	0.953	0.367
b₂₁	23.593	23.593	2284	5408	0.000	0.054	0.054	0.955	0.368
Uniform								0.923	0.816
WBATE					-0.000	0.026	0.026	0.946	0.149
LBATE					0.127	0.061	0.140	0.966	1.001

Table SA-70: Intention-to-Treat Effects (Simulations, Linear Heteroskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.538	8.538	607	820	-0.001	0.089	0.089	0.949	0.347
b₂	8.257	8.257	582	778	0.000	0.090	0.090	0.950	0.353
b₃	8.037	8.037	559	741	0.001	0.092	0.092	0.952	0.357
b₄	7.874	7.874	538	709	-0.000	0.094	0.094	0.951	0.361
b₅	7.747	7.747	518	679	-0.002	0.095	0.095	0.952	0.365
b₆	7.656	7.656	498	650	-0.004	0.096	0.096	0.951	0.368
b₇	7.576	7.576	477	618	-0.003	0.096	0.096	0.950	0.374
b₈	7.411	7.411	465	546	0.000	0.099	0.099	0.952	0.384
b₉	7.330	7.330	487	456	0.002	0.104	0.104	0.947	0.391
b₁₀	7.206	7.206	510	349	0.002	0.104	0.104	0.942	0.399
b₁₁	7.385	7.385	572	266	-0.001	0.106	0.106	0.945	0.401
b₁₂	7.278	7.278	504	395	-0.006	0.104	0.104	0.952	0.408
b₁₃	7.401	7.401	468	530	-0.002	0.105	0.105	0.949	0.406
b₁₄	7.594	7.594	480	625	-0.000	0.105	0.105	0.947	0.402
b₁₅	7.695	7.695	508	665	-0.002	0.106	0.106	0.945	0.403
b₁₆	7.855	7.855	536	705	0.000	0.105	0.105	0.944	0.407
b₁₇	8.107	8.107	566	752	0.000	0.104	0.104	0.952	0.410
b₁₈	8.468	8.468	601	809	-0.001	0.107	0.107	0.946	0.411
b₁₉	8.978	8.978	644	883	-0.001	0.106	0.106	0.946	0.410
b₂₀	9.624	9.624	692	970	0.001	0.105	0.105	0.946	0.408
b₂₁	10.338	10.338	734	1056	0.001	0.106	0.106	0.948	0.408
Uniform								0.938	0.595
WBATE					-0.001	0.031	0.031	0.946	0.121
LBATE					0.172	0.067	0.185	0.972	0.623

Table SA-71: Intention-to-Treat Effects (Simulations, Linear Heteroskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	18.343	18.343	1960	3818	-0.002	0.049	0.049	0.957	0.327
b ₂	18.051	18.051	1977	3719	-0.002	0.049	0.049	0.959	0.326
b ₃	17.855	17.855	2018	3608	-0.002	0.049	0.049	0.957	0.325
b ₄	17.785	17.785	2090	3485	-0.001	0.049	0.049	0.956	0.323
b ₅	17.710	17.710	2166	3313	0.001	0.050	0.050	0.956	0.323
b ₆	17.660	17.660	2243	3107	0.003	0.050	0.050	0.957	0.322
b ₇	17.157	17.157	2217	2745	0.004	0.052	0.052	0.953	0.331
b ₈	16.803	16.803	2205	2415	0.005	0.052	0.053	0.959	0.340
b ₉	16.462	16.462	2176	2085	0.005	0.054	0.055	0.952	0.351
b ₁₀	16.219	16.219	2146	1778	0.002	0.055	0.055	0.951	0.357
b ₁₁	16.492	16.492	2191	1581	-0.000	0.054	0.054	0.950	0.361
b ₁₂	16.520	16.520	2200	1947	-0.005	0.054	0.054	0.954	0.361
b ₁₃	17.243	17.243	2302	2481	-0.009	0.055	0.056	0.956	0.353
b ₁₄	17.877	17.877	2349	2999	-0.009	0.056	0.057	0.955	0.346
b ₁₅	18.466	18.466	2354	3483	-0.006	0.058	0.058	0.951	0.346
b ₁₆	18.673	18.673	2263	3788	-0.002	0.058	0.058	0.955	0.355
b ₁₇	18.855	18.855	2164	3992	0.002	0.058	0.058	0.961	0.365
b ₁₈	19.160	19.160	2095	4145	0.003	0.061	0.061	0.959	0.377
b ₁₉	19.705	19.705	2067	4290	0.003	0.062	0.062	0.954	0.387
b ₂₀	20.543	20.543	2067	4467	0.004	0.062	0.063	0.954	0.397
b ₂₁	21.462	21.462	2050	4606	0.004	0.065	0.065	0.960	0.409
Uniform								0.947	0.531
WBATE					-0.000	0.023	0.023	0.941	0.120
LBATE					0.080	0.050	0.095	0.976	0.617

Table SA-72: Intention-to-Treat Effects (Simulations, Linear Heteroskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.240	17.240	1807	3381	-0.002	0.037	0.037	0.947	0.272
b₂	16.756	16.756	1775	3231	-0.002	0.038	0.038	0.948	0.277
b₃	16.586	16.586	1790	3156	-0.001	0.038	0.038	0.956	0.279
b₄	16.714	16.714	1873	3125	0.001	0.039	0.039	0.954	0.277
b₅	17.111	17.111	2036	3110	0.007	0.040	0.040	0.952	0.271
b₆	17.598	17.598	2226	3064	0.014	0.039	0.042	0.952	0.266
b₇	18.024	18.024	2395	2954	0.020	0.038	0.043	0.946	0.263
b₈	18.133	18.133	2479	2728	0.021	0.037	0.042	0.946	0.266
b₉	18.233	18.233	2535	2481	0.017	0.037	0.041	0.946	0.271
b₁₀	18.423	18.423	2578	2245	0.009	0.039	0.040	0.951	0.281
b₁₁	20.110	20.110	2855	2358	-0.001	0.039	0.039	0.953	0.296
b₁₂	18.393	18.393	2574	2351	-0.008	0.038	0.039	0.952	0.275
b₁₃	18.366	18.366	2537	2735	-0.015	0.037	0.040	0.949	0.262
b₁₄	18.343	18.343	2448	3092	-0.017	0.037	0.041	0.947	0.256
b₁₅	17.668	17.668	2187	3205	-0.012	0.038	0.040	0.952	0.262
b₁₆	16.748	16.748	1880	3135	-0.004	0.039	0.039	0.951	0.274
b₁₇	16.162	16.162	1707	3017	-0.002	0.038	0.038	0.952	0.284
b₁₈	16.176	16.176	1668	2995	-0.001	0.038	0.038	0.950	0.285
b₁₉	16.736	16.736	1694	3114	-0.001	0.038	0.038	0.941	0.281
b₂₀	17.551	17.551	1727	3273	-0.003	0.038	0.038	0.941	0.274
b₂₁	18.491	18.491	1744	3422	-0.005	0.037	0.038	0.948	0.269
Uniform								0.946	0.412
WBATE					0.001	0.018	0.018	0.951	0.097
LBATE					0.032	0.029	0.043	0.990	0.424

Table SA-73: Intention-to-Treat Effects (Simulations, Quadratic Homoskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.072	17.072	1783	3314	-0.001	0.037	0.037	0.947	0.274
b₂	16.314	16.314	1705	3064	-0.001	0.038	0.038	0.948	0.282
b₃	15.415	15.415	1589	2746	-0.001	0.040	0.040	0.954	0.296
b₄	13.947	13.947	1374	2243	-0.000	0.043	0.043	0.948	0.325
b₅	12.000	12.000	1080	1642	0.000	0.050	0.050	0.950	0.377
b₆	10.000	10.000	787	1114	0.000	0.060	0.060	0.950	0.453
b₇	8.292	8.292	559	737	-0.000	0.075	0.075	0.939	0.551
b₈	7.946	7.946	537	620	0.003	0.080	0.080	0.942	0.590
b₉	7.988	7.988	582	533	0.009	0.081	0.081	0.939	0.599
b₁₀	8.071	8.071	636	435	0.008	0.082	0.083	0.941	0.617
b₁₁	8.811	8.811	788	391	0.001	0.091	0.091	0.929	0.676
b₁₂	8.058	8.058	618	477	-0.004	0.081	0.082	0.943	0.606
b₁₃	8.047	8.047	558	616	-0.001	0.078	0.078	0.946	0.580
b₁₄	8.537	8.537	590	788	0.001	0.071	0.071	0.945	0.531
b₁₅	11.200	11.200	962	1419	-0.001	0.054	0.054	0.945	0.401
b₁₆	13.954	13.954	1376	2245	-0.002	0.043	0.043	0.946	0.322
b₁₇	15.604	15.604	1614	2816	-0.002	0.039	0.039	0.951	0.291
b₁₈	16.095	16.095	1656	2963	-0.002	0.038	0.038	0.951	0.286
b₁₉	16.730	16.730	1693	3111	-0.001	0.038	0.038	0.941	0.281
b₂₀	17.551	17.551	1727	3273	-0.003	0.038	0.038	0.941	0.274
b₂₁	18.491	18.491	1744	3422	-0.005	0.037	0.038	0.948	0.269
Uniform								0.929	0.648
WBATE					-0.000	0.022	0.022	0.947	0.123
LBATE					0.090	0.053	0.105	0.978	0.823

Table SA-74: Intention-to-Treat Effects (Simulations, Quadratic Homoskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	7.553	7.553	489	640	0.001	0.082	0.082	0.946	0.315
b₂	7.341	7.341	472	613	0.001	0.085	0.085	0.942	0.321
b₃	7.267	7.267	467	605	-0.000	0.083	0.083	0.948	0.323
b₄	7.322	7.322	473	615	-0.001	0.083	0.083	0.948	0.321
b₅	7.496	7.496	489	639	-0.000	0.081	0.081	0.950	0.315
b₆	7.710	7.710	505	664	-0.000	0.079	0.079	0.951	0.308
b₇	7.896	7.896	516	673	-0.001	0.080	0.080	0.947	0.304
b₈	7.944	7.944	537	620	0.003	0.080	0.080	0.943	0.307
b₉	7.988	7.988	582	533	0.009	0.081	0.081	0.945	0.313
b₁₀	8.071	8.071	636	435	0.008	0.082	0.083	0.952	0.325
b₁₁	8.811	8.811	788	391	0.001	0.091	0.091	0.946	0.351
b₁₂	8.058	8.058	618	477	-0.004	0.081	0.082	0.946	0.317
b₁₃	8.046	8.046	558	616	-0.001	0.078	0.078	0.951	0.303
b₁₄	8.036	8.036	533	702	0.001	0.076	0.076	0.947	0.295
b₁₅	7.741	7.741	515	675	-0.001	0.079	0.079	0.947	0.303
b₁₆	7.337	7.337	476	616	-0.001	0.083	0.083	0.942	0.317
b₁₇	7.081	7.081	445	572	-0.004	0.085	0.085	0.943	0.329
b₁₈	7.087	7.087	438	562	-0.001	0.083	0.083	0.950	0.330
b₁₉	7.332	7.332	451	585	0.003	0.085	0.085	0.938	0.325
b₂₀	7.689	7.689	468	614	0.000	0.084	0.084	0.939	0.317
b₂₁	8.101	8.101	482	641	-0.002	0.079	0.079	0.947	0.311
Uniform								0.935	0.484
WBATE					0.000	0.026	0.026	0.947	0.100
LBATE					0.118	0.051	0.129	0.989	0.497

Table SA-75: Intention-to-Treat Effects (Simulations, Quadratic Homoskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	16.410	16.410	1693	3079	-0.003	0.039	0.039	0.947	0.285
b₂	16.047	16.047	1671	2986	-0.003	0.040	0.040	0.948	0.290
b₃	15.948	15.948	1694	2949	-0.003	0.041	0.041	0.956	0.292
b₄	16.089	16.089	1775	2937	-0.001	0.043	0.043	0.959	0.292
b₅	16.495	16.495	1926	2946	0.004	0.045	0.045	0.958	0.287
b₆	17.158	17.158	2138	2971	0.010	0.045	0.046	0.955	0.280
b₇	17.721	17.721	2326	2913	0.016	0.043	0.046	0.942	0.275
b₈	18.359	18.359	2512	2837	0.018	0.041	0.044	0.941	0.270
b₉	18.990	18.990	2663	2730	0.014	0.039	0.042	0.944	0.268
b₁₀	19.803	19.803	2812	2645	0.007	0.039	0.040	0.951	0.269
b₁₁	21.462	21.462	3055	2761	-0.002	0.040	0.040	0.954	0.284
b₁₂	19.640	19.640	2792	2714	-0.009	0.039	0.040	0.953	0.263
b₁₃	19.005	19.005	2650	2951	-0.015	0.040	0.042	0.954	0.260
b₁₄	18.130	18.130	2396	3075	-0.014	0.042	0.044	0.950	0.266
b₁₅	16.883	16.883	2036	3009	-0.007	0.045	0.045	0.957	0.282
b₁₆	15.616	15.616	1700	2794	-0.000	0.044	0.044	0.953	0.300
b₁₇	15.047	15.047	1543	2651	0.001	0.042	0.042	0.952	0.307
b₁₈	15.155	15.155	1525	2644	0.001	0.041	0.041	0.953	0.305
b₁₉	15.676	15.676	1553	2737	0.001	0.042	0.042	0.943	0.299
b₂₀	16.282	16.282	1566	2825	-0.000	0.043	0.043	0.945	0.295
b₂₁	16.774	16.774	1539	2833	-0.002	0.043	0.043	0.954	0.296
Uniform								0.947	0.429
WBATE					0.001	0.018	0.018	0.946	0.099
LBATE					0.038	0.033	0.051	0.991	0.461

Table SA-76: Intention-to-Treat Effects (Simulations, Quadratic Homoskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.687	17.687	1868	3552	-0.003	0.041	0.041	0.952	0.294
b₂	17.077	17.077	1823	3350	-0.003	0.041	0.041	0.955	0.297
b₃	16.696	16.696	1810	3195	-0.002	0.041	0.041	0.956	0.297
b₄	16.579	16.579	1851	3081	0.001	0.042	0.042	0.951	0.294
b₅	16.641	16.641	1943	2966	0.006	0.043	0.043	0.950	0.290
b₆	16.776	16.776	2057	2824	0.014	0.042	0.044	0.951	0.286
b₇	16.861	16.861	2154	2632	0.019	0.041	0.046	0.951	0.284
b₈	16.634	16.634	2170	2336	0.020	0.041	0.046	0.948	0.290
b₉	16.538	16.538	2193	2064	0.017	0.041	0.044	0.948	0.296
b₁₀	16.270	16.270	2159	1750	0.009	0.042	0.043	0.951	0.304
b₁₁	16.701	16.701	2235	1582	-0.001	0.043	0.043	0.952	0.313
b₁₂	16.495	16.495	2199	1901	-0.009	0.043	0.044	0.954	0.308
b₁₃	16.770	16.770	2209	2319	-0.015	0.043	0.045	0.957	0.302
b₁₄	17.120	17.120	2193	2745	-0.016	0.043	0.046	0.957	0.297
b₁₅	16.936	16.936	2037	2983	-0.009	0.045	0.046	0.950	0.305
b₁₆	16.613	16.613	1856	3091	-0.001	0.046	0.046	0.954	0.316
b₁₇	16.555	16.555	1767	3158	0.001	0.046	0.046	0.948	0.326
b₁₈	16.909	16.909	1769	3267	-0.000	0.045	0.045	0.951	0.328
b₁₉	17.755	17.755	1827	3495	-0.002	0.045	0.045	0.947	0.324
b₂₀	18.817	18.817	1881	3748	-0.004	0.045	0.046	0.946	0.321
b₂₁	20.034	20.034	1917	3997	-0.006	0.045	0.046	0.951	0.319
Uniform								0.946	0.460
WBATE					0.001	0.018	0.018	0.949	0.103
LBATE					0.042	0.032	0.053	0.985	0.479

Table SA-77: Intention-to-Treat Effects (Simulations, Quadratic Heteroskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.460	17.460	1835	3461	-0.003	0.041	0.041	0.952	0.297
b₂	16.507	16.507	1732	3136	-0.002	0.042	0.042	0.954	0.304
b₃	15.414	15.414	1590	2748	-0.001	0.043	0.043	0.953	0.317
b₄	13.916	13.916	1371	2234	0.000	0.047	0.047	0.945	0.343
b₅	11.999	11.999	1082	1642	0.001	0.053	0.053	0.939	0.392
b₆	10.000	10.000	788	1113	0.002	0.062	0.062	0.940	0.466
b₇	8.061	8.061	529	697	0.001	0.078	0.078	0.938	0.575
b₈	7.300	7.300	450	532	0.002	0.088	0.088	0.939	0.640
b₉	7.245	7.245	476	447	0.009	0.090	0.090	0.938	0.657
b₁₀	7.128	7.128	500	343	0.008	0.091	0.091	0.941	0.678
b₁₁	7.317	7.317	563	261	-0.000	0.094	0.094	0.930	0.701
b₁₂	7.227	7.227	497	390	-0.005	0.092	0.092	0.942	0.681
b₁₃	7.348	7.348	461	524	-0.002	0.089	0.089	0.943	0.665
b₁₄	8.409	8.409	573	765	-0.001	0.078	0.078	0.942	0.583
b₁₅	11.200	11.200	962	1418	-0.000	0.061	0.061	0.942	0.447
b₁₆	13.919	13.919	1371	2234	0.000	0.050	0.050	0.947	0.369
b₁₇	15.793	15.793	1640	2883	0.001	0.047	0.047	0.946	0.338
b₁₈	16.766	16.766	1748	3211	-0.000	0.046	0.046	0.951	0.330
b₁₉	17.739	17.739	1825	3488	-0.002	0.045	0.045	0.948	0.325
b₂₀	18.816	18.816	1881	3747	-0.004	0.045	0.046	0.946	0.321
b₂₁	20.034	20.034	1917	3997	-0.006	0.045	0.046	0.951	0.319
Uniform								0.917	0.711
WBATE					-0.000	0.023	0.023	0.944	0.128
LBATE					0.104	0.057	0.119	0.969	0.899

Table SA-78: Intention-to-Treat Effects (Simulations, Quadratic Heteroskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	7.749	7.749	513	675	-0.003	0.086	0.086	0.949	0.337
b₂	7.482	7.482	490	638	-0.001	0.086	0.086	0.954	0.340
b₃	7.315	7.315	474	614	-0.001	0.086	0.086	0.955	0.342
b₄	7.264	7.264	467	604	-0.000	0.089	0.089	0.943	0.339
b₅	7.290	7.290	465	602	0.001	0.088	0.088	0.943	0.335
b₆	7.350	7.350	463	600	0.001	0.086	0.086	0.948	0.331
b₇	7.387	7.387	455	589	0.000	0.086	0.086	0.951	0.330
b₈	7.287	7.287	449	530	0.002	0.088	0.088	0.948	0.336
b₉	7.245	7.245	476	447	0.009	0.090	0.090	0.945	0.343
b₁₀	7.128	7.128	500	343	0.008	0.091	0.091	0.948	0.352
b₁₁	7.317	7.317	563	261	-0.000	0.094	0.094	0.948	0.365
b₁₂	7.227	7.227	497	390	-0.005	0.092	0.092	0.951	0.356
b₁₃	7.347	7.347	461	523	-0.002	0.089	0.089	0.950	0.349
b₁₄	7.501	7.501	469	610	-0.000	0.088	0.088	0.952	0.343
b₁₅	7.420	7.420	477	619	0.000	0.092	0.092	0.942	0.349
b₁₆	7.278	7.278	468	606	0.004	0.094	0.094	0.946	0.361
b₁₇	7.253	7.253	465	601	0.004	0.097	0.097	0.942	0.370
b₁₈	7.408	7.408	475	617	0.000	0.096	0.096	0.945	0.372
b₁₉	7.779	7.779	502	661	-0.002	0.094	0.094	0.949	0.367
b₂₀	8.244	8.244	530	710	-0.001	0.093	0.093	0.948	0.362
b₂₁	8.777	8.777	556	759	-0.000	0.092	0.092	0.950	0.358
Uniform								0.933	0.535
WBATE					0.001	0.027	0.027	0.950	0.106
LBATE					0.137	0.057	0.148	0.983	0.553

Table SA-79: Intention-to-Treat Effects (Simulations, Quadratic Heteroskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	16.600	16.600	1719	3154	-0.004	0.045	0.046	0.953	0.313
b ₂	16.166	16.166	1690	3035	-0.004	0.045	0.045	0.959	0.315
b ₃	15.882	15.882	1686	2931	-0.004	0.045	0.046	0.960	0.315
b ₄	15.852	15.852	1738	2864	-0.002	0.047	0.047	0.955	0.313
b ₅	16.025	16.025	1840	2807	0.003	0.049	0.049	0.952	0.308
b ₆	16.350	16.350	1980	2738	0.009	0.049	0.049	0.951	0.301
b ₇	16.274	16.274	2037	2515	0.014	0.049	0.051	0.949	0.302
b ₈	16.244	16.244	2091	2279	0.016	0.048	0.051	0.947	0.304
b ₉	16.123	16.123	2107	2009	0.014	0.047	0.049	0.951	0.311
b ₁₀	15.915	15.915	2087	1714	0.007	0.047	0.047	0.954	0.318
b ₁₁	16.264	16.264	2148	1536	-0.002	0.047	0.047	0.955	0.329
b ₁₂	16.237	16.237	2145	1881	-0.009	0.047	0.048	0.953	0.319
b ₁₃	16.776	16.776	2209	2356	-0.013	0.047	0.049	0.960	0.308
b ₁₄	16.977	16.977	2166	2747	-0.012	0.049	0.051	0.957	0.308
b ₁₅	16.647	16.647	1996	2940	-0.005	0.053	0.053	0.956	0.321
b ₁₆	15.901	15.901	1757	2891	0.003	0.052	0.052	0.958	0.340
b ₁₇	15.588	15.588	1630	2851	0.005	0.052	0.052	0.950	0.351
b ₁₈	15.786	15.786	1612	2884	0.003	0.051	0.051	0.950	0.353
b ₁₉	16.349	16.349	1641	2994	0.001	0.051	0.052	0.952	0.352
b ₂₀	17.026	17.026	1658	3109	0.000	0.053	0.053	0.950	0.354
b ₂₁	17.757	17.757	1651	3197	0.000	0.054	0.054	0.956	0.359
Uniform								0.943	0.491
WBATE					0.001	0.020	0.020	0.947	0.106
LBATE					0.054	0.041	0.068	0.984	0.551

Table SA-80: Intention-to-Treat Effects (Simulations, Quadratic Heteroskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.940	18.940	2038	4030	-0.001	0.028	0.028	0.947	0.214
b₂	18.560	18.560	2046	3894	-0.001	0.029	0.029	0.945	0.217
b₃	18.341	18.341	2087	3764	-0.001	0.029	0.029	0.944	0.219
b₄	18.247	18.247	2164	3620	-0.001	0.030	0.030	0.945	0.220
b₅	18.265	18.265	2269	3460	-0.002	0.030	0.030	0.942	0.222
b₆	18.404	18.404	2394	3295	-0.002	0.030	0.030	0.944	0.223
b₇	18.492	18.492	2494	3084	-0.001	0.030	0.030	0.944	0.226
b₈	18.409	18.409	2536	2801	-0.001	0.032	0.032	0.948	0.233
b₉	18.445	18.445	2577	2534	0.000	0.033	0.033	0.945	0.241
b₁₀	18.570	18.570	2606	2280	0.000	0.036	0.036	0.945	0.261
b₁₁	20.183	20.183	2867	2376	-0.001	0.038	0.038	0.944	0.298
b₁₂	18.471	18.471	2589	2371	-0.004	0.035	0.035	0.945	0.251
b₁₃	18.461	18.461	2555	2759	-0.006	0.031	0.032	0.943	0.233
b₁₄	18.567	18.567	2494	3154	-0.007	0.029	0.030	0.943	0.223
b₁₅	18.485	18.485	2354	3449	-0.005	0.029	0.029	0.947	0.218
b₁₆	18.323	18.323	2178	3645	-0.002	0.029	0.029	0.948	0.217
b₁₇	18.304	18.304	2048	3779	-0.001	0.029	0.029	0.947	0.217
b₁₈	18.620	18.620	2006	3912	0.000	0.029	0.029	0.945	0.214
b₁₉	19.300	19.300	2021	4091	0.001	0.028	0.028	0.948	0.209
b₂₀	20.308	20.308	2057	4323	0.001	0.027	0.027	0.945	0.202
b₂₁	21.402	21.402	2067	4519	0.002	0.026	0.026	0.949	0.197
Uniform								0.935	0.339
WBATE					-0.001	0.015	0.015	0.950	0.086
LBATE					0.010	0.023	0.025	0.993	0.329

Table SA-81: First-Stage Effects (Simulations, Linear Homoskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.446	18.446	1967	3834	-0.001	0.029	0.029	0.947	0.218
b₂	17.301	17.301	1843	3422	-0.001	0.030	0.030	0.947	0.230
b₃	15.833	15.833	1650	2887	-0.001	0.033	0.033	0.943	0.249
b₄	13.993	13.993	1382	2255	-0.000	0.038	0.038	0.942	0.282
b₅	12.000	12.000	1081	1641	-0.001	0.044	0.044	0.940	0.331
b₆	10.000	10.000	788	1113	-0.001	0.053	0.053	0.948	0.401
b₇	8.375	8.375	570	751	-0.002	0.064	0.064	0.941	0.487
b₈	8.066	8.066	554	636	-0.002	0.070	0.070	0.931	0.520
b₉	8.081	8.081	595	543	-0.000	0.073	0.073	0.932	0.539
b₁₀	8.136	8.136	646	441	-0.001	0.078	0.078	0.931	0.564
b₁₁	8.842	8.842	794	394	-0.001	0.093	0.093	0.916	0.697
b₁₂	8.092	8.092	623	480	-0.004	0.075	0.075	0.931	0.554
b₁₃	8.088	8.088	564	622	-0.002	0.070	0.070	0.933	0.521
b₁₄	8.563	8.563	593	793	-0.002	0.062	0.063	0.939	0.471
b₁₅	11.200	11.200	961	1419	-0.000	0.047	0.047	0.936	0.353
b₁₆	13.993	13.993	1381	2257	-0.000	0.037	0.037	0.942	0.279
b₁₇	16.467	16.467	1737	3117	-0.000	0.031	0.031	0.949	0.237
b₁₈	18.090	18.090	1927	3704	0.000	0.029	0.029	0.946	0.219
b₁₉	19.259	19.259	2016	4074	0.001	0.028	0.028	0.948	0.209
b₂₀	20.308	20.308	2057	4323	0.001	0.027	0.027	0.945	0.202
b₂₁	21.402	21.402	2067	4519	0.002	0.026	0.026	0.949	0.197
Uniform								0.881	0.565
WBATE					-0.001	0.019	0.019	0.951	0.111
LBATE					0.037	0.037	0.052	0.976	0.665

Table SA-82: First-Stage Effects (Simulations, Linear Homoskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.298	8.298	578	774	-0.001	0.066	0.066	0.948	0.253
b₂	8.131	8.131	566	754	-0.001	0.066	0.066	0.944	0.256
b₃	8.035	8.035	559	741	-0.001	0.067	0.067	0.940	0.258
b₄	7.994	7.994	553	733	0.000	0.068	0.068	0.938	0.259
b₅	8.002	8.002	549	726	-0.000	0.068	0.068	0.942	0.259
b₆	8.063	8.063	546	723	-0.002	0.066	0.066	0.947	0.259
b₇	8.101	8.101	540	706	-0.003	0.066	0.066	0.949	0.261
b₈	8.065	8.065	554	636	-0.002	0.070	0.070	0.943	0.270
b₉	8.081	8.081	595	543	-0.000	0.073	0.073	0.942	0.280
b₁₀	8.136	8.136	646	441	-0.001	0.078	0.078	0.942	0.302
b₁₁	8.842	8.842	794	394	-0.001	0.093	0.093	0.940	0.359
b₁₂	8.092	8.092	623	480	-0.004	0.075	0.075	0.941	0.290
b₁₃	8.088	8.088	564	622	-0.002	0.070	0.070	0.946	0.270
b₁₄	8.134	8.134	544	719	-0.002	0.067	0.067	0.940	0.258
b₁₅	8.098	8.098	556	739	-0.000	0.067	0.067	0.939	0.254
b₁₆	8.028	8.028	556	738	0.000	0.066	0.066	0.944	0.255
b₁₇	8.019	8.019	555	737	-0.000	0.065	0.065	0.948	0.255
b₁₈	8.157	8.157	562	751	-0.002	0.066	0.066	0.940	0.252
b₁₉	8.456	8.456	580	782	-0.002	0.064	0.064	0.946	0.246
b₂₀	8.897	8.897	604	827	-0.001	0.062	0.062	0.947	0.239
b₂₁	9.376	9.376	621	865	0.000	0.061	0.061	0.943	0.233
Uniform								0.920	0.404
WBATE					-0.001	0.022	0.022	0.956	0.088
LBATE					0.071	0.042	0.083	0.989	0.394

Table SA-83: First-Stage Effects (Simulations, Linear Homoskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	18.446	18.446	1976	3846	-0.001	0.029	0.029	0.946	0.222
b ₂	18.337	18.337	2020	3819	-0.001	0.030	0.030	0.945	0.222
b ₃	18.348	18.348	2101	3775	-0.001	0.030	0.030	0.946	0.223
b ₄	18.488	18.488	2219	3716	-0.001	0.030	0.030	0.945	0.222
b ₅	18.737	18.737	2365	3637	-0.002	0.030	0.030	0.944	0.222
b ₆	18.966	18.966	2501	3501	-0.002	0.030	0.030	0.945	0.222
b ₇	19.098	19.098	2604	3300	-0.001	0.031	0.031	0.946	0.225
b ₈	19.288	19.288	2695	3089	-0.000	0.032	0.032	0.947	0.228
b ₉	19.613	19.613	2782	2892	0.000	0.032	0.032	0.945	0.233
b ₁₀	20.205	20.205	2884	2747	0.000	0.034	0.034	0.942	0.247
b ₁₁	21.806	21.806	3110	2852	-0.001	0.037	0.037	0.943	0.282
b ₁₂	19.941	19.941	2845	2797	-0.004	0.033	0.034	0.947	0.239
b ₁₃	19.520	19.520	2749	3093	-0.006	0.031	0.031	0.941	0.226
b ₁₄	19.298	19.298	2631	3407	-0.007	0.029	0.030	0.943	0.220
b ₁₅	19.009	19.009	2456	3648	-0.005	0.030	0.030	0.944	0.219
b ₁₆	18.515	18.515	2227	3730	-0.003	0.030	0.030	0.947	0.221
b ₁₇	18.115	18.115	2033	3722	-0.001	0.030	0.030	0.951	0.224
b ₁₈	18.047	18.047	1931	3705	-0.000	0.030	0.030	0.943	0.223
b ₁₉	18.280	18.280	1892	3705	0.000	0.030	0.030	0.947	0.222
b ₂₀	18.825	18.825	1878	3762	0.001	0.029	0.029	0.945	0.220
b ₂₁	19.374	19.374	1839	3766	0.002	0.029	0.030	0.946	0.220
Uniform								0.930	0.341
WBATE					-0.002	0.015	0.015	0.950	0.086
LBATE					0.012	0.025	0.028	0.992	0.350

Table SA-84: First-Stage Effects (Simulations, Linear Homoskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.488	19.488	2114	4244	-0.001	0.027	0.027	0.949	0.208
b₂	18.846	18.846	2089	4002	-0.001	0.028	0.028	0.949	0.213
b₃	18.345	18.345	2087	3766	-0.001	0.029	0.029	0.950	0.218
b₄	17.972	17.972	2110	3530	-0.001	0.030	0.030	0.951	0.223
b₅	17.684	17.684	2151	3279	-0.001	0.031	0.031	0.948	0.228
b₆	17.475	17.475	2202	3021	-0.001	0.031	0.032	0.943	0.234
b₇	17.293	17.293	2244	2747	-0.001	0.033	0.033	0.945	0.242
b₈	16.916	16.916	2229	2406	-0.001	0.035	0.035	0.943	0.253
b₉	16.732	16.732	2233	2109	0.000	0.037	0.037	0.944	0.265
b₁₀	16.448	16.448	2194	1787	0.000	0.041	0.041	0.939	0.293
b₁₁	16.857	16.857	2265	1614	-0.000	0.048	0.048	0.941	0.360
b₁₂	16.613	16.613	2223	1927	-0.004	0.039	0.039	0.942	0.277
b₁₃	16.893	16.893	2234	2351	-0.005	0.034	0.034	0.945	0.254
b₁₄	17.334	17.334	2239	2803	-0.005	0.032	0.032	0.946	0.238
b₁₅	17.565	17.565	2165	3168	-0.004	0.031	0.031	0.941	0.229
b₁₆	17.929	17.929	2101	3515	-0.001	0.030	0.030	0.939	0.221
b₁₇	18.505	18.505	2078	3852	0.000	0.029	0.029	0.944	0.213
b₁₈	19.329	19.329	2103	4188	0.001	0.028	0.028	0.947	0.205
b₁₉	20.492	20.492	2167	4569	0.002	0.026	0.026	0.949	0.196
b₂₀	21.968	21.968	2239	4998	0.003	0.025	0.025	0.946	0.186
b₂₁	23.598	23.598	2285	5410	0.004	0.024	0.024	0.945	0.178
Uniform								0.927	0.353
WBATE					-0.001	0.016	0.016	0.948	0.088
LBATE					0.011	0.022	0.024	0.993	0.347

Table SA-85: First-Stage Effects (Simulations, Linear Heteroskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.811	18.811	2015	3976	-0.001	0.028	0.028	0.950	0.214
b₂	17.441	17.441	1863	3475	-0.001	0.030	0.030	0.951	0.228
b₃	15.849	15.849	1652	2894	-0.000	0.032	0.032	0.953	0.249
b₄	13.990	13.990	1381	2255	-0.000	0.037	0.037	0.946	0.282
b₅	12.000	12.000	1081	1641	-0.000	0.043	0.043	0.945	0.331
b₆	10.000	10.000	788	1112	-0.001	0.052	0.052	0.941	0.401
b₇	8.099	8.099	534	703	-0.000	0.067	0.066	0.940	0.503
b₈	7.421	7.421	465	547	0.000	0.075	0.075	0.933	0.566
b₉	7.330	7.330	487	456	-0.002	0.079	0.079	0.935	0.592
b₁₀	7.206	7.206	510	349	-0.002	0.088	0.088	0.922	0.637
b₁₁	7.385	7.385	572	266	-0.001	0.115	0.115	0.904	0.834
b₁₂	7.278	7.278	504	395	-0.003	0.083	0.083	0.926	0.614
b₁₃	7.402	7.402	469	531	-0.000	0.075	0.075	0.934	0.568
b₁₄	8.414	8.414	574	766	0.001	0.063	0.063	0.934	0.478
b₁₅	11.200	11.200	962	1419	0.000	0.047	0.047	0.940	0.353
b₁₆	13.989	13.989	1381	2256	0.000	0.037	0.037	0.941	0.279
b₁₇	16.557	16.557	1750	3151	0.001	0.032	0.032	0.939	0.236
b₁₈	18.615	18.615	1998	3908	0.001	0.028	0.028	0.946	0.212
b₁₉	20.332	20.332	2147	4503	0.002	0.026	0.026	0.949	0.197
b₂₀	21.947	21.947	2236	4989	0.003	0.025	0.025	0.946	0.186
b₂₁	23.593	23.593	2284	5408	0.004	0.024	0.024	0.945	0.178
Uniform								0.867	0.592
WBATE					0.000	0.020	0.020	0.945	0.114
LBATE					0.046	0.045	0.064	0.970	0.759

Table SA-86: First-Stage Effects (Simulations, Linear Heteroskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.538	8.538	607	820	-0.001	0.063	0.063	0.950	0.245
b₂	8.257	8.257	582	778	-0.000	0.064	0.064	0.951	0.252
b₃	8.037	8.037	559	741	0.000	0.067	0.067	0.946	0.257
b₄	7.874	7.874	538	709	-0.000	0.069	0.069	0.943	0.262
b₅	7.747	7.747	518	679	-0.001	0.069	0.069	0.945	0.268
b₆	7.656	7.656	498	650	-0.001	0.070	0.070	0.946	0.273
b₇	7.576	7.576	477	618	-0.000	0.072	0.072	0.941	0.280
b₈	7.411	7.411	465	546	0.000	0.075	0.075	0.947	0.293
b₉	7.330	7.330	487	456	-0.002	0.079	0.079	0.944	0.308
b₁₀	7.206	7.206	510	349	-0.002	0.088	0.088	0.936	0.338
b₁₁	7.385	7.385	572	266	-0.001	0.115	0.115	0.929	0.434
b₁₂	7.278	7.278	504	395	-0.003	0.083	0.083	0.944	0.321
b₁₃	7.401	7.401	468	530	-0.000	0.075	0.075	0.945	0.295
b₁₄	7.594	7.594	480	625	0.001	0.071	0.071	0.943	0.276
b₁₅	7.695	7.695	508	665	-0.000	0.070	0.070	0.942	0.268
b₁₆	7.855	7.855	536	705	0.001	0.067	0.067	0.943	0.260
b₁₇	8.107	8.107	566	752	0.002	0.066	0.066	0.943	0.251
b₁₈	8.468	8.468	601	809	0.000	0.063	0.063	0.944	0.242
b₁₉	8.978	8.978	644	883	0.000	0.059	0.059	0.948	0.231
b₂₀	9.624	9.624	692	970	0.002	0.057	0.057	0.944	0.220
b₂₁	10.338	10.338	734	1056	0.002	0.055	0.055	0.940	0.210
Uniform								0.909	0.420
WBATE					-0.000	0.023	0.023	0.949	0.090
LBATE					0.076	0.045	0.088	0.984	0.420

Table SA-87: First-Stage Effects (Simulations, Linear Heteroskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	18.343	18.343	1960	3818	-0.001	0.029	0.029	0.947	0.224
b ₂	18.051	18.051	1977	3719	-0.001	0.030	0.030	0.950	0.227
b ₃	17.855	17.855	2018	3608	-0.001	0.031	0.031	0.948	0.229
b ₄	17.785	17.785	2090	3485	-0.001	0.031	0.031	0.947	0.231
b ₅	17.710	17.710	2166	3313	-0.001	0.032	0.032	0.948	0.234
b ₆	17.660	17.660	2243	3107	-0.002	0.032	0.033	0.942	0.238
b ₇	17.157	17.157	2217	2745	-0.001	0.034	0.034	0.943	0.250
b ₈	16.803	16.803	2205	2415	-0.001	0.036	0.036	0.945	0.262
b ₉	16.462	16.462	2176	2085	-0.000	0.039	0.039	0.944	0.277
b ₁₀	16.219	16.219	2146	1778	0.000	0.043	0.043	0.936	0.304
b ₁₁	16.492	16.492	2191	1581	-0.001	0.051	0.051	0.940	0.379
b ₁₂	16.520	16.520	2200	1947	-0.004	0.040	0.040	0.942	0.285
b ₁₃	17.243	17.243	2302	2481	-0.006	0.035	0.035	0.945	0.256
b ₁₄	17.877	17.877	2349	2999	-0.005	0.032	0.033	0.945	0.238
b ₁₅	18.466	18.466	2354	3483	-0.004	0.031	0.031	0.939	0.226
b ₁₆	18.673	18.673	2263	3788	-0.002	0.031	0.031	0.939	0.220
b ₁₇	18.855	18.855	2164	3992	-0.000	0.030	0.030	0.941	0.217
b ₁₈	19.160	19.160	2095	4145	0.001	0.029	0.029	0.945	0.213
b ₁₉	19.705	19.705	2067	4290	0.002	0.028	0.028	0.948	0.210
b ₂₀	20.543	20.543	2067	4467	0.003	0.027	0.027	0.945	0.205
b ₂₁	21.462	21.462	2050	4606	0.004	0.028	0.028	0.947	0.202
Uniform								0.923	0.367
WBATE					-0.001	0.016	0.016	0.946	0.089
LBATE					0.014	0.025	0.029	0.992	0.383

Table SA-88: First-Stage Effects (Simulations, Linear Heteroskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.240	17.240	1807	3381	-0.003	0.031	0.031	0.947	0.235
b₂	16.756	16.756	1775	3231	-0.003	0.032	0.032	0.951	0.240
b₃	16.586	16.586	1790	3156	-0.003	0.032	0.032	0.947	0.242
b₄	16.714	16.714	1873	3125	-0.002	0.032	0.032	0.946	0.241
b₅	17.111	17.111	2036	3110	0.001	0.032	0.032	0.947	0.237
b₆	17.598	17.598	2226	3064	0.004	0.031	0.032	0.945	0.233
b₇	18.024	18.024	2395	2954	0.006	0.031	0.032	0.948	0.232
b₈	18.133	18.133	2479	2728	0.007	0.032	0.032	0.946	0.235
b₉	18.233	18.233	2535	2481	0.005	0.033	0.034	0.945	0.242
b₁₀	18.423	18.423	2578	2245	0.001	0.036	0.036	0.949	0.260
b₁₁	20.110	20.110	2855	2358	-0.005	0.038	0.039	0.948	0.296
b₁₂	18.393	18.393	2574	2351	-0.007	0.035	0.035	0.947	0.248
b₁₃	18.366	18.366	2537	2735	-0.007	0.031	0.032	0.944	0.231
b₁₄	18.343	18.343	2448	3092	-0.006	0.030	0.031	0.937	0.223
b₁₅	17.668	17.668	2187	3205	-0.004	0.030	0.030	0.947	0.226
b₁₆	16.748	16.748	1880	3135	-0.002	0.031	0.031	0.943	0.234
b₁₇	16.162	16.162	1707	3017	-0.002	0.032	0.032	0.946	0.242
b₁₈	16.176	16.176	1668	2995	-0.001	0.032	0.032	0.945	0.242
b₁₉	16.736	16.736	1694	3114	-0.001	0.031	0.031	0.944	0.237
b₂₀	17.551	17.551	1727	3273	-0.001	0.030	0.030	0.950	0.231
b₂₁	18.491	18.491	1744	3422	-0.001	0.030	0.030	0.948	0.226
Uniform								0.926	0.360
WBATE					-0.001	0.015	0.016	0.948	0.088
LBATE					0.016	0.025	0.030	0.986	0.357

Table SA-89: First-Stage Effects (Simulations, Quadratic Homoskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.072	17.072	1783	3314	-0.003	0.031	0.031	0.947	0.237
b₂	16.314	16.314	1705	3064	-0.003	0.032	0.032	0.951	0.245
b₃	15.415	15.415	1589	2746	-0.002	0.034	0.034	0.947	0.257
b₄	13.947	13.947	1374	2243	-0.002	0.037	0.037	0.943	0.284
b₅	12.000	12.000	1080	1642	-0.001	0.044	0.044	0.942	0.331
b₆	10.000	10.000	787	1114	-0.000	0.053	0.053	0.943	0.400
b₇	8.292	8.292	559	737	-0.000	0.066	0.065	0.935	0.489
b₈	7.946	7.946	537	620	0.001	0.070	0.070	0.934	0.526
b₉	7.988	7.988	582	533	0.003	0.072	0.072	0.928	0.540
b₁₀	8.071	8.071	636	435	0.003	0.076	0.077	0.931	0.563
b₁₁	8.811	8.811	788	391	-0.002	0.091	0.091	0.915	0.691
b₁₂	8.058	8.058	618	477	-0.002	0.073	0.073	0.932	0.548
b₁₃	8.047	8.047	558	616	-0.000	0.069	0.069	0.930	0.514
b₁₄	8.537	8.537	590	788	0.000	0.062	0.062	0.934	0.465
b₁₅	11.200	11.200	962	1419	-0.001	0.046	0.046	0.945	0.348
b₁₆	13.954	13.954	1376	2245	-0.002	0.037	0.037	0.939	0.276
b₁₇	15.604	15.604	1614	2816	-0.002	0.033	0.033	0.944	0.248
b₁₈	16.095	16.095	1656	2963	-0.001	0.032	0.032	0.945	0.243
b₁₉	16.730	16.730	1693	3111	-0.001	0.031	0.031	0.944	0.237
b₂₀	17.551	17.551	1727	3273	-0.001	0.030	0.030	0.950	0.231
b₂₁	18.491	18.491	1744	3422	-0.001	0.030	0.030	0.948	0.226
Uniform								0.879	0.576
WBATE					-0.001	0.019	0.019	0.948	0.112
LBATE					0.047	0.040	0.062	0.962	0.675

Table SA-90: First-Stage Effects (Simulations, Quadratic Homoskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	7.553	7.553	489	640	0.001	0.072	0.072	0.942	0.279
b₂	7.341	7.341	472	613	0.001	0.074	0.074	0.944	0.284
b₃	7.267	7.267	467	605	0.000	0.074	0.074	0.941	0.286
b₄	7.322	7.322	473	615	-0.001	0.073	0.073	0.946	0.284
b₅	7.496	7.496	489	639	0.000	0.072	0.072	0.943	0.278
b₆	7.710	7.710	505	664	-0.000	0.070	0.070	0.944	0.271
b₇	7.896	7.896	516	673	-0.000	0.070	0.070	0.943	0.267
b₈	7.944	7.944	537	620	0.001	0.070	0.070	0.942	0.273
b₉	7.988	7.988	582	533	0.003	0.072	0.072	0.945	0.281
b₁₀	8.071	8.071	636	435	0.003	0.076	0.077	0.946	0.301
b₁₁	8.811	8.811	788	391	-0.002	0.091	0.091	0.939	0.356
b₁₂	8.058	8.058	618	477	-0.002	0.073	0.073	0.949	0.287
b₁₃	8.046	8.046	558	616	-0.000	0.069	0.069	0.943	0.268
b₁₄	8.036	8.036	533	702	0.000	0.067	0.067	0.940	0.257
b₁₅	7.741	7.741	515	675	-0.001	0.068	0.068	0.947	0.264
b₁₆	7.337	7.337	476	616	-0.001	0.073	0.073	0.937	0.276
b₁₇	7.081	7.081	445	572	-0.001	0.074	0.074	0.941	0.285
b₁₈	7.087	7.087	438	562	0.001	0.075	0.075	0.938	0.286
b₁₉	7.332	7.332	451	585	0.002	0.072	0.072	0.942	0.280
b₂₀	7.689	7.689	468	614	0.000	0.071	0.071	0.943	0.273
b₂₁	8.101	8.101	482	641	-0.001	0.069	0.069	0.944	0.267
Uniform								0.909	0.429
WBATE					0.000	0.023	0.023	0.946	0.090
LBATE					0.088	0.046	0.099	0.980	0.419

Table SA-91: First-Stage Effects (Simulations, Quadratic Homoskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	16.410	16.410	1693	3079	-0.003	0.033	0.033	0.945	0.248
b ₂	16.047	16.047	1671	2986	-0.003	0.033	0.034	0.950	0.253
b ₃	15.948	15.948	1694	2949	-0.002	0.034	0.034	0.945	0.255
b ₄	16.089	16.089	1775	2937	-0.001	0.034	0.034	0.947	0.255
b ₅	16.495	16.495	1926	2946	0.001	0.034	0.034	0.946	0.252
b ₆	17.158	17.158	2138	2971	0.003	0.034	0.034	0.947	0.246
b ₇	17.721	17.721	2326	2913	0.005	0.033	0.034	0.945	0.242
b ₈	18.359	18.359	2512	2837	0.006	0.033	0.033	0.947	0.239
b ₉	18.990	18.990	2663	2730	0.005	0.033	0.033	0.945	0.239
b ₁₀	19.803	19.803	2812	2645	0.000	0.035	0.035	0.947	0.249
b ₁₁	21.462	21.462	3055	2761	-0.006	0.037	0.038	0.944	0.284
b ₁₂	19.640	19.640	2792	2714	-0.008	0.033	0.034	0.948	0.239
b ₁₃	19.005	19.005	2650	2951	-0.008	0.031	0.032	0.944	0.229
b ₁₄	18.130	18.130	2396	3075	-0.006	0.032	0.032	0.937	0.232
b ₁₅	16.883	16.883	2036	3009	-0.004	0.033	0.033	0.946	0.244
b ₁₆	15.616	15.616	1700	2794	-0.002	0.035	0.035	0.940	0.258
b ₁₇	15.047	15.047	1543	2651	-0.002	0.035	0.035	0.944	0.263
b ₁₈	15.155	15.155	1525	2644	-0.001	0.034	0.034	0.941	0.260
b ₁₉	15.676	15.676	1553	2737	-0.001	0.033	0.033	0.944	0.254
b ₂₀	16.282	16.282	1566	2825	-0.001	0.033	0.033	0.946	0.250
b ₂₁	16.774	16.774	1539	2833	-0.001	0.033	0.033	0.947	0.250
Uniform								0.927	0.376
WBATE					-0.001	0.016	0.016	0.947	0.090
LBATE					0.020	0.027	0.033	0.987	0.387

Table SA-92: First-Stage Effects (Simulations, Quadratic Homoskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.687	17.687	1868	3552	-0.004	0.030	0.030	0.950	0.230
b₂	17.077	17.077	1823	3350	-0.004	0.031	0.031	0.947	0.236
b₃	16.696	16.696	1810	3195	-0.003	0.031	0.031	0.950	0.241
b₄	16.579	16.579	1851	3081	-0.002	0.032	0.032	0.946	0.243
b₅	16.641	16.641	1943	2966	0.001	0.032	0.032	0.943	0.243
b₆	16.776	16.776	2057	2824	0.003	0.032	0.032	0.944	0.244
b₇	16.861	16.861	2154	2632	0.006	0.032	0.033	0.949	0.247
b₈	16.634	16.634	2170	2336	0.007	0.034	0.034	0.944	0.256
b₉	16.538	16.538	2193	2064	0.006	0.036	0.037	0.938	0.266
b₁₀	16.270	16.270	2159	1750	0.002	0.041	0.041	0.944	0.292
b₁₁	16.701	16.701	2235	1582	-0.003	0.047	0.047	0.945	0.360
b₁₂	16.495	16.495	2199	1901	-0.006	0.038	0.039	0.945	0.275
b₁₃	16.770	16.770	2209	2319	-0.006	0.033	0.034	0.947	0.253
b₁₄	17.120	17.120	2193	2745	-0.005	0.031	0.032	0.950	0.238
b₁₅	16.936	16.936	2037	2983	-0.004	0.031	0.031	0.945	0.235
b₁₆	16.613	16.613	1856	3091	-0.003	0.031	0.031	0.947	0.236
b₁₇	16.555	16.555	1767	3158	-0.002	0.031	0.031	0.952	0.236
b₁₈	16.909	16.909	1769	3267	-0.002	0.030	0.030	0.949	0.232
b₁₉	17.755	17.755	1827	3495	-0.002	0.029	0.029	0.949	0.224
b₂₀	18.817	18.817	1881	3748	-0.002	0.029	0.029	0.940	0.216
b₂₁	20.034	20.034	1917	3997	-0.001	0.027	0.027	0.945	0.209
Uniform								0.933	0.374
WBATE					-0.001	0.016	0.016	0.949	0.090
LBATE					0.015	0.024	0.029	0.990	0.375

Table SA-93: First-Stage Effects (Simulations, Quadratic Heteroskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.460	17.460	1835	3461	-0.004	0.030	0.031	0.951	0.232
b₂	16.507	16.507	1732	3136	-0.004	0.031	0.032	0.948	0.242
b₃	15.414	15.414	1590	2748	-0.003	0.033	0.033	0.952	0.257
b₄	13.916	13.916	1371	2234	-0.002	0.037	0.037	0.946	0.284
b₅	11.999	11.999	1082	1642	-0.000	0.043	0.043	0.943	0.331
b₆	10.000	10.000	788	1113	0.001	0.053	0.053	0.939	0.400
b₇	8.061	8.061	529	697	0.001	0.067	0.067	0.934	0.501
b₈	7.300	7.300	450	532	0.002	0.078	0.078	0.921	0.569
b₉	7.245	7.245	476	447	0.006	0.082	0.082	0.922	0.593
b₁₀	7.128	7.128	500	343	0.005	0.088	0.088	0.922	0.633
b₁₁	7.317	7.317	563	261	-0.001	0.112	0.112	0.902	0.832
b₁₂	7.227	7.227	497	390	-0.002	0.083	0.083	0.921	0.609
b₁₃	7.348	7.348	461	524	0.000	0.075	0.075	0.927	0.562
b₁₄	8.409	8.409	573	765	-0.000	0.062	0.062	0.932	0.472
b₁₅	11.200	11.200	962	1418	-0.001	0.046	0.046	0.936	0.348
b₁₆	13.919	13.919	1371	2234	-0.001	0.036	0.036	0.943	0.277
b₁₇	15.793	15.793	1640	2883	-0.002	0.032	0.032	0.951	0.245
b₁₈	16.766	16.766	1748	3211	-0.002	0.030	0.030	0.949	0.234
b₁₉	17.739	17.739	1825	3488	-0.002	0.029	0.029	0.949	0.225
b₂₀	18.816	18.816	1881	3747	-0.002	0.029	0.029	0.940	0.216
b₂₁	20.034	20.034	1917	3997	-0.001	0.027	0.027	0.945	0.209
Uniform								0.862	0.603
WBATE					-0.000	0.020	0.020	0.941	0.115
LBATE					0.058	0.049	0.076	0.959	0.769

Table SA-94: First-Stage Effects (Simulations, Quadratic Heteroskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	7.749	7.749	513	675	-0.002	0.071	0.071	0.941	0.272
b₂	7.482	7.482	490	638	-0.002	0.072	0.072	0.946	0.280
b₃	7.315	7.315	474	614	-0.001	0.072	0.072	0.950	0.285
b₄	7.264	7.264	467	604	-0.000	0.074	0.074	0.944	0.286
b₅	7.290	7.290	465	602	0.002	0.074	0.074	0.944	0.286
b₆	7.350	7.350	463	600	0.002	0.073	0.073	0.944	0.285
b₇	7.387	7.387	455	589	0.001	0.074	0.074	0.944	0.286
b₈	7.287	7.287	449	530	0.002	0.078	0.078	0.932	0.296
b₉	7.245	7.245	476	447	0.006	0.082	0.082	0.937	0.309
b₁₀	7.128	7.128	500	343	0.005	0.088	0.088	0.934	0.337
b₁₁	7.317	7.317	563	261	-0.001	0.112	0.112	0.938	0.432
b₁₂	7.227	7.227	497	390	-0.002	0.083	0.083	0.941	0.318
b₁₃	7.347	7.347	461	523	0.000	0.075	0.075	0.942	0.292
b₁₄	7.501	7.501	469	610	0.000	0.071	0.071	0.947	0.276
b₁₅	7.420	7.420	477	619	0.001	0.072	0.072	0.938	0.275
b₁₆	7.278	7.278	468	606	0.002	0.072	0.072	0.938	0.278
b₁₇	7.253	7.253	465	601	-0.001	0.072	0.072	0.948	0.279
b₁₈	7.408	7.408	475	617	-0.002	0.070	0.070	0.947	0.275
b₁₉	7.779	7.779	502	661	-0.001	0.068	0.068	0.945	0.265
b₂₀	8.244	8.244	530	710	-0.001	0.067	0.067	0.943	0.255
b₂₁	8.777	8.777	556	759	-0.001	0.064	0.064	0.945	0.246
Uniform								0.905	0.445
WBATE					0.000	0.024	0.024	0.948	0.092
LBATE					0.091	0.048	0.103	0.977	0.444

Table SA-95: First-Stage Effects (Simulations, Quadratic Heteroskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	16.600	16.600	1719	3154	-0.004	0.032	0.033	0.950	0.246
b₂	16.166	16.166	1690	3035	-0.003	0.033	0.033	0.946	0.252
b₃	15.882	15.882	1686	2931	-0.003	0.034	0.034	0.953	0.257
b₄	15.852	15.852	1738	2864	-0.001	0.035	0.035	0.949	0.260
b₅	16.025	16.025	1840	2807	0.000	0.035	0.035	0.944	0.260
b₆	16.350	16.350	1980	2738	0.003	0.035	0.035	0.945	0.258
b₇	16.274	16.274	2037	2515	0.006	0.035	0.036	0.949	0.263
b₈	16.244	16.244	2091	2279	0.007	0.037	0.037	0.943	0.270
b₉	16.123	16.123	2107	2009	0.006	0.038	0.039	0.940	0.280
b₁₀	15.915	15.915	2087	1714	0.002	0.042	0.042	0.940	0.305
b₁₁	16.264	16.264	2148	1536	-0.003	0.050	0.050	0.942	0.380
b₁₂	16.237	16.237	2145	1881	-0.006	0.040	0.040	0.944	0.286
b₁₃	16.776	16.776	2209	2356	-0.006	0.035	0.035	0.944	0.258
b₁₄	16.977	16.977	2166	2747	-0.006	0.033	0.033	0.946	0.248
b₁₅	16.647	16.647	1996	2940	-0.004	0.033	0.034	0.944	0.249
b₁₆	15.901	15.901	1757	2891	-0.002	0.034	0.034	0.942	0.256
b₁₇	15.588	15.588	1630	2851	-0.002	0.034	0.034	0.950	0.257
b₁₈	15.786	15.786	1612	2884	-0.002	0.033	0.033	0.947	0.252
b₁₉	16.349	16.349	1641	2994	-0.002	0.032	0.032	0.949	0.246
b₂₀	17.026	17.026	1658	3109	-0.002	0.032	0.032	0.941	0.241
b₂₁	17.757	17.757	1651	3197	-0.001	0.032	0.032	0.945	0.239
Uniform								0.927	0.401
WBATE					-0.001	0.016	0.016	0.947	0.093
LBATE					0.021	0.027	0.034	0.987	0.420

Table SA-96: First-Stage Effects (Simulations, Quadratic Heteroskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	18.940	18.940	2038	4030	0.002	0.056	0.056	0.958	0.408
b ₂	18.560	18.560	2046	3894	0.002	0.057	0.057	0.960	0.412
b ₃	18.341	18.341	2087	3764	0.003	0.058	0.058	0.959	0.414
b ₄	18.247	18.247	2164	3620	0.005	0.060	0.060	0.960	0.415
b ₅	18.265	18.265	2269	3460	0.009	0.060	0.060	0.962	0.414
b ₆	18.404	18.404	2394	3295	0.013	0.059	0.060	0.959	0.414
b ₇	18.492	18.492	2494	3084	0.016	0.059	0.061	0.958	0.418
b ₈	18.409	18.409	2536	2801	0.015	0.060	0.062	0.956	0.427
b ₉	18.445	18.445	2577	2534	0.012	0.061	0.062	0.958	0.435
b ₁₀	18.570	18.570	2606	2280	0.007	0.062	0.063	0.967	0.447
b ₁₁	20.183	20.183	2867	2376	0.002	0.063	0.063	0.968	0.461
b ₁₂	18.471	18.471	2589	2371	-0.005	0.061	0.062	0.962	0.437
b ₁₃	18.461	18.461	2555	2759	-0.012	0.060	0.061	0.958	0.415
b ₁₄	18.567	18.567	2494	3154	-0.013	0.057	0.059	0.956	0.394
b ₁₅	18.485	18.485	2354	3449	-0.009	0.056	0.057	0.956	0.382
b ₁₆	18.323	18.323	2178	3645	-0.003	0.055	0.055	0.962	0.380
b ₁₇	18.304	18.304	2048	3779	0.001	0.052	0.052	0.963	0.378
b ₁₈	18.620	18.620	2006	3912	0.002	0.050	0.050	0.959	0.372
b ₁₉	19.300	19.300	2021	4091	0.001	0.048	0.048	0.952	0.361
b ₂₀	20.308	20.308	2057	4323	-0.001	0.048	0.048	0.955	0.347
b ₂₁	21.402	21.402	2067	4519	-0.003	0.047	0.047	0.954	0.336
Uniform								0.971	0.606
WBATE					0.002	0.027	0.027	0.954	0.148
LBATE					0.055	0.045	0.071	0.996	0.670

Table SA-97: Fuzzy Effects (Simulations, Linear Homoskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.446	18.446	1967	3834	0.003	0.057	0.057	0.958	0.417
b₂	17.301	17.301	1843	3422	0.003	0.059	0.059	0.957	0.438
b₃	15.833	15.833	1650	2887	0.003	0.063	0.063	0.954	0.474
b₄	13.993	13.993	1382	2255	0.003	0.070	0.070	0.958	0.535
b₅	12.000	12.000	1081	1641	0.001	0.081	0.081	0.961	0.629
b₆	10.000	10.000	788	1113	-0.000	0.097	0.097	0.957	0.774
b₇	8.375	8.375	570	751	0.001	0.120	0.120	0.967	0.985
b₈	8.066	8.066	554	636	0.004	0.128	0.128	0.967	1.097
b₉	8.081	8.081	595	543	0.008	0.129	0.129	0.976	1.468
b₁₀	8.136	8.136	646	441	0.008	0.133	0.133	0.972	1.622
b₁₁	8.842	8.842	794	394	0.003	0.143	0.143	0.978	28.691
b₁₂	8.092	8.092	623	480	-0.005	0.131	0.131	0.972	1.302
b₁₃	8.088	8.088	564	622	-0.002	0.126	0.126	0.970	1.010
b₁₄	8.563	8.563	593	793	0.004	0.111	0.111	0.966	0.884
b₁₅	11.200	11.200	961	1419	0.004	0.083	0.083	0.958	0.631
b₁₆	13.993	13.993	1381	2257	0.002	0.064	0.064	0.954	0.491
b₁₇	16.467	16.467	1737	3117	0.001	0.055	0.055	0.958	0.414
b₁₈	18.090	18.090	1927	3704	0.001	0.051	0.051	0.959	0.381
b₁₉	19.259	19.259	2016	4074	0.001	0.048	0.048	0.952	0.361
b₂₀	20.308	20.308	2057	4323	-0.001	0.048	0.048	0.955	0.347
b₂₁	21.402	21.402	2067	4519	-0.003	0.047	0.047	0.954	0.336
Uniform								0.978	3.166
WBATE					0.002	0.034	0.034	0.960	1.575
LBATE					0.139	0.090	0.166	0.997	23.429

Table SA-98: Fuzzy Effects (Simulations, Linear Homoskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.298	8.298	578	774	0.005	0.121	0.121	0.955	0.471
b₂	8.131	8.131	566	754	0.005	0.123	0.123	0.952	0.476
b₃	8.035	8.035	559	741	0.004	0.123	0.123	0.957	0.480
b₄	7.994	7.994	553	733	0.002	0.125	0.125	0.959	0.481
b₅	8.002	8.002	549	726	0.000	0.123	0.123	0.959	0.481
b₆	8.063	8.063	546	723	-0.002	0.124	0.124	0.955	0.481
b₇	8.101	8.101	540	706	0.000	0.125	0.125	0.957	0.486
b₈	8.065	8.065	554	636	0.004	0.128	0.128	0.959	0.495
b₉	8.081	8.081	595	543	0.008	0.129	0.129	0.963	0.506
b₁₀	8.136	8.136	646	441	0.008	0.133	0.133	0.961	0.521
b₁₁	8.842	8.842	794	394	0.003	0.143	0.143	0.963	0.551
b₁₂	8.092	8.092	623	480	-0.005	0.131	0.131	0.960	0.509
b₁₃	8.088	8.088	564	622	-0.002	0.126	0.126	0.959	0.481
b₁₄	8.134	8.134	544	719	0.004	0.118	0.118	0.958	0.458
b₁₅	8.098	8.098	556	739	0.004	0.117	0.117	0.957	0.445
b₁₆	8.028	8.028	556	738	0.001	0.114	0.114	0.954	0.440
b₁₇	8.019	8.019	555	737	-0.001	0.111	0.111	0.958	0.435
b₁₈	8.157	8.157	562	751	0.000	0.109	0.109	0.955	0.429
b₁₉	8.456	8.456	580	782	0.002	0.108	0.108	0.952	0.414
b₂₀	8.897	8.897	604	827	0.000	0.102	0.102	0.954	0.397
b₂₁	9.376	9.376	621	865	-0.002	0.099	0.099	0.953	0.383
Uniform								0.973	0.713
WBATE					0.002	0.039	0.039	0.954	0.153
LBATE					0.184	0.089	0.204	0.997	0.815

Table SA-99: Fuzzy Effects (Simulations, Linear Homoskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.446	18.446	1976	3846	0.000	0.060	0.060	0.960	0.423
b₂	18.337	18.337	2020	3819	0.000	0.061	0.061	0.958	0.422
b₃	18.348	18.348	2101	3775	0.001	0.062	0.062	0.962	0.420
b₄	18.488	18.488	2219	3716	0.003	0.063	0.063	0.962	0.418
b₅	18.737	18.737	2365	3637	0.007	0.064	0.064	0.966	0.414
b₆	18.966	18.966	2501	3501	0.010	0.064	0.064	0.958	0.413
b₇	19.098	19.098	2604	3300	0.012	0.064	0.065	0.959	0.417
b₈	19.288	19.288	2695	3089	0.012	0.065	0.066	0.959	0.419
b₉	19.613	19.613	2782	2892	0.010	0.064	0.065	0.964	0.420
b₁₀	20.205	20.205	2884	2747	0.006	0.063	0.064	0.966	0.423
b₁₁	21.806	21.806	3110	2852	0.001	0.065	0.065	0.967	0.439
b₁₂	19.941	19.941	2845	2797	-0.005	0.064	0.064	0.967	0.418
b₁₃	19.520	19.520	2749	3093	-0.011	0.063	0.064	0.960	0.405
b₁₄	19.298	19.298	2631	3407	-0.011	0.061	0.062	0.958	0.389
b₁₅	19.009	19.009	2456	3648	-0.006	0.062	0.062	0.962	0.384
b₁₆	18.515	18.515	2227	3730	-0.000	0.060	0.060	0.962	0.387
b₁₇	18.115	18.115	2033	3722	0.004	0.056	0.056	0.969	0.390
b₁₈	18.047	18.047	1931	3705	0.004	0.054	0.054	0.963	0.389
b₁₉	18.280	18.280	1892	3705	0.003	0.053	0.053	0.954	0.383
b₂₀	18.825	18.825	1878	3762	0.002	0.054	0.054	0.957	0.376
b₂₁	19.374	19.374	1839	3766	-0.000	0.054	0.054	0.956	0.372
Uniform								0.976	0.610
WBATE					0.002	0.027	0.028	0.949	0.149
LBATE					0.060	0.051	0.079	0.998	0.707

Table SA-100: Fuzzy Effects (Simulations, Linear Homoskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	19.488	19.488	2114	4244	-0.000	0.074	0.074	0.963	0.530
b₂	18.846	18.846	2089	4002	-0.000	0.075	0.075	0.960	0.534
b₃	18.345	18.345	2087	3766	0.001	0.076	0.076	0.963	0.537
b₄	17.972	17.972	2110	3530	0.004	0.076	0.076	0.960	0.539
b₅	17.684	17.684	2151	3279	0.008	0.077	0.077	0.957	0.541
b₆	17.475	17.475	2202	3021	0.012	0.077	0.078	0.964	0.543
b₇	17.293	17.293	2244	2747	0.014	0.078	0.080	0.960	0.548
b₈	16.916	16.916	2229	2406	0.014	0.080	0.081	0.959	0.562
b₉	16.732	16.732	2233	2109	0.012	0.080	0.081	0.958	0.571
b₁₀	16.448	16.448	2194	1787	0.006	0.078	0.078	0.958	0.571
b₁₁	16.857	16.857	2265	1614	0.001	0.074	0.074	0.964	0.534
b₁₂	16.613	16.613	2223	1927	-0.005	0.080	0.080	0.961	0.587
b₁₃	16.893	16.893	2234	2351	-0.012	0.084	0.085	0.957	0.584
b₁₄	17.334	17.334	2239	2803	-0.014	0.085	0.086	0.956	0.571
b₁₅	17.565	17.565	2165	3168	-0.009	0.085	0.086	0.953	0.573
b₁₆	17.929	17.929	2101	3515	-0.003	0.084	0.084	0.955	0.578
b₁₇	18.505	18.505	2078	3852	0.000	0.083	0.083	0.955	0.582
b₁₈	19.329	19.329	2103	4188	0.001	0.084	0.084	0.955	0.584
b₁₉	20.492	20.492	2167	4569	0.000	0.083	0.083	0.954	0.581
b₂₀	21.968	21.968	2239	4998	-0.001	0.083	0.083	0.958	0.575
b₂₁	23.598	23.598	2285	5410	-0.003	0.083	0.083	0.958	0.572
Uniform								0.971	0.847
WBATE					0.001	0.034	0.034	0.949	0.193
LBATE					0.085	0.060	0.104	0.996	0.934

Table SA-101: Fuzzy Effects (Simulations, Linear Heteroskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	18.811	18.811	2015	3976	0.001	0.075	0.075	0.962	0.545
b₂	17.441	17.441	1863	3475	0.002	0.078	0.078	0.958	0.572
b₃	15.849	15.849	1652	2894	0.002	0.082	0.082	0.960	0.615
b₄	13.990	13.990	1381	2255	0.002	0.090	0.090	0.954	0.687
b₅	12.000	12.000	1081	1641	0.000	0.103	0.103	0.957	0.798
b₆	10.000	10.000	788	1112	-0.002	0.123	0.123	0.961	0.966
b₇	8.099	8.099	534	703	-0.003	0.154	0.154	0.966	1.279
b₈	7.421	7.421	465	547	0.002	0.169	0.169	0.973	1.476
b₉	7.330	7.330	487	456	0.008	0.179	0.179	0.972	1.959
b₁₀	7.206	7.206	510	349	0.009	0.178	0.178	0.973	6.927
b₁₁	7.385	7.385	572	266	0.004	0.168	0.168	0.984	8383.074
b₁₂	7.278	7.278	504	395	-0.005	0.179	0.179	0.975	37.023
b₁₃	7.402	7.402	469	531	-0.002	0.178	0.178	0.972	1.650
b₁₄	8.414	8.414	574	766	-0.000	0.155	0.155	0.959	4.888
b₁₅	11.200	11.200	962	1419	-0.002	0.120	0.120	0.950	0.897
b₁₆	13.989	13.989	1381	2256	-0.001	0.098	0.098	0.952	0.733
b₁₇	16.557	16.557	1750	3151	-0.001	0.087	0.087	0.950	0.644
b₁₈	18.615	18.615	1998	3908	-0.000	0.085	0.085	0.952	0.603
b₁₉	20.332	20.332	2147	4503	-0.000	0.084	0.084	0.953	0.584
b₂₀	21.947	21.947	2236	4989	-0.001	0.083	0.083	0.958	0.575
b₂₁	23.593	23.593	2284	5408	-0.003	0.083	0.083	0.958	0.572
Uniform								0.978	625.773
WBATE					0.000	0.044	0.044	0.961	401.556
LBATE					0.201	0.125	0.237	0.998	6336.887

Table SA-102: Fuzzy Effects (Simulations, Linear Heteroskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	8.538	8.538	607	820	0.001	0.155	0.155	0.958	0.603
b₂	8.257	8.257	582	778	0.002	0.156	0.156	0.958	0.610
b₃	8.037	8.037	559	741	0.003	0.159	0.159	0.957	0.617
b₄	7.874	7.874	538	709	0.001	0.162	0.162	0.955	0.622
b₅	7.747	7.747	518	679	-0.002	0.163	0.163	0.957	0.629
b₆	7.656	7.656	498	650	-0.004	0.165	0.165	0.960	0.634
b₇	7.576	7.576	477	618	-0.003	0.166	0.166	0.958	0.642
b₈	7.411	7.411	465	546	0.002	0.170	0.170	0.960	0.656
b₉	7.330	7.330	487	456	0.008	0.179	0.179	0.954	0.669
b₁₀	7.206	7.206	510	349	0.009	0.178	0.178	0.957	0.673
b₁₁	7.385	7.385	572	266	0.004	0.168	0.168	0.967	0.632
b₁₂	7.278	7.278	504	395	-0.005	0.179	0.179	0.959	0.689
b₁₃	7.401	7.401	468	530	-0.002	0.178	0.178	0.953	0.679
b₁₄	7.594	7.594	480	625	-0.000	0.175	0.175	0.957	0.666
b₁₅	7.695	7.695	508	665	-0.003	0.174	0.174	0.953	0.663
b₁₆	7.855	7.855	536	705	-0.000	0.172	0.172	0.954	0.661
b₁₇	8.107	8.107	566	752	-0.000	0.169	0.169	0.954	0.659
b₁₈	8.468	8.468	601	809	-0.001	0.173	0.173	0.946	0.657
b₁₉	8.978	8.978	644	883	-0.002	0.168	0.168	0.951	0.648
b₂₀	9.624	9.624	692	970	0.000	0.163	0.163	0.955	0.638
b₂₁	10.338	10.338	734	1056	0.001	0.165	0.165	0.955	0.632
Uniform								0.970	0.989
WBATE					0.000	0.053	0.053	0.947	0.202
LBATE					0.268	0.131	0.298	0.995	1.125

Table SA-103: Fuzzy Effects (Simulations, Linear Heteroskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	18.343	18.343	1960	3818	-0.002	0.086	0.086	0.963	0.569
b ₂	18.051	18.051	1977	3719	-0.002	0.085	0.085	0.963	0.566
b ₃	17.855	17.855	2018	3608	-0.002	0.086	0.086	0.965	0.564
b ₄	17.785	17.785	2090	3485	0.001	0.086	0.086	0.960	0.558
b ₅	17.710	17.710	2166	3313	0.004	0.087	0.087	0.962	0.555
b ₆	17.660	17.660	2243	3107	0.008	0.088	0.088	0.967	0.554
b ₇	17.157	17.157	2217	2745	0.010	0.090	0.091	0.965	0.568
b ₈	16.803	16.803	2205	2415	0.011	0.090	0.091	0.967	0.580
b ₉	16.462	16.462	2176	2085	0.009	0.092	0.093	0.962	0.600
b ₁₀	16.219	16.219	2146	1778	0.005	0.090	0.090	0.966	0.599
b ₁₁	16.492	16.492	2191	1581	0.001	0.085	0.085	0.970	0.594
b ₁₂	16.520	16.520	2200	1947	-0.005	0.092	0.092	0.970	0.607
b ₁₃	17.243	17.243	2302	2481	-0.010	0.094	0.095	0.963	0.591
b ₁₄	17.877	17.877	2349	2999	-0.010	0.095	0.095	0.962	0.572
b ₁₅	18.466	18.466	2354	3483	-0.005	0.096	0.096	0.960	0.567
b ₁₆	18.673	18.673	2263	3788	-0.000	0.095	0.095	0.963	0.577
b ₁₇	18.855	18.855	2164	3992	0.003	0.094	0.094	0.963	0.589
b ₁₈	19.160	19.160	2095	4145	0.004	0.098	0.098	0.966	0.604
b ₁₉	19.705	19.705	2067	4290	0.004	0.100	0.100	0.959	0.615
b ₂₀	20.543	20.543	2067	4467	0.004	0.098	0.099	0.962	0.623
b ₂₁	21.462	21.462	2050	4606	0.003	0.102	0.102	0.965	0.637
Uniform								0.978	0.884
WBATE					0.001	0.038	0.038	0.944	0.200
LBATE					0.109	0.083	0.137	0.998	1.106

Table SA-104: Fuzzy Effects (Simulations, Linear Heteroskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.240	17.240	1807	3381	0.001	0.062	0.062	0.954	0.458
b₂	16.756	16.756	1775	3231	0.000	0.063	0.063	0.954	0.464
b₃	16.586	16.586	1790	3156	0.001	0.064	0.064	0.961	0.462
b₄	16.714	16.714	1873	3125	0.004	0.064	0.065	0.960	0.455
b₅	17.111	17.111	2036	3110	0.012	0.065	0.066	0.957	0.441
b₆	17.598	17.598	2226	3064	0.022	0.062	0.066	0.952	0.425
b₇	18.024	18.024	2395	2954	0.029	0.059	0.066	0.958	0.412
b₈	18.133	18.133	2479	2728	0.029	0.057	0.064	0.954	0.407
b₉	18.233	18.233	2535	2481	0.023	0.056	0.061	0.960	0.405
b₁₀	18.423	18.423	2578	2245	0.014	0.057	0.058	0.960	0.410
b₁₁	20.110	20.110	2855	2358	0.004	0.056	0.056	0.958	0.414
b₁₂	18.393	18.393	2574	2351	-0.007	0.056	0.056	0.957	0.395
b₁₃	18.366	18.366	2537	2735	-0.019	0.054	0.057	0.953	0.377
b₁₄	18.343	18.343	2448	3092	-0.023	0.053	0.058	0.953	0.364
b₁₅	17.668	17.668	2187	3205	-0.016	0.054	0.057	0.959	0.371
b₁₆	16.748	16.748	1880	3135	-0.004	0.055	0.055	0.959	0.387
b₁₇	16.162	16.162	1707	3017	-0.001	0.053	0.053	0.956	0.399
b₁₈	16.176	16.176	1668	2995	-0.001	0.053	0.053	0.956	0.397
b₁₉	16.736	16.736	1694	3114	-0.002	0.052	0.053	0.952	0.388
b₂₀	17.551	17.551	1727	3273	-0.004	0.052	0.052	0.952	0.377
b₂₁	18.491	18.491	1744	3422	-0.007	0.051	0.051	0.955	0.368
Uniform								0.966	0.616
WBATE					0.003	0.026	0.026	0.951	0.146
LBATE					0.064	0.043	0.077	0.994	0.661

Table SA-105: Fuzzy Effects (Simulations, Quadratic Homoskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.072	17.072	1783	3314	0.002	0.062	0.062	0.955	0.461
b₂	16.314	16.314	1705	3064	0.001	0.064	0.064	0.954	0.473
b₃	15.415	15.415	1589	2746	0.002	0.066	0.066	0.959	0.491
b₄	13.947	13.947	1374	2243	0.002	0.071	0.071	0.955	0.537
b₅	12.000	12.000	1080	1642	0.002	0.081	0.081	0.958	0.622
b₆	10.000	10.000	787	1114	0.001	0.096	0.096	0.962	0.752
b₇	8.292	8.292	559	737	0.000	0.117	0.117	0.966	0.937
b₈	7.946	7.946	537	620	0.004	0.123	0.123	0.970	1.013
b₉	7.988	7.988	582	533	0.012	0.123	0.123	0.964	1.032
b₁₀	8.071	8.071	636	435	0.013	0.123	0.124	0.971	1.018
b₁₁	8.811	8.811	788	391	0.005	0.130	0.130	0.968	11.321
b₁₂	8.058	8.058	618	477	-0.004	0.119	0.119	0.965	0.958
b₁₃	8.047	8.047	558	616	-0.001	0.113	0.113	0.969	2.225
b₁₄	8.537	8.537	590	788	0.001	0.103	0.103	0.963	0.797
b₁₅	11.200	11.200	962	1419	-0.001	0.076	0.076	0.955	0.581
b₁₆	13.954	13.954	1376	2245	-0.002	0.060	0.060	0.953	0.458
b₁₇	15.604	15.604	1614	2816	-0.002	0.054	0.054	0.955	0.410
b₁₈	16.095	16.095	1656	2963	-0.002	0.053	0.053	0.957	0.399
b₁₉	16.730	16.730	1693	3111	-0.002	0.053	0.053	0.952	0.388
b₂₀	17.551	17.551	1727	3273	-0.004	0.052	0.052	0.952	0.377
b₂₁	18.491	18.491	1744	3422	-0.007	0.051	0.051	0.955	0.368
Uniform								0.976	1.868
WBATE					0.001	0.032	0.032	0.953	0.752
LBATE					0.137	0.079	0.158	0.997	10.423

Table SA-106: Fuzzy Effects (Simulations, Quadratic Homoskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	7.553	7.553	489	640	0.001	0.137	0.137	0.951	0.530
b₂	7.341	7.341	472	613	0.001	0.142	0.142	0.952	0.539
b₃	7.267	7.267	467	605	0.000	0.138	0.138	0.961	0.538
b₄	7.322	7.322	473	615	-0.001	0.138	0.138	0.955	0.530
b₅	7.496	7.496	489	639	0.000	0.132	0.132	0.958	0.511
b₆	7.710	7.710	505	664	0.001	0.128	0.128	0.955	0.493
b₇	7.896	7.896	516	673	-0.000	0.125	0.125	0.957	0.480
b₈	7.944	7.944	537	620	0.004	0.123	0.123	0.955	0.476
b₉	7.988	7.988	582	533	0.012	0.123	0.123	0.954	0.474
b₁₀	8.071	8.071	636	435	0.013	0.123	0.124	0.955	0.477
b₁₁	8.811	8.811	788	391	0.005	0.130	0.130	0.954	0.495
b₁₂	8.058	8.058	618	477	-0.004	0.119	0.119	0.952	0.460
b₁₃	8.046	8.046	558	616	-0.001	0.113	0.113	0.956	0.437
b₁₄	8.036	8.036	533	702	0.002	0.111	0.111	0.952	0.424
b₁₅	7.741	7.741	515	675	-0.002	0.111	0.111	0.954	0.431
b₁₆	7.337	7.337	476	616	-0.002	0.117	0.117	0.952	0.449
b₁₇	7.081	7.081	445	572	-0.006	0.119	0.120	0.952	0.463
b₁₈	7.087	7.087	438	562	-0.003	0.117	0.117	0.956	0.461
b₁₉	7.332	7.332	451	585	0.001	0.117	0.117	0.951	0.449
b₂₀	7.689	7.689	468	614	-0.001	0.114	0.114	0.950	0.436
b₂₁	8.101	8.101	482	641	-0.003	0.108	0.108	0.957	0.425
Uniform								0.966	0.727
WBATE					0.001	0.039	0.039	0.947	0.151
LBATE					0.186	0.082	0.203	0.995	0.801

Table SA-107: Fuzzy Effects (Simulations, Quadratic Homoskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	16.410	16.410	1693	3079	-0.002	0.066	0.066	0.957	0.482
b₂	16.047	16.047	1671	2986	-0.003	0.068	0.068	0.956	0.487
b₃	15.948	15.948	1694	2949	-0.003	0.069	0.069	0.965	0.486
b₄	16.089	16.089	1775	2937	-0.000	0.071	0.071	0.966	0.482
b₅	16.495	16.495	1926	2946	0.006	0.073	0.073	0.965	0.468
b₆	17.158	17.158	2138	2971	0.015	0.072	0.074	0.957	0.449
b₇	17.721	17.721	2326	2913	0.022	0.068	0.072	0.959	0.433
b₈	18.359	18.359	2512	2837	0.024	0.064	0.069	0.955	0.415
b₉	18.990	18.990	2663	2730	0.020	0.060	0.063	0.960	0.401
b₁₀	19.803	19.803	2812	2645	0.012	0.059	0.060	0.964	0.392
b₁₁	21.462	21.462	3055	2761	0.003	0.060	0.060	0.963	0.400
b₁₂	19.640	19.640	2792	2714	-0.007	0.058	0.058	0.960	0.379
b₁₃	19.005	19.005	2650	2951	-0.017	0.058	0.060	0.956	0.373
b₁₄	18.130	18.130	2396	3075	-0.018	0.060	0.063	0.959	0.380
b₁₅	16.883	16.883	2036	3009	-0.009	0.064	0.064	0.962	0.402
b₁₆	15.616	15.616	1700	2794	0.002	0.063	0.063	0.964	0.426
b₁₇	15.047	15.047	1543	2651	0.003	0.059	0.060	0.960	0.433
b₁₈	15.155	15.155	1525	2644	0.002	0.058	0.058	0.959	0.425
b₁₉	15.676	15.676	1553	2737	0.002	0.058	0.058	0.956	0.413
b₂₀	16.282	16.282	1566	2825	-0.000	0.059	0.059	0.957	0.406
b₂₁	16.774	16.774	1539	2833	-0.003	0.059	0.059	0.960	0.406
Uniform								0.972	0.643
WBATE					0.002	0.028	0.028	0.949	0.149
LBATE					0.068	0.046	0.083	0.996	0.712

Table SA-108: Fuzzy Effects (Simulations, Quadratic Homoskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.687	17.687	1868	3552	-0.001	0.071	0.071	0.957	0.512
b₂	17.077	17.077	1823	3350	-0.001	0.071	0.071	0.955	0.510
b₃	16.696	16.696	1810	3195	0.000	0.070	0.070	0.958	0.504
b₄	16.579	16.579	1851	3081	0.004	0.071	0.071	0.961	0.492
b₅	16.641	16.641	1943	2966	0.011	0.070	0.071	0.964	0.477
b₆	16.776	16.776	2057	2824	0.021	0.068	0.071	0.958	0.461
b₇	16.861	16.861	2154	2632	0.027	0.066	0.071	0.955	0.447
b₈	16.634	16.634	2170	2336	0.028	0.064	0.070	0.955	0.445
b₉	16.538	16.538	2193	2064	0.022	0.062	0.066	0.955	0.444
b₁₀	16.270	16.270	2159	1750	0.013	0.060	0.062	0.963	0.438
b₁₁	16.701	16.701	2235	1582	0.003	0.057	0.057	0.962	0.406
b₁₂	16.495	16.495	2199	1901	-0.009	0.062	0.062	0.962	0.447
b₁₃	16.770	16.770	2209	2319	-0.019	0.063	0.066	0.961	0.442
b₁₄	17.120	17.120	2193	2745	-0.021	0.064	0.067	0.960	0.434
b₁₅	16.936	16.936	2037	2983	-0.012	0.067	0.068	0.955	0.446
b₁₆	16.613	16.613	1856	3091	-0.000	0.067	0.067	0.962	0.462
b₁₇	16.555	16.555	1767	3158	0.003	0.067	0.067	0.953	0.477
b₁₈	16.909	16.909	1769	3267	0.001	0.066	0.066	0.954	0.479
b₁₉	17.755	17.755	1827	3495	-0.002	0.066	0.066	0.952	0.472
b₂₀	18.817	18.817	1881	3748	-0.005	0.067	0.067	0.945	0.466
b₂₁	20.034	20.034	1917	3997	-0.008	0.066	0.067	0.950	0.460
Uniform								0.968	0.700
WBATE					0.003	0.028	0.028	0.953	0.156
LBATE					0.074	0.048	0.088	0.995	0.745

Table SA-109: Fuzzy Effects (Simulations, Quadratic Heteroskedastic, Smooth-Boundary Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	17.460	17.460	1835	3461	-0.000	0.072	0.072	0.958	0.517
b₂	16.507	16.507	1732	3136	0.000	0.072	0.072	0.954	0.524
b₃	15.414	15.414	1590	2748	0.001	0.073	0.073	0.954	0.540
b₄	13.916	13.916	1371	2234	0.002	0.078	0.078	0.956	0.578
b₅	11.999	11.999	1082	1642	0.002	0.086	0.086	0.961	0.653
b₆	10.000	10.000	788	1113	0.002	0.099	0.099	0.958	0.776
b₇	8.061	8.061	529	697	0.001	0.123	0.123	0.962	0.985
b₈	7.300	7.300	450	532	0.002	0.136	0.136	0.971	1.285
b₉	7.245	7.245	476	447	0.010	0.136	0.137	0.972	1.288
b₁₀	7.128	7.128	500	343	0.010	0.133	0.134	0.970	2.205
b₁₁	7.317	7.317	563	261	0.002	0.124	0.124	0.981	10.738
b₁₂	7.227	7.227	497	390	-0.006	0.135	0.135	0.966	96.646
b₁₃	7.348	7.348	461	524	-0.004	0.131	0.131	0.968	1.058
b₁₄	8.409	8.409	573	765	-0.001	0.115	0.115	0.958	0.898
b₁₅	11.200	11.200	962	1418	0.000	0.089	0.089	0.950	0.666
b₁₆	13.919	13.919	1371	2234	0.002	0.074	0.074	0.955	0.543
b₁₇	15.793	15.793	1640	2883	0.003	0.068	0.069	0.951	0.495
b₁₈	16.766	16.766	1748	3211	0.001	0.067	0.067	0.955	0.482
b₁₉	17.739	17.739	1825	3488	-0.002	0.066	0.066	0.952	0.472
b₂₀	18.816	18.816	1881	3747	-0.005	0.067	0.067	0.945	0.466
b₂₁	20.034	20.034	1917	3997	-0.008	0.066	0.067	0.950	0.460
Uniform								0.974	8.886
WBATE					0.001	0.034	0.034	0.964	5.288
LBATE					0.155	0.089	0.179	0.996	83.651

Table SA-110: Fuzzy Effects (Simulations, Quadratic Heteroskedastic, Kink-Adaptive Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b₁	7.749	7.749	513	675	-0.002	0.151	0.151	0.955	0.587
b₂	7.482	7.482	490	638	0.001	0.151	0.151	0.954	0.587
b₃	7.315	7.315	474	614	0.002	0.147	0.147	0.957	0.581
b₄	7.264	7.264	467	604	-0.000	0.148	0.148	0.954	0.568
b₅	7.290	7.290	465	602	-0.000	0.144	0.144	0.955	0.550
b₆	7.350	7.350	463	600	0.001	0.138	0.138	0.957	0.534
b₇	7.387	7.387	455	589	-0.000	0.138	0.138	0.953	0.525
b₈	7.287	7.287	449	530	0.002	0.137	0.137	0.959	0.523
b₉	7.245	7.245	476	447	0.010	0.136	0.137	0.955	0.519
b₁₀	7.128	7.128	500	343	0.010	0.133	0.134	0.958	0.513
b₁₁	7.317	7.317	563	261	0.002	0.124	0.124	0.961	0.473
b₁₂	7.227	7.227	497	390	-0.006	0.135	0.135	0.953	0.522
b₁₃	7.347	7.347	461	523	-0.004	0.131	0.131	0.959	0.512
b₁₄	7.501	7.501	469	610	-0.000	0.130	0.130	0.956	0.505
b₁₅	7.420	7.420	477	619	-0.001	0.136	0.136	0.950	0.513
b₁₆	7.278	7.278	468	606	0.005	0.136	0.136	0.955	0.527
b₁₇	7.253	7.253	465	601	0.007	0.142	0.142	0.947	0.542
b₁₈	7.408	7.408	475	617	0.001	0.139	0.139	0.949	0.542
b₁₉	7.779	7.779	502	661	-0.003	0.135	0.135	0.954	0.531
b₂₀	8.244	8.244	530	710	-0.000	0.137	0.137	0.948	0.524
b₂₁	8.777	8.777	556	759	-0.000	0.133	0.133	0.952	0.515
Uniform								0.964	0.817
WBATE					0.001	0.041	0.042	0.954	0.162
LBATE					0.211	0.091	0.230	0.995	0.895

Table SA-111: Fuzzy Effects (Simulations, Quadratic Heteroskedastic, Unknown-Kink Bandwidth, Fuzzy Bandwidth Selector).

	h_0	h_1	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
b ₁	16.600	16.600	1719	3154	-0.004	0.080	0.080	0.960	0.547
b ₂	16.166	16.166	1690	3035	-0.004	0.079	0.079	0.959	0.543
b ₃	15.882	15.882	1686	2931	-0.004	0.079	0.079	0.963	0.536
b ₄	15.852	15.852	1738	2864	-0.001	0.080	0.080	0.964	0.526
b ₅	16.025	16.025	1840	2807	0.005	0.081	0.081	0.969	0.510
b ₆	16.350	16.350	1980	2738	0.013	0.079	0.080	0.963	0.488
b ₇	16.274	16.274	2037	2515	0.019	0.079	0.081	0.956	0.480
b ₈	16.244	16.244	2091	2279	0.021	0.076	0.079	0.958	0.472
b ₉	16.123	16.123	2107	2009	0.018	0.073	0.075	0.963	0.470
b ₁₀	15.915	15.915	2087	1714	0.010	0.069	0.070	0.968	0.462
b ₁₁	16.264	16.264	2148	1536	0.001	0.065	0.065	0.969	0.458
b ₁₂	16.237	16.237	2145	1881	-0.009	0.070	0.071	0.969	0.467
b ₁₃	16.776	16.776	2209	2356	-0.016	0.072	0.073	0.963	0.452
b ₁₄	16.977	16.977	2166	2747	-0.014	0.074	0.076	0.964	0.452
b ₁₅	16.647	16.647	1996	2940	-0.004	0.079	0.079	0.964	0.471
b ₁₆	15.901	15.901	1757	2891	0.007	0.078	0.078	0.967	0.497
b ₁₇	15.588	15.588	1630	2851	0.010	0.077	0.077	0.953	0.516
b ₁₈	15.786	15.786	1612	2884	0.007	0.075	0.075	0.954	0.516
b ₁₉	16.349	16.349	1641	2994	0.003	0.076	0.076	0.955	0.512
b ₂₀	17.026	17.026	1658	3109	0.002	0.078	0.078	0.949	0.515
b ₂₁	17.757	17.757	1651	3197	0.001	0.079	0.079	0.956	0.520
Uniform								0.976	0.752
WBATE					0.003	0.030	0.030	0.948	0.164
LBATE					0.087	0.060	0.106	0.996	0.879

Table SA-112: Fuzzy Effects (Simulations, Quadratic Heteroskedastic, **rdrobust** Bandwidth, Fuzzy Bandwidth Selector).

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